Style Manual: A Springer Nature Usage Guide

Version 9 (4 December 2020)

Audience: Public, Springer Nature, Vendors

Contents

Preface

This manual has been prepared specifically for people who edit manuscripts to be published by Springer Nature. It contains the following parts:

Part I: General Guidelines. These guidelines concern editing at Springer Nature, both in general and specifically regarding journals and books.

Part II: Language and Style. This presents the specific requirements for language editing and formal style or mark-up.

Part III: Science. Science-specific editing guidelines and information.

Much of the information in this style manual is thus valid for the editing of all publications, while some of it applies to particular forms or fields of publications.

Of particular importance are the guidelines on editing in general and specifically for journals and books that are given in Part I. The first chapter is concerned with the general guidelines for editing. These are valid in principle for all types of publications (books, journals, and others). The other two chapters in the first part deal specifically with the application of this information to journals and to books and with other details specific to those forms of publications. For example, one of the major differences between books and journals is the fact that a book chapter is more a part of a larger whole than a journal article is. Since this manual is booklike, i.e., consists of parts and chapters, it is styled like a Springer Nature book.

The information in Part II applies to all subjects and all types of publications that are copy edited, in accordance with the general guidelines set down in Part I. This general validity is due to the fact that there is a fundamental similarity between a book chapter and a journal article, which form the basic units of editing.

Part III refers to information within specific disciplines.

The two immediate sources of the information collected in this manual are (1) the practice of editing as it has been conducted at Springer Nature and (2) the changes resulting from the radical technical advances taking place in publishing. Another important source is the accepted practice of writing and editing as they are conducted in the sciences, social sciences, and humanities. This knowledge has been collected in numerous excellent reviews of good editorial practice. We would like to acknowledge the most important of these sources, to which this manual owes a great deal:

– The Chicago Manual of Style

– Merriam–Webster's *Collegiate Dictionary*

– American Medical Association Manual of Style

– The ACS Style Guide

– Medical Style and Format

– ASM Style Manual

– The CBE Manual for Authors, Editors, and Publishers

Our thanks go to each of the copy editors, who have made significant contributions to both the general and the science-specific chapters, and to the many production editors who have given advice. Thanks also go to those in the editorial offices and at the other Springer Nature branches who have provided valuable comments on this style manual. We look forward to receiving further suggestions for improving this manual.

This style manual does not pretend to be complete or neutral. It is intended to describe the manner in which Springer Nature publications are prepared, without covering the procedural aspects involved, or any specifics regarding the final layout of the publications.

Part I: General Guidelines

1 General Information

1.1 What Is the Task of Copy Editing at Springer Nature?

The copy editor's task is to prepare a manuscript for publication, whether it is to be in print or online. This can consist of any of several different tasks, including:

– Language editing: this ranges from eliminating spelling and typing mistakes to correcting style or introducing a particular didactic presentation

– Formal mark-up: this concerns the presentation of the standard components of a manuscript, such as equations, tables, and figures, according to Springer Nature and general publishing standards

– Correction of formal aspects of the content according to usage in the field: examples range from the correct use of abbreviations and terminology to the verification of statistics

– Data mark-up, such as tagging, reference structuring, or correction of TeX coding.

The relative importance of each of these tasks with regard to an individual publication depends on the desired type of publication, the target group of the publication, and the planned form of production.

1.2 Publication Types and Target Groups

The publication type and the target group can each have a significant impact on the tasks assigned to the copy editor. Although the publication of all content on the Internet made some of the differences between types of publication disappear, there are still important differences between a book-style publication and a journal. Either of these types of publication, however, can be directed more at researchers and be very scientific in nature or more at professionals seeking easy access to information and news.

Scientific Publications For many scientific publications, rapid publication is the most important need of authors and readers, while for others, such as review publications, the quality of the content takes precedence. In either case, language style and the didactic nature of the presentation may be secondary, not the least because any reader of the publication will be generally familiar with the subject matter. For the copy editor this means that language changes can be limited to the correction of real mistakes. Formal aspects, such as the correct use and presentation of abbreviations and terminology, may, in contrast, be very important. This range of requirements means that it is important that the copy editor know and follow the general instructions given in this manual, especially as described in this chapter (see particularly Sect. 1.3) and in Chaps. 2 and 3, and any specific instructions given separately for a specific publication.

Professional Publications Many publications are directed at scientists or academically trained professionals who work, for example, as physicians, engineers, pharmacists, chemists, or technicians in industry or public or private organizations. Generally, it is important that the information be presented clearly, in terms of language, organization, and typography, to enable the reader to quickly find and easily comprehend a specific element of content. The copy editor may be asked to intervene and ensure that certain instructions or guidelines are applied throughout a manuscript. Textbooks are a special type of professional publication.

Academic Publications A third distinct group of publications is addressed at students and researchers working in the humanities and social sciences; disciplines include economics, sociology, psychology, philosophy, history, linguistics, and literary/media studies. The subject matter requires both language and context to be handled sensitively by the copy editor, particularly in such areas as philosophy where the author's style is key to the content (see particularly Chap. 17).

1.3 Categories of Editing

At Springer Nature, there are three levels or categories of copy editing: formal mark-up, standard editing, and intensive language editing. The level at which manuscripts should be edited depends on a number of factors, the most important ones being the quality of the original manuscript, the nature and desired quality of the publication, and market and business factors (this last factor is important since even a high-quality publication is "worthless" if it is so expensive that no one buys and reads it). For the copy editor this means: Please pay attention to your instructions. If these instructions do not seem applicable to the journal or to the book manuscript that you are to edit, consult your Springer Nature contact immediately. Do not simply edit as you see fit!

Do, however, use common sense in applying your instructions to specific manuscripts. For example, the nationality of the author or, more specifically, the general quality of the writing is an important guide to the extent of editing required (of course, *within the context of the level of editing requested*). Assume that a native speaker (or any other author of an apparently well-prepared text) has used the form of English they believe correct and acceptable. In other words, if a manuscript is generally well written, be more hesitant to make any language changes. The opposite is also true: in a poorly written manuscript you may feel free to make more changes than would otherwise be the case.

1.3.1 Category 1: Formal Mark-Up

This category of editing is generally described as formal mark-up without any language editing. The aim is to correct and mark-up the text components for further processing. Specifically it means:

1. Ensure that the manuscript is complete and put the material in the correct sequence.
2. Header:

   – Put the header material in the correct sequence.

   – Style and mark up names and addresses.

   – Mark up abstract and keywords (if present)

3. Text:

   – *Headings:* Mark up headings. Use the proper capitalization.

  – *Spaces, dashes, quotation marks:* Use the proper types of spaces (see, e.g., Sect. 6.2.1 on usage with units), dashes, and quotation marks.

  – *Numbers and Units:* Style numbers. Use standard abbreviations for units where appropriate (cf Sect.6.2.1).

  – *Spelling:* Use a standard spell checker. (It is not necessary to read the text, but correct any typos that are highlighted by the spell checker.)

4. Figures and tables:

  – Place figures and tables.

  – Style and mark up figure legends.

  – Style and mark up tables and table legends.

5. Equations: Process and mark up equations
6. References:

  – Check journal abbreviations.

  – Style and mark up references.

  – Check if all the references cited in the text have been included in the reference list and vice versa.

See additional instructions in Chaps. 2, 3, 7, 8, and 10, Sects. 6.2.1, 9.5.1, and 9.5.2.1.

1.3.2 Category 2: Standard Editing

For standard copy editing, the copy editor must have a good command of English; for a non-native speaker, this is usually demonstrated by a degree in English or at least by four years of higher education at an institution with English as the primary language. The aim of this level is to correct the author's text in terms of grammar, spelling, as well as scientific and house style. Specific tasks in addition to the ones for category 1 editing:

1. Spelling:

– Standardize to American or British English within an article or chapter (cf. A.2 Appendix for the Nature Portfolio journals).

  – Check for consistent spelling and hyphenation (use a subject-specific dictionary for names and terms)

2. Grammar and punctuation:

– Correct grammatical errors (especially if the author is not a native speaker of English) such as wrong prepositions, definite and indefinite articles, verb tenses, subject–verb agreement

– Check/correct the use of commas and other punctuation marks

3. Wording and linguistic style:

   – Correct misused words or incorrect literal translations from foreign languages

   – Correct “poor style” only if it hinders comprehension

4. Scientific and Springer Nature style: Edit scientific content in accordance with the instructions in Chap. 6, e.g., check/correct capitalization, abbreviations, numbers, special type (italicization, etc.)

5. Figures:

   – Check that the legend matches the figure

   – Check that abbreviations and symbols are identified in the legend

In addition to those listed for copy editing level 1, please refer to Chaps. 5, 6, and 9, Sects. 4.1, 4.3, “and 4.4.

1.3.3 Category 3: Language Editing

At the level of intensive language editing, the copy editor should edit as described in this manual. The copy editor must be a native speaker of English with knowledge of the subject. The goal at this level is language refinement to create a well-written text for a specific target group. (In rare cases, it may be to create a publishable text out of a poorly written manuscript.) This means in addition to the tasks listed for categories 1 and 2:

1. Correct the items mentioned in Sect. 4.2 if they affect the *readability* of the text.
2. Correct mistakes and inconsistencies in scientific style (see Part III). Apply scientific standards of the discipline and/or specific style requirements of the book or journal.

1.4 Other Tasks Related to Copy Editing

1.4.1 Contacting the Author or Springer Nature About Problems

When questions arise during editing, they must be clarified before the final files are submitted to Springer Nature for publication. In most cases, e.g., missing references or confirmation of text corrections, it is sufficient if the author receives the queries with the proofs. If the author's usage leaves doubt about whether to use American or British English and if there are no special instructions for a particular journal or book, clarify this as early as possible with the Springer Nature contact. If critical material is missing, e.g., figures or tables, or if the structure of the manuscript should be changed, the questions must be clarified before the manuscript goes to typesetting. The following categories of query must always be clarified with your Springer Nature contact before returning the manuscript:

1. Queries relating to authorship, especially consortia.
2. Missing information required by a particular publication (e.g., statements on competing interests, peer review statements)
3. Content that does not match the standard requirements of a publication (e.g., inappropriate number of figures or references for a specific content type).
4. Missing figures or tables.
5. Queries to titles or abstracts
6. Changing the structure of the manuscript (e.g., hierarchy of section heading levels or order of sections)

> In addition, clarify procedure with your Springer Nature contact if you encounter style queries not covered in this manual, or have questions about policy implementation.

All questions and remarks to the author should be concise, clear, and polite. The vast majority of questions should take a yes or no answer and include the word "please"! When you make extensive changes or are not certain that your change is correct, ask for confirmation. If you cannot understand a piece of text, ask for an explanation. However, a vaguely expressed request, such as "please clarify," often leads to an equally vague response; pinpoint the problem.

1.4.2 Technical Editing

Technical editing (pre-editing) covers the standardization of author data (data normalization), the processing of equations in MathType and LaTeX, and data structuring (tagging). It usually is the first step in the copy editing process and is always carried out by Springer Nature's full-service vendors (FSV). Copy editors usually handle files where technical and formal mark up has already been performed, and therefore should familiarize themselves with the tagging system so that they are able to carry out corrections if needed.

1.4.3 Cost, Quality, and Deadlines

It is important to find the correct mix of requirements regarding costs, quality, and speed of publication. Each of these sometimes conflicting requirements may be particularly important; an incorrect decision or the failure to follow instructions may adversely affect the value of a publication. The Springer Nature editors are responsible for setting and communicating this information. If for any reason you think that you cannot meet a deadline or that some other part of the agreed terms or instructions is inappropriate, inform your Springer Nature contact at once.

2 Journals and Articles

2.1 Organization of a Journal

A journal consists of front matter (referred to here as A pages), scientific articles (forming the primary content), and other types of documents. Traditionally, a journal differs from a book in that:

– It is sold on a subscription basis.

– Its editors are responsible for publications for a number of years or volumes.

– Each article is customarily a separate or autonomous entity, unrelated to others in a publication.

– Each article is subject to peer review, which as a rule is decisive in determining whether an article will be published or not.

A Pages The A pages of a journal consist of various elements that recur, often unchanged, from issue to issue. These elements include the cover, the table of contents, a description of the aims and scope of the journal, or advertisements. Since most of this information does not change from issue to issue or from volume to volume, it is not the object of regular copy editing. The presentation of this material in the publication is determined almost exclusively by the style rules that apply to that particular journal.

Articles The primary content of scientific and professional journals consists of articles, regardless of whether they are reports of original research (usually published after being reviewed by referees), reviews of the literature, short reports, or of another nature. These documents are often subject to copy editing; see Sect. 2.2.

Other Types of Content. In many journals, a portion of each issue is made up of miscellaneous documents that are not articles as defined above. These documents include such items as book reviews, news reports, professional or society announcements, schedules of events within a discipline, and information about advanced training; some journals also contain advertising. Not all of these are copy edited; where they are, edit according to the copy editing category indicated.

2.2 Articles

An article is made up of three main components: the article front matter, the body of the article, and the article back matter.

2.2.1 Front Matter

The front matter of an article includes numerous elements of information supplied by the author (or by the editor). Not all elements are included in all journals.

– *Category* This is the name of a part of an issue as defined by the editors of a specific journal (German, *Rubrik*). In some journals there may also be a subcategory/subrubric. Entries should always have an initial uppercase letter (i.e., follow the rules for capitalizing book headings). Examples include entries such as Original, Original Articles, Research Articles, Tips and Tricks in Radiology, Experimental Brief Reports, Technical Note, and Status Report.

– *Article title* Some journals require that this be present in more than one language. The languages that are possible are English, German, French, and Spanish. Content is captured and rendered in this sequence. A title should generally be only one sentence in length. However, if the sentence is followed by a fragment (for example, 'Does smoking cause cancer? A study'), this is usually acceptable, but cf. A.2 Appendix for the Nature Portfolio journals. The title should not end in a full stop. If the title is changed, please consult your Springer Nature contact.

– *Article subtitle* Some journals require that this be present in more than one language. The languages that are possible are English, German, French, and Spanish. Content is captured and rendered in this sequence. Subtitles may only be included in further languages if the title also appears in those languages.

– *Author names* Author names must appear on each manuscript [i.e., an author entry is required, whether individual or institutional (i.e., group) author]. The names of an individual are presented with the given name(s) first, then the surname. A given name should be left as submitted by the author, i.e., written in full or represented by initials (an initial consisting of one letter only); do not shorten an author's full given name to an initial unless specifically instructed. Academic or professional designations are not included with an author's name; there are very few exceptions to this rule for specific journals. Never use a dagger (†) to indicate that an author is dead; include the information in an ordinary footnote. If it is unclear whether a name is part of the surname or not, add a query.

– *Corresponding author*. One corresponding author is required, but more are allowed.

– *Author contact information* An e-mail address and possibly url for the purposes of contacting the author can be retained if contained in the manuscript. The e-mail of the corresponding author is usually published; for other authors, it is optional and should be retained only if given in the manuscript. NB. Private email addresses should not be included, unless specifically requested by the author. Any other information such as fax and telephone numbers as well as text and supplementary numbers (e.g., "ext. -4576) should be removed entirely.

– *Author address/affiliation* This entry may be in the language either of the country of the address or of the publication, except for the names of the city and country, which should be in the language of the publication. An entry for city and country should always be present (for affiliations that do not have a geographic location, a URL may also be used). The name of the country is not abbreviated, except for United States (USA) and United Kingdom (UK), which should always be abbreviated. Otherwise use the form of the country designation as submitted by the author (e.g., either China or People's Republic of China).

The order of presentation of information in street addresses should be that used by the country in the address (but cf. A.2 Appendix for the Nature Portfolio journals). For more information, see the web site of the (Universal Postal Union).

Within the entry for the affiliation, abbreviations may be retained where used correctly.

– *Additional addresses* The only designation permitted is "Present Address," and it should only be used when given in the manuscript. "Current Address" should be changed to "Present Address”. All other designations, such as "Contact Address," should be deleted and the address treated as a standard address. An author query should be added to flag these changes.

– *Footnotes* Information supplied in the form of footnotes in the front matter are captured as article notes in the XML. Examples include comments about prior presentation of the content (cf. A.2 Appendix for the Nature Portfolio journals), deceased authors, or equal contributions (equally contributing or jointly supervising). Check that the contents of such comments do not actually belong in specific article backmatter section, such as funding notes, detailed author contributions or ethical statements. If so, move them to the appropriate place.

– *History* The only possible entries are "Received," "Revised," and "Accepted," but there are some journals in which only one or two of these entries (or none at all) are used. Although a fourth entry ("Published Online") is published, it is added automatically shortly before publication and is thus not part of the manuscript. Another, unpublished entry is Registration Date.

– *Electronic Supplementary Material* A generic reference to the fact that ESM is part of a specific article. NB the specific layout dictates whether this occurs in the article front or back matter. The text of the reference is usually:

Supplementary Information The online version contains supplementary material available at (https://doi.org/[DOI]).

Do not forget to insert the DOI of the article (correctly!). (On usage of DOIs, see also Sect. 10.3.)

– *Abstract* The heading is always "Abstract" (not "Summary").

1. In specific journals, an abstract may (or should) be structured; structured abstracts are not permitted in other journals, and in these journals such headings should be deleted. A structured abstract has subheadings below the main "Abstract" heading, such as "Material and methods," "Results," and "Conclusion." It is not the task of the copy editor to add the headings to create structured abstracts.
2. An abstract should not be misused by putting other information in it (for example, do not include the article title in it).
3. An abstract in a second language must appear as a separate abstract and not simply as a continuation of the first. The designations of abstracts are: in German "Zusammenfassung," in French "Résumé," and in Spanish "Resumen." They should be captured and rendered in this sequence.
4. Since abstracts are often read independently of the actual article and without access to the reference list, any references that are cited must be given in adequate form. See Sect. 10.2 for instructions.

Internal and external links should be supported unless specified otherwise (but cf. A.2 Appendix for the Nature Portfolio journals).

5. If the article concerns a clinical trial, the trial registration number should be included at the end of the abstract. At final version this should be replaced with a link to the trial registration.
   The trial registry should be listed, along with the unique identifying number (e.g. Trial registration: ISRCTN73824458). Please note that there should be no space between the letters and numbers of the trial registration number.
6. Abbreviations should be spelled out in full. Only use acronyms or abbreviations in the abstract if they are more common than the full version.

– *Graphic abstract* In some journals a figure depicting, for example, a chemical structure may be included.

– *Keywords* (cf. A.2 Appendix for the Nature Portfolio journals) The heading is always "Keywords" (not "Key words"). A keyword (or the first word in a keyword phrase) is capitalized.

A list of keywords in a second language must appear as a separate set of keywords and not simply as a continuation of the first. The designations used in place of keywords in French is "Mots clés" and in German "Schlüsselwörter." They too are listed in the sequence English, German, and then French. If Spanish were included, it would appear last.

– *Classification code* In some disciplines, standardized classification systems have been developed. Examples are MSC (mathematics; see https://www.zbmath.org/classification/) and JEL (economics; see http://www.aeaweb.org/journal/jel_class_system.html). They are used in specific journals instead of or in addition to keywords. Always use the following headings: "Mathematics Subject Classification", and "JEL Classification".

– *Abbreviations* The abbreviation and definition should be shown as they appear in the text, with the exception that the initial letter of the definition is capitalized.

Springer Nature also supplies some information that traditionally is closely associated with the front matter. This metadata cannot by changed by the copy editor. It includes:

– Digital object identifier (DOI) of the article.

– *Article type*. This is the type of article as defined according to a list established by Springer Nature and valid for all its journals, irrespective of the category designation. The article types that can be used to describe a standard article, depending on length and content, are:

– Original paper. An article reporting results or work (research, case studies, theoretical work, etc.).

– Review paper. An article reviewing results reported by others.

– Brief communication. Some articles are unusually short (say, three manuscript pages); the article type makes it clear that the brevity is probably for some formal reason, such as to meet the requirements for accelerated refereeing or rapid publication without refereeing. Otherwise there is no difference from an original paper.

The other article types describe content that differs from the structure of a standard article as described in Sect. 2.2. For the other article types that can occur in a journal, see Sect. 2.3. The article type "miscellaneous" is not to be used for any article, ever! It is reserved for certain categories of non-edited material.

– *Product type*. This is the type of publication as defined according to a list established by Springer Nature. The three types that apply to journals are "archive journal" (a journal directed primarily at libraries and scientists and containing predominantly the article types mentioned in the previous item), "nonstandard archive journal" (an archive journal with special layout and/or copy editing instructions) and "magazine" (a journal directed at professionals and characterized by a more varied layout, more advertising, and a larger proportion of nonarticle content).

– *Subject*. The primary discipline of the journal according to Springer Nature's list of subject categories.

2.2.2 Body

2.2.2.1 *Headings and Sections*

In journals published by Springer Nature, one of two systems of displayed headings is used to indicate the hierarchy of sections of text: (a) the decimal system or (b) keying, an unnumbered system relying on typographical differentiation to indicate the level of the heading. The system used is always the same for all the articles in a journal. In both systems, normally no more than three levels of displayed headings are employed; run-in headings may, of course, be used at any level, as appropriate.

Very commonly, standard journal articles consist of four sections, namely an introduction, the materials and methods, the results, and the discussion. They are often followed by a conclusion. This structure frequently exists even if the wording of the headings of these sections differs somewhat.

Styling of Headings In journal articles (in contrast to headings in books), there are two standard systems for capitalizing article titles and headings. Either only the initial word and any proper nouns in a heading are capitalized, or all the major words are capitalized (for details on this, see for books Sect. 3.3.1.1).

The following items cover specific questions regarding the styling of headings.

– *Abbreviations* Although abbreviations that have been defined in the text may be used in headings, it is just as acceptable to use the written-out forms. Follow the author's usage, but aim to ensure that either written-out forms or abbreviations are used in headings throughout (although heading length may determine whether an abbreviation is used). A displayed heading normally does not consist solely of an abbreviation, exceptions being abbreviations that are very well known in a discipline.

– *Styling of Latin Phrases* Style all words of commonly used Latin phrases—i.e., those found in the main text or foreign words appendix of *Merriam–Webster's New Collegiate Dictionary*—as ordinary English words (e.g., Review of in vivo studies). Latin phrases not in *Webster's* are italicized.

– Avoid the use of *suspended hyphens*; repeat the term, as in "Success of treatment with β_2-agonists and β_2-antagonists."

– In the "lowercase" system of capitalizing headings, do not capitalize following a colon or em dash.

Decimal System In the decimal system, the first number (from the left) is the number of the respective first-level heading (see Table 2.1). The heading of the first subsection in an article is thus 1.1, that of the second 1.2, etc. Under, for example, the second primary heading, the first secondary heading would be 2.1, the second 2.2, etc. The final number in displayed headings is not followed by a period.

Keying Keying is the most frequently used and most flexible system of headings. In this system, levels of headings are differentiated in the final layout purely by typographical means. The copy editor must ensure that the appropriate level of heading is assigned. The example shown in Table 2.1 indicates how headings should be keyed in the text. Note that a level may not be skipped.

Table 2.1 The two primary systems of labeling headings

Decimal headings	Keyed headings	
	Level of heading	Heading
1 Introduction	1	Introduction
2 Materials and methods	1	Materials and methods
2.1 Model description	2	Model description
2.2 Experiments	2	Experiments
3 Results	1	Results
4 Discussion	1	Discussion
References	1	References

2.2.2.2 *Other Numbered Elements*

Within an article, certain elements are numbered. Numbering is mandatory for tables, figures, schemes, and structures; it is optional for equations or layout elements such as boxes. NB Inline figures are never numbered. See Chaps. 7, 8, and 9 for a description of these elements.

2.2.3 Back Matter

The elements of back matter in an article are listed below. Not all elements are included in all journals, but very few articles have no reference list of some sort.

- Acknowledgements
- Author Contribution
- Peer Review Information
- Funding Information
- Data, Code, and Materials Availability
- Declarations (Ethics)
- Abbreviations
- Glossary

- Appendix
- References or bibliography

2.2.3.1 *Acknowledgements*

Acknowledgements of assistance in preparation of the publication are included at the end of the article. Only general acknowledgements are included directly under this heading. Specific acknowledgements about the authors' contributions, fundings, or ethics should be given in the specific sections below (Sects. 2.2.3.2. to 2.2.3.4).

2.2.3.2 *Author Contributions*

Information on the authors themselves and their individual contributions is included in Author Contributions. All authors should be named individually using initials only with their tasks detailed, or else a general statement that all authors have reviewed the text is acceptable.

2.2.3.3 *Peer Review Information*

Some journals include a statement thanking the reviewers of the article for their peer review work.

2.2.3.4 *Funding Information*

The funding information section is used for text statements provided by the authors about any funding or grants received by authors for the preparation of the article (but cf. A.2 Appendix for the Nature Portfolio journals). Example:

CP was funded by MBI Release Early Career Researcher Award DE 54321 and Discovery Grant OP 987654. RP was funded by DCNVW and Univericon.

2.2.3.5 *Data, Materials and Code Availability Statements*

Certain journals require data, code, and/or materials availability statements. These appear as separate sections and any or none may be present. If present, the order is always data, materials, code. The headings are always 'Data availability', 'Materials availability', and 'Code availability'. A specific wording may be required.

2.2.3.6 *Declarations (Ethics)*

The ethics section is used for ethical statements relevant to the article. The heading is "Declarations"

Statements included in this section if applicable are (check with your Springer Nature contact if unsure if this applies to a specific journal):

Competing interests/Conflict of Interest Many journals request authors to disclose all information about financial support, i.e., grants and contracts with the organization(s) that sponsored the research. The heading is always "Conflict of interest" or "Competing interests". Where an author gives no competing interests, the listing must read 'The author(s) declare no competing interests'.

Ethics Approval and Consent to participate. A statement of ethical or institutional review board approval should be present if the manuscript is researching on humans. This does not include any retrospective studies, or studies that have used already published data, however if patient activity is required for the research then it should have been approved by a relevant ethical or review board approval committee or be in accordance with the Declaration of Helsinki. It does not matter which committee provides this approval, however it must be present in the manuscript. This includes questionnaires.
When research has actively used animals, then a similar statement should be present to show that the care of animals followed internationally recognised guidelines.

Consent for publication. Informed consent from patients is a mandatory requirement individual data are published (e.g., age, gender, ethnicity, hair colour, medical conditions, medical history). All case studies/reports are likely to require a statement of consent, however other manuscript types may also require it if this data is provided for specific patients.

2.2.3.7 *Appendixes*

Appendixes provide information supplementary to an article. If there is only one, it is designated "Appendix"; if there are more than one, they are designated "Appendix 1," "Appendix 2," etc.

Appendixes should be referred to in the text. The content of an appendix is contained within the sections subordinate to the major heading "Appendix." The language and styling rules for the text also apply to appendixes. The form of numbering of tables, figures, and equations in an appendix should be the same as in the body of the article, continuing the numbering used there.

References to additional files that will appear as supplementary information (electronic supplementary material) are captured within an appendix. Captions must be present for each additional file with a description of what the file contains.

2.2.3.8 *Glossary*

A glossary may be included in an article. Terms in the glossary are arranged either alphabetically or in order of occurrence in the text, each on a separate line and followed by its definition. As a rule, a glossary is part of the article back matter.

2.2.3.9 *References, Further Reading, Bibliography*

The reference list comes at the end of the article. The correct heading is "References" For information on the correct styling of reference citations and the reference list, see Chap. 10.

It is the copy editor's task to check that there are no obvious typing mistakes in the titles of works cited. Correct any evident mistakes, but do not alter spelling, etc., merely to achieve consistency with the style in the manuscript you are editing. Italics, small capitals, and boldface do not have to be added in accordance with the guidelines for editing the text (e.g., do not mark a vector to be set in bold).

In some articles, the reference list may be divided into sections. In addition to the list of references cited in the text (heading: "Cited literature"), a list of further or suggested reading may be included (heading: "Further Reading").

Occasionally, instead of a reference list, there is a bibliography, i.e., a list of publications, all or some of which are not cited in the text. Check with your Springer Nature contact if you are uncertain which heading is appropriate or permissible.

2.3 Other Article Types

Documents other than standard articles that can appear in a journal include editorial notes, book review, abstract, continuing education, interview, letter, and erratum. The general instructions on editing that apply to articles also apply to these documents if they require editing.

2.4 Copy Editing Category

Journals are also assigned to one of the three categories of editing. It is important that editing be done in accordance with these guidelines to ensure that the desired quality is achieved and that unnecessary costs are avoided.

3 Books and Chapters

3.1 Organization of a Book

The information in this section pertains to the organization of a book as a whole, concentrating on the front and back matter. More detailed information about the organization of the main text and individual chapters is given in Sect. 2.2 and on figures, tables, footnotes, etc., in Chaps. 7–10.

A book consists of front matter, the main text, and back matter. In a published book, the front matter is usually paginated with Roman numerals and the main text and back matter with Arabic numerals. The various systems used for structuring a book are covered in Sect. 3.3.

3.2 Front Matter

The front matter in Springer Nature books consists of the items listed in Table 3.1. As indicated there, some of the elements in the front matter are required, at least in the publication, while others are entirely optional.

Table 3.1 The elements that may appear in book front matter, in the order they are published. Note which elements are required in a published book or in a complete manuscript

Front matter element	Status in manuscript	Status in published book
Half-title page	Optional	Required
Frontispiece	Optional	Optional
Title page	Required	Required
Copyright page	Optional	Required
Introductory quotation	Optional	Optional
Dedication	Optional	Optional
Foreword	Optional	Optional
Preface	Optional	Optional
Acknowledgements	Optional	Optional
About this book	Optional	Optional
Table of contents	Required	Required
List of contributors[a]	Optional	Optional
List of abbreviations	Optional	Optional

[a] The entry "list of contributors" refers to edited books only.

3.2.1 Essential Items

A *half-title page* gives the main title and any series information, if applicable. Mark only for capitals.

The *title page* contains the title, subtitle, name(s) of the author(s) or editor(s) (without academic titles and addresses), number of figures (optional; rare), and the publisher's name. Mention is also made of the foreword if one is included. If an author/editor is deceased, this is not to be indicated using a "dagger" symbol, but the information may be added in a sentence in the preface. The copy editor often does not see this material; if you do, mark only for capitals, following the normal book rules for headings.

The *copyright page* is usually only prepared at the page-proof stage.

The *preface* is written by the author(s) or editor(s). Their names are always given at the end of the preface; affiliations and titles are generally not included, but the date and place of writing may be. A preface should be about the book (as opposed to the subject of the book): why it was written, who it is for, and any other pertinent information, such as its organization and the selection of contributors. It generally also contains any acknowledgements of support or assistance in preparing the book or in conducting the study or conference from which the book originated.

If an author oreditor is deceased, this is not to be indicated using a "dagger" symbol. A sentence containing the information may be added to the preface if a mark of respect is desired.

Since reviewers and potential purchasers usually read the preface, it should be well organized, easy to understand, and optimistic. Therefore edit it very carefully, even in standard and formal editing. Note that the German word for preface is *Vorwort*, which sometimes leads to it being mistakenly translated as "foreword" (in German, *Geleitwort*).

The heading for the *table of contents* is simply "Contents." Follow any book-specific instructions regarding the preparation of a table of contents (e.g., the number of levels of headings to include). In monographs that are not multiauthor edited works, with few exceptions, only the first three levels of headings are included in the table of contents. In a symposium volume or other multiauthor edited work, the table of contents generally consists of the title of each contribution followed by the authors' names as given on the chapter title pages.

If a table of contents has not been supplied, the copy editor must normally prepare one. If one has already been prepared, all headings in the table of contents must be checked against those in the text; the latter usually take precedence. Point out any major discrepancies to the author or editor.

A *list of contributors* is frequently included in multiauthor edited books. It is not the copy editor's responsibility to compile (or correct) this list unless explicitly requested; in fact, it is often not supplied with the manuscript. The list of contributors always includes the names and initials of all authors and their affiliations. Academic titles are also sometimes included. The names of any editors who have contributed chapters (or sections) to a multiauthor edited volume must also be included in the list.

When checking the list of contributors, confirm that it is complete and that authors' names in the list correspond to those at the beginning of chapters. The names and addresses of the authors in nonedited (single-author) books and of the editors in edited books are printed on the copyright page, which is generally not the responsibility of the copy editor.

3.2.2 Optional Items

A *frontispiece* is an illustration facing the book's title page. Check any caption for errors.

Any *introductory quotation* or *dedication* should be edited only for typing and language errors, respectively, not for content. Any other problems should be queried.

A *foreword* is written by a guest author whose name appears at the end of the foreword; affiliations and titles are generally not included, but the date and place of writing may be. A foreword can contain almost anything, but the information is of a background nature. Like prefaces, forewords should be edited with great care, even in standard and formal editing.

Acknowledgements are customarily included as the last part of a preface. However, they are occasionally included as a separate section following the preface if, for example, numerous people and/or companies are mentioned. Restrict corrections to spelling and grammatical errors. Acknowledgements by the author of an individual chapter appear at the end of the chapter, before the reference list.

A *list of abbreviations* may be included in a book; it may be very helpful if numerous abbreviations are scattered throughout a text. If one is to be included, abbreviations should still be defined at first use in the text. The heading is simply "Abbreviations." This list may also take the form of a list of symbols (the heading is then "Symbols").

List of headings. The book may also include a list of further elements from the book such as figures (the heading is then “List of Figures”) or tables (the heading is then “List of Tables”). These lists may be provided by the author or created by the supplier. If provided by the author, the copy editor must check the list agrees with the entries in the book

3.2.3 Metadata

Springer Nature also supplies some information that traditionally is closely associated with the front matter. This metadata is not edited by the copy editor and included here for information:

– *Digital object identifier (DOI)* of the book and of each chapter

– *Chapter type* This is the type of document as defined according to a list established by Springer Nature and valid for all its publications. The chapter types that can be used to describe a standard chapter, depending on length and content, are:

– Original paper A chapter reporting results or work (whether of research, case studies, theoretical work, etc.).

– Review paper A chapter reviewing the results reported by others.

The other chapter types describe content that differs from the structure of a chapter as described in this style manual, e.g., an erratum.

– *Product type* This is the type of publication as defined according to a list established by Springer Nature.

– *Subject* The primary discipline of the publication according to Springer Nature's list of subject categories.

3.3 Main Text

The main text of a book may be divided into chapters or into parts, each part containing one or more chapters. If a book is divided into parts, all chapters must be included in a part; only the first chapter can stand separately before Part I starts. Parts may be numbered with Arabic numbers or uppercase Roman numerals (e.g., Part I) and must be listed in the table of contents. More than one level of parts (subparts) is strongly discouraged for clarity, and should be restructured to parts if possible. Please check with your Springer Nature contact if a book is received with subpart structuring.

Chapters are the primary units of most books. Chapters are either unnumbered or numbered with Arabic numbers. If numbered, chapters are numbered consecutively throughout a book, starting from "1".

3.3.1 Headings and Sections

Springer Nature employs two major systems of headings: (a) the decimal system (see Sect. 3.3.1.2), and (b) keying, an unnumbered system relying on typographical differentiation to indicate the level of the heading (see Sect. 3.3.1.3). The choice of system to be used depends sometimes on the author's preference, but more often on the type of book. Instructions as to the system to employ are provided together with the manuscript. In both systems, normally no more than five levels of headings are employed; run-in headings that are not part of the heading hierarchy may, of course, be used in each system at any level, as appropriate. Customarily, no more than three levels of headings are included in a table of contents.

Table 3.2 Headings according to the two systems.

Heading	Level of keyed heading	Number in decimal headings
Mechanisms of Hyperlipidemia	Title	5
Introduction	1	5.1
Normal Levels of Plasma Triglycerides and Cholesterol	1	5.2
The Plasma Lipoprotein Spectrum	1	5.3
Techniques of Analysis	2	5.3.1
Separation on the Basis of Density	3	5.3.1.1
The Analytical Ultracentrifuge	4	5.3.1.1.1
The Preparative Ultracentrifuge	4	5.3.1.1.2

Heading	Level of keyed heading	Number in decimal headings
Separation on the Basis of Charge by Zonal Electrophoresis	3	5.3.1.2
Gel Filtration Chromatography	3	5.3.1.3
Precipitation with Polyanion-Metal Complexes	3	5.3.1.4
Structure and Chemical Composition of Plasma Lipoproteins	2	5.3.2

The example of decimal headings assumes that the chapter number is included, here as Chap. 5.

3.3.1.1 *Styling of Headings*

In books, in contrast to many journals, all major words in a heading are capitalized. This means: capitalize any word except articles, conjunctions, or prepositions that are four letters or less in length. The following paragraphs cover specific questions regarding the styling of headings (including run-in headings outside the section hierarchy).

Capitalization of Latin Phrases Capitalize all words of commonly used Latin phrases—i.e., those found in the main text or foreign words appendix of *Merriam–Webster's New Collegiate Dictionary*—appearing in titles and headings (e.g., Review of In Vivo Studies). Latin phrases not in *Webster's* are italicized and lowercase. Species and genus names are styled as in the text.

Capitalization of Compound Terms Always capitalize the first element of a hyphenated compound. But:

– Do not capitalize the second (or further) element(s) if one of the elements is normally only used as a combining form (e.g., Pre-urethritis and the Third Trimester).

– Capitalize both elements if the combination is temporary (e.g., Dose-Dependent Effects; Augmentation of Input-Output Devices).

– Capitalize both elements if the second is a proper noun (e.g., Study of Non-Hamiltonian Functions).

– Always capitalize the main element of a technical term, regardless of prefixes, but do not capitalize the second (or further) element(s) of a hyphenated technical term (e.g., L-Dopa and Kidney Reinfection; Analysis of threo-1-Phenyl-1-hydroxy-2-aminopropane).

3.3.1.2 *Decimal System*

The decimal system of headings is popular in both monographs and series. In the decimal system, the first number (from the left) is usually the chapter number. The first primary heading in Chap. 1 is then 1.1, the second 1.2, etc. Under, for example, the third primary heading, the first secondary heading would be 1.3.1, the second 1.3.2, etc. Neither the chapter number by itself nor the final number in section headings is followed by a period. See the example shown in Table 3.2.

3.3.1.3 *Keying*

In this system, levels of headings are differentiated purely by typographical means, which means that the copy editor must clearly indicate the level of every heading. See the example shown in Table 3.2.

3.3.1.4 *Outline System*

The outline system is no longer used, except for some Law books. In this system, main headings within a chapter are assigned capital letters; secondary headings, Roman numerals; third-level headings, Arabic numerals; fourth-level headings, lowercase letters followed by a closing parenthesis; and fifth-level headings, lowercase Greek letters followed by a closing parenthesis. Where further headings are necessary, they should be keyed. Chapters are usually numbered with Arabic numerals. If asked to use this system, please ask for confirmation.

3.3.2 Other Numbered Elements

Within chapters, certain elements such as figures, tables, equations, schemes, and structures are regularly numbered, as is described in Chaps. 2, 7, 8, and 10. These elements may not be numbered sequentially throughout a book but the numbering needs to start anew in each chapter.

If chapters are numbered (with Arabic numbers), then the figures, tables, equations, and schemes are routinely "double numbered." In the double-number system, the chapter number is put in front of the respective figure/table/equation/scheme/structure number; a period separates the two numbers. Figure 3.1 would thus be the first figure in Chap. 3, and Table 2.5 the fifth table in Chap. 2.

Thus, for example, text references to figures could take any of the forms Fig. 6 or Fig. 3.6; or Figs. 6, 7, 9 or Figs. 3.6, 3.7, 3.9.

3.3.3 Organization of a Chapter

Since a book chapter is organized in much the same manner as an article in a journal, see Sect. 2.2 for information on the internal make-up of a book chapter. The huge differences in content from one book to another make it impossible to specify absolute requirements.

A note on Appendixes: Within a chapter, an appendix is designated "Appendix" if there is only one or "Appendix 1," "Appendix 2," etc. if there is more than one. The content of an appendix is contained within the sections subordinate to the major heading "Appendix." The numbering of tables, figures, and equations continues on from that in the main text, following the same rules as in a journal article (see Sect. 2.2.3.1).

3.4 Book Back Matter

The sequence of back matter in a book is listed below. Each of the items begins on a new page, and each is optional, although very few books have no reference list of some sort.

3.4.1 Appendixes

Appendixes provide supplementary information to a book or chapter. An appendix to a chapter appears at the end of that chapter, not at the end of the book). Appendixes should be referred to in the text. The language and styling rules for the text also apply to appendixes.

In the back matter of a book, any appendix appears at the chapter level. Otherwise, follow the same procedure as for an appendix within a chapter, with one major exception. Number all tables, figures, and equations throughout all appendixes in the back matter of a book, starting with "1" in the first appendix (and continuing the numbering in any following appendixes). If the book chapters are numbered, use "A" as the "chapter" number of the appendix when numbering tables, figures, and equations (e.g., Table A.3).

3.4.2 List of Abbreviations

A list of abbreviations or symbols for an entire book routinely belongs in the front matter, although it can be included in the back matter. This may be the case for extensive lists which are subdivided into separate groupings, each with their own headings. The list may also be a list of symbols.

3.4.3 Glossary

A glossary may be included in a book (or even in an individual chapter). Terms in the glossary are arranged alphabetically, each on a separate line and followed by its definition.

A glossary always consists of terms and their explanation, whereas a list of abbreviations only contains the abbreviations and their written out forms without any further explanation.

3.4.4 References, Further Reading, Bibliography

A reference list may come at the end of the book or at the end of each chapter, depending mainly on the type of book. The correct heading is "References." For information on the correct styling of reference citations and the reference list, see Chap. 10.

It is the copy editor's task to check that there are no obvious typing mistakes in the titles of works cited. Correct any evident mistakes, but do not alter spelling, etc., merely to achieve consistency with the style in the manuscript you are editing. Italics, small capitals, and boldface do not have to be added in accordance with style guidelines.

In some books the reference list may be divided into sections with appropriate subheadings. In addition to the list of references cited in the text (heading: "Cited literature"), a list of further or suggested reading may be included (heading: "Further Reading"). There may also be primary and secondary sources, particularly in the humanities and social sciences.

Occasionally, instead of a reference list, there is a bibliography, i.e., a list of publications, all or some of which are not cited in the text (heading: "Bibliography"). Check with your Springer Nature contact if you are uncertain which heading is appropriate. If the entries in such a list are to be styled, follow the appropriate guidelines for styling references (see Chap. 10).

3.4.5 Name Index

Some books contain a separate index of the names of individuals mentioned in the text, but usually name and subject index are combined in one index. A name index is organized in a manner similar to a subject index.

3.4.6 Index

Most books contain an index. As a rule, an index is either created from entries supplied by the author together with the manuscipt or is compiled by the book's vendor. This is always indicated in the production instructions. Although it is not generally the copy editor's job to create an index, the following guidelines are intended to help those who are required to check an existing index.:

3.4.6.1 *What To Include*

An index should include important ideas, facts, names, and terms that receive significant discussion in the text. Items that are only mentioned incidentally in the text should not be included. On average, there should be two to three index entries per page of text. See also the index fragment below for examples (Sect. 3.4.6.5).

3.4.6.2 *How To Capture the Information*

Here are some general guidelines about capturing entries and subentries:

– Wherever possible, related concepts should be grouped together

– Use no more than two levels of subentries

– Information should be indexed under a main noun (or compound noun) and not under adjectives. For example, instead of capturing the terms "Numbering footnotes," "Styling footnotes," and "Verifying footnotes" as three entries, capture them as in this example:

Footnotes

numbering

styling

verifying

– Information should be listed under the term that most readers will probably look at first. Use cross-references to list variations or written-out versions and abbreviations/acronyms.

3.4.6.3 *How To Style the Entries*

Please note these general rules about styling entries and subentries:

– Main entries begin with a capital letter. Numbers and prefixes are ignored in this context.

– Subentries begin with lowercase letters unless the term is a proper noun.

– Entries are formated as in the text (e.g., italics for genus and species names, and the use of subscripts and superscripts)

– Abbreviations at the beginning of a term must be written out. For example, *A. fumigatus* should be *Aspergillus fumigatus*

– An entry is divided from the page number by a comma and a space. Any number coming at the end of a term should be marked as belonging to the term (e.g., with a nonbreaking space).

– "See" and "See also" are always rendered in italics.

3.4.6.4 *Alphabetizing the Entries*

Please note the following rules for alphabetization of entries:

– Entries should be alphabetized letter by letter (not word by word).

– Punctuation is ignored until you reach the first comma.

– Sort people's names by the family name or the prefix if one is present. Note that many exceptions are made to this rule for well-known persons commonly referred to without the prefix; see for example the biographical names section of Merriam–Webster's *New Collegiate Dictionary*.

– Ignore prefixes and symbols preceding chemical names:

5-Dehydrogenase

DH-581

Dichlorisoproterenol

2,4-Dinitrophenol

3.4.6.5 *Sample Index Entries*

The following is a fragment of an index illustrating many of the instructions in the previous paragraphs.

α-Acetylglucosaminidase in aorta, 315

Amitriptyline, 363

D-Amphetamine, 324

Androsterone, 370

Aortic lipolytic activity, 301

Apoproteins, synthesis of, 198

Ascorbic acid, 128

Atherogenicity

fructose, 217

sucrose, 217

Atherosclerosis

animal models of, 215

in baboons, 223

in rabbits

effect of fats, 218

influence of carbohydrates, 218

in rhesus monkeys

effect of fats, 222

regression of, 222

in squirrel monkeys (*Saimiri sciureus*), 223

Atherosclerotic lesions

in Cebus albifrons, 222

in Cebus fatuella, 222

in primates, 223

See also Atherosclerosis

...

Debussy, Claude, 15

Defoe, Daniel, 13

de Gaulle, Charles, 23

Dekalb, 336

De la Mare, W.J., 13

Diagnostic and Statistical Manual of Mental Disorders IV, 25–34, 56, 116–163

DSM-IV, *see* Diagnostic and Statistical Manual of Mental Disorders

3.5 Cost, Quality, and Deadlines

Books are assigned to one of the three levels of editing. It is important that editing be done in accordance with these guidelines to ensure that the desired quality is achieved and that unnecessary costs are avoided. In general, for each book there is a schedule for publication with deadlines for the completion of editing.

Part II: Language and Style

4 Editing Language

4.1 General Guidelines

Good scientific writing is direct, precise, and concise, but not necessarily elegant. The task of the copy editor is not always to turn a manuscript into "good scientific writing," let alone elegant style. The task of a copy editor working for Springer Nature is to edit a manuscript and improve it according to the instructed category of editing.

Rephrasing and rewriting must be done *in accordance with the degree of editing required* for any specific publication. If language editing in accordance with that degree of editing seems inappropriate (either insufficient or excessive), please consult your Springer Nature contact before proceeding. It may be necessary for the manuscript to be returned to the author for revision or for the project planning to be modified.

This chapter provides general guidance on language editing, comments on what may constitute bad style, examples of frequent mistakes, and an overview of differences between British and American usage. For further information on correct use of language in various fields, see Chaps. 11 (medicine and biology), 12 (chemistry), 13 (mathematics), 14 (geosciences), and 15 (physics, engineering, and computer science).

4.1.1 Rephrasing and Rewording

Copy editors often have difficulty in determining how far they should go in rewriting. With experience comes a "feel" for the appropriate amount of editing. A general guide is that if an experienced copy editor must read a sentence twice to make sense of it, something may be wrong. A novice copy editor or one with little knowledge of the field concerned must assume that much correct text will still seem strange after repeated readings.

Before making any language changes, establish precisely what is wrong with the author's sentence. Ensure that your change corrects the problem and then reread the entire sentence to make sure that you have not introduced or overlooked another error. Be guided too by the subject matter you are working with: for literature titles, for example, an author's style is key to the content. Apart from correcting typographical and grammatical errors, extreme caution must be exercised not to carry out unnecessary changes. You must constantly be aware of altering meaning and of inappropriately shifting emphasis; these are the cardinal sins of copy editing! If you are uncertain whether your change is correct, query the author.

4.1.2 Personal Preference

The language of a manuscript should *never* be changed simply because of personal preference. The copy editor *must* remain aware that someone else is the author and be able to give a cogent reason for every change made.

As far as possible, use the author's own words and phrases to reconstruct a poorly written sentence; instead of completely rewriting incorrect text in different words, transpose, reorder, and, if necessary, use a different form of the same word, e.g., an adjective instead of a noun or a verb instead of an adjective. It is very important for the author to see that their own words are still being used.

4.2 Poor Style

Bad writing style is frequently characterized by sentences or paragraphs of inappropriate length, by unnecessary or inappropriate words, and by misuse of active and passive voice. While poor style does not constitute a mistake, it can make it difficult or even impossible for an author's intended meaning to be easily understood. *The following examples are for category 3 language editing. In category 2 standard editing, poor style should only be corrected if it hinders comprehension.*

Very Long Sentences Some sentences are so long and contain so many prepositional phrases and dependent clauses that they are difficult to comprehend; such sentences should be broken down into shorter ones.

Very Short Sentences Short, choppy sentences can also hinder comprehension. Such sentences should be combined; however, avoid long strings of simple independent clauses joined by conjunctions. Dependent clauses and prepositional phrases are helpful in constructing sentences that flow smoothly.

Paragraphing A paragraph should contain information relating to one point. However, text divided into very long paragraphs or many one- or two-line paragraphs is difficult to follow. Be aware of the following guidelines:

1. Paragraphs should rarely be more than one manuscript page or less than five lines in length. A series of one-sentence paragraphs can usually be grouped into "normal-sized" paragraphs, and a single one-sentence paragraph can often be joined to the paragraph that precedes or follows it.
2. Paragraphs running to a page or more can usually logically be broken into smaller units.

Nonfunctional Words or Phrases Many words or phrases serve no meaningful purpose in a sentence and should be deleted:

Before: There are some viruses that can withstand...

After: Some viruses can withstand...

Before: As regards this species, its only representative...

After: The only representative of this species...

Extended Wording In place of a single verb, an author may use a noun form of a verb together with another verb. Such sentences should be simplified. Here before and after:

Before: Measurement of blood sugar concentration was performed.

After: The blood sugar concentration was measured.

Before: Precipitation of the substance occurred....

After: The substance precipitated....

Passive Voice and First Person Basically, the author's preference regarding use of the passive voice and first person should be respected, although some authors overuse the passive voice, particularly with "It was ...ed that" to avoid use of the first person. Editing to the active voice is often necessary to clarify ownership of actions, as shown in these examples:

Before: It could be demonstrated by our group that...

After: Our group demonstrated....

Before: It was suggested by these findings that...

After: These findings suggest that....

Personal Pronouns The use of "he" and "she" without gender-specific antecedents should be avoided where feasible, i.e., when it is possible without clumsy rewording. Unfortunately, English does not offer an easy solution, although "they" is becoming more accepted as the standard neutral option. "S/he" should not be used. "He/she," "he or she," and the English neuter personal pronoun "one" can all be clumsy, distracting, or awkward if used too often. Other solutions include rewording to avoid a personal pronoun altogether, changing to the plural, and using the passive voice. If inappropriate gender-specific personal pronouns have been used excessively, bring this to the author's attention; under no circumstances should you make wholesale changes to the text.

Grammatical contractions Always spell out grammatical contractions, e.g., change 'isn't' to 'is not' unless you are editing a magazine-type article.

Other Avoid inappropriate colloquialisms if they obscure meaning and are not central to the topic. Substitute a more "international" word or phrase. This problem occurs not only with authors whose mother tongue is not English, but also with British and American authors using an idiom specific to their own region.

4.3 Some Frequent Mistakes

4.3.1 Vague Antecedents

Vague antecedents are usually clarified by repeating the noun in place of the pronoun. Some sentences may, however, require rewording.

4.3.2 Misplaced Modifiers

Modifying words and phrases are frequently misplaced. The sentence can usually be improved simply by moving the phrase to a position immediately after the word or phrase modified.

Nonsense: Having swollen during the prolonged operative procedure, the surgeon was unable to remove all the affected tissue.

Sense: As the affected tissue had swollen during the prolonged operative procedure, the surgeon was unable to remove all of it.

A misplaced word can radically alter the meaning of a sentence. Read the following sentence, putting "only" before each word in turn: "He told me that he loved me."

NB: Be careful not to turn the placement of modifiers into an abstract principle. Sometimes it is a matter of style. There are cases, for example, when an adverb may be best placed between the parts of a verb.

4.3.3 Nouns as Adjectives

Do not use the adjectival form of a word when the sense calls for a noun form. Two examples of sensible changes are pathological study *to* pathology study and bodily odors *to* body odors.

4.3.4 Precise Use of Words

Words that have similar meanings or spellings are frequently misused. Be careful to catch them. Several examples of commonly misused words are given in Table 4.1 (see also Sect. 11.1.1 for misused words in medicine).

Table 4.1 Examples of commonly misused words

Word frequently misused	Meaning	Example of use
Affect (verb)	to produce an influence on	The weather affects her mood.
Affect (noun)	the conscious subjective aspect of an emotion	Pain is a strong affect.
Alternate (adjective)	every other, every second	She had to attend on alternate days NB do not use as a synonym of "alternative"
Alternative (adjective)	one of possible options	an alternative way of solving the problem
Determine	establish, ascertain	They wanted to determine whether this was true NB use "whether" not "if"
Effect (verb)	to bring about, cause to happen	The president will effect a complete reorganization of the board of directors.
Effect (noun)	something that inevitably follows an antecedent	Diazepam has a relaxing effect on muscles.
Classic	something that is typical	The classic symptoms of myocardial infarction are...

Word frequently misused	Meaning	Example of use
Classical	something original or historical	The classical definition of hysteria has undergone many changes over the centuries.
Compare	to examine in order to determine similarities or differences	She compared the two bunches of grapes.
Compare to	to liken one thing to another	Shall I compare thee to a bunch of grapes?
Contrast	to juxtapose dissimilar elements	He contrasted honey and vinegar.
Compose	to form by putting together	The editorial board is composed of scientists and publishers.
Comprise	to be made up of	The editorial board comprises five scientists and two publishers. NB Use "comprise" or "consist of" but not "comprised of"
Consist in	to lie in, to have as an essential feature	Liberty consists in the absence of obstruction.
Consist of	to be made up of	The editorial board consists of five scientists and two publishers. NB Use "comprise" or "consist of" but not "comprise of"
Discrete	constituting a separate entity	There were at least ten discrete black spots on each banana.
Discreet	showing discernment or good judgement	She whispered discreetly into his ear.
Dose	quantity administered at one time or total quantity administered	A 50-mg dose proved insufficient.
Dosage	the gradation and regulation of doses	The dosage was 50 mg twice daily.
Enable	to make able	His acts of cannibalism enabled him to survive.
Make possible	to render something possible	His acts of cannibalism made his survival possible.
Forgo	to do without	They decided to forgo the second round NB Do not confuse with "forego" = to go before
Historic	noted in history	This historic decision...
Historical	of, relating to, or having the character of history	Largely for historical reasons...
Identical	the same as	The second book was identical to the first NB Use "identical to" not "identical with"
Imply	to express indirectly, to suggest	He implied that she was his wife.

Word frequently misused	Meaning	Example of use
Infer	to derive as a conclusion, to deduce	I inferred from her remarks that she was single.
Implicate	to suggest or show to be involved	Viruses have been implicated in the pathogenesis of cancer.
Latter	the second of two	Of the two options, she chose the latter. NB if there are more than two, use "last" not "latter"
Less than	a smaller amount in a measured quantity	He weighed less than 200 lbs NB Use "fewer than" for individual items or people
Percent (British: per cent)	in or of every hundred	Only 20% of our patients were male.
Percentage	a number or amount stated in percent	The percentage was lower than expected.
Presently	soon	The parcel will arrive presently NB Do not use to mean "at present"
Principal (noun and adjective	chief, head	The school principal was very competent
Principle	rule	The ethical principle was not followed here
Usage	a firmly established, generally accepted practice	Artifact with an I is standard American usage.

4.3.5 Tautology

You should not delete information that the author has repeated for the sake of emphasis or nonessential words that improve the flow of the text. Redundant words that do not contribute to the flow of the sentence can, however, be deleted:

...such as oranges, apples, pears, *etc.*

The reason for this is *because of*....

4.3.6 Use of Singular and Plural Verbs

Here are a number of simple rules about subject–verb agreement:

– A collective noun takes a singular verb unless the components forming the group are emphasized:

> The crowd roars.
>
> The committee is divided on this question.

British and American usage may differ:

British: The committee agree on this question.

American: The members of the committee agree on this question.

– If the subject is a group of words conveying the idea of a number of individuals, the verb is plural even though the governing noun is singular:

A wide range of physical phenomena are involved.

A large series of items are available.

But be careful:

An average of 15 is as good as one can expect.

An average of 15 men were tested each month.

The number of children present was small.

A number of children were cured.

Note the different articles in the last two examples. The article is often a useful guide to whether a singular or plural verb is required.

– With units of measure always use the singular verb form:

Five milliliters was injected.

In our opinion, 100 km is far enough.

– "Either" and "neither" always have a singular verb unless the subject closest to the verb is plural:

Neither nausea nor irritability is....

Neither the nose nor the lips are....

Neither the lips nor the nose is....

This also applies to "and/or":

The lips and/or the nose is....

The nose and/or the lips are....

– "Each" and "none" always have a singular verb with no exceptions:

Each of the mechanisms is plausible...

– A "with" phrase that is part of a nominative phrase does not determine verb number, which must agree with the grammatical subject.

Isoprenaline given together with propranolol is effective in...

– A plural subject requires "there are," not "there is":

There were one adult and three children.

– On very rare occasions, words linked by "and" take a singular verb (e.g., bed and breakfast, dilatation and curettage).

4.3.7 Literal Translations and Non-English Words

The literal translation of technical terms may yield nonsense. Problems often occur when idioms are translated literally or when a literal translation does not yield a commonly used English phrase. Either of these may be easily overlooked, especially if you are fluent in the original language. Problems that occur frequently in English texts written by German speakers are listed in Table 4.2. Furthermore, German-speaking authors often make excessive use of articles and hyphens:

Incorrect articles: MRI makes an earlier detection of disease possible

Incorrect hyphens: risk-factors, TNM-atlas, the glucose-level

Also correct any names or terms not given in the accepted English spellings (e.g., Tschebyscheff instead of Chebyshev).

Table 4.2 Examples of frequent incorrect literal translations from German

Translation	English usage
already	as long ago as, as early as (or just the past tense)
resp., respectively	or, and (can sometimes simply be omitted)
as well as	and
until next Monday	by next Monday
...is there since 4 years	...has been there (for) 4 years
Applied Physics has been founded...	...was founded...
in the last years	in recent years, in the past few years
in this time	then, now
in the next time	soon
eventually	possibly, may be
overlooking	reviewing
both	the two (can sometimes simply be omitted)

Translation	English usage
analysis enables to determine	analysis enables *us* to determine

4.4 British–American Usage and Spelling

Both British and American English are used in Springer Nature publications. Follow any specific instructions you are given. Otherwise please observe the following general guidelines:

1. Usage in a chapter or article must be consistent.
2. Usage in an individual book or a journal can differ between chapters/articles.
3. If the author is a native English speaker, do not change the author's choice of American or British.
4. Do not change consistent usage.
5. If usage is inconsistent, standardize to American or British English according to which involves less work.

Spelling For American spelling always consult *Merriam–Webster's Collegiate Dictionary*; for British spelling refer to the *Oxford English Dictionary*, but allow the author to use spellings given in *Collins English Dictionary*. Some of the differences between British and American spelling are given in Table 4.3. For differences in hyphenation, see Sect. 5.5.3.

Table 4.3 British–American spelling differences

British usage		American usage	
Rule	Example	Rule	Example
Both -ise and -ize endings are accepted for many words, but not for all. Use either -ise or -ize throughout a manuscript, as far as possible. NB -ize is used in Nature Research for both British and American content	realise *or* realize	In most instances the -ize ending is used	realize, recognize
	recognise *or* recognize	But beware of exceptions	advertise, surmise, compromise
Use -yse	analyse (not analyze)	Use -yze	analyze (but lyse)
When the past tense, gerund, etc. is formed from a verb, the final single consonant is often doubled, regardless of stress.	preferred, occurring, labelled, marvellous, worshipper *but* benefited	A final consonant is doubled only if the stress is on the final syllable.	preferred, occurring, labeled, marvelous, worshiper, benefited, *but* transferred
Past tenses of some verbs may end in -t or -ed.	spelled *or* spelt	Use -ed in such cases.	spelled
Use -ce endings for some nouns.	defence, licence	Use -se in such cases.	defense, license

British usage		American usage	
Rule	Example	Rule	Example
Some -ce nouns take an -se ending as verbs.	practice (n), practise (v)	Both forms have the same spelling	practice (n or v)
	licence (n), license (v)		license (n or v)
The suffix -wards is more common.	forwards, backwards, towards	The suffix -ward is more common.	forward, backward, toward
Some prepositions may end in -st.	while *or* whilst	These prepositions do not end in -st.	while
	amid *or* amidst		amid, *but* in the midst of
Use ou	colour, behaviour, mould	Use o	color, behavior, mold
	but coloration, tumorous		
Use -re	centre, fibre	Use -er	center, fiber
Use ae	haematology, leukaemia, faecal	Use e	hematology, leukemia, fecal
	(*but* leukopenia)		
Generally use oe	oedema, oesophagus, manoeuvre (but fetus is preferred to foetus)	Use e	edema, esophagus, maneuver
Other examples	aluminium	Other examples	aluminum
	artefact		artifact
	fulfil		fulfill
	grey		gray
	leucocyte		leukocyte
	sceptical		skeptical
	skilful		skillful

-ic/-ical Endings The use of -ic or -ical endings depends both on the form of English being used and on the author's preference. In British English the -ical ending is more common (anatomical, histological, etc.), but the -ic ending is often acceptable. In American English both -ic and -ical endings are often accepted, but follow the usage given by *Webster's* or specialist dictionaries. Each individual term should be spelled consistently in a manuscript. Note that there is sometimes a difference in meaning between -ic and -ical forms (see some examples in Table 4.1 and Sect. 11.1.3).

Plurals In both American and British English, Latin endings are accepted for some terms but not for others:

datum–data

focus–foci (or focuses, but foci is more common)

vertex–vertices

fistula–fistulae (fistulas is also acceptable)

In American English, some anglicized plurals for Latin words are generally preferred (check *Webster's*), e.g., appendixes, indexes, formulas, antennas, apparatuses (but apparatus in British).

5 Punctuation

5.1 Period

Abbreviations Periods are generally not used after unit abbreviations (exception: "in." for inch) or in acronyms (e.g., LAD, CSF, DNA). They are used in some non-unit abbreviations, such as a.m., i.m., and i.v., and after abbreviations of parts of a publication (in conjunction with a number), such as p., Vol., Chap., Fig., Eq., Sect. (the plural forms are: pp., Vols., Chaps., Figs., Eqs., Sects.; Table and Appendix are never abbreviated). (NB In articles using the superscript numbering reference system only, it is permissible to use "ref." and "refs." followed by a superscript number in exceptional cases where needed for clarity).

Quotations In American usage periods appear before closing quotation marks (see Sect. 5.9.3), for example: "The character change in a tumor is usually called 'tumor progression.'" In British usage a period appears before ending quotation marks if it is part of the original quotation (i.e., if a complete sentence is being quoted) and after if it is not (see Sect. 5.9.3).

Footnotes and Reference Citations Periods appear before footnote citations ("...by Smith.[3]") but come after reference citations ("...by Smith [3].").

Ordinal Numbers Periods are not used with a numeral to make an ordinal number (unlike in German, where for instance "1." stands for "first"). Style ordinal numbers as first, second, third,..., tenth, 11th, 12th, etc.

Parentheses If an entire sentence appears within parentheses, the period also appears within the parentheses, e.g., "... no more experiments were conducted. (The research team left the project at this point.)"

5.2 Comma

Sensible use of commas clarifies meaning and helps to ensure that the text flows smoothly. In many cases the use of a comma is optional and reflects an author's personal style. Follow this style as long as it is not confusing, but note that in some cases the use of commas is governed by rules of grammar. In short, commas

Set off nonrestrictive, parenthetical, and miscellaneous elements.

Follow certain introductory elements, e.g., adverbial clauses.

Precede the coordinating conjunctions "and," "but," "or," "nor," "for," and the connectives "so" and "yet" if they connect main clauses.

Separate items in a series of three or more items, including before the conjunction in American usage.

See the following sections for a discussion of each of these groups.

Restrictive and Nonrestrictive Elements A restrictive (or defining) clause "completes" the subject and is consequently indispensable. By contrast, a nonrestrictive (or commenting) clause does no more than give additional information. A nonrestrictive clause should be within commas, a restrictive one should not.

> Smith, who went to school in Reading, is Britain's leading plastic surgeon. (nonrestrictive)
>
> Brenner forceps, which are used in breech presentations, were first employed in 1893. (nonrestrictive)
>
> Physicians who do not work in the United States have placed more emphasis on... (restrictive)
>
> All rats that were treated survived. (restrictive)

While "which" can introduce both restrictive and nonrestrictive clauses, "that" introduces restrictive ones only and is therefore never preceded by a comma.

Adverbial Clauses or Phrases When a dependent clause follows a main clause, a comma is not used if the dependent clause is restrictive (as in this sentence or the following examples):

> The surgeon will operate if the findings are positive.
>
> The surgeon will operate, although the patient's chances of recovery are minimal.

This distinction is not made if a dependent clause precedes the main clause, in which case a comma *may* be used:

> If the findings are positive, the surgeon will operate.

A long introductory adverbial or prepositional phrase should be set off by a comma:

> Since the day had been exhausting and the men were in very poor spirits, John...

but no comma is necessary after a short introductory adverbial phrase:

> On Saturdays he goes to town.

Within a sentence a parenthetical statement is set off by a pair of commas (or, obviously, a pair of em dashes or parentheses):

> This was the plan which, despite our objections, was implemented.

Compound Sentences and Compound Predicates A comma is used before a conjunction that joins independent clauses:

> The snow was so heavy that it covered the plants within an hour, and it remained for more than 3 weeks.

> A comma is generally not used to set off a compound predicate (two or more verbs having the same subject), but one may be used if the sentence is long and complicated:
>
> They examined the possibility of resecting the affected tissue after conducting the examination, but decided that the risk would be too great.

In other dependent clauses, a comma can be used before a conjunction to emphasize the fact that the second clause expresses an antithesis of the first:

> The neurologist examined her after admission, but not before discharge.

Series Comma In American English use a series comma before the conjunction in a series of three or more words, phrases, or clauses. In British English the series comma is rarely used, but if a British author has used it consistently, retain it; it may also occasionally be used to avoid ambiguity.

If the items in a series are phrases or clauses with internal punctuation, semicolons may be used instead of commas to separate the items.

Other Uses and Abuses Some other important cases are:

A comma should *not* be used to indicate the end of a noun phrase (i.e., separating subject and verb):

> The issue of whether resection is the treatment of choice in carcinoma of the breast[,] is still a matter of dispute.

A comma is used to replace an implied conjunction (see also Sect. 5.3), e.g., "It was malignant, not benign," and *may* also be used to replace an implied verb.

In American English, commas always precede and follow "i.e." and "e.g." and come after the words which follow in apposition ("Many compounds, e.g., potassium nitrite, have..."), but in British English there is no comma immediately following "i.e." or "e.g."

In American usage, commas appear inside (to the left of) ending quotation marks (see Sect. 5.9.3) while in British usage commas appear outside ending quotation marks.

Commas appear before footnote citations but after reference citations ("the new agent,[1] which..." but "the new agent [1], which...").

A comma never immediately precedes an opening parenthesis, bracket, or brace. If the construction calls for a comma, it follows the closing parenthesis, bracket, or brace:

> ...of Smith (and several investigators involved in other studies), who argued against it.

Commas never appear immediately before or after em dashes (even if it seems grammatically necessary).

5.3 Semicolon

A semicolon is used to join independent clauses of closely related information if a conjunction is not used:

> She broke her leg; it was a complicated fracture.

A semicolon should be used in such sentences even if the clauses are joined by a conjunctive adverb, e.g., "however," "therefore," "indeed," or "thus":

> The results of the investigations seem conclusive; however, they have been criticized on many grounds.

Semicolons may be used instead of commas to separate phrases or clauses in a series when the phrases or clauses contain internal punctuation:

> ...the first group, which had been pharmacologically treated; the second, which had received radiotherapy; and a control group.

Semicolons fall outside (to the right of) end quotation marks.

5.4 Colon

A colon is used in the following cases:

A colon may be used to introduce an independent clause, the first word of which may be capitalized: "The results led to a definite conclusion: There are no...." However, do not start with a capital if the second clause is not a complete sentence.

A colon may be used after "as follows" or "the following" if the items follow directly: "The following factors were analyzed: age, sex, weight,...."

Use a colon to introduce a series or list not introduced by "e.g.," "i.e.," "namely," etc.: "The analysis proceeded in three steps: (1) evaluation,...."

Do *not* use a colon to introduce a definition: "various imaging procedures (CT[:], computed tomography; MRI[:], magnetic resonance imaging)"

Do *not* use a colon to introduce a value: "results (mean[:] 24.6, range[:] 19.3–28.2).

Use a colon for ratios when numerals are used (e.g., the ratio was 3:1) but not between words (the patient to control ratio...).

Colons always go outside (to the right of) end quotation marks.

5.5 Hyphen

The use of hyphens depends very strongly on a writer's style. In the following, examples and guidance to recommended usage are given. More important than many of these recommendations, however, is consistency, following author usage where possible and avoiding overuse of hyphens. See also Sects. 11.2.12 and 15.3.1 for use of hyphens in scientific terms.

5.5.1 When a Hyphen Is Used

Hyphens are frequently used to connect prefixes to words if as a result the same vowel would appear twice together. However, there are numerous frequently used words for which this is not the case, e.g., coordinate or microorganism. Check *Webster's* if you are uncertain.

Compound adjectives are generally hyphenated when they *precede* a noun in the following cases:

– Combinations of adjective and past participle

hard-driven man (*but* that man is hard driven)

– Combinations of noun and past participle

handkerchief was hand-stitched

– Combinations of noun and present participle

eye-opening experience

Combinations of adverb and past participle (unless the adverb ends in "ly")

well-planned robbery, but poorly executed heist

– Compounds of degree

short-range telescope, high-energy propellant

– Numeral–unit combinations

3-ml flask, 10-kg weight, 30-year-old patient, 5-mm-diameter catheter

When using unit abbreviations with ranges, the clear style is:

3- to 10-ml flask, 20- to 30-year-old patients

but usage of the en dash is also correct:

3–10-ml flask, 20–30-year-old patients

– When there is a danger of confusion:

A "small-bowel constriction" is a constriction of the small bowel.

A "small bowel constriction" might be a small constriction of the bowel.

recover, re-cover; recreation, re-creation; unionized, un-ionized

– When a prefix or suffix modifies a two-word adjectival phrase, all three parts are hyphenated both prior to and following a noun:

non-antigen-specific reaction (in the predicate, this should read "was not antigen specific")

Hyphenate chemical names according to the *Merck Index*; otherwise leave as submitted if the style is consistent (see Sect. 13.1).

Use a hyphen with a prefix before a proper noun (e.g., non-Hodgkin's lymphoma).

Repeat the prefix in hyphenated compound terms if the prefix is constant but the main word changes (e.g., "β_2-agonists and β_2-antagonists are").

5.5.2 When a Hyphen Is Not Used

Do not put a hyphen after "-ly" adverbs (as in the fourth example in the previous list). Also, do not hyphenate compound modifiers that are commonly read as a unit: small cell carcinoma, soft tissue lesions, open heart surgery, medical school students, in vivo cultures, birth control method, amino acid solution, high molecular weight substance. The use of multiple hyphens may obstruct understanding and is usually unnecessary. Finally, do not add hyphens to proper nouns (for example, do not put a hyphen in "Monte Carlo method").

5.5.3 Differences Between British and American Usage

The above advice on hyphenation applies to both American and British English. The most common difference in usage concerns prefixes. In British usage, prefixes are sometimes hyphenated; important is that usage in a manuscript is consistent. In American, prefixes are almost always joined to the main word. Examples of such prefixes are:

ante, anti, auto, back, bi, bio, co, counter, de, di, eigen, electro, extra, ferro, giga, hyper, hypo, infra, inter, intra, iso, macro, mega, meso, meta, micro, mid, mini, mono, multi, milli, nano, neo, non, over, photo, poly, post, pre, pro, proto, pseudo, quasi, re, semi, socio, sub, super, supra, thermo, trans, ultra, un, under

Otherwise, follow the recommendations of *Webster's* for American English and of either *Collins* or *Oxford* for British English, depending on the manuscript.

5.6 En Dash

In a publication, the en dash (in German, *Bisstrich*) is longer than a hyphen. It can be used between numbers, between words, and as a placeholder. Be careful to distinguish it from a minus sign.

In Numbers The most common use of an en dash is to indicate a range (only with numerals, not with numbers which are written as words):

> The years 1960–1977 saw...
>
> measures 300–450 cm in length...

An en dash is not used with "from" or "between." Thus, change "between 75–100 kg" to "between 75 and 100 kg", and "ranged from 3,000–4,000 IU/kg body wt" to "ranged from 3,000 to 4,000 IU/kg body wt."

Do not use an en dash when a plus or minus appears with either of the numbers (e.g., use "−100 to +110 °C" rather than "−100–+110 °C").

In Text En dashes are used to link combinations or to indicate relations in adjectival compounds consisting of two or more parts of equal status, i.e., when the first part of the compound does not modify the second, but rather the noun. They often stand for "and" or "to." Examples are:

oil–water interface, dose–response curve, Bose–Einstein condensation, He–Ne laser, space–time continuum, I–V plot, solid–liquid interface, spin–orbit coupling

Do not insist on this usage if an author has consistently used hyphens since it is uncommon even though correct.

Other Uses An en dash is used to indicate a single chemical bond (CH_3–CH_2COOH). In tables, an en dash can be used to indicate that data are missing or unavailable; however, do not add dashes if the author has not provided them.

5.7 Em Dash

An em dash (in German *Gedankenstrich*) is most commonly used to set off an appositive or parenthetical expression, where it serves to emphasize the importance of the set-off phrase, in contrast to parentheses, which minimize the importance. Overuse of the em dash should be avoided because the intended effect is otherwise lost.

If a phrase being set off contains internal punctuation, use of an em dash may help avoid ambiguity.

Sometimes an author uses an em dash in place of a colon, but especially in titles and headings a colon is often more appropriate (e.g., Lupus Erythematosus: Recent Clinical Findings).

5.8 Parentheses, Brackets, and Braces

Generally, if these "fences" are used to set off mathematical content, parentheses, i.e., "(" and ")", are used first; if there are fences outside these, use brackets, i.e., "[" and "]"; if a third set, then braces, i.e., "{" and "}". In running text, brackets can be used inside parentheses, especially for reference numbers.

Specific uses of brackets are:

The word "sic" is always enclosed by brackets.

Reference numbers are set in brackets when the numerical system of referencing with numbers on the line is employed (see Sect. 10.2.1).

In chemistry, brackets are used to indicate concentration (see Sect.12.7.2). For example, $[O^2]$ means the concentration of oxygen ions.

In mathematical expressions, the use of superscript or subscript parentheses etc. does not affect the use of those on the line, e.g., $(x + y^{(n+1)})$.

5.9 Quotation Marks

5.9.1 When Quotation Marks Are Used

Quotation marks enclose a direct quotation up to several lines in length. Longer quotations are generally extracted and displayed, the quotation marks being omitted unless it would not otherwise be clear that the material is quoted.

Quotation marks (not italics) indicate uncommon or unusual usage at *first* mention. Similarly, use quotation marks to indicate that a word or phrase is being referred to as a term rather than being used in its meaning.

Quotation marks enclose the name of a published document or a part thereof when referred to in running text, e.g., an article from a journal, newspaper, etc. (the name of the journal or newspaper is italicized); a chapter or part title or a heading in a book (the book title is italicized); or a section within an article or chapter (e.g., see "Materials and methods").

5.9.2 When Quotation Marks Are Not Used

Do *not* use:

The phrase "so-called" *and* quotation marks

Quotation marks for emphasis or for foreign words (use italics, instead)

5.9.3 Differences Between British and American Usage

In American English:

Use single quotation marks only within double quotation marks.

Put closing quotation marks after (to the right of) periods and commas, before colons and semicolons, and before or after exclamation or question marks, depending on whether the mark is or is not part of the quoted material.

In British English, single quotation marks are often used instead of double quotation marks; follow the author's preference. British usage differs from American in that closing quotation marks may come before or after periods or commas, depending on whether they are part of the quoted material.

5.10 Apostrophe

Use apostrophes to indicate the possessive case. It may also be used in the plural of words or letters spoken of as such (e.g., "He uses too many *and* 's.").

Do not use an apostrophe to indicate the plurals of numbers, abbreviations, and acronyms. For example, *delete* the apostrophe in "four CSF's were isolated."

5.11 Ellipsis

When a word, phrase, or line is omitted from a quoted passage, use an ellipsis to indicate the omission. The ellipsis consists of three dots, unless the material omitted includes the end of a sentence, in which case a fourth dot is added to indicate the omitted period. Ellipses are sometimes used in mathematical expressions, where they are usually set on the line).

6 Scientific and Springer Nature Style

6.1 Capitalization

As a rule, capitalize proper nouns (names, brand names), epithets, and some geographical and historical terms as outlined below. Other terms should not be capitalized.

6.1.1 Parts of a Publication

Within the text of a chapter or article, capitalize nouns and abbreviations used to refer to a specific part of a book or a journal article:

General: Table 4, Fig. 6, Eq. 3, Sect. 4.37, Introduction, Appendix

Book specific: Contents, Preface, Foreword, Chap. 5, Vol. 4, Index

When any of these terms are used generically, they are not capitalized or abbreviated. Furthermore, do not capitalize "title page," "the table of contents," "page," and "cover."

6.1.2 Names

Proprietary or brand names are capitalized.

Valium, Xerox, Vaseline, Seconal, Teflon, Dacron, Sephadex, Plexiglas

Generally, retain the symbols ®, ©, or ™ that may come after a brand or proper name if they are provided by the author. Do not add them where they are not present. (Cf. A.2 Appendix for the Nautre Portfolio journals).

Proper names are usually capitalized when used as eponyms or possessive adjectives:

Parkinson's disease, Erlenmeyer flask, Gram's stain, Cushing syndrome

but follow the author's usage of lowercase if this is accepted alternate usage:

petri dish, gram-negative bacteria (acceptable according to *Webster's*)

Follow the *Oxford English Dictionary* or *Webster's* on the capitalization of verbs, adjectives, or units of measure based on proper nouns. Examples of usage are:

parkinsonism, pasteurize, Mendelian traits, dalton, Hamiltonian, Gaussian

For author's names, always follow the author's capitalization and spacing (e.g., particles such as "van"). If usage of other names is inconsistent, make them consistent if you are reasonably certain which version is correct.

6.1.3 Specific Terms

The practice of treating "specific" terms as names, and therefore capitalizing them, should be applied sparingly. For instance, references to "the editor," "the author," and "the editorial board" should be lowercase. Do not capitalize general terms appearing with numbers or letters designating a specific part of a series (e.g., day 1, grade II, group A) or modified by proper names (e.g., the Wistar rat). Official titles should, of course, be capitalized.

6.1.4 Historical and Geographical Terms

Capitalize historical periods and events:

the Middle Ages, the Crusades, the Industrial Revolution, the Stone Age, the Second World War (or World War II)

but do not capitalize centuries (e.g., in the eighteenth century).

Be careful when capitalizing geographical terms. *Merriam–Webster's Geographical Dictionary* is our standard reference. Here are some examples:

Central America vs central Europe

Southeast Asia vs the southeast of England

South Africa vs southern Africa, southern Asia

North Pole, North Africa vs northern Germany

the Continent (i.e., vs the UK or Ireland)

Near East, Middle East

6.2 Abbreviations

6.2.1 Unit Abbreviations

Use the accepted abbreviations of units. The use of SI units is preferred but do not make any general changes in an author's use of other units without checking the information in the science-specific chapters or consulting your Springer Nature copy editing contact. Units are written lowercase and in full when used independently of values. No changes to units should be made in quotations or historical material. They are abbreviated when used with values (the well-known exceptions regarding capitalization are Celsius, Fahrenheit, and Ångstrom). A good general source on units is http://physics.nist.gov/cuu/Units/.

When a unit measure appears with a number, use a numeral and the appropriate abbreviation:

We administered 10 mg of propranolol. (not ten milligrams)

The patient had suffered from back pain for 3 weeks. (not three)

If 10 ml of 15% dextrose solution is...

This also applies when a virgule is used:

...yields 3 mg/ml

Units are separated from each other by spaces:

5 mA h or 3 mg ml^{-1}

Units of measure are not abbreviated when they are not preceded by a numeral:

...is measured in milligrams per liter.

Never pluralize a unit abbreviation:

4 g, 300 ml, 15 mol, 3.5 M, 45 µg/1,000 ml

Unit abbreviations are always spaced from the number (as in the above examples) and are not followed by a period. This also applies to degrees Celsius and Fahrenheit (34 °C, 90 °F). Degrees, minutes, and seconds of arc are closed up (90° 45' 32") as are percent and permille (30–40%). Any exceptions to these rules are noted in the subject-specific chapters of this manual.

Unit abbreviations are not repeated in a range (e.g., 30–40%, 100–400 °C). Note: K for kelvin is not used with a degree sign (e.g., 300 K).

The item being measured should follow the correct unit: for example, either "an aqueous solution of 0.3 g NaCl/ml" or "an aqueous solution of NaCl (0.3 g/ml)." An exception can be made for drug dosages in superficial editing, such as "15 mg/kg kanamycin" (which means "15 mg kanamycin per kilogram body weight").

The use of "of" with a unit abbreviation and a quantity is not necessary but it may occasionally be used for clarity (3 g of chloride, 30,000 IU of penicillin). It is not used in phrases such as "44 mmHg." Obviously, do not delete "of" following a percentage: 30% of the patients.

Some abbreviations may be either upper- or lowercase, such as l / L, *N* / *n* and *P* / *p*. Follow the author's usage.

6.2.2 Nonunit Abbreviations and Acronyms

Generally, nonunit abbreviations should be spelled out at first use, the abbreviation following in parentheses. Some that occur very frequently need not be explained; of course, any definition given should not be deleted.

Once introduced, an abbreviation should be used throughout a manuscript. There are a number of common-sense limitations, however:

– Do not insist on the use of an abbreviation or acronym if the author has written the term out in the majority of cases.

– If a considerable amount of text separates uses, the abbreviation may be reexplained.

– An otherwise abbreviated term may also be written out in full when it is used as the first word in a major section.

– An abbreviation or acronym may also be mentioned once (but not "used" in the sense of introduced), normally at first occurrence, and then not used any more.

Abbreviations and acronyms do not need to be used consistently in every article in a journal or in every chapter in a multiauthor book; generally, each article or chapter should be viewed independently. Consistency is required within one manuscript.

Some publications may contain a list of abbreviations. The abbreviations included there may be defined in the text at first mention.

The addition of s to pluralize an acronym or nonunit abbreviation is optional.

Commonly used abbreviations of foreign words and expressions, e.g., those found in the body of *Merriam–Webster's*, are not italicized (examples are et al., etc., cf., viz., i.e., and e.g.).

Abbreviations are frequently used if another part of the same work or a specific part of another publication is referred to (see Sect. 6.1.1); examples are:

Vol., p., Chap., Sect., Fig., Eq. (Do not abbreviate "Table.")

Do not abbreviate the units "day," "week," "month," and "year." in running text (abbreviating "day" to "d" and year to "yr" within a table may be permissible) The names of months may be abbreviated. Do not abbreviate the names of countries in running text; the only exceptions are USA (noun) or US (adjective) for United States and UK for United Kingdom. Abbreviate US state names using the two-letter postal abbreviations (see Table A.1).

A genus name should be written out in full at first use, but may be abbreviated to its first letter (capital, with a period) when subsequently used in a binomial.

6.3 Numbers

6.3.1 General

In running text with nonunit quantities, use words and numerals for numbers as described here:

one to ten, 11, 12, 13,..., *n*

first to tenth, then 11th, 12th, 13th,..., *n*th (*not* 11^{th} etc.)

twofold to tenfold, then 11-fold, 12-fold, 13-fold,..., *n*-fold (*but* 4.5-fold)

one half, three tenths, 1/11, 4/12

one million, 1.3 million

Exceptions to this usage: do not change an author's usage if they consistently begin using numerals at 10 (instead of 11) or, in special contexts, consistently use numerals or writes out numbers according to another system. For example, do not change the author's usage if the numbers in a list containing numbers larger and smaller than ten have all been styled according to the highest number, as shown in these examples:

On the 1st day four rats died, on the 2nd ten, and on the 3rd 14.

On the first day 4 rats died, on the second 10, and on the third 14.

Use a numeral if the context of the number is mathematical:

a factor of 3, a multiple of 5, a ratio of 1:4 (but four times as much)

Use a comma to separate groups of three digits to the left of the decimal point (if any) for large numbers of five digits or more (i.e., all numbers larger than 9999 such as 64,000 and 1,239,999). Be sure that a comma is not used in place of a decimal point. For usage in some publications, sciences and in camera-ready copy, see Sect.13.2 and A.2 Appendix for the Nature Portfolio journals.

Exceptions:

- No changes to any numbers in title, author info, affiliation, acknowledgements, quotations
- No changes in reference section
- No changes to strings that contain 5 or more numbers, but also other characters

Numbers should always be expressed as words at the beginning of a sentence. Simple rewording may be the easier correction (e.g., "Thirty of 68 patients..." or "Of 68 patients, 30..."). Chemical names constitute an exception to this rule ("5-Methylsulfadiazine was added...").

Repeat all digits in a series or range (e.g., 1960–1975 and 29,143–29,147).

6.3.2 Units and Quantities

When a cardinal number is used with a unit, use the numeral (and unit abbreviation if applicable):

30 ml, 44%, 7 years

but with ordinals below 11 either usage is acceptable:

the fourth day or the 4th day, but the 11th day

Always use numerals with decimals:

0.03, 33.3, 4.5-fold

In decimals less than one, a zero precedes the decimal point:

0.33, 0.50

except in special cases where the quantity can never be greater than unity and customary usage allows its omission (if this is the author's preferred style):

$p < .05$ or $p < 0.05$, $r = .86$ or $r = 0.86$ (accept either lowercase or uppercase p and r)

Use numerals when referring to something that is numbered:

p. 33, step 4, day 7, point 5, group 3, phase II

A specific case is the use of numerals when referring to specific numbered parts of a publication:

Part III, Vol. 7, Chap. 4

When one number immediately follows another with no intervening punctuation, and both would normally be written in numerals, one may be written as a word for the sake of clarity:

11 40-ml flasks or eleven 40-ml flasks

In rare cases, especially in non-scientific context (e.g., an interview, direct or indirect speech), numbers and units are better written out:

We had been waiting hardly five minutes before the doctor arrived.

6.3.3 Other Uses

Time Use numerals for the time of day:

12 noon, 6 p.m., 7:55 a.m. (US) or 7.55 a.m. (UK), 1930 hours, 07:00

Dates Use numerals for years and decades:

1964, the 1960s (not the 1960's, the '60s, or the sixties)

but not for centuries, which are always spelled out and lowercase (e.g., the eighteenth century). Use numerals for the day of the month and the year, but use letters for the month:

3 January 1977 (preferred form) or January 3, 1977

3 Jan. 1977

Money Use numerals for money, specifying the country of origin where ambiguous (e.g., for dollars):

US $6.60, Australian $3 million, 100 Swiss francs, £10, 5000 € (or euros)

Names Spell out numbers that are part of a proper name (e.g., Fifth Symphony).

Product codes For numeric strings denoting specific products (e.g., grants, license numbers, machine types, software codes, library accession numbers) retain the style given.

Telephone numbers Telephone numbers should be deleted from the manuscript.

6.4 Special Type

6.4.1 Italics

Italic type is the primary format used to indicate emphasis. As a rule, the copy editor should not add emphasis. If the author makes apparently excessive use of italics for emphasis (*numerous* times per page or entire paragraphs), delete the italicization of the seemingly less important words or phrases and inform the author of your reason for doing so (e.g., overuse of special type is hard to read, thus weakening its impact and defeating the original purpose of emphasizing key words or phrases).

The author may wish to italicize a term (or even a name) at its first mention or with its definition; this is permissible if not overused.

Foreign Words Foreign words, i.e., words of foreign origin that have not come into general use in English, are italicized. Terms listed in the main section of *Merriam–Webster's Collegiate Dictionary* (this applies also for British English!) are regarded as being in general use and are not italicized. These include:

in vivo, in vitro, ex vivo, per se, in situ, ab initio, vice versa, tour de force, en route, ad hoc, laissez-faire, a priori, i.e., e.g., et al.

Also, do not italicize Latin anatomical names, foreign names, or proper nouns (e.g., Max-Planck-Gesellschaft or Max Planck Society, Deutsche Forschungsgemeinschaft).

Names of Publications In text, italicize titles of books, journals, theses, periodicals, newspapers, and plays. Do not italicize titles of articles or chapters, names of sacred writings or famous documents (e.g., Bible, Koran, Talmud, Magna Carta).

Other Uses Genus and species names are italicized. Many characters are italicized in the various subjects; see the subject area chapters for further information.

Do not italicize letters used to indicate a shape (e.g., T square, S curve, Z-plasty). It is an option to italicize a word or letter when it is used in the sense of its being the word or letter (He uses too many *and* 's.); it is, however, equally permissible to use single quotes for this case (He uses too many 'ands'; cf. A.2 Appendix for the Nature Portfolio journals).

6.4.2 Boldface

Boldface should not be used for emphasis.

There are no grammatical uses for boldface, only technical and typographic ones (e.g., in figure legends and table titles; see Sects. 7.3 and 8.3). For special uses of boldface, see the discipline-specific chapters.

6.4.3 Small Capitals

There are no grammatical uses for small capitals, only technical and typographic ones. There are a few uses where small capitals are required such as:

D- and L-dopa (to indicate dextro or levo configuration; see Sect. 12.1.5.2) .

The units AU, AMU

Oxidation states in the context of astronomy (Fe II, Cu I)

Antibody heavy and light chains in (VH and VL)

The following examples show optional uses of small capitals:

B.C., A.D. Full capital letters are now preferred, but small capitals (with or without periods) are also acceptable.

A.M., P.M. may be used in American English (but see also Sect. 6.3.3).

6.4.4 Underlining

Underlining should not be used for emphasis. There are no grammatical uses for underlining, only technical and typographic ones. Extracts or historical material that include passages which are underlined should be retained as such.

6.4.5 Special Fonts

In nonmathematical contexts, special fonts are used to render some items such as program code, phonetics, and genetic sequences. These are rendered in the nonproportional font Courier. For the use of fonts in a mathematical context, see Chap. 13.

7 Graphics and Figures

7.1 General

A graphic is an object that is drawn or photographed; it does not consist solely of characters and thus cannot be keyed (as normal text).

In Springer Nature publications, graphics are, in terms of content, normally figures, schemes, or structures. This chapter covers graphics in general and figures in particular. Outside of disciplines such as chemistry, in which structures and schemes occur frequently and are cited and numbered as such, all such graphics are referred to as figures (see also Chap. 12 on chemistry). All figures must be numbered with Arabic numerals.

In exceptional cases, equations, tables, or individual characters may technically also be produced using graphics. These items are nonetheless equations, tables, or text, and are never called figures.

If an author has labeled tabular material as a figure, it should be relabeled a table. Be sure to number it in the proper sequence, to renumber the subsequent tables and figures, and to change *all* relevant text citations (some may be far from the main citation). Nonetheless, before beginning such a task, give serious consideration to whether renumbering is really necessary (i.e., whether the criteria mentioned here are fully met).

7.2 Citation in Text

7.2.1 Entire Figure

All figures must be cited in the text by number. "Figure" is abbreviated to "Fig." (and "Figures" to "Figs.") except at the beginning of a sentence. Each figure must be cited individually; if several are mentioned together, cite them as, e.g., "Figs. 3, 6, 7, and 8".

The order of *main citations* of figures in the text, which is approximately where the figure will appear in the publication, must be sequential. Please note that extra secondary citations may occur between the main citations. This means that if a correct sequence of citations exists in the text, even if interspersed with extra citations of other figures, assume the sequence is correct and assign main citations accordingly. For instance, if the citations occur as 2, 4, **1**, **2**, 5, **3**, **4**, **5**, 7, **6, 7, 8**, take the citations marked in boldface here as the main ones. *Only* if there is no possible interpretation of sequence should one of the following options be used to achieve sequential citation:

– Insert a citation for the previously uncited figure at what appears to be the most appropriate place in order to create sequentially numbered citations.

– Renumber the figures (and all citations) to create sequentially numbered citations and figure numbering.

Adding citations, where possible, is the preferred means of making citations and numbering correspond because it avoids the renumbering of figures (i.e., of text citations, legends, and graphics). See also A.2 Appendix for the Nature Portfolio journals.

The sequential numbering applies to figures in an entire article or chapter, including in an appendix. Do *not* number figures in an appendix to an article or chapter separately (e.g., not as Fig. A1; cf. Sect. 3.4.1).

7.2.2 Parts of Figures

A figure may consist of a number of separate parts; these are normally identified by lowercase letters (for exceptions, see Sect. 7.3.2). Even if the various parts are referred to at different points in the text, there is still only one main citation of the figure as such. All the other references to either the entire figure or to individual parts are secondary citations.

If a text citation is to a specific part of a figure, the numeral and the letter denoting the part are closed up. If several figure parts are mentioned, link them with a comma or en dash as appropriate (e.g., Fig. 17b, c or Fig. 17c–e).

7.3 Legends

7.3.1 General Information

Each figure must have a descriptive legend (or caption). If the author has not provided one, devise one from the information in the text. The following describes how to capture and style the information in the legend, but not the typography (e.g., boldface); for typography details, please follow the appropriate layout instructions.

A legend should read easily; articles may be omitted and some grammatical rules broken, but a legend should not read like a telegram. It follows regular text rules for abbreviation, hyphenation, capitalization, and punctuation. It should contain enough information for the figure to be identified and understood without reference to the text. Do not begin a legend with a superfluous phrase such as "Illustration of, " "Diagram of," or "Graph of." However, such terms as "photomicrograph" and "electron micrograph" specify the method of production and should be retained.

Extensive discussion, however, is more appropriate in the text than in a figure legend. In intensive editing, consider asking the author to confirm that such information should be moved to the text if a legend is very long.

A legend begins with "Fig." followed by the number. The number is not followed by a period. A stain, magnification, and reference/source come at the end of a legend and in this order. Note that a multiplication sign comes before the number indicating magnification. A legend may have end punctuation (cf. A.2 Appendix for the Nature Portfolio journals)

There are two important qualifications to this general rule regarding the identification of figure parts.

- Other designations can be retained where they are supplied if the whole figure can be considered to be a single figure with various identified items in it. As a rule of thumb, if the image of a figure contains eight or more small parts that themselves appear to be meaningfully identified with other combinations of letters and numbers, as traditionally have been used in plates/tableaus (such as a, A, b, B, c, C, d, D etc., or analogously a, a' etc. or a1, a2 etc.), then retain these designations but do not consider them to identify figure parts, but to

label items in a figure. Such designations also include left, right, top, bottom, upper, and lower. Such alternative designations may never be used for "real" figure parts.
- As stated in Sect. 7.4, adjust the *legend* to ensure that the designations in legend and figure agree but see special instructions on capitalization.

If a legend is introduced by a general comment, use the following form:

Fig. 3 Electron micrographs of epidermis. **a** Facial region. **b** Neck. **c** Chest

If the legend does not begin with a general comment (some journals have a requirement for a figure ‘title’), a space follows the figure number; the legend as such begins with the letter identifying the first figure part (not followed by a period):

Fig. 3 a Electron micrograph of facial epidermis. **b** Electron micrograph of neck epidermis. **c** Electron micrograph of chest epidermis

If several parts of a figure are identified in a single sentence, the letters designating figure parts may follow the corresponding description in parentheses or precede the description without parentheses. Thus, the example given above could also take one of the following forms:

Fig. 3 Electron micrographs of epidermis of the facial region (**a**), of the neck (**b**), and of the chest (**c**)

Fig. 3 Electron micrographs of epidermis **a** of the facial region, **b** of the neck, and **c** of the chest

7.3.2 Symbols and Definitions

Abbreviations and symbols that appear as labels in a figure or any of its parts must be identified in the legend. Use words to describe any symbols, as these are difficult to render in a legend (even when a character is available, its rendering will probably differ somewhat from that in the graphic itself).

Fig. 3 a Formation of the layer of Chievitz (Ch). Ganglion cells (G) are still present in the outer neuroblastic layer. **b** Transformation of the layer of Chievitz into an inner plexiform layer (IPL). The arrow indicates horizontal and vertical cellular processes. INL inner neuroblastic layer, ONL outer neuroblastic layer

Fig. 4 Temporal order and duration of the various development processes in the retina of *Tilapia* (T), the chick I, and the mouse (M). Numbers 1–4 indicate the developmental stages: 1 proliferation and migration, 2 general differentiation including layering, 3 specific differentiation, 4 neurochemical differentiation. Horizontal bars indicate onset...

Clarify with your Springer Nature contact whether color figures will be printed in black and white. In that case, the caption should not refer to color. Try to refer to symbols or positions instead. If this is not possible, add "(Color figure online)" at the end of the caption.

Fig. 5 Map of the study area. Blue and green dots are the reflectors of the geodetic monitoring system; red dots indicate the position of other reflectors. The yellow star marks the position of the SLF snow height measuring station LUK-2 ... (Color figure online)

7.3.3 Sources

If a figure is reproduced from a previous publication, the source appears as the last item in the legend (see example in Sect. 7.3.1). This source should, unless otherwise specifically required by the copyright holder, be in the form of a reference citation and include wording such as "Reproduced with permission from". The original publication must be included in the reference list. If the work does not appear in the reference list, add it to the list, leaving an ordinary citation in the legend. The copyright holder may require specific wording; in such a case, follow any instructions given on accompanying permission forms.

A legend pertaining to a figure adapted from a previously published one should include wording such as "Adapted from," "Modified from," or "Redrawn from" as well as a reference, all cited within parentheses. If the source is an individual or company and the material is not copyrighted, write "Courtesy of."

7.4 Artwork

If original artwork is sent to you, treat it with great care. Never alter an original in any way. Indicate any changes on a copy. Query the author if the figure quality seems inadequate. Clarify with your Springer Nature contact whether figures will be reproduced in color.

For some exceptional publications, labeling on graphics is copy edited at the same level as the rest of the text (i.e., in such cases level 3 copy editing in the text also applies to labeling on graphics; see A.2 Appendix for the Nature Portfolio journals).

Standardly, however, only minimal changes if any are made to graphics. Unless you know the graphics will be edited fully, relettering and rearrangement of labels is to be avoided wherever possible. This means:

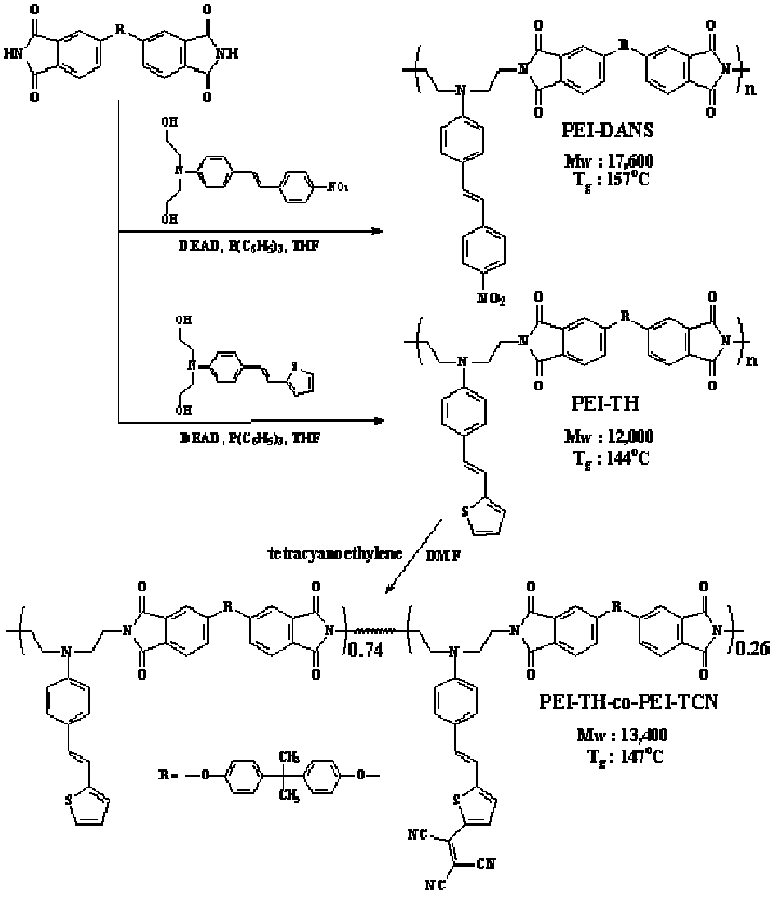
- Where there are discrepancies, change the legend not the figure to ensure consistency.
- It is not necessary to make the following consistent between graphics and text or legend: italicization, British–American spelling differences, hyphenation, and capitalization
- Specifically, do not correct figure part labels for consistency: if uppercase labels are used on the figure leave them unchanged, but use lowercase letters in the legend.
- Only correct blatant mistakes which would lead to misunderstandings. Do not correct italicization, British-American spelling differences, hyphenation, simple spelling mistakes or hyphenation errors, but do change European-style commas to points in decimal numbers or significant spelling errors which change the meaning.

7.5 Schemes and Structures

Chemical publications often contain many graphics that are not designated as a figure (and not counted as such) but are called either a scheme or a structure. Although the definitions given here are correct, do not change an author's usage if he has used the category "Figure" to designate either schemes or structures; leave the author's usage.

7.5.1 Schemes

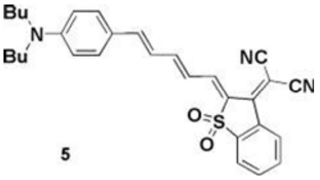
A scheme consists of a series of chemical reactions (whereas a chemical equation usually only has one). Note that there are several arrows denoting the reactions. A scheme must have a legend. For an example of a scheme, see Scheme 7.1.



Scheme 7.1 A scheme

7.5.2 Structures

A structure, as shown in Structure 7.1, often has a boldface number but never has a legend. There can be more than one structure in the graphic but they are not connected with arrows.



Structure 7.1

In contrast to figures and schemes, structures do not float, but must be placed at a particular point in the text, much like an equation.

8 Tables

8.1 Definition

A table is a compilation of data in cells, laid out as rows and columns, which is identified by a number and a title. Tables are set or keyed, not drawn, although a table may itself contain drawings, e.g., of chemical structures, within a cell or in the legend or notes. The purpose of tables is to convey information in an explicit, informative, and concise fashion. Each table must be a self-contained entity. Forms and questionnaires submitted with a manuscript should normally not be transformed into tables but be handled as facsimile figures.

8.2 Citation in Text

All tables are numbered with Arabic numerals. The order of main citations of tables in the text must be sequential. If main citations are out of order, then do one of the following:

– Insert new main citations for the tables not cited in sequential order.

– Change the numbering of the tables.

Adding citations is usually the easier way of making citations and numbering correspond. Change such phrases as "the following table" or "the table below" since the actual placement of the table in the publication will differ from that in the author's manuscript.

Any tables in an appendix of an article or chapter are standardly numbered sequentially in continuation from those in the preceding main body of text; they should *not* be called, for example, Table A1. However, there are certain journals which reset the numbering to 1 for tables in the Appendix and give them a separate designation (e.g., Extended Data Table 1). Clarify with your Springer Nature contact if you are unsure whether this applies to your publication.

8.3 Table Legend

Each table must have a descriptive legend (or title). If the author has not provided one, devise one from the information in the text. The legend begins with "Table" (not abbreviated), followed by the number. The number is not followed by a period, and the legend may have end-punctuation.

Ideally, table legends should be concise. In intensive editing, if it is obvious that some of the information in a long legend can be moved to the table notes or incorporated in the text, do so; otherwise leave the legend as it is. A table legend never includes a footnote.

If an author has specified that a table is borrowed unchanged from published material that appears in the reference list, give a reference using the same system as in the text. If the table is adapted, however, retain wording such as "adapted from," "modified from," or "after." If the work in which the table was originally used does not appear in the reference list, add it to the list, leaving a citation in the table legend. The copyright holder may require specific wording; in this case, follow any instructions given on accompanying permission forms.

8.4 Table Header and Body

The table header (all of the rows of column headings at the top of the table) and table body consist of columns, rows, and the cells where they intersect. It is vital that the precise arrangement of cells in columns and rows be adhered to.

8.4.1 Column and Stub Heads

With very few exceptions, the first row or rows in a table contain the column heads. In tables with column heads, each column must have a column head of its own (with the possible exception of the first column containing the stub, or side, heads). There may be more than one row of column heads, and major column heads may group (or straddle) two or more columns. Note, as in Table 8.3, that the column head is placed in the top row if only one of several rows of column heads is used.

Stub heads are entries in the left-hand column of a table that serve as headings grouping information that appears in that or the other columns.

Column and stub heads should be brief. If they are too long, try to footnote some information. The first word in each heading is capitalized. Column and stub heads are not end-punctuated. All headings are flush left, with subordinate entries to stub heads indented. Do not use "%" or "*n*" alone as a *major* column head; either change to "Percent" or "Number" or create an appropriate group heading (such as "Samples").

If all the data in a row or column are expressed in the same unit, the abbreviation of the unit (in parentheses) follows the head.

8.4.2 Cell Entries

The first word of each entry is capitalized (exceptions include abbreviations and gene names or symbols that should be lowercase according to accepted nomenclature). There is no end punctuation. Abbreviate all units (whether with or without numerical quantities) for which there are standard abbreviations, and use numerals for all numbers (e.g., 2 patients).

Cells may be empty. Leave any en dashes that may have been used to indicate the unavailability or inapplicability of data (providing that plus or minus signs are not otherwise used in any cells), and use zeros to indicate the value "none." Abbreviations such as NA (not available or not applicable), ND (no data), and NT (not tested) may also be used where appropriate and should be explained in the table notes. Never use ditto marks.

In intensive editing (category 3), check that the data given in tables agree with the data in the text and that there are no obvious errors in the totals and percentages.

8.5 Table Notes

Table notes may consist of the following items, presented in this sequence (but note that in many journals items 1 and 2 are placed at the end of the table legend):

1. Any comments of a general nature
2. Explanations of nonunit abbreviations in a run-in list
3. Any footnotes, arranged first by asterisk, then by letter

The list of abbreviations must include all nonstandard abbreviations. The items in such a list are separated by commas. An extensive list may be introduced by the word "Abbreviations" (followed by a colon).

Table footnotes are identified by superscript lowercase Roman letters, with the exception of footnotes detailing *p* (or *P*) values, which are usually indicated by one or more asterisks. The usual order of citation of footnotes in the table body is vertical by column, beginning at the left. However, retain any other consistent system used by the author.

8.6 Examples

The examples shown in the tables in this chapter illustrate some of the important points mentioned above. Table 8.1 shows grouped column heads. Tables 8.2 and 8.3 demonstrate complex table notes and the use of *n* and % as column heads. Table 8.4 shows information that has been submitted as a list but that is more effectively shown as a table. *Note*: The appearance of the tables in print or online will be determined by the relevant layout (lines surrounding cells or between rows, run-on footnotes). In these examples, a uniform appearance has been adopted to illustrate the points about content, rather than layout. See also A.2 Appendix for the Nature Portfolio journals

Table 8.1 Levels of classes and subclasses of immunoglobulins in mice of the H and L lines before immunization and 14 days after injection of 5,108 IU substance K

	Serum immunoglobulins (mg/ml)			
	Before immunization		After immunization	
	H line	L line	H line	L line
IgM	0.17	NT	0.40	NT
IgG1	0.78	0.31	3.60	0.24
IgG2	1.45	0.61	6.00	0.75
IgGA	0.54	0.34	0.9	0.46
Total	2.94	1.26	10.90	1.45

NT not tested

Table 8.2 Effect of coadministration of ftorafur and uracil on sarcoma 180

	FT (mg/kg)	FT (mg/kg)	Mean tumor wt.±SD (g)	Body wt. change (g)
Control	0		1.01±0.18	+7
U+FT	40	2	0.23±0.24	+4
FT alone	40	1.5	0.64±0.25	+4.5
U+5-FU	20	5	0.18±0.04	-7

The tumor cells (2×10^{7}) were inoculated into the subepidermal tissue of ICR mice on day 0. The drugs were given orally on days 1–7. On day 10, the tumor was removed and weighed
FT ftorafur, *5-FU* 5-fluorouracil, *U* uracil

Table 8.3 Effect of combination chemotherapy in DMBA-induced mammary cancer in the rat

Chemotherapy	No.	TWI on day 28	R or NC[a,b] on day 28		Change in body weight, days 0–28	Deaths[c] by day 56
		%	*n*	%	%	*n*
Control	27		0	0	+5	2
ADR[d] alone	37	52	11	30	0	3
ADR+5-fluorouracil	37	73*	23	62	+1	2
ADR+prednisolone	37	79**	30	81**	-4	2
ADR+cyclophosphamide	37	81***	20	57	0	10

Chemotherapy in this study was given alone (i.e., without adjuvant radiotherapy)
TWI tumor weight inhibition, *R* remission, *NC* no change, *ADR* adriamycin
*$p<0.05$; **$p<0.01$; ***$p<0.001$
[a] R+NC, total tumor volume <110% of initial volume
[b] Rats dying due to toxicity before day 56 were excluded
[c] Due to toxicity
[d] ADR 1.2 mg/kg day, days 1–3 and 14–16

Table 8.4 Clinical stages of acute fulminant hepatic failure

Stage[a]	Designation	Signs
I	Neurasthenic syndrome	Abnormal fatigability, depression, appropriate response, slight tremor
II	Somnolence	Drowsiness, inappropriate behavior, flapping tremor
III	Sopor	Reduced responsiveness, incoherent speech
IV	Coma	No response to verbal stimuli, decerebrate, grossly abnormal EEG
V	Deep coma	No response to pain, little or no EEG activity

[a] A comment on presentation: Although it is possible to present this information as a list (as demonstrated below), that would make the information more difficult to grasp:
Stage I Neurasthenic syndrome: Abnormal fatigability, depression, appropriate response, slight tremor
Stage II Somnolence: Drowsiness, inappropriate behavior, flapping tremor
Stage III Sopor: Reduced responsiveness, incoherent speech
Stage IV Coma: No response to verbal stimuli, decerebrate, grossly abnormal EEG
Stage V Deep coma: No response to pain, little or no EEG activity

8.7 Inline Table

An inline table (unnumbered) is not a table per se but rather a brief tabular presentation of information such as might make up a complex list. It differs from a table in that it does not have a legend or notes, is not numbered, and must appear at a specific location in the text (as does a list or an equation). It is similar to a table in that information is displayed in a combination of columns and rows (technically, in Word for example, an inline table consists entirely of cells in the table function).

9 Lists, Footnotes, Cross-References, Quotations, and Equations

9.1 Lists

Lists can be presented in text either run-in or displayed. Items should (ideally) be parallel in construction, e.g., all nouns, all prepositional phrases, or all complete sentences.

9.1.1 Run-in Lists

A list is usually run in as part of a normal sentence if it is short, the items are simple and brief, and their sequence is of no great importance. Longer lists are usually displayed.

Normally, no item identifier (e.g., number; see Sect. 9.1.3) is used; if the items in a run-in list are identified, Arabic numbers or lowercase letters (enclosed in parentheses) are usually used: "The current evidence for a beneficial effect consists of (1) the marginal value of ovarian ablation, (2) the enhancement of the value of ovarian ablation by the addition of prednisone, and (3) the observations that...."

9.1.2 Displayed Lists

If a list is long, the items are complicated, or the sequence is important, displayed presentation is preferable. This means that the list is visually set off from running text in that each entry

– Appears on a line of its own

– Consists of a label and content

– Begins with an initial capital letter

but that the list, in contrast to a table, appears at a specific point in the text (i.e., there are specific words preceding and following the list). If a list has two columns and is long or contains numerical data, consider making it into a table with a number and title (see Table 8.4).

9.1.3 Labeling of Items in Displayed Lists

Items in displayed lists may be ordered (i.e., numbered or lettered), or unordered (i.e., preceded by a symbol that does not indicate a sequence).

Items should always be numbered (preferred) or lettered if (a) the author mentions the number of items listed or (b) the list includes a sublist. A number is followed by a period; a letter is enclosed in parentheses. Each entry in a sublist should be introduced by a dash or by a lowercase letter enclosed in parentheses:

Hypermotility can be symptomatic of:

- Ischemia

  (a) Adhesions or bands

    (b) Vasculitis

        I Arteriosclerosis

    (c) Abdominal angina

        II Radiation

2. Fear and emotion
3. Carcinoid syndrome
4. Thyrotoxicosis
5. Allergic reactions

9.1.4 Style

Introduce a displayed list with a colon following the last word of the preceding text. The first letter of each item in a displayed list is capitalized. If any item comprises a full sentence, end-punctuate all items with a period; otherwise do not end-punctuate.

An item in a list may even be long (in rare cases, including several paragraphs, but not covering pages). Important words at or near the beginning of each item may then be italicized.

Certain of these factors differ between the two systems and have been the subject of a number of optimization studies:

1. The effects of photoattenuation account for one of the greatest differences between the two systems of ECT. The linear attenuation coefficient, being a function of photon energy, varies relatively little between different tissues for positron-produced photons....
2. That the number of counts in a projection element must be independent of the depth of the source in the field of view is a requirement of all conventional reconstruction algorithms. Even after allowing for attenuation, this requirement is not met by the SPC system, in which....
3. The sensitivity of a tomographic system is of considerable importance since the noise introduced by the reconstruction is approximately an inverse square function of the total number of detected events. In the SPC system....

9.1.5 Definition Lists

Definition lists are displayed lists in which the item label is a word or character explained by the text of the item. This kind of list should be used only with lists that truly provide definitions or explanations. The material shown in the footnote to Table 8.4 could be displayed as a definition list, in which the stages are defined.

The term and the explanation should be shown similar to an abbreviation list with the initial letter of the explanation capitalized.

9.2 Footnotes

In print, footnotes usually appear at the bottom of the page on which they are cited or — in some humanities, arts, and social sciences books (depending on the standard layout used—at the end of the chapter (endnotes); in digital presentations they may be presented as hyperlinked text. Footnotes may be used as frequently as required or makes sense; earlier limitations on the use of footnotes because of increased difficulty in composing pages no longer apply. Use "normal" punctuation with footnotes. Footnotes may contain bibliographic references (citations) but not the full bibliographic entries, which must appear in the reference list (cf. Sect. 10.3.9.2).

Footnotes to text, in the header, and to tables are handled differently (cf. A.2 Appendix for the Nature Portfolio journals).

– Footnotes to text. Text footnotes are cited by superscript Arabic numerals in sequential order, beginning anew in each chapter or article. Sequential numbering continues in an appendix. Footnote citation numbers appear in the text after final punctuation of a clause or sentence unless the footnote refers to a specific item in the sentence.

– Footnotes in a header. The information that is displayed as a footnote to an article or chapter header is determined by the appropriate/relevant layout.

– Footnotes to table entries. The indexes for footnotes in tables are superscript lowercase letters, although footnotes relating to statistics may be indicated by asterisks (Sect. 8.5).

– Footnotes to equations. A footnote to a numbered equation is indexed by a superscript number after the parentheses enclosing the equation number. It is numbered as one of the text footnotes.

9.3 Cross-References

A cross-reference refers to another part of the same publication, specifically to something other than one of the standard numbered objects within an article or chapter (i.e., other than a figure, table, equation, scheme, structure). The words "above" and "below" should be avoided unless the information being referred to is within the same section or short chapter. Cross-references to more distant material should include the heading or number of the section, not a page number, because the latter is unknown at the time of editing.

Cross-references to chapters and sections are styled as follows:

Decimal system. Examples:

Sect. 3.1

Chap. 3

Keying. Examples:

See "Materials and methods."

See "Classification of Results" in the chapter by R. Mustermann et al., this volume.

See the chapter by R. Mustermann et al., this volume.

9.4 Quotations

Quotations may be displayed or run-in. As a rule, they should not be edited. If you find what appears to be an error in a quotation, do *not* correct it. Ask the author whether it appears in the original, suggesting that the error be corrected unless it was of consequence. The term "[sic]" should be used sparingly. Wording not contained in the original quotation should be enclosed in brackets. For punctuation of quotations, see Sect. 5.9.

9.4.1 Displayed Quotations

Longer quotations are extracted from the text and not enclosed in quotation marks. Follow the author in deciding whether a quotation is long enough to be displayed. Do not mark for a form of type (e.g., petit) or for indentation; merely identify the quoted material. The source of a displayed quotation may be cited in the sentence introducing the quotation or at the end of the quotation.

9.4.2 Run-in Quotations

Short quotations are enclosed in quotation marks and run in with the text. The source is given in parentheses, outside the final quotation marks but within the final punctuation, unless the name is given in the introductory sentence:

Most of us would agree with Csomos (1982) when he states: "The question of aetiology or causality plays an important role in scientific research in general and in clinical research in particular."

or:

Most of us would agree with the following statement: "The question of aetiology or causality plays an important role in scientific research in general and in clinical research in particular" (Csomos 1982). However, this raises several questions...

9.5 Equations

9.5.1 Numbering

Not all equations have to be numbered; as a rule, equations should be numbered only if the author has introduced the numbering. Similarly, it is not necessary for the author to refer to every numbered equation in the text (there are no main citations of equations since their location is predetermined, i.e., their placement is not variable). Check carefully, however, that no numbers are skipped or used out of sequence; in an appendix, continue numbering equations sequentially (without a letter designating Appendix, e.g., *not* Eq. A1).

Equations are generally numbered with Arabic numbers within any chapter or article. Text references to equations take the forms "Eq. 6" or "Eqs. 6, 7, and 8" or, in physics and computer science, "(6)" or "(6–8)", i.e., in parentheses without the word "Eq." (see Chap. 15). A combination of these two forms such as "Eq. (2)" or "Eqs. (2)–(4)" is also acceptable if it is used consistently. If the reference begins a sentence, the word "Equation" is spelled out. If several equations are closely related, then the equation numbering (and any text citation) may also be, for example, Eq. 7, Eq. 7', and Eq. 7" or Eq. 7a, Eq. 7b, and Eq. 7c; follow the author's usage.

If book chapters are numbered (with Arabic numerals), the equations are “double-numbered”, i.e., the chapter number is put in front of the equation number, e.g., Eq. 1.5 (fifth equation in Chap. 1). If a journal uses the decimal system of heading numbering, the equations may also be numbered section-wise, e.g., Eq. 1.5 (fifth equation in Sect. 1). In that case (and only then), a letter is used instead of the section number for the equations in an appendix (e.g., "Eq. A.2" for the second equation in the appendix).

The equation number appears in parentheses at the margin to the right of the equation. If an equation runs over more than one line, the number appears on the same line as the end of the equation. This is in contrast to the equation number of a group of equations that shares a number; in this case, the number would be on a middle line.

9.5.2 Levels of Math Markup

For some product types, there is a defined style requirement for mathematical content, while for other products only consistency in representation is required within each book, book chapter, or journal article. As a rule, these requirements correspond to the different copy editing levels (see Sect. 1.3). Here, these levels of math markup are designated standard, extended, and intensive. See Chap. 13 for detailed information on the correct handling of math content.

9.5.2.1 *Standard Math Markup*

Standard math markup is to be performed where the copy editing level is 1 or higher. The specific tasks of standard math markup are:

- Correct handling of numbers and units
- Correct use of characters for Greek and Roman letters, e.g., β ß B (beta vs. German short double s vs. uppercase b), η n, υ v, and χ x
- Correct use of characters for symbols, numbers, and letters (e.g., x ×, l 1, and O 0)
- Correct setting of numbers, signs, punctuation, units, and recognizable functions to upright
- Retention of any special formatting (bold, italic bold, upright characters within the italic default, different fonts, e.g., sans serif) set by the author
- Correct splitting up of an equation that spreads over more than one line
- Correct equation alignment over a line break

It is not expected as part of standard math markup that context-dependent single characters (e.g., single-character functions such as the differential d or subscripts acting as labels rather than variables) will be identified and set upright. Where an author has done so, however, the formatting will be retained.

9.5.2.2 *Extended Math Markup*

Extended math markup is to be performed where the copy editing category is 2 or 3. The specific tasks of extended math markup are:

- All the items listed for standard math markup
- Correct setting of variables to italic
- Ensuring the consistency of formatting in displayed equations, inline mathematics, and symbols in the text
- Correct styling and sizing of signs, operands, fences (e.g., primes, right and left angle brackets, integral signs)

In contrast to standard markup, context-dependent single characters (e.g., single-character functions such as the differential d or subscripts acting as labels rather than variables) should be identified and made consistent or corrected.

9.5.2.3 *Intensive Math Markup*

Intensive math markup includes the highest possible standard for equations and mathematics, covering specifically those points requiring a knowledge of the subject matter, mathematics, and the context surrounding the equations. It is only possible if the text is read thoroughly in its entirety and as such is only available as part of category 3 copy editing. Communication with the author may be necessary.

Intensive math markup specifically includes the points described in Sects. 13.2, 13.3.2, 13.3.3, and 13.4. Some examples of these characteristics are:

- All the items listed for standard and extended math markup
- Correct styling of single-letter functions (e.g., differential d, exponential e)
- Correct styling of the complex number i or j
- Correct styling of all single-character label subscripts (nonvariables)
- Correct styling of particle names, e.g., e (electron) and e (electron mass)
- Correct identification and styling of vectors, matrices, tensors, and other characters with dimension

10 References

10.1 General Information

References are an integral part of scientific and academic publications. They are information provided by the author to refer the reader to further publications where more information on items mentioned in the first publication can be obtained.

"Publication" as used in this connection means any document that has been made public and thus includes journal publications, standard book publications, patent information, and so-called gray literature. Gray literature refers to:

– Publications such as presentations at proceedings of which written publications are not generally available for purchase or loan (there may only be an abstract volume)

– Personal and institutional Internet pages (as opposed to the Internet sites of science publishers), for which there is no assurance of permanence

– Publicly available prepress versions

"Publication" does not refer to information that an author has not made public (i.e., available to a larger group) or does not intend to publish. This includes references to e-mail or text messages. Mention of such information should be restricted to a parenthetical comment in the text, e.g., the name of the person supplying the information, the year of the communication, a title (optional), and a phrase such as "personal communication" or "unpublished".

The form in which references in Springer Nature publications are cited in the text has been standardized to two basic forms, just as the presentation of references in a list has been standardized across many scientific disciplines. There are nonetheless several subject areas where there are subject-specific styles of presentation. The following sections describe the forms of text citation and the styling in reference lists for the types of publication mentioned above, the latter organized by subject matter.

Please note:

– Each citation style can be combined with each reference list style. Follow the instructions for the individual publication.

– The reference list style for Internet publications has been adapted to the style for printed publications. In some cases, however, it is not identical to the style as recommended by the creators of the different reference list styles, such as APA or Chicago. Please see the following sections for details.

10.2 Text Citation

As a rule, all the references given in the list of references should be cited somewhere in the article or chapter. Of course, any reference may be cited more than once. Citation may take one of two forms:

– By number, whether sequential by citation or according to the sequence in an alphabetized list

– By name of cited author and year of publication

Only one form of citation is permitted within a publication.

Reference citations within an abstract of a journal article must be treated differently since a reader may only have the abstract (and not the text or the list of references). Regardless of the style of citation otherwise used, in an abstract a reference citation should take the form, for example, "as reported by Smith (JAMA 352:24-28, 2004)." (cf. A.2 Appendix for the Nature Portfolio journals)

If an author has prepared references as information in footnotes rather than as a list, see instructions for Notes and Bibliography below.

10.2.1 Citation by Number

Arabic numbers are used for citation, which is sequential either by order of citation or by alphabetical order of the references, whichever sequence is used in the list of references (on alphabetization, see Sect. 10.3). For most publications, the reference numbers are given in brackets and are not superscript. Superscript references without brackets are used for certain specifically designated publications; the styles may not be mixed within a journal or book. For both numbered styles, please observe the following guidelines:

– Single citation: [9].

– Multiple citation: [4–6, 9]. The numbers should be listed in numerical order.

– Sequential citation by order of citation: reference 7 cannot be cited before reference 5, for example.

– If an author's name is used in the text: Miller [9] was the first...

10.2.2 Citation by Name and Year

Citation by name and year can be given entirely in parentheses or by citing the year in parentheses after an author's name used in the text. Adhere to the following usage (cf. slight modification for APA reference style in Sect. 10.3.6):

– One author: Miller (1998) or (Miller 1998)

– Two authors: Miller and Smith (2001) or (Miller and Smith 2001)

– More than two authors: Miller et al. (1999) or (Miller et al. 1999)

– Letters are used to distinguish references whose citations would otherwise be identical (e.g., Miller 1998a, b).

– Do not repeat the names of authors of multiple citations (e.g., Miller 1998a, 2001; Miller and Smith 2001).

– The citations of several references mentioned at one position do not have to be listed in alphabetical order. Leave the order of citation used by the author.

Generally, the author names are in normal type.

10.2.3 Variants

Any pages, figures, etc., referred to specifically should be given in the text with the citations, as in these examples:

– (see p. 43 in [9]) or [9, p. 43].

– (see Fig. 4 in Smith and Jones 1997) or (Smith and Jones 1997, Fig. 4) or Smith and Jones (1997, Fig. 4).

– (see Chap. 3 in Jones 1998) or (Jones 1998, Chap. 3) or Jones (1998, chap. 3).

10.3 Reference List

The reference list is the final part of a published chapter or article or of a book. Its heading is generally "References." Some publications may contain a list of uncited works headed "Bibliography" or "Further Reading"; this may be in addition to a list of cited references. The reference list must contain entries for all the references cited in the text, except for the types of publication explicitly mentioned below.

Alphabetization Rules Entries in the list must be listed alphabetically except in the numbered system of sequential citation. The rules for alphabetization are:

1. For one author, by name of author, then chronologically
2. For two authors, by name of author, then name of coauthor, then chronologically
3. For more than two authors, by name of first author, then chronologically

In English, names are alphabetized using a prefix if one is present, i.e., J. van der Stroon is listed under "v", not "s."

General Rules of Presentation Some basic rules followed in the presentation of references are:

1. Ideally, the names of six authors should be given before et al. (assuming there are six or more), but do not delete names if more than six have been provided. One author's name and et al. are however sufficient; do not ask for additional names.

2. Always use the "core name" of a publisher without additions such as "Publishers", "Verlag", or "Press"; e.g., Wiley instead of John Wiley and Sons or Academic instead of Academic Press. Note: "University Press" belongs to the core name, e.g., Harvard University Press.
3. Give the name of only one city as the location of a publisher, using either the first supplied in the manuscript (if more than one is given) or the publisher's main location. Do not include the name of a state, province, or country.
4. Titles may be given in any language that uses Latin letters. The titles of publications using non-Latin alphabets must be transliterated. The inclusion of an English translation of the title or of the original lettering are optional; if present, they are shown in parentheses.
5. Use "in press" only in place of volume and page numbers that are not yet available, not as a substitute for the (probable) year of publication or for a journal name, both of which are required. Do not use "in press" when a DOI is present (see examples 4 and 24 in Table 10.1).
6. Do not ask an author to supply unnecessary information. The reference entry is to help a reader find the source or to support linking, not to satisfy abstract library rules. Thus, do not ask for the type of dissertation (example 13 in Table 10.1), names of additional authors (example 1), location of publisher (get it yourself where this is a requirement of the bibliography style), precise date or location of a symposium, or pages for a book publication.
7. Focus on the kind of references listed in examples 1–10 in Table 10.1. The same high quality of information is not required of references to other types of publications; do not delay publication for such references.
8. Include a DOI wherever possible.

Internet Publications As noted above, the reference list style for Internet publications has, with some minor modifications, been adapted to the different styles used for printed publications, and examples are included in the following tables. Internet publications include both unchanging content and changeable content from usually reliable sources. For the former, a DOI should be included wherever possible; for the latter, the date of viewing is given. This information is based, in part, on ISO 690-2 and on https://www.crossref.org/display-guidelines/.

Optional reference list citations The following types of reference may be omitted from the reference list or bibliography. A citation in running text of the style described below is sufficient. Any formal version of the citation provided by the author in the reference list should be retained and styled as given in the examples for each reference style, but do not ask an author to supply one if it is only mentioned in the text.

1) Article in a newspaper or popular magazine (e.g., "As Polly Toynbee wrote in the *The Guardian* on 5 May 2016 ... ")
2) Mention of a website (e.g., "Springer Nature announced on their website on January 10, 2016 ...")
3) Blog entry or comment (e.g., "In a comment posted to The XXX Blog ....")

10.3.1 Basic Springer Nature Reference Style

The basic Springer Nature reference style is used in publications in a wide range of subjects (especially in medicine, biomedicine, life sciences, geosciences, computer science, engineering, and economics). In addition to the general rules listed in Sect. 10.3:

1. Always use the standard abbreviation of a journal's name. *Index Medicus* and the ISSN *List of Title Word Abbreviations* (http://www.issn.org/2-22660-LTWA.php) are the reliable sources of the standard abbreviations used in journal titles.
2. Capitalize only the first word in the title of an article, chapter, or book and any proper nouns.

The examples listed in Table 10.1 show the presentations to be used for citations of journal articles, journal issues, book chapters, books, patents, and gray literature.

Table 10.1 Different kinds of references and their presentation in basic Springer Nature reference style. See also the basic rules listed in Sect. 10.3

Number	Type	Example
1.	Journal article	Smith J, Jones M Jr, Houghton L et al (1999) Future of health insurance. N Engl J Med 965:325–329
2.	Inclusion of issue number (optional)	Saunders DS (1976) The biological clock of insects. Sci Am 234(2):114–121
3.	Journal article with DOI (and with page numbers)	Slifka MK, Whitton JL (2000) Clinical implications of dysregulated cytokine production. J Mol Med 78:74–80. https://doi.org/10.1007/s001090000086
4.	Journal article by DOI (before issue publication with page numbers)	Slifka MK, Whitton JL (2000) Clinical implications of dysregulated cytokine production. J Mol Med. https://doi.org/10.1007/s001090000086
5.	Article in electronic journal by DOI (no paginated version)	Slifka MK, Whitton JL (2000) Clinical implications of dysregulated cytokine production. Dig J Mol Med. https://doi.org/10.1007/s801090000086
6.	Journal issue with issue editor	Smith J (ed) (1998) Rodent genes. Mod Genomics J 14(6):126–233
7.	Journal issue with no issue editor	Mod Genomics J (1998) Rodent genes. Mod Genomics J 14(6):126–233
8.	Book chapter	Brown B, Aaron M (2001) The politics of nature. In: Smith J (ed) The rise of modern genomics, 3rd edn. Wiley, New York, p 234–295
9.	Book, authored	South J, Blass B (2001) The future of modern genomics. Blackwell, London
10.	Book, edited	Smith J, Brown B (eds) (2001) The demise of modern genomics. Blackwell, London
11.	Book, also showing a translated edition [Either edition may be listed first.]	Adorno TW (1966) Negative Dialektik. Suhrkamp, Frankfurt. English edition: Adorno TW (1973) Negative dialectics (trans: Ashton EB). Routledge, London

Number	Type	Example
12.	Chapter in a book in a series without volume titles	Schmidt H (1989) Testing results. In: Hutzinger O (ed) Handbook of environmental chemistry, vol 2E. Springer, Heidelberg, p 111
13.	Chapter in a book in a series with volume titles	Smith SE (1976) Neuromuscular blocking drugs in man. In: Zaimis E (ed) Neuromuscular junction. Handbook of experimental pharmacology, vol 42. Springer, Heidelberg, pp 593–660
14	OnlineFirst chapter in a series (without a volume designation but with a DOI)	Saito Y, Hyuga H (2007) Rate equation approaches to amplification of enantiomeric excess and chiral symmetry breaking. Top Curr Chem. https://doi.org/10.1007/128_2006_108
15.	Proceedings as a book (in a series and subseries)	Zowghi D et al (1996) A framework for reasoning about requirements in evolution. In: Foo N, Goebel R (eds) PRICAI'96: topics in artificial intelligence. 4th Pacific Rim conference on artificial intelligence, Cairns, August 1996. Lecture notes in computer science (Lecture notes in artificial intelligence), vol 1114. Springer, Heidelberg, p 157
16.	Proceedings with an editor (without a publisher)	Aaron M (1999) The future of genomics. In: Williams H (ed) Proceedings of the genomic researchers, Boston, 1999
17.	Proceedings without an editor (without a publisher)	Chung S-T, Morris RL (1978) Isolation and characterization of plasmid deoxyribonucleic acid from Streptomyces fradiae. In: Abstracts of the 3rd international symposium on the genetics of industrial microorganisms, University of Wisconsin, Madison, 4–9 June 1978
18.	Paper presented at a conference	Chung S-T, Morris RL (1978) Isolation and characterization of plasmid deoxyribonucleic acid from Streptomyces fradiae. Paper presented at the 3rd international symposium on the genetics of industrial microorganisms, University of Wisconsin, Madison, 4–9 June 1978
19.	Patent. Name and date of patent are optional	Norman LO (1998) Lightning rods. US Patent 4,379,752, 9 Sept 1998
20.	Dissertation	Trent JW (1975) Experimental acute renal failure. Dissertation, University of California
21.	Institutional author (book)	International Anatomical Nomenclature Committee (1966) Nomina anatomica. Excerpta Medica, Amsterdam
22.	Non-English, Latin alphabet publication cited in an English publication. NB: Use the language of the primary document, not	Wolf GH, Lehman P-F (1976) Atlas der Anatomie, vol 4/3, 4th edn. Fischer, Berlin

Number	Type	Example
	that of the reference for "vol" etc.!	
23.	Non-Latin alphabet publication cited in an English publication. Transliterated title required. (Original title in original alphabet and English translation are optional and placed in parentheses when present.)	Marikhin VY, Myasnikova LP (1977) Nadmolekulyarnaya struktura polimerov (The supramolecular structure of polymers). Khimiya, Leningrad
24.	In press	Major M et al (2007) Recent developments. In: Jones W (ed) Surgery today. Springer, Dordrecht (in press)
25.	Publicly available preprint	Babichev SA, Ries J, Lvovsky AI (2002) Quantum scissors: teleportation of single-mode optical states by means of a nonlocal single photon. Preprint at https://arxiv.org/abs/quant-ph/0208066
26.	Online document	Doe J (1999) Title of subordinate document. In: The dictionary of substances and their effects. Royal Society of Chemistry. Available via DIALOG. http://www.rsc.org/dose/title of subordinate document. Accessed 15 Jan 1999
27.	Online database	Healthwise Knowledgebase (1998) US Pharmacopeia, Rockville. http://www.healthwise.org. Accessed 21 Sept 1998
28.	Supplementary material/private homepage	Doe J (2000) Title of supplementary material. http://www.privatehomepage.com. Accessed 22 Feb 2000
29.	University site	Doe J (1999) Title of preprint. http://www.uni-heidelberg.de/mydata.html. Accessed 25 Dec 1999
30.	FTP site	Doe J (1999) Trivial HTTP, RFC2169. ftp://ftp.isi.edu/in-notes/rfc2169.txt. Accessed 12 Nov 1999
31.	Organization site	ISSN International Centre (2006) The ISSN register. http://www.issn.org. Accessed 20 Feb 2007
32.	Data set	Ozkaya G et al (2018) Three-dimensional motion capture data during repetitive overarm throwing practice. *figshare* https://doi.org/10.6084/m9.figshare.c.4017808.

10.3.2 Chemistry Reference Style

The presentation of references in Springer Nature chemistry publications differs from the basic Springer form in that the name of the journal article or book chapter and concluding page numbers are optional (but should be retained/included wherever possible). Here are examples without article and chapter titles:

Journal article: Smith J, Jones M Jr et al (1999) Fresenius J Anal Chem 705:325

Book chapter: Hermans JLM (1989) In: Hutzinger O (ed) Handbook of environmental chemistry, vol 2E. Springer, Berlin, p 111

The other forms are the same as those in the basic Springer style (see Table 10.1).

10.3.3 Math and Physical Sciences Reference Style

The presentation of references in Springer Nature mathematics, statistics, computer science, and some physics publications differs in several regards from basic Springer Nature style. These differences include the sequence of information, use of formatting, and use of punctuation in references to journal articles and books, as shown in the examples listed in Table 10.2. The numbered citation system is usually used in these subject areas, although a few publications use the name–date system. The alpha-numeric system (e.g., [ABC99]) is tolerated, but not encouraged, in selected mathematics publications. In addition to the general rules listed in Sect. 10.3:

1. Always use the standard abbreviation of a journal's name. The ISSN *List of Title Word Abbreviations* (http://www.issn.org/2-22660-LTWA.php) is the most reliable source of the standard abbreviations used in journal titles.
2. Capitalize only the first word and any proper nouns in the title of an article or chapter, but capitalize all major words in the title of a book. For a description of capitalized headings, see Sect. 3.3.1.1.

Table 10.2 Different kinds of references and their presentation according to the math and physical sciences reference style. For the differences to this styling for other physics publications, see Sect. 10.3.4

Number	Type	Example
1.	Journal article	Hamburger, C.: Quasimonotonicity, regularity and duality for nonlinear systems of partial differential equations. Ann. Mat. Pura. Appl. **169**, 321–354 (1995)
2.	Inclusion of issue number (optional)	Campbell, S.L., Gear, C.W.: The index of general nonlinear DAES. Numer. Math. **72**(2), 173–196 (1995)
3.	Journal article with DOI (and with page numbers)	Slifka, M.K., Whitton, J.L.: Clinical implications of dysregulated cytokine production. J. Mol. Med. **78**, 74–80 (2000). https://doi.org/10.1007/s001090000086
4.	Journal article with DOI (and with article citation ID)	Hendi, S.H., Momeni, D.: Black-hole solutions in *F*I gravity with conformal anomaly. Eur. Phys. J. C **71**, 1823 (2011). https://doi.org/10.1140/epjc/s10052-011-1823-y

Number	Type	Example
5.	Journal article by DOI (before issue publication with page numbers)	Slifka, M.K., Whitton, J.L.: Clinical implications of dysregulated cytokine production. J. Mol. Med. (2000). https://doi.org/10.1007/s001090000086
6.	Article in electronic journal by DOI (no paginated version)	Slifka, M.K., Whitton, J.L.: Clinical implications of dysregulated cytokine production. Dig. J. Mol. Med. (2000). https://doi.org/10.1007/s801090000086
7.	Journal issue with issue editor	Smith, J. (ed.): Rodent genes. Mod. Genomics J. **14**(6), 126–233 (1998)
8.	Journal issue with no issue editor	Rodent genes: Mod. Genomics J. **14**(6), 126–233 (1998)
9.	Book chapter	Broy, M.: Software engineering – from auxiliary to key technologies. In: Broy, M., Denert, E. (eds.) Software Pioneers, pp. 10–13. Springer, New York (2002)
10.	Book, authored	Geddes, K.O., Czapor, S.R., Labahn, G.: Algorithms for Computer Algebra. Kluwer, Boston (1992)
11.	Book, edited	Seymour, R.S. (ed.): Conductive Polymers. Plenum, New York (1981)
12.	Chapter in a book in a series without volume titles	MacKay, D.M.: Visual stability and voluntary eye movements. In: Jung, R., MacKay, D.M. (eds.) Handbook of Sensory Physiology, vol. 3, pp. 307–331. Springer, Heidelberg (1973)
13.	Chapter in a book in a series with volume titles	Smith, S.E.: Neuromuscular blocking drugs in man. In: Zaimis, E. (ed.) Neuromuscular Junction. Handbook of Experimental Pharmacology, vol. 42, pp. 593–660. Springer, Heidelberg (1976)
14.	OnlineFirst chapter in a series (without a volume designation but with a DOI)	Saito, Y., Hyuga, H. Rate equation approaches to amplification of enantiomeric excess and chiral symmetry breaking. Top. Curr. Chem. (2007). https://doi.org/10.1007/128_2006_108
15.	Proceedings as a book (in a series and subseries)	Zowghi, D., et al.: A framework for reasoning about requirements in evolution. In: Foo N., Goebel R. (eds.) Topics in Artificial Intelligence, 4th Pacific Rim Conference on Artificial Intelligence, Cairns, August 1996. Lecture Notes in Computer Science. Lecture Notes in Artificial Intelligence, vol. 1114, pp. 157–168. Springer, Heidelberg (1996)
16.	Proceedings with an editor (without a publisher)	Aaron, M.: The future of genomics. In: Williams, H. (ed.) Proceedings of the Genomic Researchers, Boston (1999)
17.	Proceedings without an editor (without a publisher)	Chung, S.-T., Morris, R.L.: Isolation and characterization of plasmid deoxyribonucleic acid from Streptomyces fradiae. In: Abstracts of the 3rd International Symposium on the Genetics of Industrial Microorganisms, University of Wisconsin, Madison, 4–9 June 1978

Number	Type	Example
18.	Paper presented at a conference	Chung, S.-T., Morris, R.L.: Isolation and characterization of plasmid deoxyribonucleic acid from Streptomyces fradiae. Paper presented at the 3rd international symposium on the genetics of industrial microorganisms, University of Wisconsin, Madison, 4–9 June 1978
19.	Patent. Name and date of patent are optional	Norman, L.O.: Lightning rods. US Patent 4,379,752, 9 Sept 1998
20.	Dissertation, Ph.D. thesis (either text is acceptable)	Trent, J.W.: Experimental acute renal failure. Dissertation, University of California (1975)
21.	Institutional author (book)	International Anatomical Nomenclature Committee: Nomina anatomica. Excerpta Medica, Amsterdam (1966)
22.	Non-English, Latin alphabet publication cited in an English publication. NB: Use the language of the primary document, not that of the reference for "vol" etc.!	Wolf, G.H., Lehman, P.-F.: Atlas der Anatomie, vol. 4/3, 4th edn. Fischer, Berlin (1976)
23.	Non-Latin alphabet publication cited in an English publication. Transliterated title required. (Original title in original alphabet and English translation are optional and placed in parentheses when present.)	Marikhin, V.Y., Myasnikova, L.P.: Nadmolekulyarnaya struktura polimerov (The supramolecular structure of polymers). Khimiya, Leningrad (1977)
24.	In press	Holmes, R., et al.: References. In: Jones, T.C. (ed.) Science style manual. Sprint, London (2007, in press)
25.	Publicly available preprint	Babichev, S. A., Ries, J., Lvovsky, A. I.: Quantum scissors: teleportation of single-mode optical states by means of a nonlocal single photon. Preprint at https://arxiv.org/abs/quant-ph/0208066 (2002)
26.	Online document	Cartwright, J.: Big stars have weather too. IOP Publishing PhysicsWeb. http://physicsweb.org/articles/news/11/6/16/1 (2007). Accessed 26 June 2007
27.	Online database	Healthwise Knowledgebase. US Pharmacopeia, Rockville. http://www.healthwise.org (1998). Accessed 21 Sept 1998
28.	Supplementary material/private homepage	Doe, J.: Title of supplementary material. http://www.privatehomepage.com (2000). Accessed 22 Feb 2000

Number	Type	Example
29.	University site	Doe, J.: Title of preprint. http://www.uni-heidelberg.de/mydata.html (1999). Accessed 25 Dec 1999
30.	FTP site	Doe, J.: Trivial HTTP, RFC2169. ftp://ftp.isi.edu/in-notes/rfc2169.txt (1999). Accessed 12 Nov 1999
31.	Organization site	ISSN International Centre: The ISSN register. http://www.issn.org (2006). Accessed 20 Feb 2007
32.	Data set	Ozkaya, G., et al.: Three-dimensional motion capture data during repetitive overarm throwing practice. figshare. https://doi.org/10.6084/m9.figshare.c.4017808 (2018)

10.3.4 American Physical Society (APS) Reference Style

The presentation of references in some Springer Nature physics publications differs in several regards from both the basic Springer Nature style and the math and physical sciences style, following the reference style of the American Physical Society (APS). These differences include the sequence of information, use of formatting, and use of punctuation in references to journal articles and books, as shown in the examples listed in Table 10.3. The numbered citation system is usually used in the APS style. In addition to the general rules listed in Sect. 10.3:

1. Always use the standard abbreviation of a journal's name. The ISSN *List of Title Word Abbreviations* (http://www.issn.org/2-22660-LTWA.php) is the most reliable source of the standard abbreviations used in journal titles.
2. Capitalize only the first word and any proper nouns in the title of an article or chapter, but capitalize all major words in the title of a book. For a description of capitalized headings, see Sect. 3.3.1.1.
3. Note that journal article titles, chapter titles, and closing page numbers are optional (but should be retained/included wherever possible).

Table 10.3 Different kinds of references and their presentation according to the APS reference style

Number	Type	Example
1.	Journal article	S. Preuss, A. Demchuk Jr., M. Stuke, Appl. Phys. A **61**, 33 (1995)
2.	Inclusion of issue number (optional)	D.S. Saunders, Sci. Am. **234**(2), 114 (1976)
3.	Journal article with DOI (and with page numbers)	M.K. Slifka, J.L. Whitton, Clinical implications of dysregulated cytokine production. J. Mol. Med. **78**, 74–80 (2000). https://doi.org/10.1007/s001090000086
4.	Journal article by DOI (before issue publication with page numbers)	M.K. Slifka, J.L. Whitton, Clinical implications of dysregulated cytokine production. J. Mol. Med. (2000). https://doi.org/10.1007/s001090000086

Number	Type	Example
5.	Article in electronic journal by DOI (no paginated version)	M.K. Slifka, J.L. Whitton, Clinical implications of dysregulated cytokine production. Dig. J. Mol. Med. (2000). https://doi.org/10.1007/s801090000086
6.	Journal issue with issue editor	J. Smith (ed.), Rodent genes. Mod. Genomics J. **14**(6) (1998)
7.	Journal issue with no issue editor	Rodent genes, Mod. Genomics J. **14**(6) (1998)
8.	Book chapter	D.M. Abrams, in *Conductive Polymers*, ed. by R.S. Seymour, A. Smith (Springer, Berlin Heidelberg New York, 1973), p. 307
9.	Book, authored	H. Ibach, H. Lüth, *Solid-State Physics*, 2nd edn. (Springer, New York, 1996), pp. 45–56
10.	Book, edited	R.S. Seymour (ed.), *Conductive Polymers* (Plenum, New York, 1981)
11.	Chapter in a book in a series without volume titles	D.M. MacKay, in *Handbook of Sensory Physiology*, vol. 3, ed. by R. Jung, D.M. MacKay (Springer, Heidelberg, 1973), p. 307
12.	Chapter in a book in a series with volume titles	S.E. Smith, in *Neuromuscular Junction*, ed. by E. Zaimis. Handbook of Experimental Pharmacology, vol 42 (Springer, Heidelberg, 1976), p. 593
13.	OnlineFirst chapter in a series (without a volume designation but with a DOI)	Y. Saito, H. Hyuga, Rate equation approaches to amplification of enantiomeric excess and chiral symmetry breaking. Top. Curr. Chem. (2007). https://doi.org/10.1007/128_2006_108
14.	Proceedings as a book (in a series and subseries)	D. Zowghi et al., in *PRICAI '96: Topics in Artificial Intelligence*, ed. by N. Foo, R. Goebel. 4th Pacific Rim Conference on Artificial Intelligence, Cairns, August 1996. Lecture Notes in Computer Science. Lecture notes in artificial intelligence, vol. 1114 (Springer, Heidelberg, 1996), p. 157
15.	Proceedings with an editor (without a publisher)	M. Aaron, in *Proceedings of the Genomic Researchers*, Boston, 1999, ed. by H. Williams
16.	Proceedings without an editor (without a publisher)	S.-T. Chung, R.L. Morris, in Abstracts of the 3rd International Symposium on the Genetics of Industrial Microorganisms, University of Wisconsin, Madison, 4–9 June 1978
17.	Paper presented at a conference	S.-T. Chung, R.L. Morris, Isolation and characterization of plasmid deoxyribonucleic acid from Streptomyces fradiae. Paper presented at the 3rd international symposium on the genetics of industrial microorganisms, University of Wisconsin, Madison, 4–9 June 1978

Number	Type	Example
18.	Patent. Name and date of patent are optional	L.O. Norman, U.S. Patent 4,379,752, 9 Sept 1998
19.	Dissertation, Ph.D. thesis (either text is acceptable)	J.W. Trent, Dissertation, University of California, 1975
20.	Institutional author (book)	International Anatomical Nomenclature Committee, *Nomina anatomica*. (Excerpta Medica, Amsterdam, 1966)
21.	Non-English, Latin alphabet publication cited in an English publication. NB: Use the language of the primary document, not that of the reference for "vol" etc.!	G.H. Wolf, P-F Lehman (eds.), *Atlas der Anatomie*, vol. 4/3, 4th edn. (Fischer, Berlin, 1976)
22.	Non-Latin alphabet publication cited in an English publication. Transliterated title required. (Original title in original alphabet and English translation are optional and placed in parentheses when present.)	V.Y. Marikhin, L.P. Myasnikova, *Nadmolekulyarnaya struktura polimerov* (The supramolecular structure of polymers). (Khimiya, Leningrad, 1977)
23.	In press	R. Holmes et al., in *Science style manual*, ed. by T.C. Jones (Sprint, London, 2006 in press)
24.	Publicly available preprint	S.A. Babichev, J. Ries, A.I. Lvovsky, Quantum scissors: teleportation of single-mode optical states by means of a nonlocal single photon (2002), Preprint at https://arxiv.org/abs/quant-ph/0208066
24.	Online document	J. Cartwright, Big stars have weather too. (IOP Publishing PhysicsWeb, 2007), http://physicsweb.org/articles/news/11/6/16/1. Accessed 26 June 2007
25.	Online database	Healthwise Knowledgebase (US Pharmacopeia, Rockville, 1998), http://www.healthwise.org. Accessed 21 Sept 1998
26.	Supplementary material/private homepage	J. Doe, Title of supplementary material (2000), http://www.privatehomepage.com. Accessed 22 Feb 2000
27.	University site	J. Doe, Title of preprint (1999), http://www.uni-heidelberg.de/mydata.html. Accessed 25 Dec 1999
28.	FTP site	J. Doe, Trivial HTTP, RFC2169 (1999), ftp://ftp.isi.edu/in-notes/rfc2169.txt. Accessed 12 Nov 1999

Number	Type	Example
29.	Organization site	ISSN International Centre: The ISSN register (2006), http://www.issn.org. Accessed 20 Feb 2007
30.	Data set	G. Ozkaya, figshare, https://doi.org/10.6084/m9.figshare.c.4017808 (2018)

10.3.5 Vancouver Reference Style

For a full description of this reference style, including numerous examples, see http://www.nlm.nih.gov/bsd/uniform_requirements.html, as referenced from the site that is the homepage of the larger "Vancouver" system (http://www.icmje.org/). Examples of the use of this style at Springer Nature for references are given in Table 10.4. In addition to the general rules listed in Sect. 10.3:

1. Always use the standard abbreviation of a journal's name. *Index Medicus* and the ISSN *List of Title Word Abbreviations* (http://www.issn.org/2-22660-LTWA.php) are the reliable sources of the standard abbreviations used in journal titles.
2. Capitalize only the first word in the title of an article, chapter, or book and any proper nouns.

As noted in Sect. 10.1, either form of citation (i.e., name–year or numbered) may be used with Vancouver-styled references.

Table 10.4 Examples of several different kinds of references in Vancouver style

Number	Type	Example
1.	Journal article	Smith JJ. The world of science. Am J Sci. 1999;36:234–5.
2.	Journal article with DOI (and with page numbers)	O'Mahony S, Rose SL, Chilvers AJ, Ballinger JR, Solanki CK, Barber RW, et al. Finding an optimal method for imaging lymphatic vessels of the upper limb. Eur J Nucl Med Mol Imaging. 2004;31:555–63. https://doi.org/10.1007/s00259-003-1399-3.
3.	Journal article by DOI (before issue publication with page numbers)	O'Mahony S, Rose SL, Chilvers AJ, Ballinger JR, Solanki CK, Barber RW, et al. Finding an optimal method for imaging lymphatic vessels of the upper limb. Eur J Nucl Med Mol Imaging. 2004. https://doi.org/10.1007/s00259-003-1399-3.
4.	Article in electronic journal by DOI (no paginated version)	Slifka MK, Whitton JL. Clinical implications of dysregulated cytokine production. Dig J Mol Med. 2000. https://doi.org/10.1007/s801090000086.
5.	Journal article in a supplement	Frumin AM, Nussbaum J, Esposito M. Functional asplenia: demonstration of splenic activity by bone marrow scan. Blood. 1979;59 Suppl 1:26–32.
6.	Book chapter	Wyllie AH, Kerr JFR, Currie AR. Cell death: the significance of apoptosis. In: Bourne GH, Danielli JF, Jeon KW, editors.

Number	Type	Example
		International review of cytology. London: Academic; 1980. p. 251–306.
7.	OnlineFirst chapter in a series (without a volume designation but with a DOI)	Saito Y, Hyuga H. Rate equation approaches to amplification of enantiomeric excess and chiral symmetry breaking. Top Curr Chem. 2007. https://doi.org/10.1007/128_2006_108.
8.	Book, authored	Blenkinsopp A, Paxton P. Symptoms in the pharmacy: a guide to the management of common illness. 3rd ed. Oxford: Blackwell Science; 1998.
9.	Online document	Doe J. Title of subordinate document. In: The dictionary of substances and their effects. Royal Society of Chemistry. 1999. http://www.rsc.org/dose/title of subordinate document. Accessed 15 Jan 1999.
10.	Publicly available preprint	Alvarez R. Near optimal neural network estimator for spectral x-ray photon counting data with pileup. arXiv:1702.01006v1 [Preprint]. 2017 [cited 2017 Feb 9]: [11 p.]. Available from: https://arxiv.org/abs/1702.01006
11.	Online database	Healthwise Knowledgebase. US Pharmacopeia, Rockville. 1998. http://www.healthwise.org. Accessed 21 Sept 1998.
12.	Supplementary material/private homepage	Doe J. Title of supplementary material. 2000. http://www.privatehomepage.com. Accessed 22 Feb 2000.
13.	University site	Doe, J.: Title of preprint. http://www.uni-heidelberg.de/mydata.html (1999). Accessed 25 Dec 1999.
14.	FTP site	Doe, J.: Trivial HTTP, RFC2169. ftp://ftp.isi.edu/in-notes/rfc2169.txt (1999). Accessed 12 Nov 1999.
15.	Organization site	ISSN International Centre: The ISSN register. http://www.issn.org (2006). Accessed 20 Feb 2007.
16.	Data set	Ozkaya G. Three-dimensional motion capture data during repetitive overarm throwing practice. figshare. 2018. https://doi.org/10.6084/m9.figshare.c.4017808.

10.3.6 APA Reference Style for Social Sciences and Psychology

This is the style system used in psychology and the social sciences established by the American Psychological Association. For guidance, see the *Publication Manual of the American Psychological Association* and the respective web site of the Association (http://www.apastyle.org/). Springer Nature has adopted version 7 of this style. In addition to the general rules listed in Sects. 10.1 and 10.3 :

1. Do not abbreviate journal names; they are always written out.
2. Capitalize only the first word in the title of an article, chapter, or book and any proper nouns.

3. The use of notes rather than text citations is discouraged in APA, but where it has been used, refer to the general instructions in Sect. 10.3.9.2.
4. For flexibility of use, it is permitted to use either form of citation (i.e., name–year or numbered) with APA-styled references
5. In parenthetical name–year citations, include a comma before the year and an ampersand (&) instead of the word "and" between two authors, e.g. (Jones, 2019), (Brown et al., 2018) or (Miller & Smith, 2001).

See Table 10.5 for several examples.

Table 10.5 Examples of basic reference entries according to APA-based style

Number	Type	Example
1.	Journal article	Grady, J. S., Her, M., Moreno, G., Perez, C., & Yelinek, J. (2019). Emotions in storybooks: A comparison of storybooks that represent ethnic and racial groups in the United States. *Psychology of Popular Media Culture*, *8*(3), 207–217
2.	Journal article with DOI (and with page numbers)	Grady, J. S., Her, M., Moreno, G., Perez, C., & Yelinek, J. (2019). Emotions in storybooks: A comparison of storybooks that represent ethnic and racial groups in the United States. *Psychology of Popular Media Culture, 8*(3), 207–217. https://doi.org/10.1037/ppm0000185.
3.	Journal article by DOI (advance online publication and without page numbers)	Hong, I., Knox, S., Pryor, L., Mroz, T. M., Graham, J., Shields, M. F., & Reistetter, T. A. (2020). Is referral to home health rehabilitation following inpatient rehabilitation facility associated with 90-day hospital readmission for adult patients with stroke? *American Journal of Physical Medicine & Rehabilitation.* Advance online publication. https://doi.org/10.1097/PHM.0000000000001435
4.	Article in electronic journal by DOI (no paginated version)	Jerrentrup, A., Mueller, T., Glowalla, U., Herder, M., Henrichs, N., Neubauer, A., & Schaefer, J. R. (2018). Teaching medicine with the help of "Dr. House." *PLoS ONE*, *13*(3), Article e0193972. https://doi.org/10.1371/journal.pone.0193972
5.	Newspaper article	Carey, B. (2019, March 22). Can we get better at forgetting? *The New York Times*. https://www.nytimes.com/2019/03/22/health/memory-forgetting-psychology.html
6.	Book	Sapolsky, R. M. (2017). *Behave: The biology of humans at our best and worst.* Penguin Books
7.	Book chapter	Dillard, J. P. (2020). Currents in the study of persuasion. In M. B. Oliver, A. A. Raney, & J. Bryant (Eds.), *Media effects: Advances in theory and research* (4th ed., pp. 115–129). Routledge
8.	OnlineFirst chapter in a series (without a volume	Saito, Y., & Hyuga, H. (2020). Rate equation approaches to amplification of enantiomeric excess and chiral symmetry breaking. *Topics in Current Chemistry*. https://doi.org/10.1007/128_2006_108

Number	Type	Example
	designation but with a DOI)	
9.	Book, also showing a translated edition [cited edition first.]	García Márquez, G. (2003). *One hundred years of solitude* (G. Rabassa, Trans.). New York, NY: Harper. (Original work published 1967)
10.	Publicly available preprint	Babichev, S. A., Ries, J. & Lvovsky, A. I. (2002). Quantum scissors: teleportation of single-mode optical states by means of a nonlocal single photon. Preprint retrieved from https://arxiv.org/abs/quant-ph/0208066
11.	Online document	Fagan, J. (2019, March 25). *Nursing clinical brain*. OER Commons. Retrieved January 7, 2020, from https://www.oercommons.org/authoring/53029-nursing-clinical-brain/view
12.	Online database	Stein, M. B., & Taylor, C. T. (2019). Approach to treating social anxiety disorder in adults. *UpToDate.* Retrieved September 13, 2019, from https://www.uptodate.com/contents/approach-to-treating-social-anxiety-disorder-in-adults
13.	Supplementary material	Freeberg, T. M. (2019). From simple rules of individual proximity, complex and coordinated collective movement [Supplemental material]. *Journal of Comparative Psychology*, *133*(2), 141–142. https://doi.org/10.1037/com0000181
14.	Organizational report	International Organization for Standardization. (2018). *Occupational health and safety management systems—Requirements with guidance for use* (ISO Standard No. 45001:2018). https://www.iso.org/standard/63787.html
15.	Data set	O'Donohue, W. (2017). *Content analysis of undergraduate psychology textbooks* (ICPSR 21600; Version V1) [Data set]. ICPSR. https://doi.org/10.3886/ICPSR36966.v1

10.3.7 Chicago-based Humanities Reference Style

The Chicago-based reference style is used in the humanities based on the specifications given in the *Chicago Manual of Style*. In addition to the general rules listed in Sect. 10.3:

1. Do not abbreviate journal names; they are always written out.
2. Do not abbreviate the given names of the authors if they were submitted in full.
3. Follow the author regarding capitalization of titles of an article, chapter, or book but ensure it is consistent according to either the 16th edition (upper and lower case titles) or 15th edition (capitalize only the first word in the title of an article, chapter, or book and any proper nouns) of the Chicago Manual of Style. If creating a list from scratch, standardly follow the 15th edition.

As noted in Sect. 10.1, either form of citation (i.e., name–year or numbered) may be used with Chicago-styled references, and the citation of Internet publications has been modified. For the use of citations in footnotes or endnotes, see Sect. 10.3.9.2.

Table 10.6 Examples of reference entries according to Chicago style
(NB: Examples styled here according to The Chicago Manual of Style 15th edn. Retain capitalization of titles if provided by the author according to the 16th edition)

Number	Type	Example
1.	Journal article	Alber, John, Daniel C. O'Connell, and Sabine Kowal. 2002. Personal perspective in TV interviews. *Pragmatics* 12: 257–271.
2.	Article by DOI (with page numbers)	Slifka, M.K., and J.L. Whitton. 2000. Clinical implications of dysregulated cytokine production. *Journal of Molecular Medicine* 78:74–80. https://doi.org/10.1007/s001090000086.
3.	Article by DOI (before issue publication and without page numbers)	Suleiman, Camelia, Daniel C. O'Connell, and Sabine Kowal. 2002. 'If you and I, if we, in this later day, lose that sacred fire...': Perspective in political interviews. *Journal of Psycholinguistic Research*. https://doi.org/10.1023/A:1015592129296.
4.	Article in electronic journal by DOI (no paginated version)	Slifka, M.K., and J.L. Whitton. 2000. Clinical implications of dysregulated cytokine production. *Online Journal of Molecular Medicine*. https://doi.org/10.1007/s001090000086.
5.	Newspaper article	Schultz, Sean. 2011. Lessons to be learned in systems change initiatives. *Independent*, December 28.
6.	Book	Cameron, Deborah. 1985. *Feminism and linguistic theory*. New York: St. Martin's Press.
7.	Book chapter	Cameron, Deborah. 1997. Theoretical debates in feminist linguistics: Questions of sex and gender. In *Gender and discourse*, ed. Ruth Wodak, 99-119. London: Sage Publications.
8.	Book, also showing a translator	Adorno, Theodor W. 1973. *Negative dialectics.* Trans. E.B. Ashton. London: Routledge.
9.	Modern editions of the classics and classic poems and plays. Year of edition is given not original publication	Shakespeare, William. 1982 . *Hamlet*. Arden edition. Edited by Harold Jenkins. London: Methuen
10.	OnlineFirst chapter in a series (without a volume designation but with a DOI)	Saito, Yukio, and Hyuga, Hiroyuki. 2007. Rate equation approaches to amplification of enantiomeric excess and chiral symmetry breaking. *Topics in Current Chemistry*. https://doi.org/10.1007/128_2006_108.
11.	Book, also showing a translated edition [Either edition may be listed first.]	Adorno, T.W. 1966. *Negative Dialektik*. Frankfurt: Suhrkamp. English edition: Adorno, T.W. 1973. *Negative dialectics* (trans: Ashton, E.B.). London: Routledge.

Number	Type	Example
12.	Online document	Frisch, Mathias. 2007. Does a low-entropy constraint prevent us from influencing the past? PhilSci archive. http://philsci-archive.pitt.edu/archive/00003390. Accessed 26 June 2007.
13.	Publicly available preprint	Steiner, Tobias. 2018. Bron/broen, the pilot as space between cultures, and (re)negotiations of Nordic Noir. SocArXiv. May 11. https://doi.org/10.31235/osf.io/9dbw2.
14.	Online database	German emigrants database. 1998. Historisches Museum Bremerhaven. http://www.deutsche-auswanderer-datenbank.de. Accessed 21 June 2007.
15.	Supplementary material/private homepage	Doe, John. 2006. Title of supplementary material. http://www.privatehomepage.com. Accessed 22 Feb 2007.
16.	FTP site	Doe, J. 1999. Trivial HTTP, RFC2169. ftp://ftp.isi.edu/in-notes/rfc2169.txt. Accessed 12 Nov 2006.
17.	Organization site	ISSN International Centre. 2006. The ISSN register. http://www.issn.org. Accessed 20 Feb 2007.
18	Data set	Ozkaya Gizem. 2018. Three-dimensional motion capture data during repetitive overarm throwing practice. figshare https://doi.org/10.6084/m9.figshare.c.4017808

10.3.8 Standard Nature Research Style

The standard Nature Research Style is used for Nature Research Journals only. In addition to the general rules described in Sect. 10.3, please note:

1. Use the standard abbreviation of a journal's name. *Index Medicus* and the ISSN *List of Title Word Abbreviations* (http://www.issn.org/2-22660-LTWA.php) are the reliable sources of the standard abbreviations used in journal titles.
2. Reference list entries use "et al." for publications with more than five authors
3. Reference list entries for chapters in edited works with more than two editors use first editor plus "et al."
4. Text citations are always numbered sequentially
5. Reference lists are by order of citation.
6. The publisher location is no longer included in book references in Nature Research style.

Table 10.6 Examples of reference entries according to standard Nature Research style

Number	Type	Example
1.	Journal article with two authors	Gredmark, T. & Hallberg, L. Population study of women in Goteburg. *Scand. J. Soc. Med.* **6**, 51–54 (1978).

Number	Type	Example
2	Journal article with five authors	Price, R. A. Jr, Curry, N. III, McCann, K. E., Fielding, J. L. & Abercrombie, E. Jr. Analysis of obesity in twins. *Hum. Hered.* **39** (Suppl.), 121–135 (1989).
3	Journal article with more than five authors	Halpern S.D. et al. Solid-organ transplantation in HIV-infected patients. *N. Engl. J. Med.* **347**, 284–287 (2002).
4.	Article by DOI (without page numbers)	He, F. J., Marrero, N. M. & MacGregor, G. A. Salt and blood pressure in children and adolescents. *J. Hum. Hypertens.* https://doi.org/10.1038/sj.jhh.1002269 (2007).
5.	Article in electronic journal without DOI (no paginated version)	Hill, W. G., Goddard, M. E. & Visscher, P. M. Data and theory point to mainly additive genetic variance for complex traits. *PloS Genet.* **4**, e1000008 (2008).
6.	Published abstract	Feig, S. A. et al. Bone marrow transplantation for neuroblastoma. *Exp. Hematol.* **13**, abstr. 102 (1985).
7.	Publicly available preprint (with DOI - preferred)	Ganesh, R. A. et al. Immuno-phenotyping of high-grade glioma infiltrating immune cells reveals grade specific differences in cells of myeloid origin. Preprint at *bioRxiv* https://doi.org/10.1101/2020.09.26.314542 (2020).
8	Publicly available preprint (no DOI available)	Babichev, S. A., Ries, J. & Lvovsky, A. I. Quantum scissors: teleportation of single-mode optical states by means of a nonlocal single photon. Preprint at https://arXiv.org/quant-ph/0208066 (2002).
9.	Book (monograph)	Meyer, H. A. *The Role of Abdominal Fat* 2nd edn, Vol. 2 (Academic, 1970).
10.	Book (edited volume)	Diener, B. J. & Wilkinson, P. (eds) *Transplantation Techniques* (Harvard Univ. Press, 1989).
11.	Book chapter	Harley, N. H. & Vivian, L. in *Mechanisms of Disease* 4th edn, Vol. 2 (eds Sodeman, W. A. & Smith, A.) Ch. 3 (Saunders, 1974).
12.	Published conference proceedings	Smith, Y. (ed.) *Proc. 1st National Conference on Porous Sieves* (Butterworth-Heinemann, 1997).

Number	Type	Example
13.	Paper in published conference proceedings	Jones, X. Zeolites and synthetic mechanisms. In *Proc. 1st National Conference on Porous Sieves* (ed. Smith, Y.) 16–27 (Butterworth-Heinemann, London, 1997).
14.	Dissertation	Young, W. R. *Effects of Different Tree Species on Soil Properties in Central New York*. MSc thesis, Cornell Univ. (1981).
15.	Retracted article (including retraction information)	Caddy, S. G. et al. Growth of limpets on the rocky shore. *Nat. Genet.* **3,** 426–431 (1995); retraction **4**, 104 (1996).
16.	Retraction note to article	Caddy, S. G. et al. Retraction: Growth of limpets on the rocky shore. *Nat. Genet.* **4**, 104 (1996).
17.	Scientific technical report	Akutsu, T. *Total Heart Replacement Device.* Report No. NIH-NHLI-69 2185-4 (National Institutes of Health, 1974).
18.	Article with published erratum	Johnson, P., Ing-Simmons, C. H. R., Bennett, P., Adams, S. & Freeman, A. The effect of fire on the Lundy cabbage. *West. Eng. J.* **162**, 28–31 (1999); erratum **162**, 3127 (1999).
19.	Online material (Blog)	Manaster, J. Sloth squeak. *Scientific American Blog Network* http://blogs.scientificamerican.com/psi-vid/2014/04/09/sloth-squeak (2014).
20.	Software	SAS v.8 (SAS Institute Inc., 2000).
21.	Patent	Pagedas, A. C. Reusable laparoscopic retrieval mechanism. US patent 6,387,102 (2002).
22.	Newspaper article	Schultz, Sean. Lessons to be learned in systems change initiatives. *Independent* (28 December 2011).
23.	In press	Tian, D., Araki, H., Stahl E., Bergelson, J. & Kreitman, M. Signature of balancing selection in *Arabidopsis*. *Proc. Natl Acad. Sci. USA* (in the press).
24.	Database	Leiserowitz, A., Maibach, E. & Roser-Renouf, C. *Climate Change in the American Mind: Americans' Global Warming Beliefs and Attitudes in January 2010* (Yale University and George Mason University, accessed 28 October 2010); http://environment.yale.edu/uploads/AmericansGlobalWarmingBeliefs2010.pdf
25.	White paper	Belady, C., Rawson, A., Pfleuger, J. & Cader, T. *Green Grid Data Center Power Efficiency Metrics: PUE and DCIE* White Paper No. 6 (Green Grid, 2008).

Number	Type	Example
26	Data set	Ozkaya G. Three-dimensional motion capture data during repetitive overarm throwing practice. figshare https://doi.org/10.6084/m9.figshare.c.4017808 (2018).

10.3.9 Special Instructions for Style Variants

10.3.9.1 *Editing References in Law Publications*

Law texts usually include very detailed footnotes. Copy editors have to take care that none of the information included in the footnotes gets lost. These footnotes often include, in addition to textual comments, bibliographic information about books and articles and citations of court decisions, laws, and government regulations.

The aim of formal copy editing of such footnotes is to shorten the bibliographic details within the footnotes and transfer all additional bibliographic details from the footnotes into a reference list.

10.3.9.1.1 Text Citation

Citations of books, book chapters, or journal articles in the text or in footnotes should be given in a shortened form: author name(s), year and page number or paragraph (margin number). Some examples: (Miller 1991, p. 17) or (Miller 1991, para 30) or (Smit 2005, Article 5, para 7).

Do not change the styling of all other footnote information (e.g., court decisions), and follow the manuscript.

10.3.9.1.2 Reference List

The reference list must contain the bibliographic details of the cited books, book chapters, or journal articles. If the author has not submitted a reference list, it must be created by the copy editor. This means:

1. Copy the full bibliographic information from the footnotes into the reference list.
2. There should be one citation of each cited book, chapter, or article in the reference list regardless of the number of occurrences in the text or the footnotes.
3. Court decisions, laws, and regulations should not be moved to the reference list.
4. Reference list entries should be alphabetized by the last names of the first author of each work.
5. Style the reference list according to the selected Springer Nature reference style.

10.3.9.2 *Notes and Bibliography in the Humanities*

In the humanities, and especially literature, history, and the arts, many authors prefer to give bibliographic information in the footnotes/endnotes in addition to a bibliography. The advantage is that a wider range of sources can be accommodated than with a reference list and name–date citations alone. The disadvantage is that it is not possible to provide the linking features that are available with the in-text citation/reference list system. A book-end bibliography is more common than a chapter-end list in this style.

10.3.9.2.1 Citation

The reference information in the footnotes/endnotes should be left as provided by the author. There is no need to cross check between reference information in footnotes and bibliography, as there is no requirement for all to be cited there. At the highest level of copy editing, the copy editor should ensure there is sufficient information in the footnote for a reader to locate the publication as it stands.

10.3.9.2.2 Reference List

The task of formal copy editing here is to style the book-end bibliography list according to the required reference style.

1. Reference list entries should be alphabetized by the last names of the first author of each work.
2. Style the reference list according to the required reference style. The bibliography style most commonly associated with the Notes and Bibliography system is Chicago; APA is also possible although discouraged by the American Psychological Association itself.

10.4 Handling Discrepancies and Questions

We wish to publish high-quality reference information, both for its intrinsic value and to enable cross-linking between publications online. At the same time, we cannot permit missing items of otherwise unimportant information to delay the production procedure for (sometimes) long periods of time.

Edit the references as customary, including questions regarding missing information to the author, which they should resolve at proof stage. Precise formulation of the questions is important to ensure adequate answers are received. For instance:

– If a citation is present in the text but there is no entry in the list, request the author to add a reference to the list or delete the citation from the text.

– If a list entry is present but there is no citation in the text, request the author to insert a text citation if possible simply leave the entry in the list.

– If the citation and the list entry are both present but do not agree completely, use common sense to make them correspond. Usually this means changing the text to make it agree with the list, such as in the spelling of a name. Ask the author to confirm your changes are correct

– If the entry in the list is incomplete, simply leave it as is (but cf. A.2 Appendix for the Nature Portfolio journals)

Part III: Science

11 Medicine and Biology

11.1 Language

It is important that medical and scientific language be used correctly. Although not usually an expert on the subject, the copy editor must strive to eliminate ambiguity and inaccuracies as well as avoid dehumanizing language. The following points are meant to serve as a guide. Keep in mind that if an author has a good command of the English language and is precise in presenting information, what may seem to be infelicities may be exactly what the author wishes to say. In such cases, be judicious about making changes. If a manuscript is not well written, however, the copy editor has greater liberty to make alterations. But always be careful about changing meaning or emphasis.

The following sections contain a brief list of some words that are commonly misused and obstruct meaning, and lists of undesirable, acceptable, and preferred terms, and jargon.

11.1.1 Dehumanizing Language

This section highlights to copy editors potentially offensive or upsetting language that should be avoided. The copy editor should avoid any potentially dehumanizing language that refers to humans, whether in discussing clinical studies or more generally.

11.1.1.1 *Examples of Commonly Used Dehumanizing Language*

addiction, anxiety, depression	Use 'individual with addiction' rather than 'addict'; 'individual with depression' rather than 'depressive'; and 'individual with anxiety' rather than 'anxious person'.
case, patient	A case is a particular instance of disease (do not use case to mean 'affected person'; change to 'individual' or 'person' as appropriate). A patient is a person who is ill, not an unaffected individual studied (clarify if used wrongly). A case is evaluated and reported but does not have symptoms. A patient is admitted to the hospital, examined, and given medication, then recovers. Use the construction 'individuals with XXX' if people concerned are not actually patients, but use 'patients with XXX' if people concerned are clinical patients (do not use construction 'XXX individuals' or 'XXX patients'). 'Case' may be used in the context of case–control studies, but the cases must be defined at first use, e.g., 'individuals with osteosarcoma (cases)'.

diagnose, evaluate, identify	Avoid using disease names as adjectives (and adjectives related to disease, e.g., 'metastatic patients'). A patient is not diagnosed as a disease or for a disease (use 'individual with diabetes', not 'diabetic person' or 'a diabetic'), or evaluated, or identified (unless his or her identity is actually in question). A patient may be diagnosed as having a particular disease, and a condition or disease may be diagnosed. A case is evaluated. A bacterium or virus is identified.
normal, abnormal (applied to people)	A person as a whole should not be described as normal or abnormal, because adherence to/variance from the norm for a given trait is meant, which does not define the entire person. Change to e.g., 'unaffected', 'healthy', '*GENE*$^{+/+}$', 'control' (always specifying the nature of controls). Normal may be used for individual traits, e.g., 'normal myelination' or 'normal development'. 'Healthy' should not be contrasted with conditions that may constitute normal variation rather than representing a disease, such as some forms of autism; alternatives include 'neurotypical' or 'typically developing'.
Subject	An individual is not the subject of an experiment. Change to 'study participant'.
Populations of people	Care is necessary in describing populations of people studied. Avoid the term 'Caucasian' except for individuals from the Caucasus. Do not use 'colored/coloured'. The only acceptable use of these terms is when authors are reporting exactly how the data were collected, e.g., 'self-reported Caucasian'. 'White' and 'black' are phenotypic designations and hence are acceptable, but only if used adjectivally (do not use 'whites' and 'blacks'). However, more specific geographic or ethnic descriptions may be more informative and are preferred in genetic studies.

11.1.2 Some Commonly Misused Words

acute, chronic	These terms should be reserved for the description of symptoms, conditions, or diseases. Avoid the use of acute and chronic to describe patients, parts of the body, treatment, or medication. Exceptions are 'acute abdomen', 'chronically implanted electrode', 'chronic intubation', 'chronic stimulation', and 'chronic instrumentation'.
average	This word is often used imprecisely. It usually stands for the arithmetic mean, expressed statistically as the sum of values divided by the number of values. (The symbol M can also be used for the mean.) The mean should not be confused with the median (the middle value in the sense that half of the values are larger and half smaller) or the mode (the most frequently occurring score).
biologic and biological	Both mean 'pertaining to biology'. 'Biologic' but not 'biological' may be used to refer to medicinal products.

biopsy	Is a procedure, so biopsies can be done but not obtained; biopsy samples are obtained during the procedure.
content, concentration, level	Strictly speaking, 'content' is the absolute amount, in contrast to 'concentration', which refers to a ratio. 'Level' refers to any value that is relative.
evolutionary, evolutionarily	'Evolutionary' is an adjective ('molecular evolutionary analysis'), whereas 'evolutionarily' is an adverb ('both proteins are evolutionarily conserved in eukaryotes').
dose, dosage	A dose is the quantity to be administered at one time or the total quantity administered. 'dosage' implies not only an amount but also frequency of administration. 'The patient received an initial dose of 50 mg and thereafter a dosage of 25 mg three times a day until she had received a total dose of 500 mg'.
homology	(for sequence) Use if not quantified; otherwise use similarity.
negative, normal (for tests and examinations)	Examinations and most laboratory tests are neither negative nor normal. The observations, results, or findings from examinations and tests are normal or abnormal. Cultures, tests for microorganisms, tests for specific reactions, and reactions to tests may be negative or positive. e.g., Results of the physical examination were normal. Physical examinations showed no abnormalities. Findings from laboratory tests were normal. Laboratory tests showed normal values. The Wassermann reaction was negative. A tuberculin patch test was positive. Cultures were negative for *Clostridium botulinum*. Electroencephalograms, electrocardiograms, and radiographs are 'pictures' and are therefore normal or abnormal, not negative or positive.
operate, operate on	A surgeon does not operate a patient. Nor does one anticoagulate, collapse, explore, inject, obstetricate, puncture, refract, or transplant a patient. A surgeon operates on a patient. She may do an exploratory operation.

	By thoracoplasty, he may collapse a lung. He may inject a substance into a patient.
prevalence, incidence	'Prevalence' is the quality or state of being prevalent (i.e., widespread) or, more specifically, the total number of cases (old and new) at any one time in a given population (often 100,000 persons). 'Incidence' is the rate of occurrence, i.e., the number of new cases per unit time.
roentgen, X-ray	A 'roentgen' is a unit of X- or gamma-radiation. 'Roentgen ray' is an infrequently used synonym of X-ray. Both X-radiation and gamma-radiation are used in radiography. X-ray can also be used as a synonym for radiograph.
roentgenogram, radiograph	Strictly speaking, 'roentgenogram' is the correct term for a photograph made by means of X-rays. It is nowadays used much less frequently than 'radiograph', which is strictly speaking a generic term for a photograph made by any form of radiation, but is currently the usual term for a photograph made by means of X-rays. 'X-ray' is sometimes used as a synonym for radiograph.
sacrifice	Replace by kill if that is what is meant, although euthanized may be an acceptable replacement.
sensitivity, specificity	'Sensitivity' is the 'positive' accuracy of a test, i.e., the number of true positives as a percentage of those tested who actually have the disease. 'Specificity' is the 'negative' accuracy of a test, i.e., the number of true negatives as a percentage of those tested who actually do not have the disease.
sign, symptom, indication	A 'sign' is any objective evidence of disease, such as is perceptible to the examining physician, in contrast to the 'symptoms', which are subjective evidence. An 'indication' is anything (e.g., a sign) that can be used as a guide to diagnosis, treatment, etiology, etc. of a disease.
significant	Means important, notable, expressive or indicative of something; usage also in statistics (try to restrict to statistical significance in research manuscripts).
spectrum	Can be changed to range if there is a risk of confusion.
suggestive of, suspicious of	To be suspicious of is to be distrustful of. To be suggestive of is to give a suggestion or indication of. The patient's symptoms are suggestive of measles, not suspicious of measles. Good physicians are suspicious of hasty diagnosis.
toxic, toxicity	Toxic means pertaining to or caused by poison or toxin. Toxicity is the quality, state, or degree of being poisonous. A patient is not toxic. A patient does not have arsenic toxicity but arsenic toxicosis or intoxication (exception: oxygen toxicity). 'Toxic effects' can be used to describe harmful downstream effects of something that is itself not toxic, e.g., 'Paracetamol overdose is associated with hepatotoxic effects'.
transfect	Use 'transfect with'.

true negative	A correct negative result.
true positive	A correct positive result.

11.1.3 Preferred Terms

Table 11.1 gives a list of undesirable terms that should generally be changed to the preferred form.

Table 11.1 Undesirable and preferred terms

Undesirable	Preferred
expired, succumbed	died
male, female (as nouns)	man, woman, boy, girl
over (meaning in excess of)	more than
therapy of (disease)	therapy for

Table 11.2 similarly provides a list of expressions in the left column that, while acceptable, are really short or jargon forms of the expressions in the right column. They should only be used with discretion.

Table 11.2 Permitted and preferred terms

Permitted	Preferred
blood sugar	blood glucose
doctor	physician (American English)
facial paralysis	paralysis of the facial nerve
jugular ligation	ligation of the jugular vein
left heart failure	left ventricular failure
left lower lobe	lower lobe of the left lung
right heart	right ventricle or atrium
right hemothorax	right-sided hemothorax

Ages of humans are roughly divided as in the following list. The numerical age or age range is preferred to any of these designations, but do not change the author's usage unless it is ambiguous.

Embryo	about 2 to 7 or 8 weeks after fertilization
Fetus	7 or 8 weeks after fertilization to birth
Newborn, neonate	less than 4 weeks
Infant	4 weeks to 24 months
Child	2–13 years

Adolescent	13–17 years
Teenager	13–19 years
Adult	18 years and older

11.1.4 Jargon

Medical jargon is sometimes imprecise or inaccurate and often unpleasant to read. Generally, you should improve it whenever this is possible without cumbersome rephrasing.

11.1.4.1 *Adjectives*

Misused adjectives usually qualify an inappropriate noun. See several examples in Table 11.3.

Table 11.3 Jargon: adjectival use

Jargon	Correct form
Cardiac diet	Diet for patients with cardiac disease (a diet cannot be 'of the heart')
Cardiac patient	A patient may be weak or depressed, but not cardiac
Congenital heart	Congenital heart disease, defect
Left chest	Left side of the chest
Patients with maternal epilepsy	Patients whose mothers have/had epilepsy
Posterior vagina	Posterior wall of the vagina or retrovaginal wall (or septum)
Urinary infection	Urinary tract infection

Intravenous, oral, parenteral, or rectal cannot be used to modify a drug name; they should be used to modify a word like administration or changed to adverbs.

> – 'Intravenous chloramphenicol was administered' should read, for example, 'Chloramphenicol was administered intravenously'.
>
> – 'The patient received oral chloramphenicol in divided doses' should read, for example, 'The patient received chloramphenicol in divided doses per os'.

11.1.4.2 *Nouns*

-emias and -oses Words designating an abnormal condition are sometimes incorrectly used in expressions of measurement or degree.

Incorrect	Correct
hyperglycemia of 150 mg	hyperglycemia (150 mg/100 ml) or hyperglycemia with a glucose level of 150 mg/100 ml
leukocytosis of 15,000	leukocytosis (15,000 cells/µl) or a leukocyte count of 15,000/µl
4+ albuminuria	albuminuria (4+) or urine reaction for albumin was 4+

The construction 'The patient had a hyperglycemia' is frequently seen. The article 'a' should usually be deleted in such cases.

-ologies The suffix '-ology' usually denotes a science, the study of a science, or the sum of knowledge of a science. It is therefore an abstract term and should not be used as though it were a concrete, measurable entity to which such adjectives as 'normal', 'present', 'negative', etc., can be applied.

– Pathology is the branch of medicine which treats of the essential nature of disease; it is also the structure and manifestations of any given disease. A disease can have a pathology; a patient cannot.

– Cytology is the study of cells.

– Serology is the study of antigen–antibody reactions in vitro.

– Etiology is the study or theory of the causes and causative mechanisms of disease. If you are seeking the cause of a disease, you are looking for one particular pathogen or defect; if you are trying to establish an etiology, you are looking at the entirety of circumstances surrounding infection, transmission, and development of the disease.

– Symptomatology is the characteristic overall pattern of symptoms of a disease, not the pattern which appears in any particular patient.

Here are some examples of correct and incorrect usage of these ologies:

Incorrect	Correct
There was no pathology.	There were no pathological findings. *or* There were no abnormalities. *or* No pathological condition was found.
The cytology was normal.	Cytologically, the cells were normal.
Serology was negative.	Results of serological tests were normal. *or* Serological tests gave negative results.
Severe symptomatology	Severe symptoms

In the adjectival forms of such nouns, be sure that the '-ic' and '-ical' endings are used consistently for any one term. Follow the author's usage where it is not incorrect. An author may use the endings to distinguish between different meanings (as with pathologic/pathological, physiologic/physiological).

11.1.4.3 *Prepositions*

The most frequently abused prepositions are 'with' and 'for', especially the latter, which is used particularly by nonnative speakers as an undifferentiated preposition to indicate some unspecified relationship between two nouns. Be precise.

Incorrect	Correct
Patients with early admission	Patients admitted early
Results for hormone tests for testosterone for group B spoke for a role for testosterone for the development of hirsutism.	Results of hormone tests in group B indicated that testosterone plays a part in the development of hirsutism.

11.1.4.4 *Verbosity*

Many writers appear to feel that a long or formal word is 'better' or more precise than a short, less formal one. This is often not true; change when it impedes the flow of reading or is overused.

For example, 'initiate' and 'commence' are no more precise than 'begin' or 'start'. 'Experienced a weight loss' means 'lost weight', and 'received no therapy' is the same as 'was not treated'. 'Presented with' often simply means 'had', although it is correctly used when it refers to those signs and symptoms the patient has upon examination by the physician.

11.2 Style

This section contains an overview of information on the correct use and styling of scientific names and terminology.

11.2.1 Trade and Generic Names

Approved international non-proprietary names should be used. Generic or trivial (non-proprietary) names are lowercase (e.g., penicillin, brefeldin A, or paclitaxel). Trade (brand, proprietary) names are capitalized (e.g., Taxol, Retrovir, or Aspro; there are some exceptions, e.g., pHisohex). Avoid abbreviations such as BZ for benzodiazepines. See Sect 6.1.2.

The generic name should normally be used throughout a paper. This does not apply to symposia volumes or to studies of particular formulations; here it is sufficient to add the generic name at first mention and use the trade name thereafter.

If you cannot find the generic name, query the author; if no generic name is available (e.g., with a newly synthesized drug), the name of the manufacturer should be given.

If a proprietary product name must be used, provide the manufacturer's name on first use. Thereafter, use only the product name. Example: Rebif (EMD Serono or Merck Serono). Do not use ® or ™ symbols.

Although the complete name of a drug, including the salt or ester, is preferred at first mention in the text (and at first mention in any abstract or summary), use of the shorter form is acceptable. Retain the form the author has used.

Complete name	Shorter name
tetracycline hydrochloride	tetracycline
tetracycline phosphate complex	tetracycline
penicillin O potassium	penicillin O

When more than one non-proprietary name of a drug is given, the alternative should appear in parentheses at first mention in the text, and in any abstract or summary:

> paromomycin (aminosidine)

If an alternative non-proprietary name and a trade name are both given, both should appear in parentheses at first mention:

> vitamin B_{12} (cyanocobalamin, Cobamine)

11.2.1.1 *Vitamins*

The word 'vitamin' is lowercase, a capital letter is used to designate a specific vitamin, and any number or letter indicating a subgroup is subscript, for example: vitamin A, vitamin B_{12}, or vitamin B_c. Follow the author's use of the letter-number designation or the specific name of the compound: vitamin C *or* ascorbic acid, vitamin B_c *or* folic acid.

11.2.2 Taxonomy and Systematics

Many organisms that are mentioned in medical manuscripts have both a Latin and a vernacular name.

11.2.2.1 *Species and Genera*

The Latin name of an organism consists of two parts (binomial system of nomenclature), the first being the name of the genus and the second identifying the specific member of that genus, e.g., *Chlamydia trachomatis*. Names of species that are important to the article or chapter should be listed as early as possible in the text.

The specific name (e.g., *trachomatis*) cannot be used without a genus name. On first use of a binomial genus *species*, the genus must be spelled out. On subsequent uses, the first letter (capitalized and italic) of the genus name (e.g., *C. trachomatis*) can be used. If, in a chapter or paper, two genera with the same initial letter are used, the genus name should normally be written out on each occasion. If, however, the organisms are common and there is no danger of confusion, the genus names may be abbreviated as usual (e.g., *Streptococcus mutans* and *Staphylococcus aureus* on first use and *S. mutans* and *S. aureus* on subsequent mentions). Note that if the genus is abbreviated differently (e.g., for mosquitoes (*Anopheles*, *Aedes*, etc.), genus names are often abbreviated to *An*. or *Ae.*) the abbreviation should be changed to the standard, single letter format (e.g., *A*.).

The genus may be used alone (without a species name attached) when referred to as a group and also if that has become the standard usage for a particular species (e.g., *Xenopus* for *Xenopus laevis*). For subspecies, *Trypanosoma brucei gambiense*, for example, can be abbreviated to *T. b. gambiense*.

When naming systems are important, new, altered, or contentious, the name of the author should immediately follow the specific name (see Sect. 11.2.2.5). Alternative taxonomic names may also be added in parentheses: e.g., *Bos* (= *Bibos*) *gaurus* or *Hippocampus amygdalensis* (= *altmani*). Authors' names should be spelled out in full, followed by the year of authorship. The only author whose name is abbreviated is Linnaeus, written as L.

See Sect. 11.14.2 for information on *Candidatus* nomenclature and for *Salmonella* nomenclature.

11.2.2.2 *Taxonomic Rank*

Levels are styled roman with an initial capital, except for genera (italic with an initial capital) and species (italic and lowercase). One example of the various sublevels is given in the following table:

Taxonomic group	Example
Subspecies	*Bison bison bison*
Species	*Bison bison*
Genus	*Bison*
Tribe	Bovini
Subfamily	Bovinae
Family	Bovidae
Superfamily	Bovidea
Suborder	Pecora
Order	Artiodactyla
Superorder	Paraxonia
Infraclass	Eutheria
Subclass	Theria
Class	Mammalia
Superclass	Gnathostomata
Subphylum	Vertebrata
Phylum	Chordata

Taxonomic group	Example
Subkingdom	Metazoa
Kingdom	Animalia
Domain	Eukarya

For viruses, italicize approved orders, families and subfamilies, but not genus and species.

Sublevels follow the rules of the levels from which they have been derived (for example, subspecies are lowercase italic). All generic, subgeneric, species, subspecies and varietal names are italicized.

Abbreviations are always written in roman:

Abbreviation	Definition
var.	variety
cv.	cultivar
ser.	serovar, in bacteria or viruses
f.	form, in plants
sp.	species, if referring to an unspecified or unknown member of a genus: e.g., *Drosophila* sp.
spp.	more than one such species
ssp.	subspecies, in animals
subsp.	subspecies, in plants
syn.	synonymous (for alternative genus/species names)

Strains (as in bacteria and fungi) tend to be roman. Hybrids are indicated by a multiplication sign; the genus name is normally not repeated (e.g., *Hippocampus amygdalensis* × *northi*).

When using a taxonomic or systematic term that is commonly used but that defines not a natural group but rather a paraphyletic group or grade of organization, it is conventional to list the term in single quotation marks throughout: e.g., 'placoderm'. These are omitted in papers whose context is not primarily taxonomic (e.g., when 'dicotyledonous plant' or 'reptile' is used as a general descriptor).

Avoid using the terms 'higher' and 'lower' in regard to evolution ('higher eukaryotes'); instead use more descriptive terms, e.g., 'one-celled' and 'multicellular', or more exact taxonomic descriptors.

11.2.2.3 *Vernacular Names*

Vernacular names are often formed from the names of genera, and sometimes also from the names of important or common higher taxa. They are always presented lowercase roman. If a proper name is identical to the vernacular/trivial name, the italics and capitalization can be dropped (but only when the name is used in an informal way). For example, *Octopus* becomes a vulgar octopus. Specific epithets cannot be used to form vernacular names. There are no hard and fast rules governing how the names are formed, or whether the plural takes a Latin or English ending. *Dorland's* can be used as a basis for judging whether a vernacular name is in common use, but not all vernacular names are listed.

Points to note about vernacular names are:

1. More than one form of the name may exist:

    - A member of the genus *Treponema* is treponeme or treponema.
    - A member of the genus *Pseudomonas* is pseudomonas or pseudomonad.

2. In microbiology, vernacular names based on genus names, set in lowercase roman type, are often used. There are no set rules for their derivation (which can be Latin or English; sometimes both exist) or plural forms, and many also have adjectival forms (e.g., streptococcal). In some cases the plural is made using a Latin or Greek ending, in other cases the English 's' is used, and in yet others both are acceptable (in this case, one or other form should be used consistently).

Singular	Plural
escherichia	escherichiae
corynebacterium	corynebacteria
vibrio	vibrios
treponema	treponemas or treponemata
treponeme	treponemes
salmonella	salmonellas or salmonellae
bacillus	bacilli
rhizobium	rhizobia

3. Common names are not necessarily formed from a genus name:

    - Gonococcus for *Neisseria gonorrheae*
    - Pseudomonad (plural: pseudomonads) for a member of the family *Pseudomonadaceae*
    - Streptomycete for a member of the family *Streptomycetaceae*

4. Vernacular names of viruses may be formed from the name of the disease they cause:
    - Foot-and-mouth disease virus
    - Herpes simplex virus
    - Newcastle disease virus

    Others are derived from the genus or another source, and are generally written solid:

    - Coxsackievirus (genus: *Enterovirus*)
    - Herpesvirus (genus: *Herpesvirus*)
    - Papillomavirus (genus: *Papillomavirus*)
    - Poxvirus (family: *Poxviridae*)

5. A generic name that has been used formally in the past but has been replaced by another on technicalities of priority may be used as a common name. For example, the zoological name for the lancelet is not *Amphioxus* but *Branchiostoma*, the latter having priority; but amphioxus (roman, lowercase) may be used as a common name.
6. Capitals should be used only to denote proper nouns in vernacular names: e.g., Grevy's zebra, Stuart Island kiwi, African grey parrot, but lesser white-toothed shrew, great northern diver, brindled gnu, wandering albatross, yew tree.
7. At the first mention of a species in a formal communication, its zoological name must be stated; a vernacular name may also be given and can be used thereafter. For example: at first mention, 'The cranial nerves of *Triturus cristatus …*' or 'The cranial nerves of the great crested newt, *Triturus cristatus…*'; not 'The cranial nerves of the great crested newt', but can be used alone thereafter. Exceptions to this are 'human' (for which the species name need only be listed if taxonomy is at issue) and laboratory animals for which strains are listed, e.g., BALB/c mice, Wistar rats (as the strain designation implies the taxonomic name).

11.2.2.4 *Adjectival Forms*

Although a formal title of classification takes an initial capital letter, lowercase is used if the name is used adjectivally or informally: Echinodermata, but echinoderm, the echinoderms, echinodermlike. Adjectival endings vary according to the etymology of the root. The most common suffix is the '-id' suffix to denote membership of family rank (e.g., a felid is by definition a member of the family Felidae). Other common endings denoting rank include -oidea (superfamily), -inae (subfamily) and -ini (tribe). Vernacular names derived from the genus name may be used as a modifier (e.g., salmonella meningitis or chlamydia infection). If an author has been inconsistent in the use of the alternatives for a given adjective, *Dorland's* or another medical dictionary may help in determining which is more usual.

11.2.2.5 *Attribution of Authority*

The name of the author who discovered the organism and the date it was first described are often provided after the scientific name at any taxonomic level. The name and date are set in parentheses if the organism has been reclassified taxonomically since it was first described, e.g., *Myrmica rubra* (Linnaeus, 1758), while no parentheses are used if the taxonomy first described is still valid, e.g., *Formica sanguinea* Latreille, 1798. Follow the author's usage.

11.2.2.6 *New Species*

First reports of new species demand a special kind of presentation conforming to accepted zoological and botanical practice. In articles in which new species are described, the formal descriptions should be placed as early as is convenient in the paper. A ZooBank accession code, also known as a Life Science Identifier (LSID), must be included either in the Data availability section or with the formal description of the new species. Example:

Data availability KNM-NP 59050 is available for study at the National Museums of Kenya, and data analysed here are provided in the article and its Supplementary Information. Image datasets of comparative specimens can be accessed at http://paleo.esrf.eu, http://africanfossils.org, and http://morphosource.org. The species is registered in ZooBank (http://zoobank.org; LSID urn:lsid:zoobank.org:act:BE0A6575-AD3A-4415-A169-6ABC6B8280E2).

In palaeontological papers describing new species, full attention should be given to geological context, most conveniently placed between the first paragraph and any formal zoological descriptions (see Sect. 15.2). A terser version of this information should also be provided in the abstract or first paragraph, including at a minimum the locality and the horizon (geological stratum) in which the fossil or type specimen was found.

Descriptions of new taxa outside of specialist journals appear only rarely. Detailed treatments of species synonyms (where taxa are renamed, with reference to all earlier literature on naming) should be discouraged. The style set out below is necessarily condensed.

Presentation. There are two ways to present descriptions of new species: (1) where only one new species (or several unrelated species) is described; and (2) where a number of related species are described.

Describing single species

i. The first line shows the genus (in italics, centered on the line) followed by the author surname and date (in roman, separated by a comma; see above for authorship). If the generic name has subsequently been changed, the original author and date appear in brackets: *Viriophilo* (Short, 1987).
ii. The second line shows the species, again italicized, and normally with the Latin suffix 'sp. nov.' in roman. Sometimes both genus and species are new ('gen. et sp. nov.').
iii. The subsequent notes (etymology, holotype, etc.) always appear in the order shown below, and each subheading is bold and nonindented. Terms under discussion (in the following examples *tempus* and *scriptus*) are italicized.

A fictitious example of a typical presentation is given below and discussed. Hints and instructions are in square brackets:

Chiswickotherium Gee, 1988

Chiswickotherium tempuscriptus sp. nov.

Etymology. *tempus* (Latin): times; *scriptus* (Latin): writer or author. The specific name alludes to the presumed habits of the animal.

Holotype… [administrative details of the particular specimen designated by the author as representative of the species, e.g., BMNHnumberM12678]

Referred material… [other specimens assigned to the species]

Locality… [exact details of where the specimen was found]

Horizon… [exact details as to its stratigraphy, e.g., unit, member, formation, stage, etc.; required only when the holotype is a fossil.]

Diagnosis… [exact description of the holotype and therefore the characteristics that uniquely define the species].

Describing several related species

If the genus has been described elsewhere, the layout will be much as above, with a line stating the genus followed by a string of specific descriptions. But if the genus is new, it also requires definition, as in the example below.

Editorius gen. nov.

Etymology. *Editorius* (Dog-latin): an editor. The name alludes to the presumed lifestyle of the animal, scavenging fragments of text and rearranging them for financial gain. Type species *E. essexianus.*

Diagnosis. [description of generic characteristics]

Editorius essexianus sp. nov.

Etymology…

Holotype…

Referred material…

Locality…

Diagnosis… [description of specific characteristics]

Editorius perhorridus sp. nov.

[as above] [etc.]

If a monotypic species is described (a genus containing only one species), the generic diagnosis can be written for both genus and species at once, and the entire name can appear on one line:

Deccanolestes hislopi gen. et sp. nov.

Etymology…

Holotype. Right upper molar M^1 (NKIM/10).

Paratype. Right upper molar M^3 (NKIM/11).

Referred material. Right lower premolar P_3 (NKIM/12).

Horizon and locality. Terminal Cretaceous intertrappean horizon, Naskal, Pargi Taluq, Rangareddi District, Andhra Pradesh, India.

Diagnosis. Upper molars and premolars small M^1, width (W) =1.41 mm, length (L) = 0.9 mm; M^3, W = 1.3 mm, L = 0.81 mm; P^3, W = 0.9 mm, L = 0.41 mm. M^1 with transverse crown, triangular in occlusal view, broad stylar shelf, deep ectoflexus with a slightly larger anterior part and somewhat smaller posterior part, stylocone (stylar cusp B) well developed.

[Notation: M^1 and M_1 denote upper jaw and lower jaw molars, respectively—preferred to the alternative notation M/1 and M1/.]

11.3 Names of Diseases

The 'preferred' name of a disease is the one followed by the main or most complete definition in a dictionary. However, this can vary from dictionary to dictionary. Follow *Dorland's* where possible, but if an author has been consistent, follow his or her usage if it corresponds to that in any major reference work. Do not change terms not found in dictionaries without querying the author.

Distinguish between diseases and syndromes. Consult a medical dictionary for the preferred name.

A disease name normally does not include a species name without the genus, for example, 'aureus infection' should be changed to 'infection with *Staphylococcus aureus*'.

A disease name derived from the name of an organism is not capitalized or italicized, for example, schistosomiasis from the genus *Schistosoma* or candidiasis for *Candida* infection.

For some disease names that include the genus, there are two equally acceptable alternative stylings. First, the genus name, lowercase and in roman type, may be used as an adjective (check in *Dorland's* or other reference works whether the genus name can be used in this way), for example, salmonella meningitis and staphylococcus infection. Second, an adjectival form may be derived using the suffixes -ic or -al and the genus name, for example, staphylococcal (or staphylococcic) infection and streptococcal (or streptococcic) infection.

Eponyms may be used. If an eponym is not the preferred term, it should be supplemented by the preferred term in parentheses on first use. Eponyms are capitalized, but common nouns, adjectives, or words derived from eponymic names are usually not. For example: Addison (or Addison's) disease–addisonism–addisonian crisis and Parkinson (or Parkinson's) disease–parkinsonism–parkinsonian mask.

11.4 Anatomy

11.4.1 Nomenclature

Anatomical names may be in Latin (according to *Nomina Anatomica*) or in English, both forms being given in medical dictionaries. The names of many important veins, muscles, arteries, nerves, and some other structures are customarily given in English, but Latin forms are more common in some instances. Follow *Dorland's* or some other reliable source in checking names, but do not change from Latin to English or vice versa throughout a manuscript.

11.4.2 Designation of Directions and Movements

A list of frequently used designations of direction and movement is given in Table 11.4.

Table 11.4 Designations of direction and movement

Explanation	Designation
in front	anterior, frontal
to the front	anteriorly, frontally
at the rear	posterior
to the rear	posteriorly
at the back (dorsum)	dorsal
to the back	dorsally/dorsad
on the underside	ventral
to the underside	ventrally/ventrad
side	lateral (left or right)
to the side	laterally
at the midline	medial
to the midline	medially/mediad
further from the center	distal
away from the center	distally/distad
closer to the center	proximal
to the center	proximally/proximad
top	superior
to the top	superiorly
bottom	inferior
to the bottom	inferiorly
at the head, brain	cranial, cephalic
to the head, brain	cranially/craniad, cephalically/cephalad

Explanation	Designation
at the nose	rostral
to the nose	rostrally/rostrad
at the tail	caudal
to the tail	caudally/caudad

11.4.3 Designation of Planes

The following designations are used:

– side to side (vertical): frontal, coronal

– cross section: transverse, horizontal

– front to back (vertical): median, sagittal, midsagittal, parasagittal, paramedian

Note: The frontal plane passing through the center of the body is called the midfrontal or midcoronal plane. In imaging of the head an axial scan is one in the transverse plane.

The sagittal plane passing through the center of the body may be called the median plane or midsagittal plane; all other sagittal planes are then called paramedian or sagittal. Another system is to reserve the name sagittal plane for the sagittal plane passing through the center of the body and to call all the other planes parasagittal.

11.4.4 Spine

For preferred terminology consult *Dorland's* or another medical dictionary.

In reports of clinical or technical data and in tables or figures, specific vertebrae, spinal nerves, and intervertebral spaces may be abbreviated without being first written out if no confusion results. Table 11.5 shows several examples.

Table 11.5 Examples of specific vertebrae, spinal nerves, and intervertebral spaces

	cervical	thoracic	lumbar	sacral
vertebrae	C-2 or C2	T-2 or T2	L-2 or L2	S-2 or S2
spaces	C2–3	T2–3	L2–3	S2–3
nerves	C-2 or C2	T-2 or T2	L-2 or L2	S-2 or S2

For cranial nerves use roman numerals (e.g., CIII; see Sect. 11.17.1).

11.5 Cardiology

The following forms are used to designate electrocardiographic leads:

Limb (peripheral) leads	I, II, III
Augmented leads	aV_R, aV_L, aV_F
Chest leads	V_1–V_6
Right-sided chest, back, and esophageal leads	V_{3R} etc., V_B, V_{E34}, etc.

Segments, complexes, and waves should be styled as follows:

Correct	Incorrect
ST segment	S-T segment, ST-segment
ST-T segment	S-T-T segment
PR segment	P-R segment
QRS complex	Q-R-S complex
P wave	P-wave
P–R or PR interval	PR-interval
AV block	A-V block
AV junction	A-V junction

The first to fourth heart sounds and aortic and pulmonic components are abbreviated using the letters S, A, and P, and subscript numbers: S_1, S_2, S_3, S_4, S_2A, S_2P, A_2, P_2.

11.6 Imaging

Computed Tomography (CT) This is also known as computer tomography, computerized tomography, computerized axial tomography (CAT), and computerized transverse axial tomography (CTT). The preferred form is computed tomography, but follow the author's usage if it is consistent.

Magnetic Resonance Imaging (MRI) This was previously known in medicine as nuclear magnetic resonance (NMR) imaging. The preferred term is now magnetic resonance imaging (MRI). A permissible usage is to refer to the procedure as MR imaging or simply MR. If the author has consistently used 'NMR imaging', query it; but be careful, as NMR is used in chemistry and biology.

Ultrasonography This is also known as echography or sonography (but ultrasonography is preferred) and includes echocardiography and echoencephalography. The main techniques used are Doppler scan, A scan (amplitude modulated, A-mode scan), M scan (time-motion scan, M-mode scan), and B scan (brightness modulation, B-mode scan). The B scan may be either a slow B scan or a fast B scan (real-time scan). Using an A scan, gray-scale imaging may be done. The C scan (C-mode scan) is a type of B scan.

Preferred terms in ultrasonography are:

acoustic, acoustical	Acoustic is preferred over acoustical to describe properties, characteristics, or dimensions associated with sound waves.
echogenic, hyperechoic, hypoechoic	Echogenic is used to describe a structure or medium that produces echoes. Hyperechoic should be used to describe a region that is strongly echogenic; hypoechoic should be used to describe a region that is weakly echogenic.
ultrasound, ultrasonic	Ultrasound is preferred to ultrasonic, as in ultrasound examinations, ultrasound waves.
sonographer, sonologist	The sonographer is the person (technician or physician) who performs the ultrasound examination; the sonologist is the physician who interprets the results of examinations. These terms are preferred over ultrasonographer, ultrasonologist, or radiologist.

Images The systems do *not* produce a CT, an MRI, an NMR, a tomograph, or an ultrasonograph. Care should be taken with use of the word 'scan' to denote the image produced. The machine scans, and the image it produces each time may therefore be called a scan, but in dynamic CT, images are produced from several scans put together. In general, scan, picture, and image may be used, but specifically the systems produce the following:

Computed tomography	computed tomogram
	CT image
	CT scan
Magnetic resonance imaging	MRI tomogram
	MR image
	MRI scan
Ultrasonography	sonogram (In general, sonogram is preferred to ultrasonogram, echogram, B scan; change to sonogram, but ask for confirmation.)
	ultrasound image
	scan

11.7 Genetics

11.7.1 Definitions

Cytogenetics (cell genetics) is concerned with the interaction of chromosomes in eukaryotic cells and the genetic material in prokaryotes. Chromosomes are made up of genes, which code for a particular protein, and noncoding regions, which aid in the regulation of transcription. A chromosome consists of a double strand of DNA and various proteins. All the genetic material collectively forms the genotype. The phenotype refers to the physical expression of the genotype.

Molecular genetics refers to interaction at the DNA–RNA–protein product level and thus can also apply to viruses (often made up of only DNA or RNA and a protein coat) and plasmids (extrachromosomal DNA/RNA elements that are self-replicating inside a cell).

11.7.2 Nomenclature

Use the appropriate terminology and nomenclature for what the author is describing (genotype/phenotype, gene/protein) — if in doubt, add an author query to confirm correct nomenclature. Genes 'code for' or 'encode' proteins; they do not 'encode for' them.

In some journals, the Accession Numbers of any nucleic acid sequences, protein sequences or atomic coordinates cited in the manuscript should be provided, in square brackets, and the corresponding database name should be included if provided; for example, [EMBL:AB026295, EMBL:AC137000, DDBJ:AE000812, GenBank:U49845, PDB:1BFM, Swiss-Prot:Q96KQ7, PIR:S66116].

Genes, mRNAs, cDNAs, siRNAs and shRNA targets are shown in italic; proteins are shown in roman. Designations of genotype are written in italic, whereas those of phenotype are roman.

Database-approved gene abbreviations should preferably be used throughout (but do not change the author's choice if they are acceptable synonyms). See http://www.ncbi.nlm.nih.gov/gene for approved gene nomenclature, and the quick guide table below if any further information is needed; see UniProt for protein nomenclature. If the author has used an alternative gene name or abbreviation, the database-approved abbreviation should be mentioned at the first instance, e.g., insert '(also known as *XYZ*)' and add an author query.

Gene and protein abbreviations do not need to be spelled out. In general, gene abbreviations are given in italics, and protein abbreviations and full protein names are in roman. Full gene/protein names are acceptable without abbreviation (read for spelling errors and consistency), but consider using the official abbreviation (and raise an author query) if the full name is used more than three times in the text, is used in a title or heading, or is more than three words long.

The copy editor should query the author if the definition provided by the author does not seem to relate to the context of the paper, or if the handover sheet requests that a definition be provided but one cannot be easily identified from the database. The safest approach is for the copy editor to raise an author query whenever adding a gene/protein definition. Do not add definitions of genes and proteins to titles or headings. When inserting definitions, consult standard databases (see the 'Quick guide to nomenclature' below) and apply house style where necessary to entries from databases, e.g., 'c-Jun-activated kinase 1' from UniProt should be changed to 'JUN-activated kinase 1'. See the table below for standard formatting for gene and protein abbreviations:

Category	**Full protein names**	**Protein abbreviations**	**Full gene names**	**Gene abbreviations**
Greek symbols	Use Greek symbols not written out, e.g., transforming growth factor-α, not transforming growth factor alpha	Use Greek symbols, not roman letters, e.g., TGFα, not TGFA	Use Greek symbols not written out, e.g., Transforming growth factor-α	Use roman letters not Greek symbols, e.g., *TGFA,* not *TGFα*
Arabic numerals	-		Use Arabic numerals (6 not VI), e.g., Insulin-like growth factor 1, not Insulin-like growth factor I	Use Arabic numerals (6, not VI), e.g., *IGF1,* not *IGFI*
Hyphens	Remove hyphens between numbers and the end of name*, but not between Greek symbols and the end of the name, e.g., transforming growth factor-α but insulin-like growth factor 1, C-reactive protein and pentraxin 4	Remove hyphens*, e.g., TGFα, not TGF-α	Remove hyphens between numbers and the end of name*, but not between Greek symbols and the end of the name, e.g., Insulin-like growth factor 1, not Insulin-like growth factor-1	Remove hyphens*,e.g., *IGF1,* not *IGF-1*
Miscellaneous	Abbreviations match UniProt, e.g., ER for oestrogen receptor, even when UK spelling is used in the text	–	–	Use italic formatting

*Note exceptions: *C. elegans* protein abbreviations; BCL-2, etc, interleukins: IL-1/*Il1*, IL-2/*Il2*, IL-6/*Il6*, IL-12/*Il12*, etc.

Superscripts indicating the alleles should be roman (unless they are themselves the names of genes or mutations; e.g., *loxP*). For 'the Cre–*loxP* recombination system', Cre refers to the protein and *loxP* to a site on the DNA, and therefore *loxP* is italicized but Cre is not. Genotypes or similar contexts, however, would use the gene symbol *cre*.

Human alleles are italic, closed up to the gene name and separated from it by an asterisk (e.g., *D3S22*A1*, *HLADQB1*0201*). Note that there may be exceptions to the all uppercase rule for human genes, for example *CYorf15A* for chromosome Y open reading frame 15A.

Wild-type alleles may be designated by a superscript plus sign ($^+$) after the locus designation or '+' on the line if the locus is designated by the context. Diploids use the solidus to separate the genotypes of homologous (maternal and paternal) haploid genomes. Thus the wild-type homozygote for *vestigial* in *Drosophila* can be vg^+/vg^+, $vg^{+/+}$ or +/+. Use the style that the author has chosen, but ensure that it is applied consistently throughout an article.

If a promoter is named after the gene it comes from, then it should be italicized; those that are named after viruses, such as CMV, should not.

Generations are denoted by P, F_1, F_2, and so on.

Two- or one-letter species prefixes can be attached to gene symbols (e.g., At*ABC1* for *Arabidopsis thaliana ABC1*) when homologous genes of two or more species that share the same symbol are being discussed. The term 'homology' for gene sequences is reserved for genes of common evolutionary origin; otherwise use 'similarity'.

The term *cis*-active means affecting a gene on the same chromosome; *trans*-active means affecting a gene on the same chromosome or one on another chromosome. Note that the word transactivate should not contain italics.

Quick guide to nomenclature:

Species	**Gene name**	**Gene symbol**	**Protein symbol**	**Database**
Human and chick	keratin 3	*KRT3*	KRT3	Human Genome Nomenclature Committee http://www.genenames.org/ (gene names)
Mouse	fibroblast growth factor 10	*Fgf10*	FGF10 or Fgf10	http://www.informatics.jax.org/mgihome/nomen
Rat	fibroblast growth factor 10	*Fgf10*	FGF10	http://www.rgd.mcw.edu
Fruitfly	Dominant: *Deformed*	*Dfd*	Dfd	https://flybase.org/ (FlyBase)
	Recessive: *sevenless*	*Sev*	Sev	
Nematode worm	*dumpy-5*	*dpy-5*	DPY-5	http://www.wormbase.org http://www.sanger.ac.uk/Projects/C_elegans/ Extrachromosomal arrays (*Ex1*) are italicized.
Saccharomyces cerevisiae	cell division cycle 28	Dominant allele: *CDC28* Recessive allele: *cdc28*	Cdc28	https://sites.google.com/view/yeastgenome-help/community-help/nomenclature-conventions
Schizosaccharomyces pombe	benomyl resistance	*Ben*	Nothing official	http://www.genedb.org/genedb/pombe/index.jsp Kohli, J. Genetic nomenclature and gene list of the fission yeast Schizosaccharomyces pombe. *Curr. Genet.* **11**, 575–589 (1987)
Zebrafish	*cyclin B1*	*ccnb1*	Ccnb1	Database: https://zfin.org/ Nomenclature: http://zfin.org/zf_info/nomen.html
Arabidopsis thaliana	Wild-type: *ACAULIS 1*	*ACL1*	ACL1	http://www.arabidopsis.org/portals/nomenclature/guidelines.jsp Full names for protein products of genes should be written in

Species	**Gene name**	**Gene symbol**	**Protein symbol**	**Database**
				capital letters without italics (ACAULIS 1) Currently, *Arabidopsis* researchers refer to protein products of mutated genes as 'ACL1', 'the *acaulis 1* protein' or 'the protein product of *acaulis 1*'. NOT as 'acaulis 1'
	Mutant: *acaulis 1*	*acl1*	ACL1	
Xenopus laevis	cyclin B1	*ccnb1*	ccnb1	http://www.xenbase.org/gene/static/geneNomenclature.jsp
Bacteria		*aroA*	AroA	https://jb.asm.org/content/nomenclature http://ecocyc.org http://www.bacterio.net
Brassica				http://www.brassica.info/
Viruses				http://www.ictvonline.org/virusTaxonomy.asp?version=2008&bhcp=1
Neurospora spp.	*invertase*	*inv*	INV	Nonallelic genes that have the same descriptive name and symbol are distinguished from one another by numbers that are separated from the base symbol by a hyphen (e.g., *al-1*, *al-2*, *al-3*). http://www.fgsc.net/fgn46/perkins.htm
	albino-2	*al-2* (named after a mutant allele that is recessive to the wild type)	AL-2	
	Banana	*Ban* (named after a dominant mutant allele)	BAN	
	heterokaryon incompatibility-6	*het-6* (named after a codominant allele)	HET-6	

Species	**Gene name**	**Gene symbol**	**Protein symbol**	**Database**
Maize (*Zea mays*)	*shrunken1* (note no space before number)	Dominant mutant and wild type: *Sh1* Recessive mutant: *sh1*	SH1	www.maizegdb.org/maize_nomenclature.php Maize has a highly polymorphic genome, so nomenclature can get confusing (e.g., mutations arising from transposable elements). See website for details.
Chlamydomonas reinhardtii	cytochrome oxidase	*COX3* (nuclear) *cox1* (mitochondrial) *petB* (chloroplast)	COX3 COX1 CYB6	http://www.chlamy.org/nomenclature.html#p Nuclear genes are uppercase italics (*COX3*), organellar genes are lowercase followed by a number or uppercase letter (*cox1*, *petB*) Protein symbols do not always match gene symbols. Use UniProt to find correct name and write out (e.g., The *petB* gene encodes cytochrome b6 (CYB6))

11.7.2.1 *Homologues of* Drosophila *proteins*

Certain eukaryotic proteins that were first discovered in *Drosophila* take the *Drosophila* nomenclature when referring to the full protein or gene name, regardless of the organism (proteins take an uppercase first letter, and genes take an upper- or lowercase first letter depending on whether the gene is named for a dominant or a recessive allele, or a protein function; see 'Quick guide to nomenclature' table above). These proteins are Argonaute, Delta, Dicer, Drosha, Hedgehog, Helios, Lefty, Nanog, Nodal, Notch, Polycomb, Slicer, and Toll-like receptors.

In DICER/DROSHA nomenclature, styling is as follows: Dicer (general use and multiple species), DICER1 (human/mouse protein), *DICER1* (human gene), *Dicer1* (mouse gene). Drosha (general use and multiple species), DROSHA (human/mouse protein), *DROSHA* (human gene), and *Drosha* (mouse gene).

For gene and protein families:

Protein family names are written according to proteins in the organism in question (and human is the default nomenclature for eukaryotic protein families when referring to more than one species).

Gene family names are written according to genes in the organism in question in terms of use of capital and Greek letters, etc., but are roman, not italic.

Examples:

	Protein family	Protein	Gene family	Gene
Mouse	HOX	HOXA1	Hox	*Hoxa1*
Human	HOX	HOXA1	HOX	*HOXA1*

Make it clear on first mention that it is referring to a family rather than one protein or gene (e.g., 'HOX proteins', 'the HOX family of proteins').

11.7.2.2 *Bacteria*

Genotype symbols. Each locus of a given wild-type strain of bacteria is designated by an italicized lowercase three-letter symbol, usually a mnemonic for the mutant phenotype (for instance, *ara* refers to the loci in which mutations occur that affect the response of the cell to arabinose). If several loci govern related functions, these are distinguished by italicized capital letters following the locus symbol (e.g., *araA*, *araB* and *araC*). Italics are used standardly in the literature for operon names (*pap* operon) and gene families (*mod* genes or *ara* genes) in bacteria; the style 'Pap operon' is also acceptable if it is the author's preference and used consistently. Promoter, terminator and operator sites should be indicated as, for example, *lacZp*, *lacAt*, and *lacZo*. Wild-type alleles are indicated with a superscript plus (*ara*$^{+}$ *his*$^{+}$). A superscript minus is not used to indicate a mutant locus; thus, one refers to an *ara* mutant rather than an *ara*$^{-}$ strain. The use of any other superscript terms should be avoided — designations indicating amber (Am), temperature-sensitive (Ts), constitutive (Con) and cold-sensitive (Cs) mutations, and other important phenotypic properties should follow the allele number, e.g., *araA230*(Am) *hisD21*(Ts), and should be defined. Alleles are distinguished by a serial isolation number placed after the locus symbol, separated by a hyphen if the specific locus in which it occurs is not known (for example, the alleles designated *ara-1*, *ara-2* and *ara-3* were designated *araB1*, *araA2* and *araC3* after the mutant enzymes had been identified).

Phenotype symbols. Phenotypes are denoted by three roman letters (often the same as those used for the genotype), the first of which is capitalized. The nature of the phenotype abbreviated should be described in words the first time it appears: for example, Str^{R} (streptomycin resistant).

Strain designations. Strains are designated by simple serial numbers and not italicized. Do not use the genotype as a name (e.g., 'subsequent use of *leuC6* for transduction'). If a strain designation has not been chosen, select an appropriate word combination (e.g., 'a strain containing the *leuC6* mutation').

Clones, phages, plasmids and constructs. Names for these are roman, even if they include the name of a gene (for instance, pCAT contains the gene for chloramphenicol acetyltransferase). Examples of phage names: ΦX174; fdl; λ (Greek letter is preferred to the word 'lambda'); mu (here, 'mu' means mutator; it is not the Greek letter, so do not change to µ). Plasmid names have an initial lowercase 'p': e.g., pSF1040; pW2; pBR322.

Note that the **amoeba** *Dictyostelium discoideum* uses bacterial nomenclature (see: http://dictybase.org/Dicty_Info/nomenclature_guidelines.html).

11.7.2.3 Plasmodium *Species*

Plasmodium falciparum proteins and genes can sometimes take a species prefix (Pf for *P. falciparum* proteins, *pf* for genes). These prefixes are part of the official gene or protein name, usually (but not always) given to distinguish them from the host protein or gene of the same name, and should be allowed (this is not the same as the author adding a prefix to an official name, 'Blah', to say 'DmBlah' and 'HsBlah').

11.7.2.4 Drosophila *Species*

For genes on the same chromosome, separate the gene symbols with spaces (*v w f B*); for genes on homologous chromosomes, separate symbols with a solidus (*y w* /*B*); for genes on non-homologous chromosomes, separate symbols with semicolons and spaces (*bw*; *e*; *ey*). Alleles at a particular locus are represented by the same symbol (and name) but are differentiated by italicized superscripts, numbers, capitalized initials, and so on.

Gene symbols and full names begin with a lowercase letter if the gene is first named for the phenotype of a recessive mutant allele, and begin with an uppercase letter if the gene is first named for the phenotype of a dominant mutant allele. Gene symbols and full names also begin with an uppercase letter if they are first named for an aspect of the wild-type molecular function or activity of the gene product, including genes named after an orthologue or paralogue.

Protein full names and symbols take an initial capital letter.

See the 'Quick guide to nomenclature' (Sect.11.7.2) for examples, and see FlyBase for more details about nomenclature (https://wiki.flybase.org/wiki/FlyBase:Nomenclature).

When using a gene name or symbol to indicate a phenotype (rather than a genotype), the name or symbol is non-italic.

Phenotypes usually have the same abbreviations as their genes, but are usually in roman with an initial capital letter if dominant and an initial lowercase letter if recessive.

11.7.2.5 *Nematode*

Alleles are indicated in parentheses closed up to the gene symbol, with the wild-type allele designated as a '+', *dpy-5*(+). Mutant alleles are described by one or more letters representing the laboratory of origin and the mutation number, and should be italicized, *dpy-5*(*e61*). Suffixes to the laboratory/mutation number indicator may sometimes be included, and should not be italicized (for example, dominant (dm) and temperature-sensitive (ts); *unc-1*(*e1598*dm) or *fem-1*(*hc17*ts).

11.7.2.6 *Yeast*

Yeast proteins are often designated with a following 'p', Sec4p; if the author has used this style, the copy editor should remove it and alert the author via a query.

All yeasts can exist in either the haploid or diploid state, with the haploid cells being one of two sexes (called mating types), which are designated **a** (bold) and α. Fusion of **a** and α-cells yields **a**/α diploid cells. An allele in the mating-type locus in *Saccharomyces cerevisiae* determines whether a haploid cell is of the **a** or α-mating type. Note also **a** factor and α-factor.

For *S. pombe*, alleles are indicated by a suffix separated from the gene symbol by a hyphen (e.g., *ade6-M26*, *ade6-469*). Those alleles that confer resistance or sensitivity to a substance can be indicated by superscripts: *can1-1*^{r} (canavanine resistance), *cdc2-5*^{ts} (temperature-sensitive), or *top2-250*^{cs} (cold-sensitive).

Alleles are indicated by an Arabic number separated from the gene symbol by a hyphen (*his2-1*). One exception is the mating-type loci (*MAT*), with wild-type alleles *MAT***a** or *MATα* (note bold **a** but not α).

11.7.2.7 *Viruses*

Systems of viral nomenclature vary greatly. The type of mutant is designated by two lowercase letters, followed by the complementation group indicated by a capital letter, and then a mutant number, all of which are italicized:

For simian virus 40 (SV40), a temperature-sensitive (ts) mutant may be designated SV40 *tsA7*.

11.7.2.8 *Mutant Organisms*

For an example of how to describe a mutant organism, see Sect. 11.7.2.10 below.

If an organism is deficient in two proteins, spell out first in the text and then use the two genotypes. For example, 'mice that are deficient in both CXCR3 and CXCR5….'. Following this, '*Cxcr3*$^{-/-}$*Cxcr5*$^{-/-}$ mice' can be used (no space between the genotypes).

Do not use the terms 'double-deficient' or 'triple-deficient', as this can become confusing.

Keep in mind the distinction between a mutation (an alteration of the primary sequence of the genetic material) and a mutant (an organism or a strain carrying one or more mutations; or the protein resulting from a mutated gene). One may speak about the mapping of a mutation, but one cannot map a mutant. Likewise, a mutant has no genetic locus, only a phenotype. A protein can have alterations, substitutions, etc. (but not mutations, as these are in the DNA only) that make it a mutant protein. A gene is mutated, but an organism or a protein is not.

Δ may be used as a symbol of gene deletion in mutant organisms.

For altered proteins and genes, the preference is to write gene mutations in superscript (*Gene*$^{\mathrm{AxxA}}$, e.g., *Kras*$^{\mathrm{G12V}}$) (where the superscript denotes the resulting amino acid alteration) and protein variants in-line with a hyphen (PROTEIN-AxxA, e.g. KRAS-G12V). Alterations should always be in the format AxxA (A= amino acid, x=number) and not Axx (e.g., KRAS-G12V and not KRAS-V12).

11.7.2.9 *Human Mutations*

In DNA (and RNA) mutations, the nucleotide number can be preceded by 'c.', 'g.', 'm.', or 'r.' to indicate complementary, genomic or mitochondrial DNA or RNA, respectively (e.g., 'g.1234A>G').

Nucleotide substitutions are represented as the position number followed by the nucleotide change, such as '1234A>G'. Do not use italics or rightwards arrows.

Nucleotide deletions are designated by 'del' after the deleted interval, optionally followed by the deleted nucleotides, such as '1997–1999del' or '1997–1999delTTC'. Use en rules in number ranges.

Nucleotide insertions are designated by 'ins' followed by the inserted nucleotides (e.g., '1997–1998insTG'.

Nucleotide alterations (that is, mutations) can be listed in a coding region with a leading 'c.' (c.422C>G) and in a noncoding region with a leading 'g.' (g.1789032C>G); protein alterations should be listed with a leading 'p.' but can use the three-letter abbreviation for the amino acid, and should follow the format p.Xxx###Yyy, where Xxx and Yyy are the amino acid abbreviations (for example, p.Ala341Gly or p.His567Cys). Two or more changes in one individual (that is, changes in different alleles) are listed with each in square brackets separated by a '+' (according to the correct nomenclature described above as [change in allele 1]+[change in allele 2]) and a single leading 'c.' or 'p.' For example: c.[71A>C]+[71A>C] denotes a homozygous A to C change at nucleotide 71. The corresponding amino acid change can be listed as, for example, p.[Ala71Gly]+[Ala71Gly]. A heterozygous change can be listed as, for example, c.[71A>C]+[96G>T], meaning there is a change from A to C at nucleotide 71 and a change from G to T at nucleotide 96. The corresponding protein alteration can be listed as, for example, p.[Ala71Gly]+[His96Pro]. If possible, the nucleotide position of the alteration should be given in the abstract.

Variant proteins. Use the format 'Arg98Ser' (or the single-letter code) for amino acid substitutions; the format 'Thr97–Cys102del' to indicate deletions; and the format 'Thr97–Trp98ins3' (or 'Thr97–Trp98insLeuGlnSer') to indicate the number of amino acids inserted.

Trp98X means that tryptophan 98 is followed by a stop codon. Trp42fsX46 means that a frameshift at tryptophan 42 results in a stop codon after 46 more codons.

Human pedigree. For human pedigree charts, roman numerals are used for generations and Arabic numerals for individual members of each generation. Begin with I or 1 for the most senior generation or member

11.7.2.10 *Mutant Mice*

If the gene responsible for the mutations has been characterized, the mutant mouse is described as follows: mice with no CXCR5 protein can be referred to as CXCR5-deficient mice, CXCR5-null mice, *Cxcr5*-null mice, *Cxcr5*-knockout mice or *Cxcr5*$^{-/-}$ mice (but not *Cxcr5*-deficient mice). Note that null means a complete loss of function; '*Cxcr5*-mutant mice' can be used to describe mice carrying any mutation in this gene, null or otherwise. Protein symbols should not be accompanied by '$^{+/+}$', '$^{+/-}$', or '$^{-/-}$', as the solidus — which is used in genotype designations to denote the two copies of a gene in a diploid genome — is misleading in regard to proteins, which do not come on two strands. '*Cxcr5*-transgenic mice' refers to mice in which a *Cxcr5* transgene has been introduced.

If the gene is not characterized, the mouse is described using the name and symbol of the first identified mutant. Symbols of mutations are italics: those conferring a recessive phenotype begin with a lowercase letter; symbols for dominant or semi-dominant phenotype genes begin with an uppercase letter. The full name of the mutant mouse is set in lowercase roman, except when named after someone. Examples include: recessive spotting (*rs*), abnormal feet and tail (*Aft*), or Jackson shaker (*js*).

Strains of laboratory mice are roman; BALB/c, C57BL/6, for example.

There are different formats for the different types of mutations in mice: see the nomenclature guide at http://www.informatics.jax.org/mgihome/nomen.

11.7.2.11 Drosophila *Mutants*

Symbols for both mutants and chromosome aberrations in *Drosophila* mutants are italicized; they should not contain Greek letters, subscripts, or spaces. The spelled-out names of mutants are not italicized (for example, Antennapedia mutant, but *Antp* mutant). Symbols for mutant types are abbreviations of their characterizing names and should be written in full on first use.

11.7.2.12 *Genetic Complexes*

A genetic complex is made up of apparently functionally related loci that are genetically closely linked. Alternative states of complexes are referred to as haplotypes rather than alleles. Names of genetic complexes are not italicized. For example, BX-C for Bithorax complex and ANT-C for Antennapedia complex in *Drosophila*; MHC for major histocompatibility complex.

11.7.2.13 *Transposons and Insertion Elements*

Insertion elements are relatively small; it is not known whether they encode any gene products. They are called IS*1*, IS*2*, and so on. (Note that the letters are roman and the numbers italic.)

Transposons are larger than insertion elements, consisting of a bacterial gene surrounded by particular sequences. Transposons can be defined as a class (for example, TnA, which means all transposons containing an ampicillin gene) or as individuals (such as Tn*5* or Tn*901*). However, there are many types of transposon, and nomenclature is not always consistent. Other transposons include *piggyBac*, *hitchhiker*, *Sleeping Beauty* and P elements.

When a transposon or insertion element is inserted into a known gene, it is defined (for example, *lac*::TN*901* or pSC101::Tn*5*). Note the double colon.

11.7.2.14 *Chromosomes*

Use *n* for the diploid number of chromosomes.

Karyotypes may be described using a standard shorthand system of symbols. In general, this has the following order: total number of chromosomes, sex chromosome constitution, description of abnormality. For example, 46,XX,t(4;13) (p21;q32) indicates a translocation between chromosome 4 and 13, with breakpoints in the short arm (p) of chromosome 4 at band 21 and in the long arm (q) of chromosome 13 at band 32 in a female human. Symbols :: (break and join) and + or – (placed after arm or structural designation to indicate whether arm or structure is larger (+) or smaller (–) than normal, or before autosome number or group letter designation to indicate whether chromosome is extra (+) or missing (–)) are also used and do not need to be defined.

Chromosome aberrations in *Drosophila* are designated according to the elementary rearrangements: deficiency (Df), duplication (Dp), inversion (In), ring (R), translocation (T) and transposition (Tp).

A semicolon separates mutations that are on different chromosomes; a comma means they are on the same chromosome. For example, *Fgf4;Fgf9,Fgf17* indicates a triple mutant mouse — *Fgf4* is on chromosome 4 and *Fgf9* and *Fgf17* are both on chromosome 14.

Other abbreviations are given below; these are often best spelled out in full. If this is not possible (i.e., used frequently in the paper) then they should be defined:

Abbreviation	Meaning
del	deletion
dup	duplication
ins	insertion
inv	inversion
ter	terminal or end break (no reunion, as in terminal deletion)

11.7.2.15 *MicroRNAs and Small Interfering RNAs*

MicroRNAs (miRNAs) are single-stranded RNA molecules about 21–23 nucleotides in length. They are encoded by genes, but miRNAs are not translated into protein; instead, each primary transcript (pri-miRNA) is processed into a short stem–loop structure (pre-miRNA) and finally into a functional mature miRNA.

The following nomenclature rules are intended as a guideline to standard usage, but follow the author as long as consistency is maintained.

- Generally, miRNA names begin with the prefix 'mir', although the case depends on the organism, and whether it is the locus or the transcript. Exceptions include lin-4 and let-7.
- RISC is the abbreviation for RNA-induced silencing complex
- Short interfering RNAs (siRNAs) should be referred to as 'the *gene*-targeting siRNA' or 'an siRNA against *gene*'
- '*gene*-specific siRNA' is only accurate only if the siRNA is totally specific to one gene in practice.
- The miRNA gene (the locus) is given in italics.
- The stem–loop structure is given in roman.

- The mature miRNA is given in roman.
- To check the correct nomenclature for a miRNA, use miRBase (note that italics are omitted in databases): http://microrna.sanger.ac.uk/sequences.

Quick guide to miRNA nomenclature:

Organism	Gene	Stem–loop	Mature miRNA
Human	*mir-10a*	mir-10a	miR-10a
Mouse	*mir-10a*	mir-10a	miR-10a
Drosophila	*mir-4*	mir-4	miR-4
Arabidopsis thaliana	*MIR161*	MIR161	miR161

11.8 DNA and RNA

Usually, DNA is transcribed into RNA, which is then translated by a ribosome into protein. The length of a piece of DNA or RNA can be specified in base pairs (abbreviated bp) or thousands of bases (kilobases, abbreviated kb or kbp). The size can also be expressed in one of the following (equivalent) forms:

> molecular weight (abbreviated mol. wt.; no units needed)
>
> relative molecular mass (symbol M_r; no units needed)
>
> molecular mass (expressed in daltons, which may be abbreviated Da, in kilodaltons, abbreviated kDa, or in megadaltons, abbreviated MDa)

The mass or weight can be written out in full (e.g., 150,000 daltons), using a capital K for the last three zeros (e.g., 150K; no space before K), or using the abbreviated form (e.g., 150 kDa). All these forms are acceptable; follow author's usage if it is consistent. Consistency between chapters is not necessary if this involves considerable changes. There are ways to avoid this becoming cumbersome, e.g., instead of 'Murine BSF-1 has a relative molecular mass (M_r) of 20,000 (20K)', say 'Murine BSF-1 has a relative molecular mass of 20,000 (M_r 20K)'. Note that 'K' must be retained in M_r ranges: 50–100K indicates an M_r range of 50 to 100,000, not 50,000–100,000, which would be 50K–100K. The class of molecules known as 'low-molecular-weight' phosphatases keeps this designation.

The specific kinds of ribonucleic acids may be referred to in abbreviated form, but define on first use. The following prefixes may be used with RNA or DNA, if applicable:

c	complementary	cDNA
ds	double-stranded	dsDNA
hn	heterogeneous nuclear	hnRNA
k	karyosomal	kDNA
m	messenger	mRNA

n	nuclear	nDNA
r	ribosomal	rRNA
s	soluble	sRNA
ss	single-stranded	ssDNA
t	transfer	tRNA
v	viral	vRNA

DNA double helices can come in A, B and Z forms, formatted as A-DNA, B-DNA and Z-DNA.

tRNA and amino acid combinations can be described in the following forms:

glycine tRNA or $\mathrm{tRNA}^{\mathrm{Gly}}$ (tRNA that combines with glycine)

glycyl-tRNA or Gly-tRNA (tRNA attached to glycyl residue)

$\mathrm{tRNA}^{\mathrm{Ala}}$ (type 1 of tRNA that combines with alanine)

The system for indicating polymers of synthetic ribonucleotides is the same as that for polypeptides (see Sect. 11.9):

poly(A-U) or $(\mathrm{A\text{-}U})_n$

poly(A,U) or $(\mathrm{A,U})_n$

In polydeoxyribonucleotides, the position of the deoxy designation is optional:

poly(dA-dT) or poly d(A-T) or $(\mathrm{dA\text{-}dT})_n$ or $\mathrm{d(A\text{-}T)}_n$

Associated (noncovalently linked) polynucleotides are indicated with a centered dot:

poly(G)·polyI or $(\mathrm{G})_n \cdot \mathrm{I}_n$

In base-paired nucleotides, a centered dot indicates association:

G•C

a hyphen indicates sequence

G-C

and a plus sign indicates content

G+C content was 30%

Nucleotide sequences of codons (groups of three nucleotides that code for an amino acid) may be written without separating punctuation:

UAU *or* CAG

The lowercase p indicating a terminal 5′- or 3′-phosphate on a polynucleotide is roman (pA-G-A or A-U-Gp).

In polynucleotides, hyphens indicate the usual left to right direction and arrows indicate the reverse:

A-G-Gp

pG←U←A

11.9 Polypeptides

For more detailed information, refer to Sect.11.11.3.

A polymer in which the same structural unit(s) is (are) repeated may be designated by adding the prefix poly or the suffix subscript *n* to the sequence in parentheses.

poly(Ala) or (Ala)$_n$

poly(DLAla, Gly, Lys) or (DLAla, Gly, Lys)$_n$

poly(DLAla-Lys) or (DLAla-Lys)$_n$

If the composition of a polymer is known, the percentage (without the percentage symbol %) is placed superscript after the abbreviation. Note that the comma is omitted in polymers of unknown sequence (cf. above).

poly(Glu^{56} Tyr^{44}) or (Glu^{56} Tyr^{44})$_n$

poly(DLGlu^{50}-LTyr^{50}) or (DLGlu^{50-} LTyr^{50})$_n$

11.9.1 Globulins

Globulins are a class of proteins characterized by insolubility in water but solubility in saline. Pseudoglobulins are water soluble but are otherwise closely related to true globulins.

Greek letters and subscript Arabic numerals are used to designate globulins and globulin fractions (see also Sect. 11.10.9 for γ-globulin).

α-globulin, α$_1$-globulin, α$_2$-globulin

γ-globulin

β-globulin

If one of these terms begins a sentence or item in a list or table, the G in globulin is capitalized:

α_1-Globulin has been the topic of several...

11.9.2 Enzymes

Follow *Enzyme Nomenclature* for styling of all enzyme names and for recommended names. An enzyme's name nearly always ends with the suffix -ase. If a particular enzyme is a main subject of a paper, at the first mention the recommended name should be followed by the EC code number in parentheses. Thereafter use the recommended name.

Examples of restriction enzymes (or restriction endonucleases) are:

Bme 899 Bsu 1076 AccI

EcoRI (not EcoR1) Sau3A1

EcoP15 HpaI (not Hpa1) Sau3AI

Sau3A Hin1056II DpnI

The first three letters indicate the genus and species from which the enzyme was derived (e.g., EcoRI from *E. coli*). Do not use italics for the first three letters, and close up the entire name. (The changed styling from italics to roman was an IUPAC decision in 2003). Note that suffixes consisting only of Arabic numbers are separated by a space. Any other suffixes should be closed up. (See also Sect. 11.11.3).

Define non-SI rate enzyme units or add an author query if unclear. The SI rate unit is the katal (kat), equal to 1 mol substrate transformed (or product formed) per second. Michaelis constant, K_{m}; velocity of an enzyme-catalysed reaction at infinite concentration of substrate, V_{max}.

Synthetase and synthase both catalyse a biosynthetic reaction, but a synthetase acts by splitting ATP, whereas a synthase does not. The ATP-synthesizing enzyme in oxidative phosphorylation is correctly referred to as ATP synthase. Enzymic and enzymatic mean the same, but use enzymatic.

Trade names. Enzymes are sometimes given trade names, e.g., Glycanase made by Genzyme; in this case they should start with a capital and their correct name given and their function explained.

Taq polymerase used in the polymerase chain reaction is derived from *Thermus aquaticus* so is italicized.

11.10 Immunology

11.10.1 General

The cells of the immune system are generally divided into those that belong to the **innate immune system** and those that belong to the **adaptive immune system**.

The adaptive immune system comprises lymphocytes (i.e., T cells, B cells and natural killer (NK) cells), whereas the innate immune system comprises monocytes and macrophages, granulocytes (i.e. neutrophils, eosinophils and basophils) and dendritic cells (DCs). Mast cells are tissue-resident granulocytes that are closely related to basophils.

Some cell types do not fit firmly into either category and are considered 'innate-like lymphocytes' — for example, γδ T cells, NK cells and B1 cells. Innate lymphoid cells (ILCs) are a recently described immune cell type that also fit into this category. 'Group 1 ILCs' refers to NK cells and ILC1s (ILC1 being defined as an innate lymphoid cell 1 subset and not referring to all group 1 ILCs). 'Group 2 ILCs' refers to ILC2s (thus, the terms group 2 ILCs and ILC2s are currently interchangeable, but this many change if other subsets are included in this group). Group 3 ILCs refers to LTi cells, NCR$^+$ ILC3s and NCR$^-$ ILC3s.

Epitopes are the parts of the antigen that combine with the antigen-binding site on an antibody or lymphocyte receptor. Antigen-presenting cells (APCs) present foreign antigens to T cells.

11.10.2 Abbreviations and Terminology

Immunology papers tend to contain enormous numbers of abbreviations. The copy editor should aim to reduce them as much as possible by avoiding unnecessary use (e.g., of 'mAb' and 'Ig') or writing the term out each time if an abbreviation is used fewer than three times. Hyphens should be used carefully to clarify ambiguities: antibody-binding site (site binds antibody) versus antibody binding site (site on antibody).

Quick reference I: Common abbreviations

Abbreviation	Definition
APC	antigen-presenting cell (plural APCs)
β_2m	β_2-microglobulin
CDR	complementarity-determining region
CML	cell-mediated lympholysis
conA	concanavalin A (note lowercase first letter)
CSF	colony-stimulating factor
CTL	cytotoxic T lymphocyte (plural CTLs)
Fab	antigen-binding fragment
Fc	crystallizable fragment

Abbreviation	Definition
Fh	hinge fragment
FITC	fluorescein isothiocyanate
FK506	(immunosuppressant; closed up)
Flag epitope	[EYKEEEK]$_2$ 14-amino acid C-terminal extension
HLA	human leukocyte antigen
HRP	horseradish peroxidase
IFNγ	interferon-γ
Ig	immunoglobulin; is not allowed to stand alone, but always with designating letter, e.g., IgA
IL-1, IL-2, etc.	interleukin-1, 2, etc.
LFA	lymphocyte function-associated antigen
LPS	lipopolysaccharide
mAb	monoclonal antibody
Mϕ, MΦ	macrophage (abbreviation permitted only in figure label)
MHC	major histocompatibility complex
NLR	NOD-like receptor (when referring to the protein family)
NLRP1, 2...	NOD-, LRR- and pyrin domain-containing 1, etc.
NLRC4	NOD-, LRR- and CARD-containing 4
PBL	peripheral blood lymphocyte
TCR	T cell receptor
T_D	delayed-type hypersensitivity T cell
T_H	T helper cell
T_S	suppressor T cell
TNF	tumour necrosis factor
TNP	trinitrophenyl

For a fuller list of accepted immunological abbreviations, see the AAI list at http://www.jimmunol.org/info/abbreviations.

Quick reference II: Notes on correct use of terminology

Terminology	Comment
brefeldin A	Ensure 'b' is lowercase
C57BL/6	Can be abbreviated to B6, but define on first use
CCR5Δ32	This is an allele of CCR5 with a 32-bp deletion

Terminology	Comment
$\mathrm{CD45}^{\mathrm{lo–neg}}$	This styling is correct. Do not allow the incorrect $\mathrm{CD45}^{\mathrm{lo/–}}$. Note that an individual cell cannot be $\mathrm{CD45}^{\mathrm{lo–neg}}$, but a group of cells can.
CD8αα	This styling is correct. Do not allow a hyphen.
cell dimensions (crystallography)	Use italics for $a = b = c = \beta$
common γ-chain	Common γ-chain is sometimes abbreviated γ_c but avoid if possible as this abbreviation is ambiguous
immunosuppressed	Do not use immune suppressed
immunoregulatory	Do not use immune-regulatory
Langerhans	Do not use an apostrophe
$\mathrm{LT}\alpha_1\beta_2$	This means lymphotoxin with one α-subunit and two β-subunits. Note the subscripting.
mRFP	Clarify ambiguity. The 'm' can mean monomeric or mouse. Clarify and always spell out.
$\mathrm{p65}^{Lck}$	This means p65 protein from *Lck* gene
pancreata	This is the correct plural form of pancreas
rIL-2	Clarify ambiguity. This can mean recombinant or rat. Always spell out.
SCID, *scid*	SCID is the phenotype (strain); *scid* is the genotype
sIgX	Use for 'soluble' or 'secreted' only; revise to mIgX (membrane IgX) if 'surface' is meant
thymi	Correct plural form of thymus
tolerize, tolerogenic	These terms are acceptable
type I or II interferons	Write out. Do not use Type 1 or 2 IFNs.
V_α	Numbers that follow subscript Greek letter are NOT subscripted (e.g., $\mathrm{V}_\alpha 5$)
'vaccinal','activatory'	Incorrect. These terms do not exist and must be changed.
T cell, B cell and NK cell	T cell, B cell and NK cell are never hyphenated, even when used adjectivally
DP, DN and SP	These cell subset abbreviations are specifically defined as follows: DP, double-positive (e.g., $\mathrm{CD4}^+\mathrm{CD8}^+$); DN, double-negative (e.g., $\mathrm{CD4}^-\mathrm{CD8}^-$); SP, single-positive (e.g., $\mathrm{CD4}^+$ or $\mathrm{CD8}^+$). Clarify if they seem to be used with another meaning (e.g., other markers).
IL-1R	Receptors for growth factors are indicated by an R after the acronym, e.g., IL-1R
$\mathrm{T}_{\mathrm{reg}}$ cells, Foxp3 proteins	Pluralize the words 'cell', 'protein', or 'value', not the designation, i.e., $\mathrm{T}_{\mathrm{reg}}$ cells, not $\mathrm{T}_{\mathrm{regs}}$; Foxp3 proteins, not Foxp3s, M_{r} or pK_{a} values, not M_{r}s, pK_{a}s.

Terminology	Comment
poly(I:C)	Refers to RNA
poly(dI:dC)	Refers to DNA

11.10.3 Nomenclature for the NLR Family of Receptors

The NOD-like receptor (NLR) family of intracellular pattern recognition receptors has various names, including NODs, NALPs, NOD-LRRs, NATCH-LRRs, CARDs, PYPAFs and Caterpillers. Specifically:

- NLR is used to refer to the family.
- NLRP1, 2, 3, etc. is used *exclusively* for NOD-, LRR- and pyrin domain-containing 1, etc.
- NALP1, NALP2, NALP3, etc. are the former designations for NLRP1, 2, 3, etc. Where these are used, include 'also known as NALP3' in defining at first mention, e.g., from NALP3 to 'NLRP3 (NOD-, LRR- and pyrin domain-containing 3; also known as NALP3)', and use NLRP3 thereafter.
- NLRC4 is used for NOD-, LRR- and CARD-containing 4.
- NOD stands for nucleotide-binding oligomerization domain.
- LRR stands for leucine-rich repeat.

All abbreviations should be defined at first mention, but NOD and LRR do not need to be defined within those definitions.

11.10.4 B cells

The standard notation for B cells is to use B1 cells, B2 cells.

11.10.5 Chemokines

The standardized form for chemokines is CCL3, CCL4, CXCL3, CXCL2, CX_3CL1, CXCL1, etc. Standard names should be used. Where alternative names are used by the author, it is preferable for them to be accompanied by the standard names, but follow the author and be consistent. Examples of common names still in use:

- CCL5 (also known as RANTES)
- CXCL2 (also known as MIP2)
- CX_3CL1 (also known as fractalkine)
- CCR2 (also known as MCP1R)

11.10.6 Nomenclature for T cells

- T cells that express αβ T cell receptors (TCRs) are considered to be conventional T cells. Some T cells express γδ TCRs.
- There are two main populations of conventional T cells: $CD4^+$ T cells and $CD8^+$ T cells.

- $CD8^+$ T cells recognize peptides on MHC class I molecules. They can differentiate into cytotoxic T lymphocytes (CTLs).
- $CD4^+$ T cells recognize peptides on MHC class II molecules. They can develop into different T helper (T_H) cell subsets, which are characterized by the expression of unique transcription factors and cytokines. The main T_H cell subsets are the T_H1, T_H2, T_H9 and T_H17 cell subsets.
- T cells that are resident at epithelial surfaces are referred to as intraepithelial lymphocytes (IELs).
- CTLs and T_H cells are collectively referred to as 'effector T cells' or T_{eff} cells.

Regulatory T (T_{reg}) cells are T cells that are associated with the expression of the transcription factor FOXP3 and negative regulation of inflammatory responses. There are two major populations of T_{reg} cells that are classified based on whether they emerge in the thymus or in peripheral tissues. Thymus-derived T_{reg} cells (tT_{reg} cells) are sometimes referred to as natural or naturally occurring T_{reg} cells. Peripherally induced T_{reg} cells (pT_{reg}) cells are sometimes referred to as 'induced' or 'adaptive' T_{reg} cells. Use the abbreviations tT_{reg} cells and pT_{reg} cells.

Change 'T_H1 response' to 'T_H1 cell response' 'or T_H1-type response' as appropriate to context. . It is now common in the literature to refer to e.g. 'type 2' responses when discussing T_H2-type biological responses. The distinction is that a type 2 immune response is characterized by T_H2-type cytokines, but sometimes the cytokines are produced by non-T cells (hence the need to use 'type 2' rather than 'T_H2-type').

For invariant natural killer T cells, use the style iNKT cells.

T cell antigen receptor subtypes have no hyphen or space (TCRαβ)

Quick reference guide: T cell abbreviations

Abbreviation	Definition
T_H1	T helper 1 cell
T_H2	T helper 2 cell
T_H9	T helper 9 cell
T_H17	T helper 17 cell
T_{FH}	T follicular helper cell (follicular helper T cell also acceptable
T_{reg}	regulatory T cell
T_{CM}	central memory T cell
T_{EM}	effector memory T cells
T_{RM}	tissue resident memory T cell
T_{VM}	virtual memory T cells

11.10.7 CD Numbers

Immunologists often denote the phenotype of a cell by indicating which proteins are expressed at the surface, for example, $CD4^{+}CD8^{-}$ (not $CD4^{+}8^{-}$) T cell

Use superscript $^{+}$ or $^{-}$ (minus) when describing cells as positive or negative for a surface marker (e.g., $CD4^{+}$). The words 'positive' or 'negative' must written in full if the cell-surface molecule name is a complete word (e.g., 'vimentin'). There should be no space between separate surface markers, e.g., $CD4^{+}CD8^{-}$.

Letters appended in protein names are always in lowercase, e.g., CD11a (where human gene = *CD11A*; mouse = *Cd11a*).

Where '+' is divided into 'low', 'mid' and 'hi', e.g., $CD4^{+}CD25^{hi}$ T cells, ensure consistency of terminology, and — if inconsistent — change as follows:

'intermediate/int' to 'mid'

'bright' to 'hi'

'dim' to 'low'

11.10.8 Histocompatibility Antigens

11.10.8.1 *Mice*

Skin graft rejection in mice is controlled by the histocompatibility (H) gene families, *H1*, *H2*, *H3* and *H30*. *H2* is the most important because grafts with an *H2* mismatch are rejected much more rapidly than grafts in which *H2* matches. *H2* is therefore called the major histocompatibility complex (MHC); the rest are minor histocompatibility loci.

The *H2* family has several loci, for example *H2-K*, *H2-D* and *H2-L*. The genes at these loci vary among mouse strains. The haplotypes of several common strains of mouse have names: the BALB/c strain of mice has the haplotype $H2^{d}$, and each locus is designated with superscript d, e.g., $H2\text{-}K^{d}$. A cross between two strains of mice with different haplotypes may give rise to a strain of, for example, $H2^{d/k}$ mice. **Note**: When describing haplotypes, phenotypes or proteins, use roman; when talking about genes, use italics.

The *H2* loci encode molecules that fall into three classes, I–III. These must be referred to as MHC class I, II, or III, never simply as 'classes I, II or III' to avoid confusion with other molecule classes.

Immune response (*Ir*) genes are linked to the *H2* complex. For nomenclature, follow http://www.informatics.jax.org.

11.10.8.2 *Humans*

The histocompatibility complex in humans is called the human leukocyte antigen (HLA) region. It has classes I–III; class I molecules are subdivided into HLA-A, HLA-B and HLA-C, and are expressed in association with β_2-microglobulin (β_2m). The genes vary among individuals as they do between mouse strains, and are named, for example, *HLA-B7*. The notation HLA-DQA1*0102 means HLA locus, DQA1 gene, 0102 allele. Class II genes are found in the HLA-D region, which is divided into DP, DR and DQ (roman) for historical reasons.

Note that H-2, I-A/E and HLA-A take hyphens.

When antigens are recognized by T cells, they are 'seen' in the context of the MHC or HLA antigens.

11.10.9 Antibodies

Antibodies (immunoglobulins) are proteins made by plasma cells (which develop from B cells) and are directed against antigens. The preferred wording is 'antibody to X', not 'antibody against X'. Immunoglobulin-γ, the first to be discovered, is termed IgG and contains several subclasses (IgG1–IgG4 in humans). Other antibodies are IgM, IgA1 and IgA2, and IgD and IgE. Immunoglobulins are made of two types of polypeptide chain: the heavy (H) and light (L) chains. The class of the immunoglobulin depends on the type of heavy chain. Heavy chains are: γ1–γ4 (IgG(1–4)); μ (IgM); α (IgA); δ (IgD); or ε (IgE). IgG is composed of two H and two L chains; IgA has four of each and IgM has ten. The chains are arranged in pairs and are linked by disulfide bridges. Light chains are classed as either κ or λ (preferably, refer to the protein as 'immunoglobulin κ-chain' or 'immunoglobulin λ-chain' at first use, then drop 'immunoglobulin' thereafter (i.e., λ-chain) if used many times in an article).

Define 'immune exclusion' as 'the process by which immunoglobulin-coated bacteria are prevented from traversing the gut epithelium'.

Antibodies can be sliced up by various proteases, yielding Fab (antigen-binding fragment), Fc (crystallizable fragment), Fh (hinge fragment), or F(ab′) (two Fabs plus Fh, making a divalent antigen-binding fragment).

Antibodies are very diverse in structure because they are made from a selection of several different types of gene. All heavy and light chains have a region that is called 'constant' — the μ heavy chain of IgM has a Cμ constant region (similarly, γ and Cγ, δ and Cδ). Thus the C region of the heavy chain of an antibody is made up of constant region, which is why it can be crystallized; the rest of an antibody is 'variable' (V). The variable portions from a mixture of antibodies (polyclonal antibodies) are sufficiently different that they do not crystallize.

Between the C and V regions of the antibody are the diversity (D) and joining (J) segments. A heavy chain may use any one of several JH segments (JH1–JH4) and any one of several D regions (D1–*n*), and there are also several VH regions to choose from. This is the basis of antibody diversity. The segments are rearranged as DNA, by recombination in the switch region encoding the D and J segments, giving a gene with one of each type of segment separated by introns that are spliced out after the gene is transcribed. Generally, put Greek subscripts in 'alphabetical' order when closed up: Vα4.1Vβ6.2 (not Vβ6.2Vα4.1). Genes for C, V, D and J are italic. The segments are as follows: V, variable; D, diversity; J, joining; C, constant; S, switch; I, intervening.

Light chains are much the same as the heavy chain but have no D regions. Certain parts of antibodies are even more variable than the above scheme predicts. These are the hypervariable regions, also called the complementarity-determining regions (CDRs: CDR1, CDR2 and CDR3). The rest of the antibody is composed of framework regions (FRs).

These names can be applied to the antibody protein or to the DNA that codes for that piece of antibody.

For immunoglobulin structure, do not subscript domains α1, α2.

When referring to an antibody raised against a particular antigen, the prefix anti- is allowed. The abbreviation α- is not allowed (only consider using in figures and tables to save space). Do not change 'anti-x' to 'x-specific' (the antibody may be x-reactive without being x-specific).

11.10.10 V(D)J Recombination, Class-Switch Recombination and Somatic Hypermutation

Most content on **V(D)J recombination, class-switch recombination and somatic hypermutation** refers to gene segments (e.g., Cm), which are not italicized as they are only *parts* of genes.

V(D)J recombination generally takes parentheses because it refers to two antibody chains: the heavy chain, in which V, D and J are joined, and the light chain, in which only V and J segments are joined.

11.10.11 T Cell Differentiation Antigens

11.10.11.1 *Mice*

THY1 is found on all T cells in two forms: THY1.1 and THY1.2. L3T4 (CD4; see Sect.11.10.11.2) is an antigen found on some T cells; subsets with this antigen are termed L3T4$^+$. The antigens LY2 and LY3 (also called LYT2/CD8A and LYT3/CD8B1; the T stands for T cell) are linked (all LY2$^+$ cells are LY3$^+$), but L3T4 (CD4) and LY2, LY3 (CD8) are mutually exclusive in peripheral blood lymphocytes. LY1 (or LYT1) is expressed on either subset in varying amounts. L3T4 cells are (usually) T_D or T_H cells; LY2$^+$ cells are CTLs or T_S cells.

11.10.11.2 *Humans*

The names of the molecules that have the same characteristics in humans vary; T4 (CD4) corresponds to L3T4, and T8 (CD8) is the counterpart of LY2, LY3. The CD nomenclature is now used in both humans and mice, as the historical names tend to obscure the fact that the molecules are to all intents and purposes the same.

CD3 (or T3) is present on both CD4 and CD8 subsets and is closely associated with the T cell receptor (TCR) on the surface of the T cell. Note that TCR–CD3 complex takes an en rule (per style for complexes).

The TCR is composed of two subunits, each of which is rather similar to an antibody heavy chain in structure (each has C, J, D and V regions); the genes for these are rearranged like those of antibodies. There are at least two types of heterodimer, αβ and γδ, which are disulfide-linked, giving αβ and γδ T cells. The nomenclature for each chain separately is TCRα, TCRβ, TCRδ, or TCRγ. 'TCR α-chain', etc. may be used if appropriate in context. The whole TCR protein is TCRαβ or TCRδγ; the types of TCR are αβ TCR and δγ TCR.

CD3 contains several subunits: two are also named γ and β, but these should not be confused with the antibody-like γ-chain and β-chain of the T cell receptor.

11.11 Biochemistry

11.11.1 Lipids and Carbohydrates

For carbohydrates, see: https://www.qmul.ac.uk/sbcs/iupac/2carb/

For lipids, see: http://www.lipidmaps.org/data/classification/LM_classification_exp.php

11.11.2 Oligosaccharides

For oligosaccharides, follow nomenclature as outlined in *Essentials of Glycobiology* (eds Varki, A. et al.; online version can be found here: http://www.ncbi.nlm.nih.gov/books/NBK1908/). For example, two glucose residues can join together in more than one way, giving maltose (Glcα1–4Glc) and gentiobiose (Glcβ1–6Glc). If branches and sub-branches must be represented, use parentheses for the former and brackets for the latter. Examples: 'GalNAcα1–4GalNAcα1–4(Glcβ1–3)GalNAcα1–4GalNAcα1–4GalNAcα1–3Bacβ1–NAsn,' 'Galα1–6Glcα1–2βFru'.

11.11.3 Proteins

Nomenclature generally follows the author. UniProt (http://www.uniprot.org) may be referred to as a guide.

Amino acids and their residues can be referred to by their three-letter abbreviation, the first letter always being uppercase, or by the standard single-letter notation (no definition is necessary):

Amino acid	Abbreviation	Symbol
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid*	Asp	D
Cysteine	Cys	C
Glutamine	Gln	Q
Glutamic acid*	Glu	E
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I

Amino acid	Abbreviation	Symbol
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

*Asp and Glu are spelled out as aspartate and glutamate (not aspartic acid and glutamic acid, unless their side chains become protonated).

Styling. *Do not use the single-letter code* if the peptide is shorter than six residues, or if there is danger of confusion with single-letter nucleotide codes. Do not include a space between residue and its number, e.g., Thr80 (it is helpful to not have to hyphenate, e.g., the Thr-80 mutant). Point mutants with position numbers should preferably be listed as single-letter code (e.g., 'R12L' instead of 'Arg12 to Lys' or 'Arg12→Lys'). Using superscript for alterations (Cdk9^{D167N}) is also OK (but see also Sect.11.7.2.8). If no position number is specified, 'arginine-to-lysine alterations' is OK in the text.

Protein names should generally be lowercase (e.g., actin, laminin) with the following exceptions:

- The nomenclature committees or conventions for some organisms, notably *Drosophila*, specify that protein names should start with an initial capital letter, and proteins that were first identified in these organisms may be capitalized by convention when they appear in other species.
- Proteins for which the full name matches the gene symbol may be capitalized to match the gene symbol if desired (e.g., Ras, Rac, or Rho); follow the author. Others derived from abbreviations may also be capitalized by convention (e.g., EphA2).
- Capitalization is permissible if it contributes to clarity or consistency. For example, protein names that might otherwise be misread as common English words may be capitalized to avoid confusion.

Stereoisomers of amino acids are distinguished by D and L (small caps).

Copy editors should avoid the term 'primary sequence' and instead use 'primary structure'. In polymers or sequences of amino acids, the three-letter abbreviations should be joined by hyphens if the sequence is known, or separated by commas if it is not. So, for example, Gly-Ile-Gly-Phe-(Gly, Tyr, Val, Ser)-Leu-Val-Ala represents an undecapeptide composed of four amino acids of known sequence, followed by four whose sequence is unknown and then three in known sequence.

In a sequence where 'X' denotes any amino acid, standardize as uppercase. Amino acids that are stereochemical neighbours, as in a serine-protease triad, are separated by two full-point dots: Asp..His..Ser. For other amino acid substituents and modifications, use lowercase symbols, placed before amino acid symbol (fMet) or after position number.

For protein domains, allow a solidus without definition in α/β or β/α domains (which denote β-α-β units in which the α-helices and β-folds are impacted or 'wound'). Do not change solidus to 'and', 'and/or', or hyphen. Allow α+β notation (which denotes mixed α and β) without definition. Do not change plus to 'and' or hyphen.

The N-terminal (amino) end of the sequence carries the free amino group in the peptide chain; the C-terminal (carboxy) end carries the free carboxyl group. Note N terminus (no hyphen), but N-terminal (adjectival). Spell out amino and carboxy terminus on first use, then use N and C.

Alpha-carbon atoms are denoted as α-carbon or Cα.

Information for secondary, tertiary and quaternary structure:

- Secondary structure. α-helix, α-helical; parallel, or antiparallel β-sheet.
- Tertiary structure. van der Waals forces (no apostrophe); disulfide bridges (not S–S bridges); hydrogen bonds (not H-bonds, unless the term is used frequently), hydrogen-bonded (hyphenated).
- Quaternary structure. Subunits and chains are denoted as α-chain, β-subunit, or αβ heterodimer (no hyphen), for example. (Do not confuse with protein subtypes, e.g., interferon-γ.)

Protein motifs bind DNA and are common to prokaryotes and eukaryotes. Examples are: helix–loop–helix (en rules), HLH; basic-HLH, bHLH; zinc-finger (hyphenated even as a noun); leucine zipper, Z.

For ATPase pumps, see Sect.11.15.5. For GTPases, see Sect. 11.13.15.

Restriction enzymes. DNA can be cut with restriction enzymes, which are named by contraction of the name of the species from which they are isolated. These contractions use upright for the letters that are italicized in the source strain followed by roman numerals: EcoRI is the first enzyme isolated from *E. coli* strain R; HindIII the third enzyme from *H. influenzae* strain d. Other examples include HpaI, HinfI, SmaI, AvaI, AluI, BamHI, EcoB, HaeI and PstI, but note Arabic numbers for Sau3A, Bal 31. See Roberts R. J. et al. A nomenclature for restriction enzymes, DNA methyltransferases, homing endonucleases and their genes. *Nucleic Acids Res.* **31,** 1805–1812 (2003); http://nar.oxfordjournals.org/content/31/7/1805.full. Also note that the genes encoding these enzymes are italicized. (See also Sect. 11.9.2)

For kinases such as p70$^{\mathrm{S6K}}$ or S6K, note roman superscripts as these are not genes.

11.11.3.1 *Fusion Proteins*

To visualize, track or purify a protein of interest, this protein can be fused (by linking gene sequences) to a marker protein. Fusion proteins are linked by an en rule, e.g., CREB–GFP. Common marker proteins, which do not need to be defined, include:

- GFP, green fluorescent protein (eGFP, enhanced)
- GST, glutathione *S*-transferase
- HA, haemagglutinin (UK); hemagglutinin (US)
- MYC (use this version rather than C-MYC or CMYC), avian virus myelocytomatosis (full name unlikely to be used by authors).

11.11.3.2 *Oncoproteins*

Oncoproteins are encoded by oncogenes. Copy editors are likely to encounter the following oncoproteins in primary research content: the protein kinases SRC, YES, ABL1, EGFR, ERBB2 and MOS; the nuclear proteins (transcription factors) NMYC, MYB, FOS, JUN, RB and ETS1; the p21 proteins (GTP-binding proteins with GTPase activity) HRAS, KRAS, NRAS, or $\mathrm{p21}^{ras}$. Note that a GTPase-activating protein (GAP; examples are p120 and the *NF1* gene product) regulating *ras* protein is designated RasGAP, closed up.

11.11.3.3 *Coding Motifs*

Genes containing the homeobox motif encode proteins containing a homeobox domain or homeodomain. A POU protein contains a POU-specific domain and a homeodomain. Zinc-finger proteins often have a series of zinc-fingers each with a $\mathrm{Cys_2}$-$\mathrm{His_2}$ motif. Each finger is encoded by a zinc-finger motif. For more information on protein domains, see InterPro: http://www.ebi.ac.uk/interpro/.

11.11.3.4 *Hormones and Receptors*

Where the hormone name is a single word (or two reasonably short words) the copy editor should not use an abbreviation but write out, e.g., insulin, progesterone, erythropoietin, and retinoic acid. For receptors, the word is abbreviated as for the hormone and a capital roman R suffix is added: AChR for acetylcholine receptor; RAR for retinoic acid receptor (subtypes: RAR-α, RAR-β, RAR-γ), and RXR for retinoid-X receptor; ER, oestrogen receptor; EPOR, erythropoietin receptor.

Abbreviation	Definition
ACTH	adrenocorticotropic (not -trophic) hormone
EPO	erythropoietin (EPO should be avoided)
FSF	follicle-stimulating factor
GH	growth hormone (spell out unless used frequently)
MSH	melanotropin (melanocyte-stimulating hormone)
STH	somatotropic hormone

11.11.3.5 *Growth Factors*

Abbreviations for growth factors must be defined at first use:

Abbreviation	Definition
CSF	colony-stimulating factor
EDRF	endothelium-derived relaxing factor
EGF	epidermal growth factor
FGF	fibroblast growth factor (prefix a, acidic; b, basic)
GM-CSF	granulocyte–macrophage CSF
IGF	insulin-like growth factor (IGF-I, IGF-II…)
IL	interleukin (IL-1, IL-2…)
LIF	leukaemia inhibitory factor
NGF	nerve growth factor
PDGF	platelet-derived growth factor
TGF	transforming growth factor (TGFα, TGFβ)
TNF	tumor necrosis factor (NB no longer known as TNFα)
VEGF	vascular endothelial growth factor

11.11.3.6 *Transcription Factors*

Use roman for Sp1, AP-1, TFIIIA, NF-κB, GF-1 (also called NF-1E1, Eryf-1), Ets (but ETS domain) and PU.1, σ^{54}.

Use TATA-binding protein (TBP) for the TFIIDτ subunit. TAFs are TBP-associated factors.

CBP/p300 are transcriptional adaptors targeted by E1A.

11.11.3.7 *Cell-Adhesion Molecules (CAMs)*

Cell-adhesion molecules are typically single-pass transmembrane receptors. Examples include the immunoglobulin superfamily (e.g., neural N-CAM), integrins (heteromers of α and β chains, subtypes, e.g., α4β7 integrin), selectins (L-, E- and P-selectins), and cadherins (prefixes: neural, N-; placental, P-; brain, B-; epithelial, E-).

11.11.3.8 *Common Proteins*

Below is a table listing common proteins; copy editors should take note of formatting, abbreviations and usage advice:

Protein	Notes
adenylyl cyclase	not adenylate cyclase or adenyl cyclase
apolipoprotein B-100	apoB-100
ATPases	Na^+/K^+-ATPase; Ca^{2+}-ATPase; H^+/K^+-ATPase; F_0F_1-ATPase
bone morphogenetic proteins (BMPs)	BMP1, etc.
carboxypeptidase A	NA
casein kinase II	NA
chlorophyll *a*, *b*	note use of italic lowercase letters
cytochrome *c*, c_1, *a*, a_3, *b*, *b*-562, P450	note use of italic lowercase letters
cyclooxygenase	no hyphen
DNA (RNA) polymerase I, II, III (*E. coli*); mammalian, α β γ δ	can be abbreviated to PolI, PolII, etc.
elongation factors (protein synthesis)	EF1, EF2, EF3, etc.
F-actin, G-actin	note hyphenation

Protein	Notes
glutathione *S*-transferase	abbreviated as GST; fusion proteins take en rule; note use of italics
glycophorin A	NA
guanylyl cyclase	not guanylate or guanyl cyclase
haemoglobin	HbA, HbF, HbS, etc.
heat shock protein 70	HSP70
histones	H1, H2A, H2B, H3, H4
initiation factors (protein synthesis)	IF1, IF2, IF3, etc
interferon-α, -β, -γ	IFNγ, etc. (encoded in humans by *IFNG*, etc.)
leptin	preferred name for OB, human *obese* protein
lipoprotein (*a*)	Lp(*a*)
OB (humans and mice)	use leptin; OBR, OB receptor
phospholipase A_2	NA
phospholipase C; phospholipase Cγ	avoid phosphoinositidase C

Protein	Notes
phosphorylase *a*, *b*	note use of italic lowercase letters
RB	humans and mice
RNase	ribonuclease
S1 nuclease	NA
T4 ligase	NA
troponin C	NA
apo and holo	closed up when attached to spelled-out protein names but hyphenated when attached to abbreviations or special characters (including protein symbols); they may also stand alone as adjectives: e.g., 'apotransferrin', 'apo-STING', 'holoenzyme', 'the apo state of the protein', 'apo structure'
70S ribosomes	prokaryotic ribosomes with sedimentation coefficient of 70 Svedberg units (closed up)
5S, 18S and 28S rRNA	rRNAs often used as size markers on gels, although sized by sedimentation
northern and western blots but Southern blot	note use of uppercase or lowercase
ligand	avoid 'ligand' as a verb or inflected verbal, instead use 'ligand-bound protein' instead of 'liganded protein' and 'ligand-free protein' instead of 'unliganded protein'

11.11.3.9 *Post-Translational Modifications*

Post-translational modifications of proteins include acetylation, methylation, phosphorylation, ubiquitylation (often called ubiquitination), sumoylation and ADP-ribosylation. Post-translational modifications have been extensively studied in histones. Covalent thioester linkages are preferably denoted by an en rule, but the symbol ~ is allowed if used consistently by the author, e.g., 'E2~Ub complex' or, 'Ubc9~SUMO'. Describe each histone at first mention, e.g., 'histone H4' and subsequently 'H4' may be used.

Some examples of post-translational modifications are trimethylated histone H4 K20 (H4K20me3); histone H4 trimethylated at K20 (H4K20me3); K20-trimethylated histone H4 (H4K20me3). Special case: phosphorylated histone H2AX (γ-H2AX). These guidelines follow the Brno nomenclature (https://www.nature.com/articles/nsmb0205-110) where abbreviations refer to only the modified amino acids but not to the modification 'mark' or the process of modification. Therefore, copy editors should reword to avoid inaccurate wording such as 'H3 K4 trimethylation (H3K4me3)' or 'the H3K4me3 mark.'

11.11.4 Nucleic Acids

The symbols for the primary nucleobases, which can be used without definition, are A (adenine); G (guanine); C (cytosine); T (thymine); U (uracil); N (unspecified).

The equivalent nucleosides (base+sugar) are adenosine, guanosine, cytidine, thymidine and uridine.

In styling nucleic acids, set position numbers on the line, placing negative position numbers in parentheses, e.g., C(–2).

A nucleotide is a phosphorylated nucleoside. X is used to denote unspecified purine, Y an unspecified pyrimidine. 2′-deoxyribonucleosides are designated by the same symbols preceded by d, e.g., dA, dC. Sugar carbon positions are distinguished from base carbons by primes after the numbers, for example, NMP nucleoside 5′-phosphate; NDP nucleoside 5′-diphosphate. The term 'oligonucleotide' should be used instead of 'oligo'; 'decanucleotide' or '10-base polymer/oligonucleotide' should be used instead of '10-mer'.

The following table shows common nucleic acid types, which should be defined at first use unless marked with an asterisk here. NB Genomic DNA and viral DNA/RNA are not abbreviated:

Abbreviation	Definition
cAMP	cyclic AMP*
cDNA	complementary DNA*
GTP-γS	non-dissociating analogue of GTP
cRNA	antisense RNA
hnRNA	heterogeneous nuclear RNA, pre-mRNA
mtDNA	mitochondrial DNA
mRNA	messenger RNA*
poly(A)$^+$ RNA	processed mRNA
(+)RNA, (–)RNA	positive RNA, negative RNA (applies to virus genome)
rRNA	ribosomal RNA*

Abbreviation	Definition
snRNA	small nuclear RNA
scRNA	small cytoplasmic RNA
snRNP	small nuclear ribonucleoprotein particles (not snurps)
scRNP	small cytoplasmic ribonucleoprotein particles (not scurps)
siRNA	short interfering RNA
snoRNA	small nucleolar RNA
sRNA	small RNA (generic term encompassing the various classes of short/small RNAs)
slRNA	spliced leader RNA
tRNA	transfer RNA*
RNAi	RNA-mediated interference

Sequences. The phosphodiester bond is indicated by a hyphen, e.g., d(A-T), deoxyadenylyl(3′-5′)thymidine; in a sequence the hyphen can be omitted, as in pGAUG, where p indicates a 5′ terminal phosphate. No hyphen is used in triplet codons and anticodons: UUU, AUG. Hydrogen bonding is indicated by a median point, e.g., A•T base pair; and miscellaneous hydrogen bonds are represented by three tiny median points in a row H···S. Base composition: 45% A+T content, not 45% AT content; (A+T)-rich.

Nucleotide substituents. For nucleotide substituents use lowercase symbols (placed before the nucleotide), superscript locant numbers and subscript multipliers, e.g., m_2^6A = N^6-dimethyladenosine. However, allow 5mC, 5hmC, etc., if used and defined by the author. Nucleotide linkages other than the 5′→3′ phosphodiester linkage should be specified with locant numbers and a 'p' for each phosphate, with no hyphens; e.g., $m^7G5'ppp5'N$ is the 5′ cap on processed mRNA, where a 7-methylguanosine is attached 5′→5′ by a triphosphate to the first nucleotide.

Introns. IVS, intervening sequence. Convention is to use square brackets for group I introns, as [5′], [3′].

11.11.5 tRNAs and rRNA

Below is a table listing tRNA style; copy editors should take note of formatting, abbreviations and usage advice:

tRNA	Notes
Nonacylated	alanine tRNA or $tRNA^{Ala}$
Aminoacylated	Ala-tRNA or $Ala\text{-}tRNA^{Ala}$
Isoacceptors (two or more tRNAs accepting the same amino acid)	$tRNA_1^{Ala}$, $tRNA_2^{Ala}$ (superscript and subscript symbols should be aligned)
Initiator tRNAs: formylmethionine (prokaryotes only)	$tRNA^{fMet}$, $Met\text{-}tRNA^{fMet}$, $fMet\text{-}tRNA^{fMet}$

tRNA	Notes
Initiator tRNAs: eukaryotes	allow Met-$tRNA^{Met}_i$ without definition
$tRNA^{Leu}$	tRNA recognizing triplet codon for leucine
Leu-tRNA	tRNA with its amino acid loaded at its 3′ end
$poly(A)^+$ mRNA	mRNA with 20–250 adenosine residues at 3′-hydroxy terminus

For ribosome binding states, use solidus notation to denote tRNAs binding the aminoacyl (A), peptidyl (P), exit (E) and initiator (I) sites at the ribosome's small and large subunit. There are three binding states that use different nomenclature:

- 'Classically bound' tRNAs bind in P/P and A/A pretranslocation states (in which the first letter represents the binding site on the small subunit, and the second letter represents the site on the large subunit).
- 'Hybrid-bound' tRNAs bind in P/E and A/P states; for example, 'hybrid-state peptidyl (P)-site and exit (E)-site (P/E) tRNA'.
- Chimeric hybrid states occur en route to the post-translocation state. In the ap/P state, 'ap' indicates contact between the tRNA anticodon stem–loop and the A site of the small-subunit head and the P site of the small-subunit body, whereas 'P' indicates contact of the tRNA acceptor end with the large-subunit P site. In the pe/E state, 'pe' indicates contact between the tRNA anticodon stem-loop and the P site of the small-subunit head P and the E site of the small-subunit body, whereas 'E' indicates contact of the tRNA acceptor end with the large-subunit E site.

Allow the author's notation for rRNA helices as 'H' and 'h' (for large and small subunits, respectively) in text and figures, without changing case. For example, helix H95 of 28S rRNA and helix h14 of 18S rRNA (Brimacombe nomenclature).

11.11.6 Compound Numbers

Important small-molecule compounds are sometimes assigned a boldface Arabic numeral. At first mention in both the text and display items, the compound number is defined, and thereafter the compound may be referred to by either its name, abbreviation, or the boldface numeral (as long as there is internal consistency for each compound in the manuscript). Do not use both the compound's name (or abbreviation) and its boldface numeral after the initial assignment of the numeral (e.g., if a boldface numeral is assigned to leucine, i.e., 'leucine (**5**)', do not refer to this compound as 'leucine (**5**)' afterwards; use either 'leucine' or '**5**').

11.11.7 Polymers

Below is a table listing polymers and their definitions:

Polymer	Definition
poly(A)	homopolymer of A (no space after prefix)
poly(A)$^+$	used for processed mRNA
poly(dA-dT)	alternating single-stranded co-polymer of dA and dT
poly(dA)•poly(dT)	base-paired dA homopolymer and dT homopolymer (note median point)
poly(dA-dC)•poly(dT-dG)	base-paired chains of alternating co-polymers
poly(A, G, C)	random single-stranded co-polymer of A, G, C

11.11.8 Sulfur Chemistry

Place [] around sulfur complexes that include stoichiometric numbers: e.g., [2Fe–2S]. For the simplest case, [Fe–S], include brackets if any more complex sulfur complexes are also mentioned; in articles with more general content this can be omitted: Fe–S.

11.12 Cancer

11.12.1 General Styling

The following proteins are commonly encountered in papers covering the area of cancer. Copy editors should take note of indicated rules on formatting, abbreviation requirements and any useful information.

- There is no need to define mTOR, PI3K, PTEN, ERK, MAPK, MEK, RAS, RAC, BRCA1 or BRCA2, BCL-2 (protein), *BCL2* (gene), DMBA and TPA (the carcinogens), ARF, MET, SRC, p53 (also *TP53* and *Trp53*), BCR–ABL1, or BCR–ABL. RB does not require definition if it means the protein (the gene encoding RB is *RB1*) but it should be spelled out (retinoblastoma) when the disease is meant.
  p53 is the protein encoded by the genes *TP53* (for human) and *Trp53* (for mouse).
- Many proteins have several isoforms. Where it is not specified whether a particular isoform or the whole family is referred to, clarification is required in the following cases: RAS (specifically NRAS, KRAS or HRAS, or the whole family), AKT

(specifically AKT1 or AKT2), BRCA (BRCA1 and BRCA2, or just one), VEGF (whole family or just VEGFA). Similarly, if MAPK is used without specification, query the author as to whether a specific MAPK (usually ERK, sometimes p38) or all MAPKs are meant.

- For the gene/protein ERBB2/HER2/NEU, ERBB2 is the official and preferred name; HER2 is acceptable if qualified by 'also known as ERBB2'; and *Neu* or *neu* is often used when the rat gene is meant, but *Erbb2* (or *Her2*) is OK if used consistently.
- 'Glioblastoma multiforme' is no longer a recognized classification, and so this should be changed to 'glioblastoma'. Authors sometimes use 'GBM' to mean either glioblastoma multiforme or glioblastoma; however, the 'M' is usually interpreted as meaning 'multiforme' and therefore the preference is to avoid it.

Below is a table listing general style to be used when handling cancer content; copy editors should take note of formatting, abbreviations and usage advice:

Protein	Notes
HDAC inhibitor	this is abbreviated to HDACi (plural HDACis)
any proteins with the 'c' prefix (e.g., c-SRC or c-MET)	remove 'c-' so it is just 'SRC', 'MET', etc.
$p38^{MAPK}$	should be changed to p38 (on first mention, 'the p38 MAPK' is acceptable)
IAP proteins	keep IAP nomenclature and add '(also known as BIRC1, etc.)'
AP1	'activator protein 1'; this is a dimeric transcription factor complex that contains members of the JUN, FOS, ATF and MAF protein families
CML	chronic myeloid leukaemia (preferred over 'myelogenous')
HR	keep 2-letter abbreviation (for 'homologous recombination'); it is recognizable to the field and used very frequently

11.12.2 Cyclin-Dependent Kinase Inhibitor

Cyclin-dependent kinase inhibitor (CDKN) nomenclature:

Human gene	Human protein	Mouse gene	Mouse protein
CDKN1A	p21 (WAF1)	*Cdkn1a*	p21
CDKN1B	p27 (KIP1)	*Cdkn1b*	p27
CDKN1C	p57 (KIP2)	*Cdkn1c*	p57
CDKN2A	Either INK4A (p16) or ARF (p14) (both are encoded by *CDKN2A*), or CDKN2A if discussing both transcripts and protein products. ARF does not need to be defined.	*Cdkn2a;* mouse models in which either INK4A or ARF (but not both) are deleted should be written as $Cdkn2a^{ARF}$ or $Cdkn2a^{INK4A}$	Either INK4A (p16) or ARF (p19 — note this is different from human ARF): both are encoded by *Cdkn2a*. Use CDKN2A if discussing both transcripts and protein products.
CDKN2B	INK4B (p15)	*Cdkn2b*	INK4B (p15)
CDKN2C	INK4C (p18)	*Cdkn2c*	INK4C (p18)
CDKN2D	INK4D (p19)	*Cdkn2d*	INK4D (p19)

11.12.3 Fusion Genes/Proteins

BCR–ABL1 or BCR–ABL is a fusion oncoprotein (or oncogene) that causes chronic myeloid leukaemia. The fusion is often referred to as the 'Philadelphia chromosome', which is OK to retain. Technically, ABL1 is correct, but ABL is OK if used by the author. Although this term does not need to be defined, in this fusion 'BCR' means 'breakpoint cluster region' and not 'B cell receptor'.

11.12.4 Cell Cycle Phases

Cell cycle phases are G0, G1 (G standing for gap), S (synthesis phase, during which the DNA is replicated), G2 and M (mitosis). The first event in the cycle is called Start and takes place in G1; the decision or commitment to undergo another cell cycle is made at Start. Cell cycle phases (such as M phase, S phase, etc.) do not take a hyphen.

11.12.5 Apoptosis

Apoptosis is regulated, for example, by the CD95/FAS/APO1 receptor (CD95 is the preferred term), which requires activation of the cysteine protease ICE (for interleukin-1β-converting enzyme), which is homologous to the *C. elegans ced-3* product CED-3. FAS-mediated apoptosis is inhibited by BCL-2 (*C. elegans* homologue, CED-9).

11.13 Cell Biology

11.13.1 General Styling

The *Oxford Dictionary of Biochemistry & Molecular Biology* is a useful cell biology dictionary.

Below is a table listing general style to be used when handling cell biology content; copy editors should take note of formatting, abbreviations and usage advice:

Term	Notes
Aurora kinases	take a capital (for example, Aurora A kinase)
APC/C	define as: APC/C (anaphase-promoting complex, also known as the cyclosome)
BiP	use lowercase 'i' in the abbreviation for the ER chaperone immunoglobulin-binding protein BiP
PARP	define as poly(ADP-ribose) polymerase; PARylation as poly(ADP-ribosyl)ation
MARylation	define as mono(ADP-ribosyl)ation
SUMO	but use lowercase for sumoylation
eIF	eukaryotic translation initiation factor (note lowercase 'e').
AP1 and AP-1	there is a difference between AP1 (protein involved in actin cytoskeleton reorganization) and AP-1 (family of transcription factors that bind to AP-1 sites in DNA)
murine	means 'mice and related rodents' (including rats), subfamily Murinae; use 'mouse' when referring specifically to mice
SWI/SNF, the ARP2/3 (actin-related protein 2/3) complex, Ser/Thr kinase and ENA/VASP (enabled/vasodilator-stimulated phosphoprotein)	a solidus is allowed in these terms

11.13.2 Amino Acids

The three-letter code is preferable to the single-letter code except in long sequences or motifs, where the one-letter code is preferred. There is no need to spell out the full name (see also Sect.11.11.3).

11.13.3 Embryonic Stem Cells

For stem cells, the copy editor should use the following abbreviations: embryonic stem cells are abbreviated as 'ES cells'; mouse embryonic stem cells are defined as 'mouse ES cells'; haematopoietic stem cells are defined as 'HSCs'; induced pluripotent stem cells are abbreviated as 'iPS cells'.

11.13.4 Phosphatidylinositols

Phosphatidylinositols should be styled as: phosphatidylinositol 3-phosphate (PtdIns3P); phosphatidylinositol 4,5-bisphosphate ($PtdIns(4,5)P_2$); phosphatidylinositol 3,4,5-trisphosphate ($PtdIns(3,4,5)P_3$).

11.13.5 Cytoskeleton

Actin. The monomeric form of actin is G-actin, present in eukaryotic cells (but not present in prokaryotes). The polymerized form is F-actin, which forms the thin filaments of muscle and microfilaments of the cytoskeleton. F-actin and myosin together form a contractile complex called actomyosin.

Microtubules are hollow protein tubules comprising tubulin molecules that are one of the main structural components of the cytoskeleton. They are involved in intracellular transport and movement of organelles, and also form the mitotic spindles. Microtubules also comprise the core of eukaryotic flagella and cilia and are responsible for their motility.

Motor proteins. Myosin, dynein and kinesin are motor proteins, which can move along actin filaments or microtubules. Motor proteins cause muscle contractions, intracellular vesicle transport, cytoplasmic streaming, and the movement of eukaryotic cilia and flagella, and are powered by an intrinsic ATPase activity.

Spindle microtubules attach to the kinetochore during meiosis or mitosis. The kinetochore itself comprises specialized proteins that bind to the centrosomal DNA and associated proteins.

Mechanotransduction involves the transformation of mechanical signals such as stretching (at tissue or cellular level) into biochemical signals. It can be initiated from inside or outside the cell.

11.13.6 Endoplasmic Reticulum

The endoplasmic reticulum (ER) is involved in the synthesis, glycosylation and transport of membrane proteins and secreted proteins. The ER makes contact with other cellular organelles such as mitochondria and with the plasma membrane, affecting their mutual function.

11.13.7 Histology

Below is a table listing general histology information; copy editors should take note of formatting, abbreviations and usage advice:

Term	Notes
physiological salines and histological fixatives	if named after a person, they take a capital (Ringer's solution, Bouin's fixative); otherwise they are lowercase (paraformaldehyde fixative)
dyes and stains	these take an uppercase letter if they are trade names (e.g., Lucifer yellow, Coomassie blue); otherwise they are lowercase (methyl blue)
calcium-indicator dyes	Fura-2 (note hyphen)

Structures that react with antibodies raised to neurotransmitters, peptides or other enzymes are usually referred to as immunoreactive (because an antibody reaction alone cannot identify the chemical nature of the antigen, particularly in fixed tissue).

11.13.8 Intracellular Protein Transport

Style rules and abbreviations for proteins and processes involved in intracellular protein transport:

Term	Notes
NSF	*N*-ethylmaleimide-sensitive fusion protein
SNAPs	soluble NSF attachment proteins
SNAP-25	synaptosome-associated protein (relative molecular mass 25K), different from NSF-associated SNAPs
SNARE	SNAP receptor
v-SNARE	vesicle SNARE
t-SNARE	target SNARE
VAMP	vesicle-associated membrane protein (= synaptobrevin)
endocytosis	the typical form of endocytosis is clathrin-mediated endocytosis, but there is also clathrin-independent endocytosis, of which there are various forms, depending on which proteins mediate them

11.13.9 Autophagy

Autophagy is the intracellular degradation system. There are distinct types of autophagy: 'basal autophagy', which occurs under normal (resting) conditions; 'starvation-induced autophagy', which is a general self-degradation pathway that can affect all cellular content; and 'selective autophagy', which affects specific organelles, such as mitophagy (degradation of mitochondria), pexophagy (peroxisomes) and reticulophagy (ER; do not use ERophagy); cellular components, such as protein aggregates, chaperone-mediated autophagy and lipids (lipophagy); and pathogens (xenophagy).

The components of the autophagy pathway are isolation membranes (phagophore), autophagosomes and autolysosomes.

11.13.10 Intracellular Signaling

Intracellular signaling is mediated by second messengers that operate in the pathway that couples the activation by a first messenger of a cell-surface receptor to effectors inside the cell. Several pathways involve participation by G proteins (see below). In photoreceptors, the second messenger is often called a transmitter — this should be changed to 'messenger'. Several of these systems regulate intracellular calcium concentration, $[Ca^{2+}]_i$.

11.13.11 Transmembrane Signaling Systems

Transmembrane signaling systems involve pathways with several components. The pathways below provide an overview of those most likely to be encountered:

Cyclic AMP (cAMP) pathway

Receptor→G protein→adenylyl cyclase→cAMP→protein kinase A (PKA)

Cyclic GMP pathway

Receptor→G protein→guanylyl cyclase→cGMP

Phosphatidyl inositide pathway

Receptor→phospholipase C (PLC) → diacylglycerol→activated protein kinase C (PKC)
→inositol 1,4,5-trisphosphate (Ins(1,4,5)P3)→opens calcium channels

Inositol 1,4,5-trisphosphate (Ins(1,4,5)P$_3$) is produced from phosphatidylinositol 4,5-bisphosphate (PtdIns(4,5)P$_2$). This inositol lipid receptor mechanism involves the generation of two second messengers. The first messenger binds to the receptor and activates phospholipase C (PLC) (also called phosphoinositidase C, but do not use this form) to produce the second messengers diacylglycerol and inositol 1,4,5-trisphosphate (abbreviated to Ins(1,4,5)P$_3$) from phosphatidylinositol 4,5-bisphosphate (abbreviated to PtdIns(4,5)P$_2$). If it is clear from the context that Ins(1,4,5)P$_3$ is meant, the abbreviation InsP$_3$ can be used. However, ensure there is no potential confusion with other inositol trisphosphates (see below). Diacylglycerol activates protein kinase C (PKC); InsP$_3$ opens calcium channels.

PtdIns-3-OH kinase or phosphatidylinositol 3-OH kinase can be written as PI3K.

Intermediates formed in the recycling of inositol trisphosphate include inositol 1,3,4,5-tetrakisphosphate (Ins(1,3,4,5)P_4), Ins(1,3,4)P_3, Ins(3,4)P_2, Ins(3)P and Ins(4)P. There are also higher derivatives: $InsP_5$ and $InsP_6$. For lower derivatives, the abbreviation should include the phosphate positions on the inositol ring. Note: in bisphosphates, phosphates are in different positions; in diphosphates, phosphates are linked.

11.13.12 Protein Kinase C

Protein kinase C (PKC) is a family of proteins, the members of which have different patterns of tissue expression and intracellular localization. Subspecies are α, β_I, β_{II}, γ, δ, ε and ζ (on the line, as PKCα, etc.). Protein kinase C is stimulated by phorbol esters such as TPA (also known as PMA).

11.13.13 Ras Protein Cascade

Below is a table listing useful information on the Ras superfamily of monomeric GTPases:

Protein	Notes
Ras	Proteins that relay signals from receptor tyrosine kinases to the nucleus to stimulate cell proliferation/differentiation
Rab	Proteins that help regulate traffic of intracellular transport vesicles
GAPs	(GTPase-activating proteins) increase the rate of GTP hydrolysis by Ras; an example of a rasGAP (note lowercase 'r' and closed up to indicate type of GAP) is $p120^{GAP}$
GNRPs	guanine-nucleotide-releasing proteins stimulate exchange of GDP for GTP on Ras
GTP-bound Ras	activates a Ser/Thr kinase cascade (MAPKKK, e.g., Raf→MAPKK→MAPK)
Sentence or upper case (e.g., RHO or Rho)	both allowed — follow author usage but be consistent within the article

11.13.14 Arachidonic Acid Cascade

See http://www.nature.com/nrrheum/journal/v10/n5/full/nrrheum.2014.2.html#f1 for a useful Review of the arachidonic acid cascade.

Arachidonic acid is released from esterified stores in the cell membrane by cytosolic phospholipase A_2 ($cPLA_2$), and is oxidized to eicosanoid second messengers. These are local hormones that can pass across cell membranes and so can function as intracellular or extracellular messengers. They are products of the action of either cyclooxygenase (prostaglandins or thromboxanes) or lipoxygenase (leukotrienes, lipoxins or hydroxyeicosatetraenoic acids) on arachidonic acid.

For prostaglandins, PGA–PGI denotes class; subscript (PGE_2 for example) indicates the number of double bonds outside the ring.

11.13.15 G Proteins

G proteins are guanine-nucleotide-binding proteins that transduce signals across the plasma membrane. Different types are distinguished by subscripts:

G protein	Definition
G_s	stimulates adenylyl cyclase
G_i	inhibits adenylyl cyclase
G_o	function unknown ('other')
G_p	couples receptor to phospholipase C
G_t	(transducin) activates cGMP phosphodiesterase

Authors sometimes use idiosyncratic names for G proteins, which need to be defined: e.g., G_K, a G_i protein that controls the K^+ channel in heart muscle.

Subunits of G proteins (G-protein subunits, hyphenated) are α, β and γ. These dissociate into α and $\beta\gamma$. If only one subunit is referred to, then use $G\alpha_i$ or just α_i according to context. Subunit subtypes are $G\alpha_{i1}$, $G\alpha_{i2}$, $G\alpha_{i3}$.

The GTP/GDP-bound forms of the subunits are represented as α•GDP, α_t•GTP (note median points) to avoid the confusing '$G\alpha$•GDP' format.

G proteins may be sensitive or insensitive to pertussis toxin or to cholera toxin. G protein is hyphenated only when used adjectivally (e.g., G-protein-coupled receptor).

11.13.16 Prefixes

Prefixes that qualify the source of the factor are lowercase roman: recombinant (r); human (h); and so on, closed up to the factor they describe. Thus, recombinant human growth hormone is rhGH.

11.13.17 Common Abbreviations

The table below lists common abbreviations used in cell biology papers:

Abbreviation	Definition
AMP-PNP	(inactive ATP analogue), β-γ-imidoadenosine 5′-phosphate
8-Br-cAMP	8-bromoadenosine-3′,5′-cyclic monophosphate
$[Ca^{2+}]_i$	intracellular calcium concentration
ChIP–seq (note en rule)	chromatin immunoprecipitation followed by sequencing
cryo-ET	cryo electron tomography
DAG	diacylglycerol
dibutyryl cAMP	cell-permeable analogue of cyclic AMP
Fura-2	Ca^{2+}-sensitive fluorescent dye (capital F)
GAP	GTPase-activating protein; rasGAP works on Ras
GNRP	guanine-nucleotide-releasing protein
GPI	glycosyl phosphatidylinositol (anchor)
HETE	hydroxyeicosatetraenoic acid, etc.
MAPKK	MAP-kinase kinase
IBMX	3-isobutyl-1-methylxanthine, phosphodiesterase inhibitor
$Ins(1,4,5)P_3$	inositol 1,4,5-trisphosphate (IP_3 also allowed as an abbreviation)
$Ins(1,3,4,5)P_4$	inositol 1,3,4,5-tetrakisphosphate
PKC	protein kinase C
PI3K	phosphatidylinositol 3-OH kinase
PLA_2	phospholipase A_2
PLC	phospholipase C (Greek letters for subtypes, on line: phospholipase Cγ or PLC-γ, with hyphen)
$PtdIns(4,5)P_2$	phosphatidylinositol 4,5-bisphosphate
PTX	pertussis toxin
STATs	signal transducers and activators of transcription

11.14 Microbiology

11.14.1 General

Avoid the term 'microbe' — it is mostly used by non-microbiologists and is not the generally accepted term, so it should be changed to microorganism. Exceptions include terms in which the use of microorganism is unwieldy; for instance, plant–microbe interactions, host–microbe interactions etc. 'Microbial' is an acceptable adjective.

'Prokaryote' refers to bacteria and archaea. Some authors use it incorrectly when they mean bacteria only, so add an author query to clarify whether they are referring to both and change to 'bacteria' if this is what they actually mean.

The terms 'microbiota' and 'microbiome' are often used interchangeably in the literature, but strictly speaking they are two distinct terms. Microbiota refers to the total collection of microorganisms at a particular site, whereas microbiome refers to the total genetic content of the microbiota at a particular site. For instance, gut microbiota refers to all the microorganisms in the gut, whereas gut microbiome refers to the genetic content of all of the gut microorganisms.

11.14.2 Nomenclature and Taxonomy

Species names should be checked in the NCBI Taxonomy database (http://www.ncbi.nlm.nih.gov/taxonomy) for spelling. Synonymous names are OK to use if the author requests it — such as for species with recent name changes or that are commonly known by more than one name (*Caulobacter crescentus* is also known as *Caulobacter vibroides*).

Write genus and species out in full (e.g., *Escherichia coli*) the first time it is used in the text, but abbreviate the genus (*E. coli*) thereafter. Always spell out the genus name on first mention when referring to different species: e.g., spell out genus *Salmonella* in *Salmonella gallinarum* even if *Salmonella pullorum* had been used in the text earlier on.

For the domain names, use Eukarya (not Eukaryota), Bacteria and Archaea. These are referred to as domains, not superkingdoms (both mean the same thing but 'domain' is more widely used).

General rules to follow for capitalization/italicization:

Taxonomic rank	Rule
phylum	not italicized; capitalized (for example, Proteobacteria; but lower case if used adjectivally (e.g., proteobacterial))
class	not italicized; capitalized (for example, Alphaproteobacteria)
order	not italicized; capitalized
family	not italicized; capitalized
genus	italicized and capitalized if singular, or if referring to multiple species of the genus (e.g., *Sodalis* spp.); not italicized nor capitalized if plural (for example, yersiniae, streptococci, campylobacters)
species	italics
subspecies	italics
strain	roman

Taxonomic rank	Rule
serovar	roman
pathovar	italics

For viruses, italicize approved orders, families, and subfamilies, but not genera and species (see also Sect. 11.14.3).

If an author is clearly referring to all species of a genus (e.g., all *Streptomyces* species), then just *Streptomyces* is OK. However, if a paper sometimes refers to one particular species and sometimes to all species of a genus, then it is better to use *Streptomyces* spp.

When a new strain of a previously mentioned bacterial or archaeal species is introduced, the genus name does not need to be spelled out in full again. Thus, if *Escherichia coli* K12 was introduced, introduction of the B strain would be *E. coli* str. B. The same goes for subspecies and pathovars. Preferentially use abbreviations 'str.', 'subsp.', 'pv.', etc., to save space.

For Gammaproteobacteria, Alphaproteobacteria, etc., write out the Greek word (gamma) and don't use the symbol.

It is convention (currently) to refer to *Prochlorococcus*, *Synechococcus* and SAR11 just as genus names, without adding 'spp.'. They have only recently been discovered and occur mainly in metagenomics studies, in which they are identified only by the presence of gene sequences; they are usually described as 'ecotypes' or equivalent, not species.

11.14.2.1 Candidatus *Nomenclature*

Strictly, the category *Candidatus* should be used for describing prokaryotic entities for which more than a mere sequence is available but for which the detailed characteristics required for description are lacking. Names included in the category *Candidatus* should be placed in single quotation marks and written like so: '*Candidatus* Nitrosopumilus maritimus' (see http://www.bacterio.net/-candidatus.html). If *Candidatus* names are used extensively in an article, *Candidatus* can be abbreviated to *Ca*. — note, however, that each new name needs to be written in full the first time it is mentioned.

11.14.2.2 Salmonella *Bacteria*

Salmonella bacteria have a complicated nomenclature. The *Salmonella* genus has three species: *enterica*, *bongori* and *subterranea*. *S. enterica* can be further broken into six subspecies: *enterica*, *salamae*, *arizonae*, *diarizonae*, *houtenae* and *indica*. *Salmonella enterica* subsp. *enterica* (the subspecies most often arising in research and medicine) is further broken into 2,500+ serotypes, or serovars, typically written in roman text starting with an uppercase letter (e.g., Typhimurium, Typhi), with the full name being *Salmonella enterica* subsp. *enterica* serovar Typhimurium. To save space, *S.* Typhimurium can be used once the full name has been listed once. salmonellae (not capitalized and no italics) is ok to refer to all *Salmonella* species as a group.

11.14.3 Virus Nomenclature

Names of viruses are defined at first use and should then be abbreviated. The following table gives the preferred abbreviations for some common viruses. See http://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi?mode=Undef&id=2&lvl=3&keep=1&srchmode=1&unlock for more details, and check names and taxonomy in the International Committee on Taxonomy of Viruses (ICTV) (https://talk.ictvonline.org/taxonomy/).

For viruses that begin with a Greek letter, write them out and close up the space between the letter and the rest of the word; e.g., Betaherpesvirus.

Abbreviation	Definition
AEV	avian erythroblastosis virus
ALV	avian leukosis virus
A-MuLV	Abelson-murine leukaemia virus
ARV	AIDS-associated retrovirus
BMV	brome mosaic virus
BPV	bovine papilloma virus
CMV	cytomegalovirus or cauliflower mosaic virus
FeLV	feline leukaemia virus
FeSV	feline sarcoma virus
FMDV	foot and mouth disease virus
HBV	hepatitis B virus
HCMV	human cytomegalovirus
HIV	human immunodeficiency virus (HIV-1, HIV-2)
HPV	human papilloma virus (HPV-11, HPV-16)
HSV	herpes simplex virus (but herpesvirus)
HTLV	human T-lymphotrophic virus (HTLV-I, HTLV-II etc.)
LAV	lymphadenopathy-associated virus
MMTV	mouse mammary tumour virus
MoMuLV	Moloney murine leukaemia virus
PV	polyoma virus (do not abbreviate)
RSV	Rous sarcoma virus
SIV	simian immunodeficiency virus
SFV	Semliki forest virus
SSV	simian sarcoma virus
STLV	simian T-lymphotrophic virus (STLV-1)
SV40	simian virus 40

Abbreviation	Definition
TMV	tobacco mosaic virus
TRSV	tobacco ringspot virus
VSV	vesicular stomatitis virus
WNV	West Nile virus

A few virus abbreviations and alternative names are not present in ICTV but are listed in NCBI Taxonomy (http://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi?mode=Undef&id=2&lvl=3&keep=1&srchmode=1&unlock); these should not be changed if the alternative is in common usage.

- Some viruses are written as one word: herpesvirus, poliovirus (see http://www.onelook.com/?lang=all&w=*virus&scwo=1&sswo=1)
- cowpox virus, monkeypox virus, etc. (two words)
- HIV strains should be specified as HIV-1 or HIV-2, unless HIV is discussed in a general sense
- HIV/AIDS is fine

11.14.3.1 *Variants and Strains*

A strain is an isolate of a virus that resembles the type virus in the main properties that define the type, but differs in minor properties such as vector specificity. In practice it is difficult to distinguish between variants, strains and distinct types.

11.14.3.2 *Influenza*

Refer to http://www.cdc.gov/flu/about/viruses/index.htm for information on types of influenza viruses and details on influenza biology.

Specific influenza strain isolates are identified by a standard nomenclature that virus type, geographical location where first isolated, sequential number of isolation, year of isolation and HA and NA subtype (e.g., A/New York/269/2003 (H3N2)). See *Nature* http://dx.doi.org/10.1038/nature04239 (2005).

11.14.3.3 *HIV and Related Viruses*

The retrovirus that causes AIDS (acquired immunodeficiency syndrome; no need to define) in humans by killing T-helper lymphocytes is called the human immunodeficiency virus (HIV, previously known as HTLV-III). There are several forms of HIV: HIV-1 and HIV-2. Isolates are identified by subscripts indicating their source, as in HIV-1_{ROD}.

HIV genes are *gag*, *pol*, *env*, *tat*, *rev*, *vif*, *vpr*, *vpx* (only in HIV-2) and *nef*. Many have previously been given other names, but these should not be used. Protein products are written in roman with the first letter uppercase, e.g., Gag, Tat and Nef.

Simian immunodeficiency virus (SIV) causes a disease in monkeys that is similar to AIDS in humans. Simian viruses have the name of the species of monkey (three-letter abbreviation) as a subscript after the virus name: e.g., $\mathrm{SIV}_{\mathrm{mac}}$ (mac, macaque; agm, African green monkey; smm, sooty mangabey).

SIV isolates should be identified either as, e.g., *Macaca mulatta* isolate 251 or $\mathrm{SIV}_{\mathrm{mac251}}$, but the isolate itself should not be referred to as Mm 251. The animal from which the isolate is derived can be described as Mm 251 if comparing between species, but 251 alone is preferable.

SHIVs (simian-human immunodeficiency viruses) are synthetic chimeric viruses containing parts of both HIV and SIV genomes, generally used to generate more realistic models of HIV infection (as opposed to SIV only) in nonhuman primates.

11.14.4 Antiretroviral Agents

Retrovir (uppercase first letter), also known as zidovudine or 3′-azido-3′-deoxythymidine (AZT), is a nucleoside analogue used against HIV; other examples are DDC (2′,3′-dideoxycytidine) and ddI (2′,3′-dideoxyinosine). Acyclovir (acycloguanosine) is used against herpes simplex virus. Ganciclovir is 9-(1,3-dihydroxy-2-propoxymethyl)-guanine and is used to treat infection by cytomegalovirus.

11.14.5 Microbial Genes and Proteins

See Sect. 11.7.

11.15 Neuroscience

11.15.1 Neuroanatomy

Latin names for parts of the CNS are in roman (not italic): pars intermedia of pituitary. But brain nuclei and tract names are usually anglicized: facial nucleus not nucleus facialis. An exception is locus coeruleus.

The table below lists general rules for neuroanatomy formatting:

Term	Notes
intracortical	within the cortex (usually one hemisphere)
intercortical	between left and right hemispheres
V1	primary visual cortex
cortical area MT	cortical area defined by context
motion-sensitive area MT	cortical area defined by context
hippocampal area CA1	cortical area defined by context
Brodmann area 17	cortical area defined by originating authority or literature reference
bregma	no uppercase for initial letter

Term	Notes
spino–bulbar tract	tracts run from (adj.) to (adj.)
uppercase neurons: Purkinje, Golgi, Mauthner, Rohan-Beard	names of neurons take lowercase (basket, stellate, granule, pyramidal) unless named after someone
V1–V5	areas of the visual cortex in monkeys
mollusc neurons	most mollusc neurons have a letter and a number; eg, L15, C1

11.15.2 Neuropathology

11.15.2.1 *Alzheimer Disease*

The plaque protein in extracellular amyloid fibrils is amyloid-β; this is a fragment of β-amyloid precursor protein (APP), a transmembrane glycoprotein.

11.15.2.2 *Spongiform Encephalopathies*

Spongiform encephalopathies (including scrapie and BSE) are caused by prions, which are infectious self-reproducing protein structures.

Term	Notes
BSE	bovine spongiform encephalopathy
CJD	Creutzfeldt–Jakob disease (note en rule)
$\mathrm{PrP}^{\mathrm{C}}$, $\mathrm{PrP}^{\mathrm{sen}}$	normal α-helical, protease-sensitive form of the cellular prion protein
$\mathrm{PrP}^{\mathrm{Sc}}$, $\mathrm{PrP}^{\mathrm{res}}$	pathological β-sheet, protease-resistant, aggregated prion
vCJD	new variant CJD

Note the unusual nomenclature of yeast prions, which have square brackets: e.g., the [URE3] determinant is the prion form of the *URE2*-gene-encoded protein Ure2. Strains with the prion form of Ure2p have the [URE3+] phenotype.

11.15.3 Chemical Transmission

Table showing transmitter chemicals and related enzymes/toxins:

Term	Notes
ACh	acetylcholine
AChE	acetylcholinesterase
ChAT	choline acetyltransferase
6-OHDA	6-hydroxydopamine (also 5-OHDA)
DOPA	3,4-dihydroxyphenylalanine (note L-DOPA)
GABA	γ-aminobutyric acid (abbreviation first)
5-HT	5-hydroxytryptamine or serotonin
amino acids	glutamate, glycine, aspartate
adrenaline	(US epinephrine)
dopamine	(do not abbreviate)
noradrenaline	(US norepinephrine; do not abbreviate)

11.15.3.1 *Peptides*

Peptides can be named arbitrarily (e.g., substance P, substance K); by the physiological effect first described for them (if the name is long or is several words, use an all-capital abbreviation, otherwise spell out: vasoactive intestinal peptide, VIP; cholecystokinin, CCK); and by the single-letter amino acid code for the sequence, e.g., FMRF amide (do not spell out unless the peptide is newly discovered). But modifications/variants such as leucine and methionine enkephalins are written as Leu-enkephalin, Met-enkephalin.

11.15.3.2 *Receptors*

Receptors have a confusing nomenclature as each transmitter historically followed a different system. In general follow the author by default (but see most common list below); note IUPHAR has a unified system for reference http://www.guidetopharmacology.org/nomenclature.jsp.

Acetylcholine receptor	AChR	
	nicotinic subtype	nAChR
		subdivisions: α7-nAChR, etc.
	muscarinic subtype	mAChR
		subdivisions: M1, M2, M3
Noradrenaline (US norepinephrine) adrenoceptors	subtypes: α_1-AR, α_2-AR, β-AR	subdivisions: α_{1A}-AR, α_{1B}-AR etc.
Dopamine receptors	subtypes: D1, D2, D3, D4	
Opioid receptors	subtypes: α, δ, χ	
Histamine receptors	subtypes: H_1, H_2	
Serotonin (5-hydroxytryptamine, 5-HT) receptors	subtypes: 5-HT_1, 5-HT_2, 5-HT_3	
	subdivisions of 5-HT: 5-HT_{1A}, 5-HT_{1B} etc.	
Purinoceptors:	$P2X_2$, P2Y, P2U	
GABA receptors	subtypes: $GABA_A$, $GABA_B$. (Note: GABA receptors contain a separate site that binds benzodiazepine drugs, termed the benzodiazepine receptor.)	
Glutamate receptors	kainate (K) receptors (spell out)	
	AMPA (α-amino-3-hydroxy-5-methyl-4-isoxazole propionic acid) receptors	
	N-methyl-D-aspartate (NMDA) receptors	

Selective antagonists for NMDA-type glutamate receptors: AP5 (not APV): 2-amino-5-phosphonovalerate (or -valeric acid, -pentanoic acid, -pentanoate). To specify stereochemistry, prepend any one of the following: DL- or D-, (±) or (–) (closed up), (*RS*)- or (*R*)-. AP7: 2-amino-7-phosphonoheptanoate (or -heptanoic acid). Stereoisomers as above.

11.15.4 Electrical Transmission

Transmitters bind to receptors (see above) that may alter the flow of ions through channels in the plasma membrane. Ion flow causes a change in the potential across the membrane that may lead to an action potential. The passage of ions is measured with intracellular electrodes by voltage clamp (holding voltage steady and measuring current) or current clamp techniques. Voltage clamp and related experiments can also be performed with membrane-surface electrodes in the patch clamp technique.

11.15.4.1 *Channels*

Channels are usually named after the main species of ion they carry: Na^+, K^+, Ca^{2+} and Cl^- channels. The S-channel is a particular potassium channel. Voltage-sensitive channels should not be abbreviated to VS channels. Gap junctions are formed from transmembrane proteins called connexons, which are made up of six subunits called connexins (Cx; closed up to their number).

11.15.4.2 *Currents*

Currents (I) are usually named after the ion that carries the current: potassium current or I_K, although there are other currents such as I_A (also a potassium current). The subscript can be used to designate particular properties: calcium-dependent potassium current, $I_{K(Ca)}$; voltage-sensitive chloride current, $I_{Cl(V)}$. Do not use + or − in subscripts.

11.15.4.3 *Conductances*

Conductances (G) are named after the ion that carries the main current: K^+ conductance, G_K. Conductances are measured in siemens: pS or fS, not to be confused with s for seconds.

11.15.4.4 *Membrane Potentials*

Membrane potentials (V) are measured in millivolts (mV). In voltage-clamp experiments, the membrane is held at a fixed voltage, the holding potential V_H. Membranes have different equilibrium potentials (E) for each ion: E_K or E_{Cl}; E_R or V_R for resting potential. Note en rule for I–V, current–voltage relationship.

There are several types of membrane potential:

- action potential (do not abbreviate to AP)
- excitatory postsynaptic potential, EPSP (plural, EPSPs; miniature, mEPSP) and the resulting excitatory postsynaptic current (EPSC; mEPSC)
- inhibitory postsynaptic potential, IPSP; likewise mIPSP, IPSC, mIPSC
- endplate potential, EPP (miniature endplate potential, mEPP)

11.15.5 Pumps and Transporters

Pumps are ATPases that move ions actively against a concentration gradient, often exchanging one ion for another: the sodium pump exchanges Na^+ for K^+ and is described as Na^+,K^+-ATPase; also, Ca^{2+}, Mg^{2+}-dependent ATPase; Ca^{2+}-ATPase; H^+, or K^+-ATPase. Transporters (also called cotransporters) are subdivided into symporters (in which the transported molecule and cotransported ion move in the same direction) and antiporters (or 'exchangers', in which they move in opposite directions); e.g., Na^+/Ca^{2+} antiporter, Cl^-/H^+ antiporter (also called Cl^-/H^+ exchanger), or Na^+/glucose symporter. See *Nature* https://www.nature.com/articles/nature03720.

11.15.6 Computational Neuroscience

For computational models, it is permissible to use the present tense for timeless properties of the simulation or algorithm: 'When there is local uncertainty, the model predicts horizontal motion'. But use past tense for particular instantiations of the model: 'When we applied the Gaussian noise assumption, the model predicted vertical motion'.

11.15.7 Abbreviations for Neuroscience

Abbreviations that should be defined at first mention:

Abbreviation	Definition
AD	Alzheimer disease (Alzheimer's also acceptable; abbreviate only if used frequently throughout article)
BOLD	blood oxygenation level-dependent (contrast)
CNG	cyclic-nucleotide-gated (channels)
CNQX	6-cyano-7-nitroquinoxaline-2,3-dione
DRG	dorsal root ganglion
ENS	enteric nervous system
fMRI	functional MRI
i.c.v.	intracerebroventricular
LTD	long-term depression
LTP	long-term potentiation
MS	multiple sclerosis (abbreviate only if used frequently throughout article)
PD	Parkinson disease (Parkinson's also acceptable; abbreviate only if used frequently throughout article)
PET	positron-emission tomography
PNS	peripheral nervous system
TTX	tetrodotoxin

11.16 Hyphenation

Some combinations of words are commonly read as a unit. As nouns or as modifiers, these combinations should not be hyphenated:

Active site	Amino acid chain
Bone marrow biopsy	Base excision repair
Cell culture	Cerebrospinal fluid cells

Electron density	Free radical
Heavy atom	Hydrogen bond
Lead compound identification	Leaving group
Main chain	Major satellite sequence/repeat
Medical school students	Natural product
Nuclear pore complex	Nucleic acid
Nucleotide excision repair	Public health officials
Relaxation dispersion NMR	Schiff base
Side chain	Social Security benefits
Steady state	Stereo view
Structural biology	Tissue culture
Urinary tract infection	White blood cell count

Do not hyphenate names of disease entities used as modifiers.

Basal cell sarcoma
Grand mal seizure
Hyaline membrane disease
Myocardial infarction invalidism
Sickle cell anemia

An exception is an adjectival phrase such as 'multiple-sclerosis-like'.

Prefixed Greek letters are preferably hyphenated, but follow the author's usage.

Terms not hyphenated include:

Angiotensin I	Beta blocker (or beta-blocker, β-blocker, not β blocker)
AV block	Beamline
Colocalize	Counteranion
Countercharge	Counterion
V domain (variable domain of gene)	Factor III
G protein	Interferon alpha/alpha interferon
Type I interferon (IFNα, β; type II is IFNγ)	T lymphocyte, B lymphocyte
T cell, B cell	QT prolongation
C region (constant region of gene)	D_2 receptor, S_2 receptor
ST segment	Student's *t* test
Carboxy, amino terminus	*V-D* unit (variable-diverse unit of gene)
p value, *P* value	P wave

Terms hyphenated include:

Acetyl-CoA (acetyl coenzyme A)	α_1-Adrenoreceptor/adrenoceptor
α-Fetoprotein	*ara-1* (gene with unknown locus)
β-Blocker or beta blocker (or beta-blocker)	Band-pass filter

C-2, T-2, L-2, S-2 vertebrae (or C2, T2, L2, S2)	C2–3 etc. (vertebral space)
C-3 (third carbon atom, vs C_3 for a chain of three carbon atoms)	Chemo-enzymatic
γ-globulin	Gly-Lys-Ala (amino acid chain)
HSV-1 (herpes simplex virus 1)	HTLV-III (AIDS virus)
Interferon-α, (but IFNα, IFNγ_{2b})	Interleukin-2, IL-2TNF-α (tumor necrosis factor alpha)
P-R interval, Q-T interval (ECG items; hyphens are acceptable)	L-DOPA
L-*methyl*-[^{14}C]methionine	winged helix-turn-helix motif
X-ray	C- or COOH-terminal (do not hyphenate with 'terminus')
N- or NH$_2$-terminal (do not hyphenate with 'terminus')	

En dashes are used only where required for general reasons, i.e., in

– Ranges (e.g., stages 1–3, P–R interval, the range t_1–t_2)
– Word or name combinations (see Sect. 5.6)
– Chemical bonds (see Sect.12.1.3)

11.17 Numbers

For use of units and numbers, see Sect. 6.3.

11.17.1 Roman Numerals

In general, avoid the use of roman numerals except when part of formally established terminology; use Arabic numbers instead. Examples where roman numerals are used are:

Blood clotting factors (I–XIII)
Cancer staging (I–IV)
Cranial nerves (I–XII; (abbreviated, for example, 'CIII', but note the phrase 'the third cranial nerve')

11.17.2 Arabic Numbers

Examples:

Complement components of serum (1–9)	C1, C2a
ECG leads	V$_1$ (but limb leads I, II, and III)
Heart murmurs (1–6)	grade 3 murmur
Platelet factors	platelet factor 3
TNM classification of cancer	T3, N1, or M0 (but together: T3N1M0)
Vertebrae	

cervical (1–7)	C-2, C2
thoracic (1–12)	T-2, T2
lumbar (1–5)	L-2, L2
sacral (1–5)	S-2, S2
Vertebral spaces	C2-3 etc.
Virus or organism types	echovirus 30, coxsackievirus B2
Vitamins	vitamin B_{12}

11.17.3 Zero

A zero should generally be used before a decimal point. This also applies for values for s.d., SD, t, χ^2, and F, but is not obligatory (but is preferred) for P or r.

11.17.4 Time

Many scientific data must be reported with time using the 24-h clock (e.g., 0915 hours, 2230 hours). Follow the author's usage in reporting time, correcting any inconsistencies.

11.18 Reference Works

The following reference books can serve as reliable guides.

For medicine:

Dorland's Illustrated Medical Dictionary
Butterworths Medical Dictionary (for British English)
For biological nomenclature:
Council of Biology Editors Style Manual
Bergey's Manual of Systematic Bacteriology
Oxford Dictionary of Biochemistry & Molecular Biology
Webster's Third New International Dictionary
https://www.qmul.ac.uk/sbcs/iubmb/
For pharmaceutical nomenclature, trade and generic names:
The Merck Index
Pharmacological and Chemical Synonyms (Excerpta Medica)
Martindale: The Extra Pharmacopoeia
For medical styling:
American Medical Association Style Manual
American Psychological Association Style Manual
Council of Biology Editors Style Manual
Medical Style and Format (Huth)
ASM Style Manual (American Society for Microbiology)

12 Chemistry

12.1 Chemical Names

12.1.1 Spelling and Reference Books

Chemical names should be checked for spelling and styling. The major reference works containing such information are *The ACS Style Guide* (American Chemical Society, but not their reference style!), *Quantities, Units, and Symbols in Physical Chemistry (IUPAC)*, *Compendium of Macromolular Nomenclature (IUPAC)*, *The Merck Index*, *Martindale's*, and *Pharmacological and Chemical Synonyms*. See also http://goldbook.iupac.org. Some common names are so well established — for example, formaldehyde, chloroform, acetic acid — that the 'correct' name would confuse, so it is acceptable to use the common name instead. Note that there are a few differences between American and British spelling, such as aluminum (US) vs aluminium (British), sulfur, sulfate, sulfide (US) vs sulphur, sulphate, sulphide (British) although the Royal Society for Chemistry now advises using f instead of ph and this is preferred Springer Nature style, estradiol (US) vs oestradiol (British), as well as different names, such as adrenaline/epinephrine.

12.1.2 Element Symbols and Names

A list of the symbols for elements is contained in *Webster's* under "elements." These symbols are set in roman type (except when they appear as locants in a chemical name; see Sect. 12.1.5.

Names and symbols (e.g., iron, Fe) may both be used in running text, and the extent to which the copy editor should attempt to achieve consistent use of one or the other depends greatly on the individual manuscript. The following points should be considered:

1. When many elements or compounds are mentioned or discussed together, it is often clearer and saves space to use the symbols.
2. When ions are being discussed, the symbol is sometimes more precise and usually shorter (e.g., iron ions, Fe^{2+}, Fe^{3+}).
3. When elements or compounds are mentioned relatively infrequently, use of symbols interferes with the flow of the writing and the spelled out name is preferable.
4. If the name is being used as part of a formula (CuO superconductors) the symbols can be used.
5. If an element is rare and has an unusual name (such as praseodymium), define the symbol at first use in non-chemistry journals.

'Shorthand' formulae such as Me, Bu, Pr, Ph, OBn, OAc can be used. Acceptable abbreviations in figures, schemes, methods include, Ph, Ac, Bn, Me, Et, *i*-Pr, Bu, *n*-Bu, *t*-Bu (or *t*Bu or ^{t}Bu for tertiary butyl; follow the author's style for branched alkanes as long as it is consistent), Fmoc, Boc, Bz, Cbz, PG, LG, R, Nu, E (or E^{+}), X. Pi and P in a circle are acceptable in reaction schemes.

6. See http://www.chem.wisc.edu/areas/reich/Handouts/acronyms.pdf for a more exhaustive list.
7. Abbreviations for complicated organic chemicals (MOPS, HEPES, BAPTA buffers, EDTA) are acceptable and do not need to be defined at first use.

8. Superscripts denoting isotopes should be used only with the element symbol: ^{99m}Tc, ^{131}I, ^{14}C etc. With the element name, style as iodine-131 etc. Similarly for the other superscript and subscript positions surrounding an element symbol: Na^{+}, but sodium ion; Na$_2$, but disodium.

12.1.3 Chemical Bonds

Use en-dashes for chemical bonds:

CH_3–COOH

Note: In protein structures featuring the three-letter amino acid abbreviations, hyphens are used, not en rules (see Sect. 12.3.2).

12.1.4 Addition Compounds

In the formula for an addition compound, a centered dot is used and should be marked as such.

$Na_2SO_4 \cdot 10H_2O$ $BH_3 \cdot NH_3$

Also use a centered dot for hydrated crystals (e.g. $CuSO_4 \cdot 6H_2O$).

12.1.5 Prefixes

Note that the first letter of a chemical name after a prefix, such as shown in Sects. 12.1.5.1 and 12.1.5.2, should be capitalized at the beginning of a sentence or in a heading.

12.1.5.1 *Locants*

Symbols for elements that occur as locants (i.e., the portion of a chemical name that designates the position of an atom or group in a molecule) in chemical names are set in italics. This rule covers any single capital letter except D or L (these refer to configurations; see Sect. 12.1.5.2) that appears within a chemical name. But note in N-heterocyclic, the N is not italic because there it means the N is the heteroatom in a ring.

O-methyltyrosine

N,*N*-dimethylurea

S-benzyl-*N*-phthalocysteine

hexahydro-2*H*-azophin-2-one

When a chemical name does not appear, the element symbol remains roman.

N-linked glycan

O glycosylation

N-terminal, N-terminus or N terminus

12.1.5.2 *Rotation/Configuration*

Symbols for configuration or rotation precede the name of the compound and are joined to it by a hyphen. The symbols are *d*, *l*, and *dl* (italics) or (+), (−), and (±) for optical rotation; D, L, or DL (small capitals) for configuration in carbohydrates and amino acids; and *R* and *S* (italics) for absolute configuration.

d-6-hydroxytryptophan

emodin-*l*-rhamnoside

(+)-6-hydroxytryptophan

D-(+)-alanine

(1*R*,3*R*,5*S*)-[(1*S*-*sec*-butoxyl)]-3-chloro-5-nitrocyclohexane

Molecular species with planar chirality are described as having an R_p or S_p configuration.

12.1.5.3 *Other Prefixes*

Other prefixes indicating configuration that should be marked for italics include:

allo- cis- trans- ortho- para- meta- syn- amphi- myo- sn-

erythro- xylo-meso- endo- exo- threo- ribo- xylo-

o m p n sec s t tert

These prefixes may also be used alone in the text without being attached to a specific chemical name and should also be marked for italics.

...encoding a *trans*-acting product

...*ortho* coupling

...via the *meta* pathway

...in *cis*

Use roman for bis, cyclo, iso, neo, tris, epi, nor.

12.1.5.4 *Latin and Greek Prefixes*

These are set in Roman type and closed up to the word.

Latin: di, tri, tetra, penta, hexa, hepta, octa, nona, deca

Greek: bis, tris, tetrakis, pentakis, hexakis, heptakis, octakis, nonakis, decakis

12.1.6 Polymer Nomenclature

In polymer nomenclature, *co*, *alt*, *b*, *g*, *r*, and *m* are set in italic type as are *block*, *graft*, *cross*, *inter*, and *blend*. Generally follow the author, making sure it is consistent within the article or chapter; e.g., PMMA can be written polymethylmethacrylate; poly(methylmethacrylate); poly(methyl methacrylate) — there is no confusion as to what the polymer is with any of these formats.

The abbreviation M_w means weight-averaged molecular mass; M_n means number-averaged molecular mass. Does not need to be defined in chemistry texts.

12.1.7 Electron Orbitals

These are designated with the italic letters *s, p, d, f, g, h* and do not take a hyphen unless used adjectivally. For orbital splitting, use e_g, $t_{2\mathrm{g}}$ etc.

12.1.8 Capitalization

The first letter of the chemical name should be capitalized at the beginning of a sentence. When a prefix is at the beginning of a name that starts a sentence, capitalize the first letter of the chemical name itself.

S-Benzyl-*N*-phthalocysteine was added...

12.2 Ions, Nuclides, and Other Species

12.2.1 Ionic Charges

Ionic charges are indicated by superscript numbers and plus and minus signs.

Na^+

Cl^-

Zn^{2+} (*not* Zn^{++} or Zn^{+2})

$\mathrm{PO}_4{}^{2-}$

Note that the superscript should be placed outside the subscript, not aligned with it.

12.2.2 Nuclides and Isotopes

The ionic charge (upper right), number of atoms (lower right), atomic number (lower left), and mass or atomic weight number (upper left) characterizing a nuclide are indicated as follows:

$^{32}_{16}S^{2+}_{2}$

In this case, the superscripts and subscripts are not staggered.

The atomic weight often appears alone as a superscript on the left.

^{235}U ^{14}C

It should be hyphenated to the right of a spelled-out element name.

uranium-235 carbon-14

For usage of the symbol versus written-out name, see Sect. 12.1.2. See also Sect. 12.4.2 for styling of labeled compounds.

12.2.3 Elementary Particles

There are many types of subatomic or elementary particles (listed in the *McGraw-Hill Dictionary of Scientific and Technical Terms*); the most common are:

Electron: e^-

Positron: e^+

Neutron: n

Proton: p

12.2.4 Oxidation Numbers

If the oxidation number or level is given with the name of an element, it should be in Roman numerals, in parentheses, and closed up to the name.

iron(II) chloride

The following styles are also acceptable in running text:

Fe(II) *or* Fe^{II}

When used in conjunction with element symbols in a formula, the Roman numeral should be a superscript without parentheses.

Pb_2^{II} $Pb^{IV}O_2$ $Fe^{II}Cl_2$

An excited electronic state may be indicated by an asterisk.

He* NO*

12.2.5 Radicals (Free Radicals)

In the formula for a free radical, the unpaired electron is indicated by a centered dot.

$H_3C\cdot$ $HO\cdot$

Cation and anion radicals are indicated by a bold superscript dot, followed by a minus or plus sign.

$R^{\bullet+}$ $C_6H_5NO^{\bullet3-}$ $O_2^{\bullet-}$ $HO_2^{\bullet}$

12.2.6 Physical State

The letters c, g, l, s, and aq may be used to indicate the state (crystal, gas, liquid, solid, aqueous) of a compound. They should be in parentheses, on the line, and closed up to the element symbols.

Li(c) O_2(g)   Mg(l)   Ca(s)   Cl(aq)

12.3 Organic Chemistry

12.3.1 Carbon Atoms

A C_{18} acid is an acid containing 18 carbon atoms. C-3 denotes the carbon atom numbered 3. The form C3 is an accepted alternative to C-3, but if the author uses the form C3, ensure it is not used to denote a C_3 compound. Query the author if it is ambiguous. Cα indicates the α position of the atom relative to another.

12.3.2 Amino Acids

See Sect. 11.13.2.

12.4 Labeling

12.4.1 Isotope Labeling of Natural Constituents

If an isotope of an element normally part of the compound is used to label the compound, the symbol for the isotope is enclosed in brackets immediately before the name of the compound or specific part of the compound.

[^{32}P]phosphate   (*not* ^{32}P-phosphate)

[^{32}P]ATP   (*not* AT[^{32}P]P or AT^{32}P)

[^{14}C]glucose   (*not* ^{14}C-glucose)   [methyl-^{14}C]glucose

sodium [^{14}C]formate   (*not* [^{14}C]sodium formate)

$\delta^{18}O$ (not $\delta\ ^{18}O$)

If a chemical is identified by its formula instead of its spelled out name, the isotope number appears as a superscript preceding the symbol for the labeled element.

$^{14}CO_2$ (*not* $[^{14}C]CO_2$)

$H_2{}^{35}SO_4$

Abbreviations that are used in the text should have the isotope symbol in brackets at the beginning.

$[^{32}P]$ATP (*not* $AT^{32}P$)

12.4.2 Labeling Through Addition of an Isotope

If an isotope of an element *not* normally part of the compound is added to label the compound, or if labeling does not refer to a specific compound, the isotope symbol together with "labeled" is usually inserted before the name, e.g.,'^{14}C-labeled' indicates ^{14}C has been added to a substance that does not contain carbon naturally.

^{131}I-labeled albumin	*not* $[^{131}I]$albumin, because albumin in its native state does not contain iodine
^{14}C-labeled steroids, ^{14}C-labeled amino acids (also, ^{14}C-amino acids)	*not* $[^{14}C]$steroids or $[^{14}C]$amino acids, because they are classes of compound
^{3}H labeling	*not* [3H]labeling or ^{3}H-labeling

An alternative permissible styling used by some authors to refer to labeled classes of compounds and compounds where the label is not a normal constituent of the compound is simply to use a hyphen: ^{14}C-steroids, ^{131}I-albumin. This form is not preferred, but if an author has used this style consistently allow it to stand. The position of isotope labeling is indicated in square brackets and before the symbol of the element, connected by a hyphen: for example, $[1\text{-}^{14}C]$aniline, $[\gamma\text{-}^{32}P]$ATP.

Uniform labelling is sometimes denoted with U- before the isotope (e.g., D-$[U\text{-}^{14}C]$glucose is selectively labelled, and the isotope is uniformly distributed among the six positions in the ring). If this notation is used, define 'U' at first use in the main text, e.g., uniformly labelled ^{13}C ($U\text{-}^{13}C$). General labeling may be denoted with a 'G'.

12.5 Spectroscopy

For frequency, the standard abbreviation is ν (discourage the use of f). Wavelengths (λ) are in meters (m) and SI submultiples thereof (mm, µm, nm etc); the ångström (10^{-10} m) is also often used.

The energy of a photon is $h\nu$ (h is Planck's constant, ν is Greek lowercase nu). In SI the energy of a photon is measured in joules (J), but more often electron-volts (eV) are preferred
(1 eV = 1.602×10^{19} J). Both are acceptable.

Spectral regions. Familiar names for spectral regions, such as visible, infrared and ultraviolet (can use UV with no definition), X-ray and γ-ray, microwave and radio, are allowable. Optical radiation encompasses some of the UV, the visible, and some of the IR portion (about 320 nm to 1 µm). More particular familiar references such as hard X-ray, UVA, S-band and far infrared should be used only when the context removes the inherent ambiguity. In the radio spectrum, UHF and VHF have a meaning defined by the ITU. The main spectral regions are (in order of increasing energy):

Spectral region	Description
Radio	Wavelength, 3 km–3 m; frequency, 105–108 Hz; affects nuclear spin
Microwave	30 cm–3 mm; 109–1011 Hz; affects molecular rotation and electron spin
Infrared	3 mm–3 µm; 1011–1014 Hz, affects molecular vibrations
Visible/ultraviolet	600 nm–30 nm; 1015–1016 Hz; affects valence electron excitation.
X-ray	30 nm–30 pm; 1016-1019 Hz; affects core electron excitation
γ-ray	30 pm–300 pm; 1019–1022 Hz; affects nuclear processes

Notation of atomic states Use *s*, *p*, *d*, *f*, etc., e.g., $1s^2$, $3p^6$

Spin angular momentum. The spin angular momentum of an electron is specified by the quantum number m_s that can assume two values +½ or –½, sometimes abbreviated as ↑ and ↓, respectively; both are acceptable.

Symmetry operators: In symmetry operators such as C_n, C_{2h} and D_{nd}, the letters should be italic, but not the numbers.

Electronic states of molecules. For example, the ground state of CH is $\mathrm{X}^2\Pi_{1/2}$.

X is the energy state (for ground or lowest energy state).

Π is the term symbol that describes the symmetry of the state: for diatomics, the Greek letters used are Σ, Π, Δ, Φ. A left superscript integer 1, 2, 3, refers to the 'multiplicity' of the spin state. A right subscript gives the total angular momentum and will be integral or half integral.

For polyatomics, the energy label has a tilde (~) over it to avoid confusion with the term symbol. In the term symbol, uppercase roman letters signify the symmetry group of the state, for example, A_{1g}, E_{1u}, T_{2g}. The multiplicity left superscript is still used. For example, benzene's ground state is $\tilde{\mathrm{X}}\,{}^1\mathrm{A}_{1g}$.

Rotational levels are denoted by uppercase italic letters, e.g. Λ, Σ, Ω, M_L, M_l, M_J, M_S, *K*, *J*.

In *visible/UV spectroscopy,* both absorbance and optical density (OD) are allowed.

For *spectroscopic states* use roman type, e.g., $^7\mathrm{F}_0$, $^5\mathrm{I}_8$.

12.6 Crystallography

Axes. In preference use *a–b* plane (en rule), but allow *ab* plane if consistent. Different types of brackets are used for axes and planes:

- () refers to a particular plane, e.g., (100) plane, while { } refers to a set of planes, e.g., {100} signifies the set (100), (010), (001).
- [ ] signifies a particular axis, e.g., [110] while < > signifies the general axes, i.e., all the axes pertaining to it, e.g., <110> includes [110] and [011] and so on.

Space groups. In crystallographic space groups, letters are italic and the numbers are not, e.g. $P4/mm$. Space groups should be defined or made clear, whichever is more practicable (e.g. orthorhombic).

If there are two protein chains, a prime is used to differentiate, e.g. a chain and a′ chain. If there are more than two chains, follow the author regarding nomenclature, but ensure it is defined or explained.

The following table shows some standard abbreviations that should be defined at first use (unless indicated here).

Abbreviation	Definition
bcc	body-centered cubic
fcc	face-centered cubic
hcp	hexagonal close-packed
F_o, F_c	No definition necessary, but note styling (italic variable F and roman 'o' and 'c' subscripts for observed and calculated, respectively). F_obs and F_calc are allowed alternatives. Follow author usage.

12.7 Units

12.7.1 Molecular Weight

The term "molecular weight" is short for relative molecular mass (symbol M_r). Since this is a ratio (of the mass of one molecule of a substance to 1/12 of the mass of an atom of ^{12}C), it is a pure number and has *no units*. Structures like ribosomes and viruses are not single molecules and should not therefore be described as having a molecular weight.

A dalton is a unit of mass equal to 1/12 of the mass of an atom of ^{12}C. It is therefore incorrect to say that something has a molecular weight of 72,000 daltons, although this is often done. In a manuscript that is undergoing intensive copy editing, change to either "a mass (or molecular mass) of 72,000 daltons" or "a molecular weight of 72,000."

12.7.2 Concentration

Concentration is the amount of dissolved substance (solute) present in proportion to the amount of solvent or total solution. It is usually given in moles per liter (abbreviated mol l^{-1}) or as a molarity (symbol M). Of the two, use of moles per liter is preferred, but either form is acceptable provided it is used consistently within a manuscript.

Concentration is sometimes also given as a mass or volume in proportion to the amount of solvent or solution (e.g., milligrams per liter, mg l^{-1}; volume per cent, vol%). While these forms are, in general, not preferred, do not query the author or attempt to change them unless they have been used inconsistently within a manuscript.

For particular species, use square brackets but define first: $[Cl^-]$. Note that in Sect. 15.2, concentrations are denoted simply with the chemical symbol, e.g. Fe, Mg/Mg+Fe. To insist on square brackets here would result in a cumbersome text, so try to spell out the representation in words.

Also note, pH is roman, but pK_a.

12.7.3 Special Styling Conventions for Metrology

For metrology in chemistry, SI units and ISO norms must be followed in all cases, even where this contradicts Springer Nature style (e.g., insert a space between number and %).

12.7.4 Additional Notes on Units

Unit	Definition	Comment
AMU	atomic mass units	Usually equivalent to Hartree units
a.u.	atomic unit	Atomic unit is a unit of distance, being equivalent to the Bohr radius, or 0.529 Å. If used, define in appropriate SI units.
c.c.	cubic centimeter	Do not use. Replace with ml. For cc STP: use the form cm^3 STP or cm^3 at STP.
ppm	parts per million	No definition necessary
ppb	parts per 10^9	Define at first mention
ppt	parts per 10^{12}	Define at first mention
‰	per mille	Do not use "per thousand". Replace with ‰.
per mole, mol^{-1}	--	Both are acceptable. (per mol is not).
K	Kelvin	No degree symbol is used. Style with a space between number and K, e.g., 273 K.

12.8 Miscellaneous

The following tables list some specific abbreviations and styling for nomenclature in chemistry.

Abbreviation	Definition
d.r.	Diastereometric ratio
e.e.	Enantiomeric excess
(aq)	Aqueous species, e.g. S_4^{2+}(aq) (affices are staggered)
p, P	Pressure. Preferred usage is *p*, unless there is confusion with other symbols.

Topic	Styling
Hydrogen bond(ing)	H-bond not generally used, but follow specific publication guidelines where appropriate. Hyphenation is only inserted when the phrase is used adjectivally or as a verb.
Double oxides	Use the 'double' form when spelling out (calcium titanium oxide for $CaTiO_3$, not calcium titanate). The exception is barium titanate, which is well established in this form in electrochemistry.
Emission lines	Hα, Hβ(Balmer); Lyα, Lyβ(Lyman); Cu Kα etc. (see Sect. 16.2.4.8).
Hydrocarbons	Refer to alkanes instead of paraffins. Olefins are alkenes; acetylenes are alkynes. All terms are in common usage and acceptable.
Structure symbols	Structure symbols for ZIF (zeolitic imidazolate framework) are lowercase: sod, cag, mer etc. Names of zeolites are caps: SOD, CAG, MER etc
Superconductor formulae	The order of the elements is Tl-Ca-Ba-Cu-O; Bi-Ca-Sr-Cu-O. e.g. $YBa_2Cu_3O_8$. En rules are used where the stoichiometry is not known or need not be specified (in referring to a general class of compounds). Both δ and x are used to denote stoichiometry e.g. $YBa_2Cu_3O_{7-x}$ or $Bi_2Sr_2CaCu_2O_{8+\delta}$: ($\delta$ denotes a small departure from stoichiometry and x an arbitrary one) Follow the author
Structures	Use R^1, R^2 for generic groups (note superscript)

13 Mathematics

13.1 General

This chapter deals with the representation of mathematical content in Springer Nature's scientific, technical, and medical books and journals. It does not specifically cover the style requirements of the Springer Nature book and journal programs in the discipline of mathematics.

Mathematics is a language unto itself. Every mathematical object that exists has a clear and unambiguous meaning. When choosing a style of notation for depicting mathematics the most important consideration is how to retain the clarity of meaning of the mathematical objects. In plain language, the symbols we use often have more than one meaning. The sentence "Virginia is in America" may refer to both a place "Virginia" and a person "Virginia." With no further information the sentence is ambiguous. When reading a mathematical expression there must be no room for such confusion. Though the objects of mathematical discourse have unambiguous meanings the long history of the subject has led to the use of many different notations. In many instances there is no international standard for mathematical notation. Springer Nature has adopted a best-use policy and has chosen a style that we hope aids in the clear presentation of mathematics. When copy editing mathematics as with other disciplines, the central question is: "Is the author consistent? Is the meaning clear?"

For some product types, there is a defined style requirement for mathematical content, while for other products only consistency in representation is required within each book, book chapter, or journal article. For the bulk of the products, we define two levels of math markup: standard and intensive (see Sect.9.5.2), which correspond to the copy editing levels described in Sect. 1.3. The present chapter provides detailed information on the correct handling of this content.

13.2 Numbers

As described in Sect. 6.3.1, preferred Springer Nature style is to use words for the numbers "one" to "ten" when not used in conjunction with units, but larger numbers (e.g., 11, 12, etc.) should always be digits. A common alternative in technical disciplines such as mathematics and physics is to use words for "one" to "nine" and numerals thereafter ("10," "11," etc). This also applies to ordinals, e.g., "tenth" or "10th." A mixture of styles should be resolved.

Decimals The decimal point should be represented using a period, so correct any numbers in the non-English-language form, e.g., change "10.000,00" to the "10,000.00" form. The center dot is not recommended for decimal points.

As an exception to the general rule given in Sect.6.3.1, commas or thin spaces can be used in TeX-based products for separating each bank of three digits to the left of the decimal point. In such products, separators can be retained if used for numbers of four digits and up.

Commas	9,999	10,000	10,000,000
Spaces	9 999	10 000	10 000 000

Exponential Notation The "×" sign is preferred to center dots "·" in exponential (or "scientific") notation, e.g., 2.99×10^8 not $2.99\cdot10^8$. The 1.2E03 form (sometimes 1.2e03) is not preferred but is acceptable. A mixture of styles should be resolved, but an author is allowed to use the 1.2×10^3 form in the text and the 1.2E03 form in tables to save space.

Logarithms Use the following forms:

$\log_a x$	log to the base a of x
$\ln x$ or $\log_e x$ or $\log x$	natural log of x
$\log x$ or $\log_{10} x$	common log of x

If a single variable the format is $\log x$; if more than one variable then brackets are needed to clarify what the log is being applied to, e.g. $\log (x/y)$. Log numbers have no units; it is the quantity that the logarithm is applied to that has units, and so brackets are used to clarify, for example $\log[t \text{ (min)}]$. The square brackets are used so that the unit can have the usual parentheses and preceding space. The word 'log' never takes an uppercase L, even at the start of a sentence or figure label.

Extra Zeros Be very careful when inserting or removing zeros, whether to the right or left of decimal points; they often contain information. For example, the values 6 mV, 6.0 mV, and 6.00 mV are not equivalent, they imply different measurement precisions. Although, ideally, comparable entries in a column of a table should usually have the same precision, i.e., the same number of digits after the decimal point, leave the usage provided by the author. When the origin of a graph is zero, use 0 rather than 0.00.

Billions Use the "American" system, where "billion" is 10^9 and "trillion" is 10^{12} (i.e. not the old "British" system's 10^{12} and 10^{18}, respectively); see *Merriam–Webster's* under "number." The spelled-out versions are discouraged because of the potential ambiguity; advise the author to change to numerals.

Number Systems Various number systems appear in scientific texts:

> **Binary** Use spaces to separate bytes or nibbles (half-bytes), e.g., 00111010 or 0011 1010. In long lists or tables, advise the author to use nibble separation. An author may use a base subscript, e.g., 00111010_2; leave it even though it may be redundant.
>
> **Hexadecimal** Impose consistency using a recognized scheme; the "0×" prefix is preferred. Use upright uppercase letters A–F. Example:
>
> $$0\times 003\mathrm{A} = 3\mathrm{A}_{16} = 3\mathrm{A\ H} = 3\mathrm{A\ Hex} = 58_{10}$$
>
> **Octal** Use a base-8 subscript. For binary equivalents, use banks of 3 digits. Example:
>
> $$100\ \ 101 = 45_8 = 37_{10}$$

SI Prefixes Use the standard prefixes for decimal multiples of units. These are written in upright font and are closed up to the unit.

Positive			Negative		
Value	Prefix	Abbreviation	Value	Prefix	Abbreviation
10^{24}	yotta	Y	10^{-1}	deci	d
10^{21}	zetta	Z	10^{-2}	centi	c
10^{18}	exa	E	10^{-3}	milli	m
10^{15}	peta	P	10^{-6}	micro	µ (upright)
10^{12}	tera	T	10^{-9}	nano	n
10^{9}	giga	G	10^{-12}	pico	p
10^{6}	mega	M	10^{-15}	femto	f
10^{3}	kilo	k	10^{-18}	atto	a
10^{2}	hecto	h	10^{-21}	zepto	z
10^{1}	deca (or deka)	da	10–24	yocto	y

13.3 Formatting

Mathematical expressions are frequently complex with many components, for example in a given expression some symbols may represent operators, some represent functions and some represent variables. The most common question when typesetting mathematical notation is, "Should symbol "x" be set to upright or set in italic?"

13.3.1 Upright Characters

Do not italicize numbers, punctuation, parentheses, common function names (see Sect.13.3.3), units, or mathematical signs (e.g., "+") wherever they occur (regardless of whether or not they appear in an equation or as a superscript or subscript).

13.3.2 Variables

Use the following rule of thumb:

– If a symbol in an equation takes a value that needs to be substituted into it before the result of the equation can be calculated, then it should be set in *italic*. This symbol is a container for a value. The contents of the container can either be a constant value or it can vary, but the important point is that the symbol, at the final reckoning, will be replaced by something else.

– If the symbol is a label, either for the name of a particle for example, or attached to a variable giving information about that variable, then it should be set in upright. This label does not act as a container for a value.

Here are some examples. When talking about the mass of the electron we use m_e. The "m" will eventually be replaced by the number for the electron mass, the "e" simply labels this mass as being that of the electron. When we talk about the electronic charge of the electron we use "e". Here, when calculating, "e" will be replaced by the number for the electronic charge.

If the author is talking about some point x, some distance d, or some time t, all of these are set to italic in the expectation that later they will be replaced by the value of the coordinate, distance or time respectively. If authors wish to label these as being in some positive region of some space then they might mark them as x_pos, d_pos, or t_pos. Here the "pos" is set to be upright as it denotes "positive."

If a variable has a running index such as x_i, where i runs between 0 and 3 for example, then this index i, though a label, is set to italic. In this instance the label i is a container for a set of values and in each instance x_0, x_1, x_2, and x_3 each take an independent value.

The numbers themselves do not act as containers for other values and so are set to upright. The complex number "i" we consider to be a creature of the same genus as the numbers and so we set it upright. Often it is represented as "j." Upright presentation is also a great aid in differentiating the complex number "i" from "i" used as a running index. The mixed use of "i" is very common.

The Roman and lowercase Greek alphabets provide the most common examples of variables. The use of upright and italic display for Greek letters follows the same convention as for Latin letters. Sometimes the Hebrew alphabet is used and sometimes the Frankfurter alphabet is used (mathematicians and physicists are fond of using many symbols).

An exception to these rules is made for a multiletter abbreviation. It is usually written upright rather than in italics even when it represents a value, for example, IP (for ionization potential) and BER (for bit error rate). (Note that "Re" commonly denotes the "real" part of a complex number.)

Constants generally follow the same typographic pattern as variables: e.g., h, k_B (Boltzmann constant), k_a (association constant). Exceptions are the constant π and logarithmic e, which are set roman.

Here are some examples of commonly used combinations:

μ_B ("Bohr"), T_C ("Curie"), m_p ("proton"), m_e ("electron"), e (value of the elementary charge), N_A ("Avogadro"), k_B ("Boltzmann"), c or c_0 (speed of light in vacuum), h and $\hbar = h/2\pi$ (Planck's constant and reduced Planck's constant), μ_0 and ε_0 (magnetic and electric constants), n-ary, p-adic, etc. Note "e" is upright when it stands for the word "electron."

The square root of −1 is denoted i. Coordinates are styled '(x, y)'. Fractal dimension is denoted d_f.

13.3.3 Operators

For total differentiation, the 'd' is roman: e.g., dx/dy. For partial differentiation, use ∂ (curly d).

Use ∝, not ~, to signify 'varies as'; ~ may be used for 'is distributed as', but this should be defined. Note “an ~5%”, not “a ~5%”, matching the pronunciation. Where it is necessary to make the operator explicit, use the multiplication symbol '×', not an asterisk or bullet point, to denote standard multiplication (although see below regarding vector multiplication).

13.3.4 Functions

In an equation with an arbitrary function the function has to be replaced by the operation that it represents before a final value for the equation can be obtained. General functions are therefore set in italic. In the equation $\mathrm{y}=f(x)+g(x)$, before calculating the value of y we need to know what the functions $f(x)$ and $g(x)$ are. These functions act as containers for some proper given operator on x and therefore are set in italic. When the function is named explicitly, it does not act as a container for some arbitrary operation but tells us exactly what the operations to be performed on the argument is. In this case these function names should be set upright.

The most common functions that are set upright are:

c.c. for complex conjugate

cos for cosine

cot for cotangent

csc, cosec for cosecant

curl for curl

d for differential

Δ (Greek "D")

det for determinant

div for divergence

erf for error function

exp or e for exponential

grad for gradient

h.c. for Hermetian conjugate

Im for imaginary part

log for logarithm

ln for natural logarithm

max for maximum

min for minimum

mod for modulo

$\prod$ for product (the mathematical form of the greek letter "P")

Re for real part

sgn for sign

sec for secant

$\sum$ for summation (greek "S")

sin for sine

sqrt for square root

tan for tangent

tr for trace

Here are several equations containing such functions:

$$\int_{k_0-\Delta k}^{k_0+\Delta k} c(k)\mathrm{e}^{\mathrm{i}[kx-\omega(k)t]}\,\mathrm{d}k$$

$$e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n$$

$$\sin x = \sum_{r=0}^{\infty}(-1)^r\frac{x^{2r+1}}{(2r+1)!}$$

$$\frac{\mathrm{d}^n x}{\mathrm{d}t^n}$$

13.4 Vectors, Tensors, and Matrices

These are multidimensional objects. They have more than one component and so some way is needed to distinguish them from simple objects.

Vectors Vectors are represented differently in different fields. For Springer products in the physics program, use bold and italics for characters representing vectors (e.g., ***v***). In the mathematics program, use bold and upright (e.g., **v**). Vectors are often represented using other styles, such as an overhead arrow, $\vec{v}$; this is acceptable but deprecated. Any style mixtures should be corrected. For all disciplines in Nature Research, use bold and upright. For other disciplines, check with your Springer Nature contact. Vector components should be in italics, not in bold, e.g., $\mathbf{v} = (a,\ldots,d)$. For the norm of a vector, follow the author's usage of $\|\mathbf{v}\|$ or $|\mathbf{v}|$, or simply **v**. Vector variables are set in italics, as are tensors and matrices. The magnitude of a vector is also set in italics. A unit vector is set boldface with a circumflex ('hat': $\hat{\mathbf{v}}$). The vector operator ∇ (nabla) is set roman (it is not a variable). For the gradient of a function *f*, use ∇f not grad(*f*). For the divergence of a vector-valued function **v**, use $\nabla \cdot \mathbf{v}$ (center dot) not div(**v**). For the curl of a vector-valued function **v**, use $\nabla \times \mathbf{v}$ (multiplication sign) not curl(**v**). For the scalar, or 'dot', product of vectors **r** and **x**, use $\mathbf{r} \cdot \mathbf{x}$ (center dot). For the vector, or 'cross', product of vectors **r** and **p**, use $\mathbf{r} \times \mathbf{p}$ (multiplication sign).

Matrices The characters representing matrices may be bold, bold and italic, or italic. Matrix elements should be in italics and nonbold. For the trace of a matrix *A*, use tr(*A*) (note lowercase 't'). The superscripts "T" or (rarely) "t" (transpose) and "H" (Hermitian) should be upright. Full matrices should be written as displayed equations. For matrix dimensions use "×," e.g., "a 3×3 matrix" or "an $n \times m$ matrix." Matrix determinants can be represented using straight lines, e.g., |B|, or as "det *B* ." Follow the author's style for representing omitted elements (e.g., use ellipses). Examples:

$$M = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}$$

$$\det B = |B| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Tensors The characters representing tensors are generally displayed using a sans serif font (e.g., Arial). For tensor product use ⊗.

13.5 Displayed Equations

13.5.1 General

Copy editors are, as a rule, not responsible for making improvements with regard to issues such as page breaks and equation positioning. They are responsible, however, for some matters of presentation and for the text. For example, phrases such as "the following equation" or "the equation below" in the text may frequently be removed as redundant or superfluous. For citation rules see Sect. 9.5.1.

The *alignment* of parts of an equation may be important. The analogy of "verbs" and "conjunctions," such as described in the *Chicago Manual of Style*, may be helpful in this context. "Verb" characters include

$$= \; \neq \; < \; > \; \leq \; \geq \; \supset \; \subset \; \in \; \notin \; \cong \; \equiv \; \approx$$

while "conjunctions" are

$$+ \quad - \quad \times \quad \pm \quad \cup \quad \cap$$

"Verbs" in successive lines should be aligned beneath each other. A "conjunction" at the start of a line should be aligned under the first element to the right of the "verb" in the line above it. Example:

$$\begin{aligned} x_1 &= a+q+r \\ x_2 &\geq a+(b+c) \\ &\quad +(w-z) \end{aligned}$$

A "verb" or "conjunction" should not be repeated across a linebreak; it should appear once only as the first element of the following line. Example:

$$\begin{aligned} x_2 &\geq a+(b+c) \\ &\quad +(w-z) \end{aligned} \qquad \text{not as} \qquad \begin{aligned} x_2 &\geq a+(b+c)+ \\ &\quad (w-z) \end{aligned}$$

13.5.2 Punctuation

It is useful to assume that a displayed equation is part of a sentence and that punctuation follows normal language rules, which is demonstrated in the following examples showing combinations of text and equations (see Chap. 5). Unless otherwise instructed, follow this usage unless an author has been consistent in not punctuating displayed equations.

Equation ends the sentence:

... where z is fixed and

$$x = y + 1. \tag{13.1}$$

From this relation ...

Sentence continues after equation:

... where z is fixed,

$$x = y + 1, \tag{13.2}$$

and *w* takes the value 1.

Combined example:

... because

$$x = y + 1, \qquad (13.3)$$

where x refers to ..., and

$$w = z + 4. \qquad (13.4)$$

In this example, the addition of punctuation would be incorrect:

... where

$$a = \sum_{i=1}^{N} x_i$$

and *b* are known.

Punctuation within an equation environment:

... because

$$a = b + 1, \qquad y = w + 5, \qquad (13.5)$$

and

$$x_i = z_i + 4, \qquad i = 1, 2. \qquad (13.6)$$

Sometimes it is useful to consider equations to be the items in a list. Here is such an example:

... our list of results:

$$x = y + a + 1 \qquad (13.7)$$

where *x*, *y*, and *a* refer to ..., respectively,

$$w = z + 4 \qquad (13.8)$$

where *w* and *z* are defined elsewhere, and

$$q = r + s. \qquad (13.9)$$

As in any list, be careful not to misplace commas or other punctuation.

13.6 Inline Equations

Grammatically, an inline equation is a normal part of a paragraph and sentence. It follows normal punctuation rules (see Sect. 13.5.2 and Chap. 5). It is also never numbered; if the author has numbered an inline equation, either it must be changed to a displayed equation (to retain the numbering) or the numbering must be dropped. Because of the consequent renumbering of following equations, such an equation should as a rule be changed from inline to displayed.

You may suggest to an author that certain mathematical content be changed from inline to displayed for better presentation. Examples are fractions and content with "tall" symbols. Inline fractions should employ a solidus instead of a built-up construction wherever sensible, for example using $(a + b)/(c - d)$ instead of the built-up construction

$$\frac{a + b}{c - d}$$

If tall symbols are used inline, suggest to the author that the more convenient inline forms be used; for example, use $\sum_{i=1}^{n} x^i$ instead of $\displaystyle\sum_{i=1}^{n} x^i$, or use $\sqrt{x/y}$ for $\sqrt{\frac{x}{y}}$.

RPN, or reverse Polish (postfix) notation, cannot be changed to infix form; do not change "a b +" to "a + b." RPN is sometimes used in texts dealing with microprocessors, stack-based programs, etc.

13.7 Theorems, Definitions, and Proofs

The heading of definitions, propositions, lemmas, theorems, corollaries, remarks, proofs, and similar items should be run in. If needed, the number follows the word, i.e., Lemma 1. Both the consecutive and section-wise numbering systems are accepted. Such items are sometimes numbered sequentially (e.g., Definition 1, Theorem 2, Proposition 3, Definition 4); although not preferred usage, this is acceptable.

The style of the overall text and run-in headings will be determined by the layout and should not be marked up by the copy editor. In general, the text of propositions, lemmas, theorems, corollaries, etc. is set italic. That of definitions, remarks, proofs, examples, etc. is upright. Mathematics, numbers, and symbols should be styled as they are in regular text. Retain any emphasis for terms as supplied by the author.

Theorem Style In styling a theorem, the word "Theorem" and the number are in bold, and the following text in italics. Example:

Theorem 12 (Fermat's Little Theorem [28]) *Let p be a prime and assume that* $\gcd(a, p) = 1$, *then*

$$a^{p-1} \equiv 1 \pmod{p}.$$

Numbers, punctuation, parentheses, etc. should not be in italics.

Edit the text if the author has used a "dangling" theorem:

> ... according to the following
>
> **Theorem 12** Let x be a prime ...

Proof Style In styling a proof, the word "Proof" is in bold italics and the text follows in upright. An end-of-proof box may be used, the preferred position being flush right. Example:

> ***Proof*** From (12.9) we know that a polynomial in *x* with
>
> ... and the rest follows by induction. □

Definition Style In styling a definition, the word "Definition" is in bold, the text is upright, and the terms being defined are italic.

Other Authors may use the theorem, definition, or proof styles for algorithms, assumptions, axioms, cases, claims, conjectures, corollaries, demonstrations, examples, exercises, hypotheses, lemmas, notes, problems, properties, propositions, questions, remarks, rules, solutions, etc.

13.8 Other Issues

Functions and Symbols For clarity, functions and operands are usually distinguished using parentheses, e.g., "$f(x)$" and "$\sin(\varphi+\theta)$." A symbol may act as the verb in a sentence where it can be translated into an English-language phrase such as "is equal to" or "is an element of." Examples:

Here the function $f(x) = g(x) + 1$

The modulus is large and $g \in B$

Multiplication Symbol Center dots (·) or line dots (.) are often used for multiplication. In many cases they are redundant and can be removed, i.e., "a·b" or "a.b" be changed to the "ab" form. For a text involving vectors (see also Sect. 13.4), an author may have used the center dot to indicate scalar (inner) multiplication ($\boldsymbol{u} \cdot \boldsymbol{v}$) as distinct from vector multiplication $\boldsymbol{u}\times\boldsymbol{v}$, and in such cases the center dot must be retained. Use the "times" (×) symbol, rather than an asterisk (*) or the letter "x," in constructions such as "10 mm×0.7 mm×0.3 mm," "10,000×g centrifuge," and "×100 lens."

Similar Symbols There are many symbols that are very similar in meaning. Be extremely careful in changing an author's usage of such symbols, but query any apparent misuse. For example, some scientists distinguish between "≈" meaning "is approximately equal to" and "~" meaning "is similar to," "is of the order of," or "is asymptotically equal to," while others consider "≈," "~," "≅," and "≃" to be interchangeable, each meaning "is approximately equal to." The symbol ≈ is used for 'approximately equal to' in an equation; ~ is allowed in the mathematically correct sense (random variables, asymptotic similarity). The symbols "≅" and "≡" are often used to mean "is approximately congruent to" and "congruent to" a specific mathematical relation. Many references avoid defining these symbols because they vary in usage, even within disciplines. It is the author's responsibility to use the correct symbols consistently. Take into account the scale and measurement precision implied in the text and query the author if you suspect a mistake or inconsistent usage.

Equation References On general usage, see Sect. 9.5.1. It may sometimes be useful to retain specific qualifiers such as "congruence" or "inequality" but remove normally redundant qualifiers such as "equation," "relation," or "expression." Some people, however, distinguish between a "relationship" and an "equation" while others distinguish the term "relationship" (e.g., "in this chapter we examine the relationship between x and y") and a specific instance of a relationship at some point in time, a "relation" (e.g., "here the relation $x = y + 1$ applies"). Accept an author's usage; do not change it.

Derivatives An author may use schemes such as dots and double dots above characters to represent derivatives, e.g., double derivative $\ddot{x}$.

Ellipses Centered dots are preferred for "=" and for operators such as "+," "–," and "×," and line dots are preferred for commas. Alternatively, line dots may be used throughout. Symbols and commas should precede and follow ellipses. Examples:

$$x_1 + x_2 + \cdots + x_n$$

$$a_i, \quad i = 0, 1, \ldots, n$$

Parentheses In general, apply the hierarchy $\langle\{[(\ldots)]\}\rangle$, e.g.,$[(1 + x)^2 + y]^3$. If necessary, this nesting hierarchy may be repeated in bold, working outwards. Alternatively the parentheses may be nested using "taller" symbols, i.e., $\Bigg\{\Bigg[\bigg(\bigg\{\bigg[\Big(\ldots\Big)\bigg]\bigg\}\bigg)\Bigg]\Bigg\}$. Parentheses of different style are not required in unambiguous cases, e.g.,$\left((1 + x)^2 + \frac{y}{z}\right)$. Unmatched ("lonely") parentheses in equations often reveal a typographical error; sometimes the required correction is obvious, but more often it's safer to ask the author to correct the problem. Parentheses should be "taller" than the enclosed content and most software packages automatically handle this; such issues are usually more efficiently corrected by the author or typesetter, so where possible use "global" instructions with specific examples. The "less than" and "greater than" signs ("<" and ">") should not be mixed up with angle brackets ("⟨" and "⟩").

Note any special cases. In many scientific treatments parentheses have strict meanings, and the style should not be changed, e.g., {111} vs ⟨111⟩ vs (111) in solid-state physics. Some styles involve deliberately asymmetrical notation. Examples: the mathematical intervals $(a,b]$ and $[a,b)$ have different meanings; bra, $\langle\psi_1|$, versus ket, $|\psi_2\rangle$, in Dirac "bra-ket" notation; and in astrophysics the distinction between forbidden and semiforbidden lines, e.g., "[O III]" vs "O III]." Query the author if you suspect an error.

Signs In descriptive treatments, terms such as "greater than" are preferred. For example, use "the number of patients was greater than 30," not "... was > 30." Where the text is more "mathematical" than descriptive, or there are numerous relations, signs are more efficient for the reader and should be used.

Prime A prime should appear adjacent to the relevant character. In terms of meaning, the character and its prime form a unit, which itself may, for example, have a superscript attached to it. The usages ${A'}_i$ and ${A_i}'$ are not equivalent.

Special Fonts Some fonts have specific mathematical meaning and usage. Examples include number sets ($\mathbb{R}, \mathbb{Z}, \mathbb{C}, \mathbb{Q}, \mathbb{N}$, , , , , etc.) and symbols with defined or "local" meaning.

Abbreviations The term "iff" (if and only if) is in Merriam–Webster's and does not require explanation. For abbreviations that do not appear in this dictionary, such as "wlog" (without loss of generality) and "wrt" (with respect to), add a first-use definition. Confirm proper use using a suitable dictionary. The abbreviations for Hermitian conjugate ("h.c."), Hermitian adjoint ("h.a."), and complex conjugate ("c.c.") should be upright, as should terms such as "const." for "constant".

14 Statistics

14.1 Statistical Tests

Statistical tests fall into two main categories, parametric and non-parametric. If in doubt about the test used or the definition of a statistic, query the author to define it. Statistical variables follow the same italicization rules as other variables (see Chap.13). Some statistical variables and terms are listed below (all should be defined except those marked with an asterisk).

Predetermined statistical significance cut-off for P-value	α
Regression coefficient of population	β
*Statistical datum derived from chi-squared test	χ^2
Cramer coefficient	C
Coefficient of variation	c.v.
*Degrees of freedom	d.f.
*Statistical datum derived from an F-test (variance ratio)	F
*Statistical datum derived from a G-test	G
Mean of the population	μ
*Number of individuals or variants	n or N
*Probability of wrongfully rejecting null hypothesis	P
*Coefficient of correlation	r
Coefficient of multiple correlation	R
Pearson's product–moment correlation coefficient	ρ or r
Root mean square	r.m.s.
Sample variance	s^2
*Standard deviation	s.d. or σ
*Standard error of the mean	s.e.m.
Population variance	σ^2
*Statistical datum derived from Student's t-test	t
*Statistical datum derived from Mann–Whitney U-test	U
*Arithmetic mean of a sample	$\overline{x}$
*The probability of wrongfully rejecting the null hypothesis, i.e., of obtaining the observed result by chance	P

14.2 Statistical Distributions

The statistical distributions commonly referred to include binomial, normal (or Gaussian), Poisson, Weibull. Symbols denoting distributions are normally roman unless referring to a variable: N distribution, Γ distribution (or gamma distribution—note that this does not take an initial capital), but *F* distribution (referring to the variable).

14.3 Nomenclature and Abbreviations

The word 'significance' should generally be used to refer to statistical significance or 'clinically significant'. There are several valid ways to denote significance levels: *P* values can be given as exact numbers (e.g., $P = 0.00145$) or described as being above or below some critical value (e.g., $P > 0.05$; $P < 0.01$; $P \leq 0.01$) or lying between two levels of significance (e.g., $0.01 < P < 0.05$). To define *P* values tersely, rather than using the usual sequence of footnotes, authors may instead use increasing numbers of symbols (e.g., *, **, ***) to denote increasing degrees of significance (decreasing *P* values). This is acceptable but not required; authors may also follow conventions (e.g., $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$), even if this results in 'unused' symbols (that is, the paper might contain * and *** but not **). Degrees of freedom for statistics are subscript with no parentheses: $F_{2,14} = 6.05$. Omit commas in subscripts of ≥4 digits.

It is also valid to express confidence levels in terms of standard deviations (e.g., '1 s.d.', '2σ'); note the number is closed up to the symbol.

Authors must define error bars as s.d., s.e.m., etc. If a figure contains error bars, check for a definition in the figure legend, and add a standard query asking the author to provide one if missing.

For box plots, all elements (e.g., center line, box limits and whiskers) must be defined. Pearson's product–moment correlation coefficient (PPMCC, or Pearson's *r*) is commonly denoted ρ when applied to populations and *r* when applied to samples. The term 'Wilcoxon test' is ambiguous and must be clarified: it can refer to either the Wilcoxon signed-rank test or the Mann–Whitney (a.k.a. Wilcoxon/Mann–Whitney) *U*-test (a.k.a. Wilcoxon rank-sum test).

The following common abbreviations must be defined at first use:

Abbreviation	Definition
ANCOVA (not ANCOVAR)	analysis of covariance
ANOVA (not ANOVAR)	analysis of variance
lod (score)	log odds ratio
MANCOVA	multiple analysis of covariance
MANOVA	multivariate analysis of variance
ND	not done, not determined
NS	not significant

Definitions of other common statistical abbreviations can be found in *The Cambridge Dictionary of Statistics* by B. S. Everitt (Cambridge Univ. Press, Cambridge, 1998) and *Applied Statistics: A Handbook of Techniques* by L. Sachs (Springer, New York, 1982).

15 Geosciences

The geosciences cover a variety of fields, e.g., mineralogy, environment (including chemicals and water), petrology, volcanology, and hydrogeology. Since copy editors are not usually experts on the subject, good reference works, e.g., a glossary of geology and/or an Internet website, are recommended.

Geologic Time Periods All geologic periods are capitalized: e.g., Cambrian, Precambrian, Pliocene, Eocene, Palaeocene, Quaternary, Cenozoic, Tertiary, etc. to name a few. Note that for time periods that have recognized subdivisions of Late, Middle, Early, then they should have an initial capital letter. For full details, see Sect. 15.2.5.

The time divisions AD (anno Domini), BC (before Christ), and BP (before present) should be written in small capitals.

Equations Geologists often use mathematical equations and/or equations expressing relationships between minerals, elements, and/or compounds. In both displayed and inline equations, variables should be italic, constants upright, and vectors boldface. As in all manuscripts, it is the task of the copy editor to make terms that are explained (defined) in the text correspond with those given in the equations, i.e., variables in equations and text should both be in italics.

The labels of minerals, elements, and/or compounds should not be in italics, e.g., Na, N, Fe, K, H_2O, H_2SO_4, etc., whether used in an equation or not.

En Dashes Often combinations of minerals and elements are mentioned, e.g., Fe–Al–K analyses or Na_2O–CaO–FeO system. When an equivalence in terms is intended, then an en dash should be used. In adjectival combinations, such as Na-rich, then a normal hyphen is used. Also, an en dash is correct in the term P–T (pressure–temperature) conditions.

15.1 Atmospheric Sciences

15.1.1 General

Atmospheric chemists use the term NO_x as an abbreviation for nitrogen oxides such as NO and NO_2 (nitric oxide and nitrogen dioxide). NO_y is used to denote total reactive nitrogen (NO + HNO_3 + PAN + PPN…). Total inorganic carbon is the preferred text wording for $CO_2 = \Sigma CO_2 = [CO_2] + [HCO_3^-] + [CO_3^{2-}]$. The pascal (Pa) is the SI unit of pressure, and the WMO (World Meteorological Organisation) has decreed that the hectopascal (hPa) is now the preferred unit for the measurement for meteorological purposes. However, many atmospheric scientists use millibar (mbar) where 1 hPa = 1 mbar = 100 Pa; author use of mbar can be retained. Where meteorologists refer to measurements made at a number of mbar, it should always be made clear that this means the level in the atmosphere (e.g., 500-mbar level in the atmosphere).

Ozone is measured in Dobson units and the standard abbreviation is DU. Concentrations of dissolved substances in seawater should be expressed in units of mol per kg of seawater not per litre. Meteorologists often refer to “thickness charts”, which are effectively charts of the mean temperature between different pressure levels — this should be explained where necessary in the text. The copy editor should not hyphenate 'greenhouse gas emissions', and other similar uses of 'greenhouse gas'.

The following abbreviations should be defined at first use:

Abbreviation	Definition
CCN	cloud condensation nucleus, or nuclei
CFC	chlorofluorocarbon (use CFC-22 etc. when referring to a specific one)
CFM	chlorofluoromethane
CLAW	Charlson, Lovelock, Andreae and Warren (spell out at first use)
CTD	conductivity–temperature–depth
DIC	dissolved inorganic carbon
DOC	dissolved organic carbon
ENSO	El Niño/Southern Oscillation
ERBE	Earth radiation budget experiment
EUV	extreme ultraviolet
GCM	general circulation model or global climate model
IPCC	Intergovernmental Panel on Climate Change
ITCZ	intertropical convergence zone
LGM	Last Glacial Maximum (note caps)
LWC	liquid water content
MWP-1A	meltwater pulse 1A (note hyphen)
NADW	North Atlantic Deep Water (note caps)
NAT	nitric acid trihydrate
NLTE	non-local thermodynamic equilibrium
NMC	National Meteorological Center
NOAA	National Oceanic and Atmospheric Administration
n.s.s.	non-seasalt
NWP	numerical weather prediction
OLR	outgoing long-wave radiation
PBL	planetary boundary layer
POC	particulate organic carbon
POM	particulate organic matter
PPN	peroxypropionyl nitrate
PSC	polar stratospheric cloud
PTN	particulate total nitrogen

Abbreviation	Definition
PWV	precipitable water vapour
QBO	quasi-biennial oscillation
SAR	synthetic aperture radar
SAT	surface air temperature
SLR	satellite laser ranging
SLP	sea-level pressure
SMOW	standard mean ocean water
SMMR	scanning multichannel microwave radiometer
SST	sea surface temperature
SWR	short-wave radiation
TOMS	total ozone mapping spectrometer
w.e.	wet equivalent (for snow)

15.2 Earth Sciences

15.2.1 General

Latitude and longitude: 24° 39′ N, 36° E (space before letter) northeast, north-northeast etc., northeastern. Figures that have maps need latitudes and longitudes on the *x* and *y* axes. It should be clear that these are latitudes and longitudes and should not just look like random numbers i.e. each number on the axes should have the direction, e.g., ° E or ° W etc., after it, or give the axes an appropriate title, e.g., Longitude (° E). Similarly, the Equator should be 0° (not E or Eq.). Note, 180° should have no label, i.e. it is neither E nor W.

Directions: Avoid using a three-digit number to measure angle clockwise from north, so N70 E, not 070.

Samples: ODP 1234, CMP 1243, MGS 8348 (no commas). Review article *Nature* 392, 461 (1998).

15.2.2 Seismology

Seismic waves. P and S waves are roman (they are named P and S because they are the first ('primary') and second arrivals on a seismogram). They are both body waves (as opposed to surface waves); P waves are compressional and S waves are shear waves. Surface waves are divided into Love and Rayleigh waves (caps), depending on the direction of particle motion (see any geophysics textbook). Do not use 'P velocity distribution' but 'P-wave velocity distribution'; 'upper-mantle velocity distribution' is more correct as 'velocity distribution in the upper mantle'. Use an en rule in S–P and note that SV and SH refer to vertical and horizontal polarization, respectively. v_P, P-wave velocity.

Seismic waves are named according to the path they follow through the Earth; for example the core phases PcP, PKP and PKIKP are, respectively, reflected from the core–mantle boundary (CMB), transmitted through the outer core, and transmitted through outer and inner core. Shear waves cannot traverse the liquid outer core as shear waves, but may convert to P waves (and back again) at the boundaries of the outer core. An important phase in explosion seismology is P_n — the wave refracted along the Moho.

Seismic moment is M_0; moment magnitude is M_w; body-wave magnitude is M_b; and surface-wave magnitude is M.

Explosion seismology experiments are divided into refraction and reflection surveys. If there are many receivers, the term 'multichannel' (not hyphenated) applies. The vertical axis on a seismic section is usually two-way travel time (sometimes abbreviated 'TWTT', but it should always be spelled out first). Converting two-way travel time to depth (migration) is non-trivial.

Discontinuities. According to the standard model of the Earth iasp, main mantle discontinuities are at 660 km depth (not 650, 670 km) and 410 km (not 400 km).

15.2.2.1 *Common Abbreviations*

The following abbreviations must be defined at first use:

Abbreviation	Definition
CMP	common midpoint
DSDP	Deep Sea Drilling Project
MCS	multichannel seismic (line numbers)
ODP	Ocean Drilling Program (site numbers)
IODP	Integrated Ocean Drilling Program; from 2013 International Ocean Discovery Program (site numbers)

Note that the numbers used for these datum points are labels and therefore do not require commas, e.g., ODP Site 1264, or, as per more recent system, IODP Site U1334. Note, use 'Site' at first use, but thereafter it can be dropped, e.g. ODP 1264; go with the author's use.

15.2.3 Marine Geology and Geophysics

The Mid-Atlantic Ridge and the East Pacific Rise are both mid-ocean (not 'mid-oceanic') ridges. Strictly speaking, the term mid-ocean ridge refers to the topographic feature, whereas 'spreading center' refers to what we now know goes on at ridges; however, in practice the two terms are used interchangeably. Similarly, the term 'fracture zone' refers to the topographic expression of a transform fault. If an author uses the terms interchangeably, ensure 'transform fault' (or 'transform') is used only for the active segment, between two sections of ridge. 'Fracture zone' may be used either for a 'fossil' transform, or for the topographic expression of an active transform.

Follow the author's capitalization for geographical features (deserts, mountains, rivers, etc.) and undersea features (ridges, rises, fracture zones, basins). Note: oceanic crust (to match continental) but ocean floor. Seawater and groundwater are both used as nouns and adjectives. Names of ships are italicized (e.g., RV *Challenger*).

The following common abbreviations should be defined at first mention:

Abbreviation	Definition
deval (not caps)	deviation from axial linearity (a sort of mini-OSC)
EPR	East Pacific Rise
ESP	expanding-spread profile (a particular type of two-ship seismic experiment)
MAR	Mid-Atlantic Ridge
OSC	overlapping spreading centre

15.2.4 Geochemistry

15.2.4.1 *Abbreviations*

Common abbreviations of mineral names include the following:

Abbreviation	Definition
qz	quartz
mt	magnetite
ol	olivine
opx	orthopyroxene
cpx	clinopyroxene
amph	amphibole
dol	dolomite
cc	calcite
ilm	ilmenite
fsp	feldspar

Components of solid solutions are abbreviated with an uppercase first letter:

Abbreviation	Definition
En	enstatite
Fs	ferrosilite
Fo	forsterite

Abbreviation	Definition
Wo	wollastonite
Ts	tschermakite

15.2.4.2 *Styling and Nomenclature*

Use parentheses around elements that substitute for one another in a solid solution: $(Mg,Fe)SiO_3$. The comma indicates unspecified amounts of Mg and Fe.

The 'Mg number', abbreviated Mg#, is defined as 100Mg/(Mg+Fe). The # (upright hash symbol) is on the line, not a superscript.

Fugacities (f_{O_2}) and partial pressure (p_{CO_2}) are both styled as italic lowercase variables with gases as subscripts.

The oxygen fugacity is often expressed with reference to specific buffers: some examples are:

EMOG	enstatite–magnetite–olivine–graphite
IW	iron–wüstite
QFM	quartz–fayalite–magnetite

For helium concentrations, exponents are restricted to multiples of three: 10^{-3}, 10^{-6}, 10^{-9}, 10^{-12} cm^3 STP g^{-1}.

Soil chemistry. For analysis of exchangeable cations, units are $cmol_c\,kg^{-1}$ for cmol of charge per kg soil; for soil solutions, concentrations are expressed in $\mu mol_c\,l^{-1}$ or mmol of charge per litre.

Isotopes. Avoid using δ or ε notation in the first paragraph of a Letter or heading of an Article. For example, "higher $\delta^{13}C$" can be replaced by "higher $^{13}C/^{12}C$ ratio", or even "higher ^{13}C content". δ should be defined at first use, preferably in a table or figure legend:

$$\delta^{13}C = [(^{13}C/^{12}C)_{sample}/(^{13}C/^{12}C)_{standard}] - 1$$

Do not allow authors to include '×1,000' or '×100' in the definition if the units ‰ or % are used, since this is unnecessary (and wrong). The fractionation of isotopes between two samples (1 and 2) is described by $\Delta = \delta_2 - \delta_1$. In Sm–Nd systematics, 'epsilon notation' is commonly used to compare the Nd isotope ratio of a sample to that of the 'bulk Earth'. Here the '$\times 10^4$' is appropriate, because ε is given as a simple number, without units; for example, $\varepsilon^{205}Tl=4$, $\varepsilon Nd=0$.

15.2.5 Geological Time

The Geological Society of America's timescale can be used for general reference (https://www.geosociety.org/documents/gsa/timescale/timescl.pdf). Avoid “the Phanerozoic”, “the Archaean” as these are adjectives, the relevant noun should be used at first mention (the Phanerozoic eon; the Cretaceous period); after that, it is OK to lapse into common usage. Note that the periods have standardized abbreviations that must be used; for example 'T' for Tertiary, 'Tr' for Triassic; note K (not C) for Cretaceous, C for Cambrian. Use British or American spelling as required by the publication: (Br. Eng) Archaean, Palaeozoic, (Am. Eng) Archean, Paleozoic, but note Br Eng. Cenozoic, not Caenozoic. British spelling is used in the table below. Note that time periods that have recognized subdivisions of Late, Middle, Early, should have an initial capital letter.

Stratigraphic boundaries take a solidus, not an en rule (Cretaceous/Tertiary, K/T boundary). If time periods are used adjectivally they should be hyphenated, e.g. 'end-Cretaceous extinction'

There are three, substantially independent timescales, based respectively on isotope dating, fossils and magnetic reversals, and they are tied together at only a few points, so one cannot simply replace a stratigraphic term (such as Aptian) or a magnetic polarity “chron" (such as Brunhes) by an equivalent absolute age range. Nevertheless, for content read by non-specialists it is acceptable to request that the author include the approximate age range in parentheses (see Table below of geological timescales or add a standard query to check) after the stratigraphic term.

Names of ages (Maastrichtian, Campanian, etc.) can be used to refer to time intervals or to rock units (in which case they are stages, not ages). Thus, the climate may have been cool in the Maastrichtian age, but at a given site the rocks in the Maastrichtian stage are limestones. Normally the distinction does not arise — just write “the Maastrichtian" except when you have to subdivide a unit. Then, use Upper, Middle and Lower (upper case) for rocks; Early, Middle and Late for time (periods; but eras, lower case — see Table).

For dating and chronology, use ^{40}Ar–^{39}Ar, and K–Ar ages (that is, use en rules). But, use solidus for ^{87}Sr/^{86}Sr and ^{39}K/^{40}K ratios.

Time before present and duration (define the 'years ago' units at first use):

kyr	thousand years
ka	thousand years ago
Myr	million years
Ma	million years ago
Gyr	billion years
Ga	billion years ago

Era	Period	Epoch	Approximate time to present (Myr)
CENOZOIC	Quaternary	Holocene (Recent)	
		Pleistocene	2
	Tertiary	Pliocene	7
		Miocene	26
		Oligocene	38
		Eocene	54
		Palaeocene	63–64
MESOZOIC	Cretaceous		136
	Jurassic		190–195
	Triassic		225
PALAEOZOIC	Permian		280
	Carboniferous		345
	Devonian		410
	Silurian		440
	Ordovician		530
	Cambrian		540
PROTEROZOIC ARCHAEAN	Precambrian		>2,500

15.3 Lunar and Planetary Science/Meteoritics

Antarctic meteorites are numbered, because there are so many of them. Close up letters and numbers: ALHA0001, EETP0055. Lunar KREEP basalts are so called because they are rich in potassium (K), rare-earth elements (REE) and phosphorus (P).

15.4 Terminology and Styling

anticlockwise (not counterclockwise. Except in chemistry, particularly for rotating molecules and chirality the standard term is often CCW, so allow the author's use)

antiferromagnetic (no hyphen)

boron isotope ratios

breakup

deep-sea (adj.)

deforestation

deglaciation

de-water (hyphenate) (to be avoided)

Dupal anomaly (not DUPAL) — from Dupré and Allégre

dyke (not dike)

endmember

Equator (capital E for Earth, but not for Mars etc.)

GCM (general circulation model)

geographical

groundwater (noun and adjective)

haematite (but maghemite)

hotspot (one word)

hydrologic

interhemispheric

island arc basalt (no hyphen)

ITCZ (intertropical convergence zone)

lherzolite (with an h)

mainshock

mid-ocean-ridge basalt (MORB)

Mollweide projection (MAPS)

multichannel

night-time (but daytime) (adj.)

ocean island basalt (OIB), but ocean-floor sediments

offshore

palaeo-oceanography (not paleoceanography)

PDB standard (PeeDee Belemnite)

protosolar (no hyphen)

pseudopotential (no hyphen)

rainfall (one word)

rainforest

rare-earth elements (REEs — try to avoid abbreviation)

redeposition (no hyphen)

runoff

sea floor (n.); sea-floor (adj.)

sea surface temperature (no hyphen)

seawater (n. and adj.)

semi-major

steady-state (adj.)

stepwise

strewn field (n.)

subfragment; subtropical

sub-plinian

sub-solidus (hyphen)

subsurface, but sub-continental

time series (n.); time-series (adj.)

timescale (one word)

trace element composition

trade wind (n.)

wall rock

wildfire (one word)

wüstite

16 Physics, Engineering, and Computer Science

16.1 General

This chapter addresses issues specific to books and journals in Springer Nature's physics, engineering, and computer science programs (and may also be useful as a guide to the representation of related material in other Springer Nature products), both covering some specific issues that have not been touched on elsewhere in the style manual and offering some subject-specific examples for physics, engineering and computer science on matters of style.

Most products in these disciplines involve mathematical representations, and the relevant style issues are covered in Chap. 13. Refer also to the instructions in Chap. 12 for chemistry-related content.

Adherence to most of the conventions described here is required for many journals, book series, and textbooks, while in other cases, because of the significant time and cost implications of style issues, only consistency within a book, book chapter, or journal article is required. For example, you may be asked to apply certain conventions in the case of intensive copy editing projects only. Check with your Springer Nature contact on a per project basis.

16.2 Scientific Style

16.2.1 Italics and Upright Presentation

The following issues are applications of the general rules on the use of upright and italic display, as outlined in Sects. 6.4.1 and 13.3, to content specific to these subjects.

Symbols that represent a variable (or constant) are generally italic; accordingly, superscripts and subscripts are italic only if they refer to a variable (or constant). For example, T_m, roman 'm' for melting point, but C_p, italic 'p' for pressure.

Two-character variables are roman (in fluid flows the dimensionless ratios are roman: Re, Reynolds number; Ra, Rayleigh number; Pr, Prantl; Pe, Peclet). These should be defined the first time they are used.

Script letters are acceptable for, e.g., Hamiltonian and Lagrangian.

Elements and Compounds When element names are not written out they should be represented using the standard one-, two-, or three-letter abbreviations in upright font, e.g., "O" and "Na". Stoichiometry variables should be in italics, e.g., $\mathrm{Si}_{1-x}\mathrm{Ge}_x$.

Elementary particles are italic; follow the style on the Particle Data Group site, http://pdg.lbl.gov. Long-lived elementary particles: photon (γ); leptons (e, μ, τ, ν); hadrons, which are either mesons (π, Δ, Ξ, κ) or baryons (n, p). Others: u, s and d quarks (also c, t, b); α-particle, γ-ray (note α and γ are upright).

Semiconductors The terms "n-type" and "p-type" refer to negatively and positively doped material, and the "n" and "p" should be upright. The values of the related doping concentrations should be referred to using italics, "n" and "p."

Subatomic Particles Use upright display for normal or Greek letters, e.g., electron, "e," proton, "p," pion, "π," and muon, "μ," even in particle "equations," e.g., $\mathrm{p} \rightarrow \pi^0 + \mathrm{e}^+$. Use italicized "*e*" to refer to the value of the electronic charge.

Crystals The lattice parameters "*a*" and "*b*" take values and should be written in italics. The letters in "fcc" and "bcc" stand for "face-centered cubic" and "body-centered cubic," respectively, and should be upright. Use italics for "*P6/mmm*," "*I4/mcm*," and "C_{2v}" groups. Miller indices are written in italics, "*hkl*."

Lines and Points Use upright text, e.g., "point A" and "line BC."

Orbital electron shells are roman: K, L, M, N, O, P. The subshells *s, p, d* etc. are italic and are not hyphenated unless used adjectivally (*d* orbital but *d*-orbital energy).

For electronic configuration, use italic orbital plus superscript number, e.g. d^8

Quarks Quark flavors are represented using "u" (up), "d" (down), "c" (charm), "s" (strange), "t" (top), and "b" (bottom). There are sets of fixed quantum numbers behind these letters, and they should be represented using upright letters, but they are often represented using italics.

16.2.1.1 *Miscellaneous*

- **Isotope mass number** is at the top left-hand corner of the element symbol: ^{14}C not C-14. Use hyphen when element name is spelled out: carbon-14
- **Atomic mass** A, atomic number Z.
- **Critical current density** (no hyphen).
- **Space-time** (hyphen not en rule).
- For semiconductor physics, use **bandgap** (not band gap) and upright (not italic) **p (or n) doping** (for positive and negative).
- **Lasing** is ok as a verb.

16.2.2 Units

The basic SI units and the standard derived units are preferred. Refer to the *AIP Style Manual* (Table IV and Appendix C) or other well-established references for lists of standard units in physics and related disciplines (see Sect. 16.5). In addition, please follow any style instructions in this guide. Query the author on any nonstandard units that appear without definition. See also Sect. 6.2.1.

Units often appear in combinations. The "$\mathrm{m\ s^{-1}}$" form is preferred, and is the format required for Nature Research; for other products the solidus form "m/s" is acceptable, but is often ambiguous for long lists of units. Inconsistencies should be resolved.

Between groups of units a space is preferred, e.g., use "$\mathrm{m\ kg\ s^{-2}}$" rather than the "$\mathrm{m{\cdot}kg{\cdot}s^{-2}}$" form.

Here are some specific uses:

Astrophysics Use italic font for the mass of Earth, $M_\oplus$, and the Sun's mass, $M_\odot$, radius, $R_\odot$, and luminosity, $L_\odot$. Use "AU" (preferred Springer Nature style) or "A.U." for astronomical units.

General Use "a.u." for "atomic units." In texts where there is no potential ambiguity you may use "a.u." for "arbitrary units," but "arb. units" is preferred. Use "ML" (upright) for monolayers, "atm." for "atmosphere," and "wt%" and "at.%." Either "L" or "l" is acceptable for "liter" but a mixture should be resolved. Preferred Springer Nature style is 'l'.

Non-SI Units These are not preferred but are acceptable. Do not change to SI equivalents unless this is specifically requested. For example, "in." is common for wafer size in the semiconductor industry, and "lb." or "lb" and "ft." or "ft" are also still used.

Computer Science Use "b" and "B" for "bits" and "bytes," respectively.

Unit Prefixes in Data Processing and Data Transmission Some unit prefixes used in computer science, telecommunications, and electronic engineering texts are problematic (see http://physics.nist.gov/cuu/units/binary.html). Engineers in some contexts have used the character "K" to represent 1024 (2^{10}), as distinct from "k" (10^3), e.g., "KB" means 1024 bytes, and "100K" (often without a space when it's not followed by a unit; make consistent) is 100×1024. However, "MB" is either 1,000,000 (10^6) bytes, 1,048,576 (2^{20}) bytes, or 1,024,000 (1000×1024) bytes, and "Mbps" is either 1,048,576 (2^{20}) or 1,000,000 (10^6) bits per second. Similarly, a "GB" is ambiguous, it's either 2^{30} or 10^9 bytes, etc.

To avoid this ambiguity, a IEC/IEEE scheme has been proposed, as shown in Tables 16.1 and 16.2. Do not change the author's scheme; determine the usage and check for consistency. Query the author on any ambiguities and suggest a glossary or definition at first use.

Table 16.1 The IEC/IEEE scheme for prefixes

Value	Prefix	Abbreviation	Meaning
2^{10}	kibi	Ki	kilobinary
2^{20}	mebi	Mi	megabinary
2^{30}	gibi	Gi	gigabinary
2^{40}	tebi	Ti	terabinary
2^{50}	pebi	Pi	petabinary
2^{60}	exbi	Ei	exabinary
2^{70}	zebi	Zi	zettabinary
2^{80}	yobi	Yi	yottabinary

Table 16.2 Examples of the IEC/IEEE scheme for prefixes

2^{10} Scheme			10^3 Scheme		
Unit	Value	Value	Unit	Value	Value
1 Kibit	2^{10}	1024	1 kbit	10^3	1000
1 MiB	2^{20}	1,048,576	1 MB	10^6	1,000,000
1 GiB	2^{30}	1,073,741,824	1 GB	10^9	1,000,000,000

16.2.3 Computer Code and Algorithms

Program code and algorithms are usually represented in a sans serif font such as Arial. The corresponding grammar, spelling, and indentation should not be changed to conform with the rules for general text. Algorithms may be presented in figures, tables, structured text, etc. If you suspect an error in content or form, check with the author before making corrections. If there are language mistakes and/or typographical errors in comments, these should be corrected.

In books it is generally also acceptable for an author to develop and use a representation with a strictly local meaning, i.e., one independent of general language rules or Springer Nature's conventions, using a mixture of fonts (sans serif, italics, bold, etc.), operators, parentheses, etc. Determine the author's style and check that it is used consistently. Prompt the author to provide a key if you think one is necessary in this connection. *Example*:

$$\frac{P\ \mathbf{believes}\ Q \overset{Y}{\rightleftharpoons} P,\ P\ \mathbf{sees}\ \langle X\rangle_Y}{P\ \mathbf{believes}\ Q\ \mathbf{said}\ X} \qquad \mathsf{seq}(\mathrm{func}_a(x)) \quad (15.1)$$

Style names of programs roman, following the author's capitalization. For file formats and sizes, allow either lower- or uppercase, e.g., '.ext files' or 'EXT files' but be consistent within an article or chapter. Use KB, MB, GB, TB for file sizes; use b for bit, Mb etc.

Set names of path elements roman and separate with arrows, with a space (not '>') on either side. (Alternatively, in articles where italics are used copiously for other purposes, e.g., gene symbols, file and path names can be set off with single quotes.) e.g.:

> Open all maps that should be included by selecting *File → Open.* Next, select *File → Create Website.*

Capitalization should ideally match the way the commands actually appear, so defer to the author, but query if capitalization is inconsistently applied. Style computer code (i.e., actual commands in programming languages, not file/menu/path/tab nomenclature) in italics.

When appropriate, include the command prompt, usually the symbol '>' (greater than) or sometimes '$' (dollar sign). Format any quotation marks as straight quotes (") rather than slanted or 'smart' quotes (“,'), and leave spacing exactly as provided by the author, with no added extra spaces around operators or symbols). Query en-rules, as these are probably autocorrect artefacts (from one or two hyphens) and should be verified.

In discussions of computational models, the usual convention of discussing previously established or published fact in the present tense and the author's actions in past tense can become confusing. In such cases, it is permissible to use present tense for timeless properties of the simulation or algorithm: "When there is local uncertainty, the model predicts horizontal motion." But use past tense for particular instantiations of the model: "When we applied the Gaussian noise assumption, the model predicted vertical motion."

For logical operations, use uppercase font: AND, OR, NOT, NAND, NOR, XOR, CNOT (controlled NOT) for Boolean operators, switches, and types of sensory neurons. Use 'on', 'off' for states of a system (with single quotes at first mention or wherever necessary to avoid confusion).

16.2.4 Astronomy

16.2.4.1 *Capitalization*

Capitalize our Galaxy or Galactic Centre, but use lowercase for other galaxies. Similarly, capitalize our Universe, but use lowercase for theoretical universes. Furthermore, capitalize Solar System, Earth, Moon (but lowercase for the moons of Jupiter etc.), Sun (but lowercase for type I and type II supernovae or sunspot), Martian, Jovian, Saturnian. Orion nebula (but lowercase for solar nebula). Extrasolar planets (exoplanet is acceptable for these): hot Jupiter, super-Earth. γ-ray and X-ray source are hyphenated. The A ring of Saturn, but the A-ring gap.

16.2.4.2 *Symbols*

Symbol	Definition
$M_\odot$	mass of Sun
$R_\odot$	radius of Sun
$L_\odot$	luminosity of Sun
$M_\oplus$	mass of Earth
$M_\bullet$	mass of a black hole
$\oplus$	Earth sign
Z	Redshift
H	Hubble constant
H_0	current value of H; constant in velocity–distance relation
H	multiplier in formulae to accommodate uncertainty in H
Ω	cosmological density parameter (1 if average density insufficient to halt the expansion of the Universe)

Symbol	Definition
Q	deceleration parameter
Λ	cosmological constant
C	velocity of light
G	gravitational constant

16.2.4.3 *Coordinates*

Right ascension (RA or α) given by: 20 h 30 min 37.620 s. Note that the hour angle should always be two digits (06 h rather than 6 h).

Declination (dec. or δ) given by: 40° 47′ 12.85″ (not 12″.85); galactic **latitude** *b* and longitude *l*.

Use J (2000) system coordinates.

16.2.4.4 *Classifications*

Stars: Main spectral types are O, B, A, F, G, K, M, L, T: for example, O stars. Sub-classes are denoted by a numeral from 0 to 9. Luminosity classes are denoted by roman I to V (e.g. F5II star). Lowercase letters may be added to spectral classes for unusual features. Note, α Ursae Minoris (no hyphen). Exoplanets/brown dwarfs are also sometimes characterized by a spectral type (L, T, Y).

Galaxies: The Hubble scheme is the most common, thus:

elliptical	E0–E7
spiral	Sa–Sd
barred spiral	SBa, SBd
nucleus+disk	S0
irregular	Irr

16.2.4.5 *Nomenclature*

We follow the IAU Recommendations for Nomenclature, http://cdsweb.u-strasbg.fr/iau-spec.html. For example, NGC 205, PSR 0855+27, 3C 196. Note also Cyg X-1. The number zero and the capital letter O should not be used interchangeably, make sure each instance is correct. Give names in full at first use, but thereafter use shorter alternatives e.g. Cyg not Cygnus, AM Her for AM Herculis. There is no hyphen between a Greek letter and the name, e.g. α Ursae Minoris.

Supernovae: SN 1987A or supernova 1987A. SNR, supernova remnant. Types, I, II etc. Do not use SNe (supernovae) and avoid SNs (spell out).

Satellites are roman, e.g. Einstein, Rosat, COS-B.

Comets: Retain the prefix where given (for example, 73P/Schwassmann-Wachmann 3) in research papers, but in magazine articles after first use they can be omitted, e.g., comet P/Halley (note lowercase 'c' in comet), then comet Halley.

Asteroids should use the IAU designation: e.g., 4 Vesta or 433 Eros. Before its official naming an asteroid may be designated like this, e.g., 2002 AT_4.

Exoplanets. Exoplanet names should be written according to the currently accepted convention (check https://www.iau.org/public/themes/naming_exoplanets/). In the simple case (a planet around a star) the structure is [name of star][space][lowercase letter designating the planet (if more than one planet around a system, from b onwards usually in order of discovery)].For example, 55 Cancri d. Some planets have an official personal name, but if present on a paper, the original name should also be mentioned the first time (e.g. 55 Cancri d for Lipperhey). In case of a binary system, the structure is [name of star][space][uppercase letter indicating which stellar component the planet is orbiting around][lowercase letter designating the planet]. For example, HD 41004 Ab; 16 Cygni Bb. The 'A' can be omitted as by default a planet orbits the main star of the system. The star designation is necessary when more than one star in the system has its own planetary system, such as in the case of WASP-94 A and WASP-94 B.

16.2.4.6 *Brightness*

Brightness is described as the magnitude of an object, e.g., bolometric magnitude *M*. Particular wavelengths are symbolized by letters as follows, e.g.:

U	ultraviolet	360 nm
B	blue	420 nm
V	visual	540 nm

Magnitude differences are often discussed in the form $B-V$ (minus sign) (color) plotted against $U-B$. Brighter objects have smaller magnitudes, e.g. a 23 mag star is fainter than a 3 mag star which is fainter than a star of magnitude −1. Use 19 mag rather than 19. B and V are not italic unless they represent numerical values.

M	absolute magnitude
m	relative magnitude and apparent magnitude
$m_{\mathrm{B}}-m_{\mathrm{V}}$	note the upright subscript.

16.2.4.7 *Time*

The following examples are all acceptable formats for time:

- 1987 February 23.401
- Julian day (number of days since Greenwich noon, Jan 1, 4713 BC), such as JD 2440953.375
- 1987 February 23, 13 h 14 min 22 s.

The Julian date (JD) of any instant is the Julian day number for the preceding noon plus the fraction of the day since that instant. Julian dates are expressed as a Julian day number with a decimal fraction added. For example, the Julian Date for 00:30:00.0 UT January 1, 2013, is 2,456,293.520833

16.2.4.8 *Spectra*

Ionization states are set in roman small caps: H I, H II. Forbidden transition is set within square brackets [O II], while semi-forbidden transition has just the opening square bracket [C II. For emission lines, use Lyman α, Lyα; Paschen α, Paα; Bracket γ, Brγ, etc. (see also Sect. 12.5).

16.2.4.9 *Units*

AU	astronomical units (1 AU = Sun–Earth distance; note small caps)
Arcsec	seconds of arc (also arcmin)
Mas	Milliarcseconds
μas	Microarcseconds
light yr	light year
sr	steradian
pc	parsec
Jy	flux unit (Jansky) (radio)

Other wavelength bands have idiosyncratic flux units (Uhurus, milli-Crabs and so on). There is a case for insisting that SI units be supplied as well or instead.

Note that Fe(III) is used to indicate iron in the formal 3+ oxidation state; Fe^{3+} means a cation. In astronomy, H I means a neutral hydrogen atom, which normally occurs in a particular region of space with lots of other similar atoms (an H I region, or cloud); and similarly H II means a singly ionized hydrogen atom, which chemists would call H^+.

16.2.5 Quantum Physics

16.2.5.1 *Notation*

A quantum state Ψ is written in ket notation as $|\Psi\rangle$. The right angle bracket symbol, '⟩', is character code 241 in Symbol font and can be produced in MathType using the LaTeX command '\rangle'.

Operators, such as H (the Hamiltonian, measuring energy) or S (measuring spin) or the creation operator $b_i^\dagger$ (with a dagger) and annihilation operator b_i (no dagger) should be defined at first use. They may have hats (for example $\hat{H}$) if the author wishes, but there is no need to add them. Note that 'Hamiltonian operator' is ambiguous, because there are many operators in the Hamiltonian formulation of quantum mechanics. Other operators appear frequently in quantum mechanics, such as the Lagrangian (not Lagrangean) and the Laplacian (∇^2). Various types of unitary operators can also appear, and are often denoted as U.

An operator turns one state into another: $O|\Psi\rangle$ is also a state.

$|\Psi\rangle$ can be thought of as a column vector:

$$|\varPsi\rangle = \begin{pmatrix} \varPsi_1 \\ \varPsi_2 \\ \varPsi_3 \\ \vdots \end{pmatrix}$$

$\langle\varPsi|$ is the quantum state $\varPsi$ expressed in bra notation, and is the row vector you get from $|\varPsi\rangle$ by transposing it and changing all entries to their complex conjugates: $\langle\varPsi| = (\varPsi_1^*, \varPsi_2^*, \varPsi_3^*, \ldots)$. The left angle bracket symbol, '⟨', is character code 225 in Symbol font and can be produced in MathType using the LaTeX command '\langle'.

Note that ket and bra notation itself indicates this vector nature: do not make the contents of either the bra or the ket bold (write $|\varPsi\rangle$ and $\langle\varPsi|$, not $|\boldsymbol{\varPsi}\rangle$ and $\langle\boldsymbol{\varPsi}|$).

$\langle\varPhi|\varPsi\rangle$ and, more generally, $\langle\varPhi|O|\varPsi\rangle$ are the complex numbers gained from matrix multiplication of the column vector and the row vector:

$$\langle\varPhi|\varPsi\rangle = (\varPhi_1{}^* \quad \varPhi_2{}^* \quad \varPhi_3{}^* \quad \cdots) \begin{pmatrix} \varPsi_1 \\ \varPsi_2 \\ \varPsi_3 \\ \vdots \end{pmatrix} = \varPhi_1{}^*\varPsi_1 + \varPhi_2{}^*\varPsi_2 + \varPhi_3{}^*\varPsi_3 + \cdots$$

Tensor products often appear in quantum mechanics: for example, a quantum state is a composite of various quantum states, which is usually denoted by the symbol ⊗. Tensors can be thought of as multidimensional arrays which represent relations between vectors. That means that often complicated indexing and summations over the indices appear. These summations can be expressed without writing the sum explicitly. A common convention in this regard is the Einstein summation convention, whereby a repeated index is assumed to be summed over: $a_{ij}X^{ki} = \sum_{i=1}^{3} a_{ij}\, X^{ki}$.

Sometimes, second quantized notation is used in quantum mechanics, especially to describe many-body systems and quantum field theory. To describe these systems in a compact way, creation and annihilation operators are used, which create and annihilate particles in the system.

16.2.5.2 *Interpretation*

Quantum mechanics was devised at the beginning of the twentieth century because experimental observations, like the spectrum of black body radiation and the photoelectric effect, were incompatible with the laws of classical mechanics. One of the most basic features of quantum mechanics is wave–particle duality, which means that quantum mechanical objects like electrons and photons in some cases have to be described as particles and in other cases as waves. Heisenberg's uncertainty principle means that in quantum mechanics there are complementary variables — for example, position and momentum — that cannot be determined simultaneously with arbitrary precision.

$|\Psi\rangle$, in its most general form, contains all of the information about the quantum mechanical object it describes: its position, momentum, spin and any other degrees of freedom it might have. Usually a single aspect of this, such as spin, is of interest at one time. It can be thought of as a vector in a multidimensional space (Hilbert space, more or less) whose coordinate axes correspond to each possible value that each of the object's degrees of freedom could have (i.e., a big space). In focusing on one degree of freedom (such as spin) one only looks at the component of the vector that occupies the corresponding subspace of that big space.

If $|\Psi\rangle$ describes an electron and you are interested in its spin, then the subspace is two-dimensional: the electron spin can be only either 'up' (one coordinate axis, $|\uparrow\rangle$) or 'down' (the other coordinate axis, $|\downarrow\rangle$). Neglecting the other degrees of freedom, we can write

$$|\Psi\rangle = a|\uparrow\rangle + b|\downarrow\rangle = a\begin{pmatrix}1\\0\end{pmatrix} + b\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}a\\b\end{pmatrix}$$

where a and b are complex numbers that are related to the probability of the spin being up or down, and we have written $|\uparrow\rangle$ and $|\downarrow\rangle$ more explicitly as basis vectors spanning the two-dimensional spin subspace. To find a and b, we calculate $\langle\uparrow|\Psi\rangle$ (which equals a) or $\langle\downarrow|\Psi\rangle$ (which equals b).

If you are interested in position rather than spin, then in general you would have to write $|\Psi\rangle$ in terms of the infinite number of coordinates ($|x_1\rangle$, $|x_2\rangle$, ...) corresponding to each possible position of the electron, and to find the probability that the electron is at point x you would calculate $\langle x|\Psi\rangle$. Because x is a continuous variable (position) it does not make sense to tabulate individual values of $\langle x|\Psi\rangle$ the way we do for spin (where those values are only a and b): Instead we keep x as a variable and treat $\langle x|\Psi\rangle$ as a function of x. This is written $\Psi(x)$, and is the wavefunction. The position is generally a vector (in ordinary space, not Hilbert space), in which case we write $\Psi(\boldsymbol{x})$. For general rules on vectors, see Sect. 13.4.

In these examples, the spin and position will generally be functions of time, which dependence is encoded in the coefficients that tell us the probability of finding the electron with a certain spin or at a certain position. Thus, generally $a = a(t)$ and $b = b(t)$ in the spin case, and for the wavefunction we write $\Psi(\boldsymbol{x}(t))$, or just $\Psi(\boldsymbol{x}, t)$.

One of the most important equations in quantum mechanics is the Schrödinger equation:

$$H|\Psi\rangle = E|\Psi\rangle$$

Here H is the Hamiltonian (a quantum operator) and E is the energy (a real number). Written in terms of the wavefunction, this becomes

$$H\Psi(\boldsymbol{x},t) = E\Psi(\boldsymbol{x},t)$$

where, like $|\Psi\rangle$, H is expressed in the 'position basis' ($|x_1\rangle$, $|x_2\rangle$, …) and is no longer a quantum operator but a classical differential operator. The second equation is thus a normal differential equation.

16.2.5.3 *Useful links*

For an introduction to the foundations of quantum mechanics, see: http://aeon.co/magazine/nature-and-cosmos/our-quantum-reality-problem/.

For a simple explanation of Bell's theorem, see:
http://www.upscale.utoronto.ca/GeneralInterest/Harrison/BellsTheorem/BellsTheorem.html.

For a general introduction to quantum versus classical computing, see:
http://www.scientificamerican.com/article/the-limits-of-quantum-computers/.

16.2.6 Symbols for Physical and Chemical Quantities

16.2.6.1 *Mechanical and related quantities*

fugacity	f_{O_2} ($_{O_2}$ is subscript)
partial pressure	p_{CO_2} ($_{CO_2}$ is subscript)
viscosity	η
surface tension	γ
sedimentation coefficient in water at 20 °C, extrapolated to zero concentration	$S^0_{20,W}$ (with 0 over 20)

16.2.6.2 *Molecular and Related Quantities*

activity	a (as in a_{Na^+})
relative molecular mass	M_r
number-averaged molecular mass	M_n
weight-averaged molecular mass	M_w
molar concentration of substance B	c_B or [B]

16.2.6.3 *Thermodynamic and Related Quantities*

molar gas constant	R
internal energy	U or E
Entropy	S
Helmholtz free energy	A or F
Enthalpy	H

Gibbs free energy	G
heat capacity	C
activity coefficient of substance B: mole fraction basis	f_B
activity coefficient of substance B: concentration basis	y_B
electrode potential	E
Boltzmann constant	k_B or k (k_B preferred)

16.2.6.4 *Chemical Reactions*

equilibrium constant	K
dissociation constant	K_d
Michaelis constant	K_m
inhibition constant	K_i
rate constant	k
binding constant	K_b
velocity of enzyme-catalysed reaction at infinite concentration of substrate	V_max

16.2.6.5 *Electricity and Magnetism*

permittivity	ε
permeability	μ
magnetic field strength	H
magnetic susceptibility	χ
impedance	Z

16.2.6.6 *Light*

luminous intensity	I
transmittance (I/I_0)	T
absorbance ($-\log_{10}T$)	A
molar absorption (extinction) coefficient	ε
reflectance	R

16.2.6.7 *Miscellaneous*

wavelength	λ
Hubble constant	H
redshift	z
atomic number	Z
international units	IU
Planck's constant	h
reduced Planck's constant	$\hbar = h/(2\pi)$
median lethal dose	LD_{50}
median effective dose	ED_{50}
retardation factor (chromatography)	R_{F}
parts per billion	ppb

16.3 Language Style

The technical disciplines generally follow American style regarding the use of punctuation, numbers, and spelling (see Sects. 4.4 and 5.5.3). The information given here concerns common usage in these disciplines including some examples of specific usage that disagrees with some of the rules given earlier.

16.3.1 Hyphens and en Dashes

It is acceptable to use *hyphens* in phrases such as "*z*-plane" and "*x*-axis" where used adjectivally and if used consistently. Further examples of usage from these fields are "high-voltage low-pressure chamber," "first-, second-, and third-order systems," "high-frequency terms," "10-nm-wide layer," "512-bit-long moduli," and "finite-dimensional groups."

Prefixes are customarily written without a hyphen. Examples are: ante, anti, auto, back, bi, bio, co, counter, de, di, eigen, electro, extra, ferro, giga, hyper, hypo, infra, inter, intra, iso, macro, mega, meso, meta, micro, mid, mini, mono, multi, milli, nano, neo, non, over, photo, poly, post, pre, pro, proto, pseudo, quasi, re, semi, socio, sub, super, supra, thermo, trans, ultra, un, and under.

It is acceptable to use *numbers* in expressions such as "3-fold" (instead of "threefold") and "1D" (instead of "one-dimensional") but a mixture should be resolved.

Regarding en dashes, follow the instructions in Sects. 5.5.1 and 5.6, i.e., use an en dash for ranges, combinations, compounds, and relations. Examples are "525–535-nm range," "electron–hole," "He–Ne laser," "space–time continuum," "I–V plot," "UV–Vis," "solid–liquid interface," "spin–orbit coupling," and "P–T–X curves." An en dash should not be used within a single double-barreled name such as "Murray Gell-Mann." Do not restyle terms with fixed representations to make them adhere to our conventions; for example, although "Hewlett-Packard" combines the surnames of the company's founders, it is not represented using an en dash. A division sign should not be used in ranges; it should be replaced by an en dash.

16.3.2 Acronyms and Abbreviations

As in other disciplines, abbreviations and acronyms should be defined at first use unless they are defined in the document's list of notation/abbreviations or are so well known within the field that a definition is redundant. Query the author regarding the use of a nonstandard abbreviation. Repeat definitions in each chapter of a multiauthor book.

Acronyms are generally presented in uppercase letters, while their definitions are lowercase if they do not include proper nouns. Some acronyms are a mixture of lower- and uppercase letters, such as "QoS" and "IPv6". For all acronyms, follow the author usage if it is consistent. The names of some computer languages are not acronyms and, as a result, should usually be capitalized, not uppercased, for instance, "Pascal" vs "FORTRAN." UNIX commands (e.g., "mkdir") should be in lowercase. Finally, some acronyms have come to be treated as normal words in dictionaries, such as "modem" and "pixel" in *Merriam–Webster's*.

The abbreviated form of the name of organizations may be used (e.g., IEEE, ISO). Be careful when expanding a reference to such an abbreviation.

Logical operators are presented using uppercase letters, e.g., "AND," "OR," "NAND," and "XOR."

16.3.3 Terminology

See Sect. 6.1.2 on questions of *capitalization*. In contrast to the usage described there, words derived from names may be capitalized, for example, "Abelian," "Cartesian," "Eulerian," "Euclidean," "Gaussian," "Hamiltonian," "Hermitian," "Ohmic," "Newtonian," and "Wronskian." Follow an author's usage if he has consistently used the noncapitalized form of a word for adjectival use and/or the capitalized form for noun use, e.g., " ... the beam's gaussian shape ..." vs "... the Gaussian showed a peak at ...." Here are some other special cases:

– The form "Boolean" is usually used in physics and electronics, while "boolean" is common in computer science; either is acceptable, but mixed use should be resolved.

– It may suit the context to consider terms such as "Big Bang" and "Fermat's Last Theorem" as proper names and to use the appropriate capitalization.

– "Sun," "Moon," and "Earth" are sometimes deliberately capitalized, for example, to indicate our local "Moon" rather than the "moon" of another planet.

– "Internet," "Net," "World-Wide Web," and "Web," are proper names, while a company's "net" and "intranet" are not. This form is also used when these terms are used as an adjective.

Remember that nonproper nouns should not be capitalized: spelled-out units (e.g., "volt," "pascal," "ohm," "newton"), elements (e.g., "fermium"), minerals (e.g., "perovskite"), and particles (e.g., "boson").

Correct the improper wording of any common terms or expressions that have been awkwardly presented; for instance, change "number of Avogadro" to "Avogadro's number."

Some terms used in these fields may not appear in dictionaries. For example, "satisfiability" is not in *Merriam–Webster's* but it appears often in mathematics and theoretical computer science texts. In most such cases the meaning is obvious from the context or the word's root or components; otherwise, you may prompt the author to supply a short explanation or a bibliographical reference.

16.4 Other Issues

Style issues applied in the main text must also be applied in the figure captions, e.g., italics vs upright font. If the author refers to color, e.g., "the red line in Fig. 4.3," in a figure, figure caption, or in a main text passage relating to a figure, advise your Springer Nature contact. In figures for non-Nature Research content correct only content errors rather than, for example, style inconsistencies (Nature Research corrects for all); if you are in doubt about how to implement this instruction, please check with your Springer Nature contact on a per project basis.

Style issues applied in the main text must also be applied in the table captions and in the table entries, e.g., italics vs upright font. Some inconsistencies for the sake of efficiency are acceptable, e.g., the use of "1.2E03" in a table vs "1.2×10^3" in the main text.

If the author has provided a glossary of terms, list of symbols, list of abbreviations, or equivalent, copy edit it before starting work on the main text, picking up on or correcting the author's representation style. If the author has not produced a glossary and complications in representation suggest that one would be of benefit to the reader, or at least of benefit during copy editing, ask your Springer Nature contact for advice.

16.5 Sources on Symbols and Units

In general, follow the author on symbols, units, abbreviations, and acronyms. There is usually no need to overrule the author's choice of symbol or unit. If there are inconsistencies or you suspect an error, consult an authoritative source, but follow our conventions on style. Examples of useful sources of standard physics units and symbols are:

"Quantities, Units, and Symbols," Symbols Committee, The Royal Society, London, 1975, and Addenda, 1981.

"AIP Style Manual," 4th ed., American Institute of Physics, New York, 1990 (http://www.aip.org/pubservs/style/4thed/toc.html/).

17 Humanities, Arts, and Social Sciences

17.1 General

These publications are addressed at students and researchers working in the humanities, arts and social sciences. Disciplines include economics, sociology, psychology, philosophy, history, linguistics, literature, drama, dance and media studies.

17.1.1 Humanities and the Arts

In the humanities and arts, it is necessary to be sensitive to language and context, particularly in areas such as philosophy, where the nuances of language are key to the meaning. There is a general principle of the copy-editor maintaining respect for the "authorial voice" and, in consequence, stylistic and other more extensive changes should be suggested rather than introduced.

17.1.2 Non-technical Texts

Although the general copy editing guidelines given in parts I and II of this Manual are valid for these disciplines as well, do not apply strict scientific styling such as for numerals and units in non-technical texts (e.g., do not change "twenty-four hours" to "24 h"). Follow the author's usage and be consistent. In quotations and historical material, no changes should be made.

17.1.3 References

Particular attention must be paid to references and citations, as errors and discrepancies are common. Styling of references, where applicable, needs to be done with particular care, as it is easier to inadvertently introduce errors (e.g. changing New Labour to new labour) than with the sciences.

17.1.4 List of /Notes on Contributors

Edited humanities books always contain a list of contributors, which can take one of two forms. A list containing only names and affiliations should be headed "List of Contributors"; lists containing biographical data in addition to the names and affiliations should be headed "Notes on Contributors". If no list of contributors or notes on contributors list is provided, please request one via your Springer Nature contact.

Appendix

A.1 Regional Abbreviations

Table A.1 is a list of US state names and abbreviations, from which the two-letter postal abbreviation should be used, as noted in Sect. 6.2.2. For your information, Tables A.2 and A.3 give the Canadian and Australian postal codes.

Table A.1 US state names and abbreviations. See http://www.usps.com/ncsc/lookups/usps_abbreviations.html for any updated information.

State Name	Traditional Abbreviation	Postal Abbreviation
Alabama	Ala.	AL
Alaska	Alaska	AK
Arizona	Ariz.	AZ
Arkansas	Ark.	AR
California	Calif.	CA
Colorado	Colo.	CO
Connecticut	Conn.	CT
Delaware	Del.	DE
District of Columbia	D.C.	DC
Florida	Fla.	FL
Georgia	Ga.	GA
Hawaii	Hawaii	HI
Idaho	Idaho	ID
Illinois	Ill.	IL
Indiana	Ind.	IN
Iowa	Iowa	IA
Kansas	Kan.	KS
Kentucky	Ky.	KY
Louisiana	La.	LA
Maine	Me.	ME
Maryland	Md.	MD
Massachusetts	Mass.	MA
Michigan	Mich.	MI
Minnesota	Minn.	MN
Mississippi	Miss.	MS
Missouri	Mo.	MO
Montana	Mont.	MT

State Name	Traditional Abbreviation	Postal Abbreviation
Nebraska	Neb.	NE
Nevada	Nev.	NV
New Hampshire	N.H.	NH
New Jersey	N.J.	NJ
New Mexico	N.M.	NM
New York	N.Y.	NY
North Carolina	N.C.	NC
North Dakota	N.D.	ND
Ohio	Ohio	OH
Oklahoma	Okla.	OK
Oregon	Ore.	OR
Pennsylvania	Pa.	PA
Puerto Rico	Puerto Rico	PR
Rhode Island	R.I.	RI
South Carolina	S.C.	SC
South Dakota	S.D.	SD
Tennessee	Tenn.	TN
Texas	Tex.	TX
Utah	Utah	UT
Vermont	Vt.	VT
Virginia	Va.	VA
Washington	Wash.	WA
West Virginia	W.Va.	WV
Wisconsin	Wis.	WI
Wyoming	Wyo.	WY

Table A.2 Abbreviations of Canadian provinces and territories

Province	Abbreviation
Alberta	AB
British Columbia	BC
Manitoba	MB
Newfoundland	NF
New Brunswick	NB
Northwest Territories	NT
Nova Scotia	NS

Province	Abbreviation
Nunavut	NT
Ontario	ON
Prince Edward Island	PE
Quebec	QC
Saskatchewan	SK
Yukon Territories	YT

Table A.3 Abbreviations of Australian states and territories

State/territory	Abbreviation
Australian Capital Territory	ACT
New South Wales	NSW
Northern Territory	NT
Queensland	QLD
South Austalia	SA
Tasmania	TAS
Victoria	VIC
Western Australia	WA

A.2 Appendix for the Nature Portfolio journals

This section contains special notes for copy editors handling the Nature Portfolio journals (specifically applying to Nature, Research Journals, Reviews, Nature Communications, Communications titles, NPJs and AJs).

British/American English (section 1.3.2). For level 3 copy editing journals, whether to use American or British English is determined on a journal level. Please confirm with your Springer Nature Contact which to use.

Title length (section 2.2.1). Article titles for Nature Portfolio journals should only be one sentence (enforced on level 3 copy editing journals).

Affiliations (section 2.2.1). Nature Portfolio journals only use Institute, City and Country for affiliations.

Article Notes (section 2.2.1). Nature Portfolio journals don’t include comments about prior presentation of the content as footnotes.

Abstract (section 2.2.1). For level 3 copy editing journals, external links are avoided in the Abstract where possible, but are included for clinical trials.

Keywords (section 2.2.1). Keywords are not used on Nature Portfolio journals.

Funding Information (section 2.2.3.4). The Funding Information section is not yet rolled out across all journals. Until full rollout is complete funding information can be added to the Acknowledgements section. Check with your Springer Nature contact if unsure.

Names (section 6.1.2). Use ‘f’ instead of ‘ph’ for sulfur, in line with The Royal Society for Chemistry.

Trademark symbols (section 6.1.2). For level 1 and 2 copy editing journals, retain the symbols ®, ©, or ™ that may come after a brand or proper name if they are provided by the author. Do not add them where they are not present. These symbols should be removed for level 3 copy editing journals.

Number separator (section 6.3.1). For level 1 and 2 copy editing journals, use a comma to separate groups of three digits to the left of the decimal point (if any) for large numbers of five digits or more (i.e., all numbers larger than 9999 such as 64,000 and 1,239,999). Be sure that a comma is not used in place of a decimal point. For level 3 copy editing journals, this rule should apply for large numbers of four digits or more (i.e., all numbers larger than 999 such as 6,400 and 1,239,999).

Italicization (section 6.4.1). Do not italicize letters used to indicate a shape (e.g., T square, S curve, Z-plasty). For level 1 and 2 copy editing journals, it is an option to italicize a word or letter when it is used in the sense of its being the word or letter (He uses too many *ands*.); it is, however, the style of level 3 copy editing journals not to italicize, but to use single quotes for this case (He uses too many 'ands').

Order of figure citations (section 7.2.1). Follow the figure citation guidance for section 7.2.1, but add an author query on the proof whenever figure citations are out of order to ask the author to confirm the intended citation order; occasionally authors do wish to refer to a later figure, without it being the formal citation of it.

Figure legends (section 7.3.1). For all journals, a legend ends with a full stop. If a descriptive legend is not provided, follow specific journal instructions and either query the production editor or add an author query on proof.

Figure parts (section 7.3.2). Level 3 copy editing journals use commas after panel labels in the following format (figure titles are required for levels 2 and 3):

> **Fig. 3 Electron micrographs of epidermis. a**, Facial region. **b**, Neck. **c**, Chest.

Some level 3 copy editing journals do not allow panel labels to be cited in the figure title, or if included, all panel labels must be cited in the title. If unsure, check with your Springer Nature contact. For level 2 and 3 copy editing journals, add an author query if the figure title is missing.

Artwork (section 7.4). For levels 1 and 2 copyediting journals that do not relabel figures, the vendor copyeditors check for matching panel names between the artwork and the figure legend, and for any symbols in the artwork that need to be explained in the legend (and query if no explanation is provided), and vice versa. Figures are not proof read at the copyediting stage to identify typos or change the decimal points to style. The vendor will correct figure part labels for consistency: i.e., if uppercase labels are used on the figure the legend will be changed to uppercase letters to match.

Cell entries (section 8.4.2). For levels 1, 2 and 3 copy editing journals, if a particular alignment of data in a table is necessary for data interpretation, copy editors/proofreaders/authors should specify alignment requirements for the typesetter to follow.

Table footnotes (section 8.6). For level 3 copy editing journals, table footnotes take the following format for defining abbreviations (non-italic font and use of commas; all journals also use a full stop at the end of the footnote):

> The tumor cells (2×10^7) were inoculated into the subepidermal tissue of ICR mice on day 0.
> The drugs were given orally on days 1–7. On day 10, the tumor was removed and weighed.
> FT, ftorafur, 5-FU, 5-fluorouracil, U, uracil.

Note that in addition to non-run-on table footnotes, some Nature Portfolio journals allow run-on table footnotes (check with your Springer Nature contact if unsure):

> The tumor cells (2×10^7) were inoculated into the subepidermal tissue of ICR mice on day 0. The drugs were given orally on days 1–7. On day 10, the tumor was removed and weighed. FT, ftorafur, 5-FU, 5-fluorouracil, U, uracil. *P<0.05; **P<0.01; ***P<0.001. [a]R+NC, total tumor volume <110% of initial volume. [b]Mice dying due to toxicity before day 56 were excluded. [c]Due to toxicity. [d]ADR 1.2 mg/kg day, days 1–3 and 14–16.

Footnotes (section 9.2). Nature journals that use superscript reference citations do not use footnotes other than for Tables

Reference citations in abstracts (section 10.2). For level 1, 2 and 3 copy editing journals, reference citations within the abstract of an article do not need to be treated differently to reference citations in the main text.

Incomplete references (section 10.4). In addition to the information in section 10.4, if the reference citation information is incomplete for an entry in the reference list, for levels 1, 2 and 3 copy editing journals either find the information yourself or add an author query.

A.3 Change History

This version of the Style Manual contains a number of changes versus the previous one (version 8.8, dated July 2019). All changes are marked yellow.

Table A.4 Change history of this version of the Style Manual versus the previous one

Section	Change
Parts I and II	Major revision with multiple changes.
	Main goal: Clarify the use of the Springer Nature Style Manual for copy editing. Restrict instructions to content editing and remove references to specific layouts or procedures which are not universally applicable. Align options for all Springer Nature portfolios.
10.3.6	Introduction of APA reference style version 7 to replace Springer APA. Version 7 should now be used universally where APA style is required (no other versions).
A2	Appendix with specific instructions for Nature Portfolio journals

Document History

Version	**Date**	**Authors/ Reviewers**	**Comment**
7.0	2012-01-15	Michael Wilson, Ilse Wittig	
8.0	2014-04-17	Michael Wilson, Ilse Wittig, Alison Hepper	Changes made to allow automatic processing by FSV
8.1	2014-04-30	Alison Hepper	Clarification of the changes made in the previous version
8.2	2015-04-02	Alison Hepper	Added details from BioMed Central Style Guide adopted for general Springer use
8.3	2016-06-15	Alison Hepper/Jonathan Lewis	Rebranded to Springer Nature for general Springer Nature use. Major revision to use of notes system with Chicago reference style.
8.4	2016-08-17	Alison Hepper/Paul Fletcher	Added Nature Journals reference style and other minor updates
8.5	2016-10-08	Alison Hepper/Paul Fletcher	Further alignment with Nature journals.
8.5a,b	2016-11-05	Alison Hepper	Corrected technical document errors. Minor update for metrology in chemistry
8.6	2017-07-25	Alison Hepper/Jonathan Lewis/Paul Fletcher	Added details for humanities and social sciences. Update to use of notes and bibliography system. Other minor updates
8.7	2018-04-10	Alison Hepper/Ruben Kramers	Clarifications of editing levels
8.71	2018-07-05	Alison Hepper	Minor wording changes. Adding Data Availability to article back matter.
8.8	2019-07-22	Alison Hepper/Paul Fletcher/Jane Morris	Major revision to update Part III science chapers. Minor further clarifications in Parts I and II (see change history for details)
9	2020-12-04	Alison Hepper/Stefan van Dijl/Paul Fletcher/Jane Morris/Francine McNeill/Lian Evans/Kate Neil	Major revision to update Parts I and II to align all Springer Nature portfolios. Change to APA version 7. Addition of Appendix for Nature Portfolio Journals.

For any concern or problem with this document, please contact alison.hepper@springernature.com.